AccDSL

Company Histories, Accounting Interpretations, and Scenario Simulation

Product and method proposal

Version 2.38 (proposal candidate) 12 October 2026

This document specifies a system that translates company disclosures into traceable evidence, represents several compatible economic and accounting histories, and simulates their future journals and financial statements under explicit economic scenarios.

The proposed data contracts, methods and acceptance cases specify the intended implementation. Current source and domain limitations remain explicit; this candidate is not a completed product or a claim of full proposal readiness. Proposed libraries can be replaced while preserving those contracts. Sector rule packages require their own accounting review before they can be described as compliant with a reporting standard.

Research materials, source qualifications and review records accompany this document in `docs/design/research` and `docs/design/review`.

Contents

1 The product decision **1**

1.1 Why journals are necessary but insufficient 1

1.2 Users, questions and deliverables 2

1.3 What constitutes a useful first implementation 3

2 A complete company and a real disclosure **4**

2.1 What a balance must remember 4

2.2 Recovering movements without inventing invoices 5

2.3 The complete historical statements 5

2.4 A correction changes what was knowable 6

2.5 One economic scenario, several complete futures 7

2.6 Apple: source labels do not identify exposure 7

2.7 Apple: revenue recognition and cash collection 10

2.8 Apple: commitments and the cash needed to fulfil them 12

2.9 Apple: resources available for payment 14

2.10 Sales geography is not a map of costs or assets 17

2.11 From geographic sales to legal entities 20

2.12 Legal proceedings beyond the booked liability 23

2.13 When later disclosures change the historical query 25

2.14 Claims against another party can still affect Apple 27

2.15 A regulation can change the feasible operating plan 28

2.16 Demand, disruption and recovery follow different paths 29

2.17 Product transitions change the relevant history 31

2.18 A supply agreement does not establish available output 32

2.19 Available output can still create losses and obligations 33

3 Representing a company's historical record **36**

3.1 The record is a collection of assertions 36

3.2 Different names, different measurements 37

3.3 Time, revisions and information availability 38

3.4 The company boundary changes . . . 38
3.5 Further problems that the representation must handle . . . 39
3.6 Account for a disclosure before translating it . . . 39
3.7 Mappings must establish a financial relationship . . . 40
3.8 Selecting original and revised assertions . . . 40
3.9 What an accepted reading permits . . . 41
3.10 Three meanings of disclosure coverage . . . 41
3.11 Displayed precision does not determine a rounding rule . . . 42
3.12 Public availability and system possession are different views . . . 42

4 What the literature contributes 44
4.1 Accounting structure and company dynamics . . . 44
4.2 Incomplete information and provenance . . . 45
4.3 Inverse problems, identification and probability . . . 46
4.4 Languages for rules and contracts . . . 46
4.5 Extraction and financial reasoning . . . 47
4.6 Solvers and probabilistic execution . . . 48
4.7 The remaining design problem . . . 48

5 Architecture and reuse 49
5.1 One representation, several authoring methods . . . 49
5.2 Why a DSL, and when another approach is better . . . 50
5.3 Open-source components . . . 51
5.4 Responsibilities that AccDSL must implement . . . 52

6 Historical compatibility and future simulation 53
6.1 Define what is observed before inverting it . . . 53
6.2 Why published statements cannot identify all transactions . . . 54
6.3 The accounting state and its invariants . . . 54
6.4 Simulate events from economic state . . . 55
6.5 Preserve sets and distributions separately . . . 55
6.6 Resolution and computational limits . . . 56
6.7 A solver formulation for ordinary rounded disclosures . . . 57
6.8 Building the family with explicit templates . . . 57
6.9 Which details may safely be combined . . . 58
6.10 A finite probability model that can actually be evaluated . . . 58
6.11 Numerical precision is part of a financial method . . . 59
6.12 Updating a law without counting evidence twice . . . 60
6.13 A continuous example whose posterior can be checked . . . 60

6.14 A stress is not an observed condition or an intervention 61

7 The accounting language 63

7.1 Scope of the first language 63
7.2 Types that protect meaning 63
7.3 A portable record has an exact meaning 65
7.4 The first company profile and its authoring boundary 66
7.5 A small model with a continuous interpretation family 67
7.6 Posting rules and explicit exceptions 68
7.7 Alternative readings, assumptions and updates 68
7.8 Modules and stable intermediate representation 69
7.9 Errors are part of the interface 70

8 Reconstruction and execution procedures 71

8.1 Ingest, reconcile and construct histories 71
8.2 Condition a run without rewriting history 72
8.3 Schedule and commit events 72
8.4 From an economic event to permission to post 73
8.5 A lost acknowledgement must not create a second sale 74
8.6 Funding, tax and other simultaneous relations 75
8.7 Worked example: identical history, different future journals 75
8.8 Worked example: a bond has several relevant amounts 76
8.9 Reporting and explanations 76
8.10 The corporate reference package as operations 77
8.11 Scenario meaning and information access 78

9 Accounting methods for an operating company 79

9.1 A common core with distinct accounting books 79
9.2 Non-financial companies 80
9.3 A matched inventory case across reporting profiles 80
9.4 Operating events 81
9.5 Apple: asset balances do not identify asset movements 83
9.6 Lease and tax integration 85
9.7 Apple: recognising tax and settling the obligation 86
9.8 Returning to the company forecast 88
9.9 Employee benefits 89
9.10 Impairment and provisions 90
9.11 Share-based compensation 90
9.12 Fair value and capitalization boundaries 91

9.13 Rule selection . . . 91
9.14 IFRS and incorporating jurisdictions . . . 92
9.15 US GAAP proposed methods and authority limitation . . . 92
9.16 ASBE lease adoption and deferred tax . . . 93
9.17 Matched-case outcomes and decision . . . 95

10 Financing, ownership and the group boundary 96
10.1 Debt and hybrid financing . . . 96
10.2 Equity and ownership . . . 99
10.3 A compound bond exposes classification dependencies . . . 102
10.4 Schedules and initial classification . . . 103
10.5 Modification, extinguishment and default . . . 104
10.6 Equity and compound instruments . . . 104
10.7 Control and an elimination book . . . 105
10.8 Perimeter and consolidation state . . . 105
10.9 Acquisition and ownership case . . . 106
10.10 Intercompany inventory and tax . . . 106
10.11 Functional currency versus group presentation . . . 107

11 Credit losses and the bank balance sheet 108
11.1 Banks: accounting capital is not regulatory capital . . . 108
11.2 A bank's cash, allowance and capital in one path . . . 114
11.3 State and event order . . . 115
11.4 Expected credit loss . . . 115
11.5 Full loan and allowance roll-forward . . . 116
11.6 Funding, capital and decision policies . . . 118

12 Insurance service, finance and claims 119
12.1 Insurance: cohorts and future service . . . 119
12.2 Insurance service and reinsurance in a complete cohort . . . 120
12.3 Group state and general measurement . . . 121
12.4 Complete two-year general-measurement case . . . 121
12.5 Nonzero discount and service allocation . . . 122
12.6 PAA, reinsurance and valuation methods . . . 123
12.7 Adoption dates and the restatement of 2022 . . . 125
12.8 Interfaces, exclusions and acceptance cases . . . 127

13 Implementation contracts 128
13.1 Reference implementation and component boundaries . . . 128

13.2 Persistent records . . . 128
13.3 Public API and result types . . . 130
13.4 Versioned requests and independent results . . . 131
13.5 Correction changes a view without rewriting its history . . . 132
13.6 Durability and migration have observable obligations . . . 132
13.7 Plugin interfaces . . . 133
13.8 Human review and model-assisted extraction . . . 134
13.9 Failure recovery and reproducibility . . . 134
13.10 Review decisions have consequences for later results . . . 134
13.11 Identity must survive harmless reordering . . . 135
13.12 Adapters exchange typed records . . . 135
13.13 A successful extraction can still lose the fact . . . 136

14 Acceptance and implementation sequence 140
14.1 Three different questions require different evidence . . . 140
14.2 Core conformance cases . . . 140
14.3 Extraction and reconstruction evaluation . . . 142
14.4 Evaluate the DSL against the strongest practical baseline . . . 142
14.5 Forecasting evaluation comes later . . . 143
14.6 Capability contracts for sector packages . . . 144
14.7 Implementation milestones and dependency order . . . 144
14.8 Decisions and replaceable assumptions . . . 145
14.9 Concrete conformance and corpus protocol . . . 146
14.10 Workload limits and release evidence . . . 147

A Portable operation reference 148
A.1 source.prepare@1 . . . 148
A.2 review.propose@1 . . . 148
A.3 review.commit@1 . . . 148
A.4 time.select@1 . . . 148
A.5 model.compile@1 . . . 149
A.6 history.construct@1 . . . 149
A.7 history.refine@1 . . . 149
A.8 history.query@1 . . . 149
A.9 uncertainty.evaluate@1 . . . 149
A.10 uncertainty.admit@1 . . . 149
A.11 uncertainty.select@1 . . . 149
A.12 scenario.project@1 . . . 150

A.13 accounting.assess@1 . . . 150
A.14 accounting.authorize@1 . . . 150
A.15 accounting.propose@1 . . . 150
A.16 accounting.commit@1 . . . 150
A.17 report.build@1 . . . 150
A.18 record.explain@1 . . . 150
A.19 work.resume@1 . . . 151
A.20 store.export@1 . . . 151
A.21 store.import@1 . . . 151
A.22 store.migrate@1 . . . 151
A.23 Closed accounting records . . . 151
A.24 Method and extension definitions . . . 152

B Core grammar, records and execution details 153
B.1 Preserved Core 0 teaching grammar . . . 153
B.2 Required core record definitions . . . 154
B.3 Typing and partial-information rules . . . 155
B.4 Operational rule selection . . . 156
B.5 Example run request and response . . . 156
B.6 Requirement traceability . . . 157

C Research basis and outstanding review 158
C.1 Source collection and reading scope . . . 158
C.2 Version and access qualifications . . . 159
C.3 Companion records and status . . . 160

Chapter 1

The product decision

A company's accounts describe the financial consequences of its activities, but they leave much of the activity itself unobserved. Revenue combines many sales, a borrowing balance combines contracts with different terms, and a provision combines estimates about uncertain future payments. Aggregation is necessary for reporting. It also makes the inverse task—recovering economic history from published reports—fundamentally different from entering known transactions into a ledger.

AccDSL will preserve this distinction. Its first product is an evidence-backed representation of a company's historical record. Its second is a set of compatible histories and current economic states. Its third is a collection of conditional simulations, each explaining how a particular history, accounting treatment and business response produces future journals and reports. These products share identifiers and provenance, so an analyst can move from a forecast balance to the assumption or disclosure responsible for it.

The recommended design is a small, typed accounting language over a shared intermediate representation, combined with an evidence database, contract calculators and replaceable inference engines. A language earns its place by making recurring accounting decisions explicit and checkable: which entity and period an amount belongs to, whether it is known or assumed, which rule applies, and how an event changes a balanced ledger. The language does not supply missing facts or determine a company's future behaviour.

1.1 Why journals are necessary but insufficient

Consider a company with a borrowing of 100 currency units and annual interest expense of 5. Assume that the entire principal was outstanding throughout the year, interest was paid at year-end, and there were no fees, defaults, hedges or foreign-exchange effects. Both a fixed-rate loan at 5% and a floating-rate loan whose rate happened to be 5% generate the same entries:

Account	Debit	Credit
Cash, on initial borrowing	100	
Borrowing liability		100
Interest expense, at year-end	5	
Cash		5

Even the full historical journal would not distinguish these two contracts. Now impose a scenario in which the floating rate resets immediately to 7% for the whole next year. The fixed

loan produces interest of 5; the floating loan produces interest of 7. If the disclosure is also compatible with any fixed-rate fraction $\alpha \in [0, 1]$, future interest is

$$I(\alpha) = 100[0.05\alpha + 0.07(1 - \alpha)] = 7 - 2\alpha. \tag{1.1}$$

The historical accounts are compatible with the whole interval $[5, 7]$ of future interest. This is a range of feasible amounts. It becomes a probability distribution only when a distribution for α is supplied. Neither the number of generated interpretations nor a language model's confidence provides that distribution.

Journals remain essential because simulated economic events must have coherent accounting consequences. However, the simulated state must also contain contract terms, maturities, customer or borrower cohorts, collateral, unrecognised commitments, and management decisions. These determine what the journals will be under a changed economy. Reconstructing every historical transaction is unnecessary when a smaller state preserves the responses needed for the intended scenarios. Establishing that smaller state is an explicit modelling task, examined in Chapter 6.

The analogy with competing legal interpretations is useful at the level of representation. Several readings may survive the evidence and require separate reasoning. In accounting, some alternatives concern the meaning of a disclosure; others concern undisclosed facts or future behaviour. The system must identify which kind of uncertainty is involved. A known contractual term must not become an arbitrary interpretive choice merely because another choice is convenient for a simulation.

1.2 Users, questions and deliverables

An accounting analyst needs to inspect a number in its original table, verify its entity, currency and period, and see how it reconciles with the notes. A financial modeller needs to distinguish a reported balance from assumptions about its composition. A language engineer needs deterministic semantics and tests. A researcher needs to compare explanations under the same scenario, including explanations that fit history equally well.

The first release must answer the following complete workflow. An analyst loads a set of reports and specifies an information cutoff. AccDSL preserves the original documents and extracts source-linked observations. It identifies reconciliation failures and constructs an explicitly bounded family of compatible histories. The analyst declares additional assumptions where simulation requires them. The system executes each requested scenario and returns statements, journal traces, explanation-specific distributions or bounds, and unresolved limitations. When a later filing arrives, the analyst can compare the revised interpretation with the earlier information set.

Requirement	Required behaviour
R1 Evidence	Every observed amount or narrative assertion retains its document version, location, extraction method and original representation.
R2 History	Queries distinguish economic time, reporting period, publication time and database ingestion time; restatements remain recoverable.
R3 Meaning	Entity, perimeter, dimensions, currency, measurement basis, standard version and accounting concept accompany values.

Requirement	Required behaviour
R4 Ambiguity	Discrete alternatives and continuous unknowns retain their joint constraints across accounts and periods. Missing values never mean zero.
R5 Accounting	All committed journals balance in the book's functional currency, and balances reconcile to explicitly classified movements.
R6 Economics	Contracts, exposures and behavioural assumptions drive events. Accounting identities alone never determine scenario responses.
R7 Results	Conditional results survive by interpretation and scenario. Weighted results require declared, normalised probability measures.
R8 Explanation	A report cell can be traced to journals, events, rules, assumptions and evidence, including consolidation adjustments.
R9 Extensibility	Standard, industry and company packages extend shared types through versioned interfaces, with explicit unsupported cases.
R10 Reproduction	A saved run records source versions, model versions, parameters, calendars, seeds and numerical conventions.

These are acceptance requirements. Python, SQL storage, a particular parser, and a particular constraint solver are implementation choices. Initial support will cover a non-financial company with cash, receivables, payables, inventory, fixed assets, ordinary debt and ordinary equity. Banks and insurers must be representable in the core data model from the outset, but their executable measurement and behavioural packages follow dedicated milestones. Until a package supports an instrument, the result is explicitly incomplete.

1.3 What constitutes a useful first implementation

The first demonstrator should reconstruct three reporting dates and the intervening annual and interim flows for one company, including at least one restatement or revised comparative. It must preserve a borrowing-composition ambiguity and execute the fixed/floating example through both journals and all affected statement lines. A synthetic company supplies known hidden history; a public company supplies realistic disclosure and extraction problems. Their roles differ: the synthetic case tests preservation and rejection of known alternatives, while the public case tests traceability and practical coverage.

Success is a reproducible chain of evidence and computation, with identified gaps. A convincing-looking complete ledger reconstructed from incomplete disclosures would fail this product contract if its assumptions were hidden. Forecast accuracy requires a later evaluation with historical information cutoffs and suitable baselines; it is not established by the first demonstrator.

Chapter 2

A complete company and a real disclosure

2.1 What a balance must remember

Northstar Components is a synthetic seller of one homogeneous good. Its disclosures, transactions and contract terms are specified together, so we can check whether reconstruction preserves a known possible history. All amounts in this example are currency units, not thousands. The company has no foreign currency, subsidiaries, leases, returns or other comprehensive income. These are declared features of the example, not assumptions to impose on an imported company. Its source package and expected results are supplied in `fixtures/northstar.json`.

At the beginning of 2024, Northstar holds cash of 60, receivables of 20, inventory of 30 and plant costing 100 with accumulated depreciation of 20. It owes suppliers 20 and lenders 100. Share capital is 50 and retained earnings are 20. Thus assets of 190 equal liabilities of 120 plus equity of 70. Inventory consists of six units costing 5 each. The existing plant depreciates by 10 a year, with no residual value. The loan remains outstanding throughout 2024; both possible contract types charge 5% during that year.

The six months ending June and December provide two successive observation periods. Here the chosen accrual convention assigns one half-year to each. Cash-flow dates within a half-year are not recovered from a period aggregate; the synthetic history supplies a feasible ordering in which inventory is purchased before delivery and customer receipts precede the payments they fund. More detailed liquidity questions require more detailed dates.

Movement	First half	Second half
Delivered sales on credit	110	150
Customer collections	110	135
Inventory purchases on credit	70	85
Supplier payments	65	80
Cost of goods delivered	75	90
Payroll paid	20	22
Plant purchased at period end	10	12
Depreciation	5	6
Interest accrued and paid	2.5	2.5
Tax expense	1.5	5.9
Tax paid	3.5	3.9
Dividends paid	4	5

Each row has an economic meaning before it has a posting. Delivery of the good transfers control and creates an unconditional receivable; invoicing is assumed simultaneous in this example. Collection settles that receivable. Purchases increase inventory and trade payables; paying the supplier settles the payable. Delivery separately removes inventory at cost. Plant purchases are capital expenditure, while depreciation consumes its service potential. The addition of 10 enters service after June and has a five-year life, producing one unit of second-half depreciation. The addition of 12 enters service after December and has a twelve-year life.

Tax is deliberately simple: 20% of positive profit before tax, with all these expenses deductible and no deferred tax. The first-half payment exceeds its tax expense by 2, leaving prepaid tax; that prepayment is used in the second half. A genuine corporate tax package would need jurisdiction, tax basis, losses, timing differences and payment schedules. The example's tax rule governs its own numbers only.

2.2 Recovering movements without inventing invoices

Suppose the annual report discloses sales of 260, cost of sales of 165 and the opening and closing working-capital balances, while the collection and purchase totals are hidden. Under Northstar's explicit no-write-off, no-acquisition and no-currency-effect scope, the three bridges are

$$35 = 20 + 260 - C, \tag{2.1}$$

$$20 = 30 + P - 165, \tag{2.2}$$

$$30 = 20 + P - S. \tag{2.3}$$

Here C denotes collections, P purchases and S supplier payments, all in currency units over the same year. Substitution gives $C = 245$, $P = 155$ and $S = 145$. These are identifiable aggregate movements within this model. They do not identify which invoice was paid on which day. If acquisitions or write-offs are admitted, their movement variables enter the same bridges and these totals may cease to be unique.

The loan remains different. Write F and V for fixed and floating principal. The source imposes $F + V = 100$, $F, V \geq 0$ and $0.05F + 0.05V = 5$. The last equality adds no information about the split. Both endpoints and every intermediate allocation survive. Northstar therefore has identified working-capital totals and an unidentified financing exposure in the same joint historical family. A solver must not turn its convenient witness $F = 0$ into the company's recovered contract.

The model's historical slots are one delivery, collection, purchase, supplier settlement, payroll, plant addition, depreciation, interest, tax payment and dividend per half-year. Each slot represents a classified aggregate event. The interim filing supplies the information for allocating the annual movements between the two half-years; the annual bridges alone identify only their totals. The template for collection contributes $+C$ to cash and $-C$ to receivables; its state effect reduces the unpaid receivable pool by C. The purchase template adds inventory quantity $P/5$, inventory cost P and a payable P. A cash-balance identity by itself cannot substitute for either template.

2.3 The complete historical statements

The corrected statements follow from those movements. Temporary income and expense accounts close to retained earnings at each half-year boundary; annual income is the sum of the two periods' classified movements, excluding closing entries.

Balance	Opening	June	December
Cash	60	65	74.6
Receivables	20	20	35
Inventory	30	25	20
Plant, net	80	85	91
Prepaid tax	0	2	0
Total assets	190	197	220.6
Trade payables	20	25	30
Debt	100	100	100
Share capital	50	50	50
Retained earnings	20	22	40.6
Liabilities and equity	190	197	220.6

Annual revenue of 260 less cost of sales 165, payroll 42, depreciation 11 and interest 5 gives profit before tax of 37. Tax expense is 7.4 and profit is 29.6. Other comprehensive income is zero under the declared scope, so total comprehensive income is also 29.6. Equity moves from 70 to $70 + 29.6 - 9 = 90.6$ after dividends.

The cash-flow statement supplies an independent reconciliation. Collections 245 less supplier payments 145, payroll 42, interest 5 and tax payments 7.4 produce operating cash of 45.6. Investing cash is -22 and financing cash is -9, giving a cash increase of 14.6. The indirect route gives the same operating cash: profit 29.6 plus depreciation 11, minus the receivables increase 15, plus the inventory decrease 10 and the payables increase 10. The prepaid-tax balance is zero at both annual endpoints. The two cash routes use different information and therefore provide a useful check on classification.

2.4 A correction changes what was knowable

The original interim filing reported revenue of 120 and receivables of 30. Ten units of revenue had been recorded before delivery. Its tax expense was 3.5, profit 14 and equity 80. The annual filing subsequently corrected the interim revenue and receivable by 10 and recognised the excess tax payment as prepaid tax of 2. Corrected interim profit is 6 and equity is 72, as above. The original statement remains an accurate record of what was published, even though its revenue recognition was wrong under the stated delivery rule.

For the temporal fixture, the original report becomes available on 1 August 2024 and is ingested on 3 August. The correction and annual report become available on 2 March 2025 and are ingested on 5 March. An investor-information query at 1 September 2024 00:00 UTC returns 120. For this synthetic example, take UTC as the reporting timezone and use an instant cutoff of 3 March 2025 at 00:00 UTC, after the correction's entire publication-date interval has ended. That query returns 110 when evaluated using the subsequently acquired archive. A query that additionally asks what the database contained on 4 March still returns 120. Publication and ingestion answer different questions.

The annual total 260 and corrected interim total 110 share revision group `v2`, so second-half revenue is $260 - 110 = 150$. Subtracting the original 120 would yield 140, but that result mixes incompatible revisions. The query must reject the subtraction unless a reviewed compatibility relationship supports it. A later date alone does not establish such a relationship.

2.5 One economic scenario, several complete futures

For 2025, assume demand for 30 units at price 10, unit purchase cost 5 and a target closing inventory of eight units. Opening inventory is four units, so the purchase decision is $30+8-4=34$ units, costing 170. Deliveries cost 150, leaving inventory of 40. Collect 90% of current sales and 10 of old receivables: $270+10=280$, leaving receivables of 55. Pay old suppliers 30 and 130 of the new purchases, leaving payables of 40. Payroll is 50, capital expenditure of 20 occurs at year-end and dividends are 8.

These decisions are inputs to a conditional forecast. They are not estimates of demand or optimal management responses. No new borrowing is available. Under the existing asset schedules, depreciation is $10+10/5+12/12=13$. The new year-end asset begins depreciation in the next period. Let $\alpha = F/100$. With the floating rate resetting to 7%, interest is $I = 7-2\alpha$. Profit before tax is $300-150-50-13-I = 80+2\alpha$, giving tax $16+0.4\alpha$ and profit $64+1.6\alpha$.

2025 statement amount	All fixed	All floating
Revenue	300	300
Interest expense	5	7
Tax expense	16.4	16
Profit and comprehensive income	65.6	64
Operating cash flow	48.6	47
Investing cash flow	−20	−20
Financing cash flow	−8	−8
Closing cash	95.2	93.6
Receivables / inventory / net plant	55 / 40 / 98	55 / 40 / 98
Assets	288.2	286.6
Payables / debt	40 / 100	40 / 100
Equity	148.2	146.6

The joint symbolic result is cash $93.6+1.6\alpha$, profit $64+1.6\alpha$ and equity $146.6+1.6\alpha$, with the same $\alpha \in [0,1]$ throughout. Selecting the smallest cash and largest equity independently would describe no feasible company. A requested distribution requires a measure over that same historical family; the accounting equations do not supply one.

The later method schedules use the floating share x rather than the fixed share α used here: $x = 1-\alpha$. Both parameterisations describe the same joint family. Its 2025 opening cash is 74.6 and equity is 90.6; the forecast spends 20 on period-end capital expenditure, distributes 8 and settles its stipulated tax in the same year. These teaching assumptions are retained when the later lease example introduces a different tax-payment schedule.

2.6 Apple: source labels do not identify exposure

The real reference package is Apple's 2024 Form 10-K and first-quarter 2025 Form 10-Q, for fiscal dates 28 September and 28 December 2024 respectively (Apple Inc., 2024, 2025). The annual balance sheet also supplies 30 September 2023. The annual filing is archived as the company's PDF from AnnualReports; the interim filing is archived from SEC EDGAR with its inline XBRL. Apple's Note 1 describes a 52/53-week year ending on the last Saturday of September. Calendar-year assumptions would therefore misstate the periods.

The package inventory preserves every annual PDF page and every interim top-level section, all thirteen annual notes and all ten interim notes. It records internal cross-references and externally

incorporated material, including the proxy statement, as distinct obligations. Statement and debt facts below are structured; policies, other notes and exhibits retain their source locations and specific modelling obligations. This is complete inventory coverage of the acquired files, with partial semantic formalisation. It does not claim to reconstruct every Apple contract or to have acquired every incorporated external document.

Apple's annual balance sheet reports assets 364,980, liabilities 308,030 and equity 56,950, in millions of US dollars. Note 9 reports term-debt principal 97,341, unamortised discount/premium and issuance costs of −321, and hedge accounting adjustments of −358. Thus its carrying amount is $97{,}341 - 321 - 358 = 96{,}662$, split into current 10,912 and non-current 85,750. Fair value is separately disclosed as 88.4 billion. None of these three measurements may replace another merely because a table calls it debt.

Note 9 identifies the notes as fixed-rate, then states that interest-rate swaps effectively convert a portion to floating rates. Note 4 reports interest-rate contract notional of 14,575 at the annual date. A notional is neither a liability carrying amount nor sufficient information for an exact interest sensitivity: reset timing, remaining periods, curve conventions and hedge coverage still matter. The cross-reference connects the derivative to the borrowing without erasing either contract.

At the interim date, total term-debt carrying amount is $10{,}848 + 83{,}956 = 94{,}804$. Cash is 30,299, total assets 344,085, liabilities 277,327 and equity 66,758. The interim debt note's rounded "94.8 billion" and the statement's 94,804 million can coexist; their precision differs. Treating them as conflicting exact numbers would create an artificial infeasibility.

These balances leave two practical financing questions open: how much cash can fund a payment, and when the payment falls due. The annual cash and cash equivalents balance of 29,943 includes 2.6 billion held in escrow and restricted from general use for the State Aid tax obligation. Note 4 states the inclusion explicitly, and Note 7 connects the escrow to that obligation (Apple Inc., 2024, Notes 4 and 7, printed pp. 37 and 39). Subtracting the reported restricted component gives $29{,}943 - 2{,}600 = 27{,}343$ million as an indicative remainder at the displayed amounts. The restricted component is rounded to tenths of a billion, so this subtraction does not establish an exact freely available balance. The earlier case annotation that described the reported cash balance as excluding restricted cash was wrong relative to this footnote. Correcting that interpretation changes funding calculations without changing the issuer's reported number. It also requires revisiting any liquidity result that relied on the exclusion.

The debt movement supplies a second reason to preserve a calculation's missing terms. The first-quarter cash flow statement reports repayments of term debt of 1,009 million, while carrying amount falls by $96{,}662 - 94{,}804 = 1{,}858$ million. At the displayed amounts, the reconciliation is

$$94{,}804 = 96{,}662 - 1{,}009 - 849.$$

The 849 million is the residual after the reported cash repayment; it does not identify an exchange loss, an income-statement charge or a particular journal. Explaining it requires the missing movement information, including applicable currency, discount/premium and hedge-accounting effects. The quarterly cash flow statement still reconciles independently: operating inflow 29,935 plus investing inflow 9,792 less financing outflow 39,371 gives the reported cash increase of 356, from 29,943 to 30,299 million (Apple Inc., 2025, statement of cash flows and Note 6). A balanced cash bridge therefore does not identify the debt history.

The annual filing's securities exhibit adds terms that a debt-total query cannot supply. Exhibit 4.1 describes two euro-denominated series bearing the year 2025 in their names. Its reported outstanding principal is dated 18 October 2024, after the annual balance-sheet date. The description concerns Apple Inc. as issuer, rather than every subsidiary in the consolidated group (Apple Inc., 2024, Exhibit 4.1, pp. 3–4).

Series	Principal (EUR m)	Fiscal year	Scheduled maturity
0.875% 2025 Notes	1,250	2025	24 May 2025
0.000% 2025 Notes	1,000	2026	15 November 2025

Apple's disclosed 52-week fiscal 2025 ends on 27 September 2025 under its last-Saturday convention. The November maturity therefore belongs to fiscal 2026 even though the bond's name says 2025. The annual Note 9 principal-payment schedule uses fiscal years, whereas the instrument's stated maturity is a calendar date. Joining those fields on their year number would put a billion euros of scheduled principal into the wrong fiscal bucket. Neither the later holding date nor the euro principal may be silently substituted for a September US-dollar consolidated balance.

Suppose the disclosed faces remain outstanding until scheduled maturity, with no early redemption, default or additional withholding-related payment, and euro remains the payment currency. The annual coupon on the 0.875% series is then the reported face multiplied by the stated rate:

$$1{,}250 \times 0.00875 = 10.9375 \quad \text{million EUR.}$$

Its final scheduled principal plus annual coupon is consequently 1,260.9375 million EUR; the zero-coupon series has 1,000 million EUR of scheduled principal and zero stated coupon. Both scheduled maturity dates fall on a Saturday. Exact settlement requires the applicable payment-adjustment provision and business calendar. Exhibit 4.1 is expressly a summary qualified by the 2013 and 2018 indentures; its payment-currency exceptions and redemption rights also matter. The retained 2021 indenture cannot replace those earlier contracts. The calculation establishes conditional scheduled amounts, while their exact settlement dates and subsequent holdings remain unresolved.

Those missing facts have different consequences for the financial statements. Principal and coupon payments reduce cash, but settling principal differs from recognising interest expense. The latter also depends on carrying amount, effective-interest accrual and the treatment of discounts and hedges; Note 9 expressly includes applicable amortisation and hedging adjustments in effective rates. Exchange rates, hedge designations and tax treatment can affect profit, other comprehensive income, retained earnings and closing debt differently. The cash flow statement additionally needs the payment classification and any associated hedge settlements. A usable answer thus carries the conditional euro schedule together with these unresolved dependencies. A dollar company forecast must supply consistent holdings, exchange, hedge and tax assumptions across all affected statements.

To demonstrate a source-to-forecast calculation, freeze the interim balance sheet and apply a *marginal* financing stress for a following 13-week period: a 200-basis-point rate increase, no principal change, no offsetting hedge beyond the explicitly selected exposure, immediate cash settlement and no tax effect. An assumed effective floating exposure $E = 14{,}575$ million produces $0.02 \times (13/52)E = 72.875$ million additional interest. This exposure uses the annual notional as a conditional input; it is not asserted to be the interim net sensitivity. Under that condition, debit interest expense and credit cash by 72.875. The resulting marginal statements have cash 30,226.125, assets 344,012.125, liabilities 277,327, equity 66,685.125, profit and comprehensive income of -72.875, and operating cash flow of -72.875 relative to the frozen baseline. The balance-sheet difference equals the equity difference exactly.

If exposure is instead only restricted to $0 \le E \le 14{,}575$, the conditional added interest lies in $[0, 72.875]$ before monetary rounding. That restriction is a stress assumption and is not claimed to bound every real Apple exposure. This real case verifies evidence, context, mappings and a coherent marginal propagation. Northstar supplies the complete-company forecast; a complete Apple forecast remains dependent on its unmodelled operations and undisclosed terms.

2.7 Apple: revenue recognition and cash collection

An iPhone sale can create revenue today and an obligation to provide services later. Apple's annual revenue note separates delivered hardware and bundled software from the right to receive product-related services such as iCloud, Siri and Maps. It allocates the price and related discounts by relative stand-alone selling prices. For the bundled services those prices are estimates, because observable separate prices are unavailable. The hardware portion is recognised when control transfers, generally on shipment; the service portion is deferred and recognised straight-line over the estimated period of provision (Apple Inc., 2024, Note 2, printed p. 35). A historical model consequently needs the allocation and service cohort as well as the sale date and amount. Neither the product label nor the customer payment supplies those missing terms.

The note describes collection shortly after control transfer or the start of service delivery. It does not specify one collection lag for every contract. Products revenue also reflects expected returns, price protection and customer incentives. Revenue excludes taxes collected for remittance to governments (Apple Inc., 2024, Notes 1–2, printed pp. 34–35). Together these qualifications prevent a simple conversion of reported revenue into gross customer receipts.

Presentation creates a second distinction. Where Apple obtains control of a third-party product before sale, it reports the gross amount billed. For third-party applications sold through the App Store, its stated treatment records only the retained commission in Services revenue. Services also already includes recognition of the deferred value of services bundled with products (Apple Inc., 2024, Note 2, printed p. 35). Adding that recognition to reported Services a second time would double count it; treating Services as gross App Store customer spending would change the measured quantity.

The product and geographic tables describe different divisions of the same sales. The five product/service rows sum to 391,035 million dollars in fiscal 2024 and 124,300 million in the quarter ended 28 December 2024. The geographic segment sales sum to those same totals. They must not be added together. Apple says the product mix is generally similar across segments, except that iPhone represents a moderately higher share in Greater China. That qualitative statement does not identify a product-by-region matrix (Apple Inc., 2024, Notes 2 and 13, printed pp. 35–36 and 46–47) (Apple Inc., 2025, Notes 2 and 10, printed pp. 6 and 11).

2.7.1 The movement that the note leaves unidentified

At 28 September 2024 Apple reports total deferred revenue of 12.8 billion dollars. At 28 December it reports 13.2 billion, and says that quarter's net sales include 3.7 billion recognised from the deferred balance at the September date (Apple Inc., 2025, Note 2, printed p. 6). The balance sheet's December current deferred-revenue line is 8,461 million, with 8,249 million at September (Apple Inc., 2025, printed p. 3). Those are current liabilities; the note's larger amounts are totals. The precision differs too: the total is displayed to tenths of a billion, whereas the current line is displayed in millions.

Let D_0 and D_1 denote opening and closing total deferred revenue. During the quarter, let A be additions to deferred revenue, R_o revenue released from the opening balance, and R_n revenue released from amounts added during the quarter. Let M collect other signed movements, positive when they increase the liability. These symbols describe a balance reconciliation; M is not an assertion that a particular adjustment occurred. In billions of dollars,

$$D_1 = D_0 + A - R_o - R_n + M, \qquad A - R_n + M = D_1 - D_0 + R_o \simeq 13.2 - 12.8 + 3.7 = 4.1. \tag{2.4}$$

The source identifies R_o, not the whole release $R_o + R_n$. The 4.1 billion residual therefore

measures net additions after new-origin releases and other movements at the displayed amounts. Calling it gross new deferrals, billings or cash collected would be wrong relative to what has been calculated.

Rounding also matters. If each of the three displayed inputs is rounded to the nearest 0.1 billion, a closed outer envelope for that residual is $[3.95, 4.25]$ billion: the lower endpoint uses $13.15 - 12.85 + 3.65$, and the upper uses $13.25 - 12.75 + 3.75$. This is a conditional precision calculation, not a verified issuer rounding rule or a bound on gross collections. Its endpoints leave the treatment of exact rounding ties unspecified.

Two constructed histories make the missing movement visible. For this illustration alone, use the displayed amounts as exact inputs and set $M = 0$. Keep the reported quarter's total revenue at 124.3 billion. The table supplies two different choices for A and R_n; other recognised revenue is the remainder needed to match that total.

Quarterly amount (USD billion)	History 1	History 2
New deferrals, A	4.1	5.1
Release from those new deferrals, R_n	0	1.0
Release from the opening balance, R_o	3.7	3.7
Other recognised revenue	120.6	119.6
Closing deferred revenue	13.2	13.2
Total recognised revenue	124.3	124.3

Both histories satisfy the selected note totals, yet their new deferrals differ by one billion. They are witnesses within this aggregate model, not reconstructed Apple contracts or a fit to every financial-statement constraint. Additional contract, cohort, return and movement information can rule out or refine them. No probability attaches to either merely because two examples were written down.

The corresponding recognition entry debits deferred revenue and credits revenue when service is provided. Receipt or billing of consideration that creates the deferral instead credits deferred revenue and debits cash or, where an unconditional right to payment exists, a receivable. Collection of that receivable changes cash and receivables without creating revenue a second time. For example, hold History 1's other revenue cash-settled and omit costs, taxes, refunds and other movements solely to isolate this subsystem. Receiving its 4.1 billion of new consideration gives revenue-side cash inflow 124.7 billion. If that new consideration is receivable at quarter end, cash inflow is 120.6 billion and receivables increase 4.1 billion. In both cases the asset increase is 124.7, matched by a 0.4 increase in deferred revenue and 124.3 of revenue before omitted expenses. These conditional entries explain the identification problem; neither inflow is an estimate of Apple's actual cash collections.

2.7.2 A dated recognition expectation is not a cash forecast

At both September and December, Apple expects 64% of the then-reported total deferred revenue to be realised in less than a year, 25% within one to two years, 9% within two to three years and 2% after three years (Apple Inc., 2024, Note 2, printed p. 36) (Apple Inc., 2025, Note 2, printed p. 6). These are two dated expectations about two balances. Multiplication at the displayed December values gives $13.2 \times 0.64 = 8.448$ billion in the first bucket. The corresponding September calculation gives 8.192 billion. Their difference does not measure revenue recognised during the intervening quarter: the dates, remaining horizons and contract populations have changed.

The percentages divide an expected amount by recognition horizon. They do not assign probabilities to alternative histories, fix collection dates or forecast all future sales. The 8.448 billion calculation also cannot replace the reported 8.461 billion current liability. Rounded amount and percentage disclosures do not justify an exact equality between those figures, and the query must retain the source's distinct description of each quantity. A future calculation that requires monthly service or receipts needs finer timing and payment evidence.

The note leaves another part of future service unquantified. For certain long-term arrangements, Apple has not yet delivered the service, has no right to bill for it and attributes the unbilled consideration entirely to that undelivered service. It reports neither revenue nor an amount for those obligations in this passage (Apple Inc., 2024, Note 2, printed p. 35). A query must retain an unknown amount; setting it to zero or equating it with recorded deferred revenue would erase the distinction the company makes.

These disclosures therefore yield several different results. The total and current deferred balances are reported observations. The 4.1 residual is a derived net movement with unresolved components. The recognition buckets are management's dated expectations, and the excluded consideration is unquantified. Contract-level allocations, service lives, new-origin releases and collection facts remain necessary for a particular future journal path. The annual and interim periods also stay separate: the December 2024 quarter follows the September 2024 fiscal year, so subtracting it from that annual revenue cannot produce the year's final nine months. The complete revenue-note reading makes these consequences explicit while the wider Apple disclosure package remains only partly formalised.

2.8 Apple: commitments and the cash needed to fulfil them

The same distinction between recognition and cash arises when Apple buys capacity and components. A lease liability measures one part of an existing obligation; a purchase commitment describes goods or services still to be acquired. A funding projection needs the payments arising from both, together with new orders and operating decisions. Adding every disclosed amount would mix measurements and could count the same requirement twice.

2.8.1 From a lease liability to future payments

At 28 September 2024, Apple's lease note reports the following amounts in millions of dollars (Apple Inc., 2024, Note 8, printed p. 42). The maturity table first totals undiscounted payments and then deducts imputed interest to reach the recognised liability. The right-of-use assets are reported separately.

Amount (USD million)	Operating	Finance	Total
Undiscounted lease payments	13,518	1,272	14,790
Less: imputed interest	1,984	376	2,360
Recognised lease liabilities	11,534	896	12,430
Right-of-use assets	10,234	1,069	11,303

Thus $14{,}790 - 2{,}360 = 12{,}430$ reconciles the maturity and liability measurements. The recognised liability is already included in the balance sheet; it is not an extra cash requirement to add to 14,790. Its current components total $1{,}488 + 144 = 1{,}632$ million, whereas the fiscal-2025 maturity bucket is 1,991 million. A current carrying amount and a gross payment bucket answer different questions. The right-of-use asset likewise need not equal the remaining liability; these

reported balances do not disclose the complete amortisation, prepayment and modification history.

Apple also reports 849 million of fixed payments under additional leases that had not commenced. It expects them to commence in 2025–2026, with terms from less than one year to twenty-one years. Adding this separately disclosed amount to the maturity total gives $14{,}790 + 849 = 15{,}639$ million. That sum is consistent, under a nearest-0.1-billion rounding assumption, with the 15.6 billion of fixed lease obligations in management's cash-requirements discussion (Apple Inc., 2024, printed pp. 25 and 42). Numerical compatibility does not establish a contract-by-contract reconciliation. It neither adds 849 to the recognised liability nor identifies its commencement and payment dates.

The remaining lease information cannot supply those dates mechanically. The note describes original terms typically no longer than ten years, with renewal options, some reasonably certain to be exercised. Its weighted-average remaining term of 10.3 years describes a different quantity. The reported 3.1% discount rate, generally based on estimated incremental borrowing rates, does not supply every contract's rate or the timing hidden in "thereafter". Applying that one rate to invented year-end payments would introduce new assumptions. The policy combining lease and nonlease components also has a stated scope: corporate, data-centre and retail facilities (Apple Inc., 2024, Notes 1 and 8, printed pp. 34 and 42).

Fixed and variable costs add another distinction. For 2024, Apple reports 2.0 billion of fixed operating-lease cost, 1.9 billion of fixed operating-lease cash payments and 1.0 billion of noncash right-of-use assets obtained for lease liabilities. Variable costs across its leases were 13.8 billion; payments of that kind primarily depend on purchases of output from the leased assets (Apple Inc., 2024, Note 8, printed p. 42). Historical expense is therefore not a fixed future cash schedule. Forecasting it requires output, price and settlement assumptions. Adding it to manufacturing commitments without identifying their relationship could double count procurement requirements.

2.8.2 Why a fall in commitments does not identify cash spent

The purchase note and management discussion disclose different views. Note 12 lists 11,226 million of future payments under unconditional purchase obligations whose remaining terms exceed one year. Its fiscal-2025 row is 3,206 million (Apple Inc., 2024, Note 12, printed p. 45). The term restriction selects contracts; it does not place every payment beyond the coming year. The note describes these as off-balance-sheet commitments, primarily involving supplier arrangements, intellectual property, content and distribution rights.

Management's discussion provides additional near-term evidence. At the same September date it reports manufacturing purchase obligations of 53.0 billion, of which 52.9 billion is payable within twelve months. Other purchase obligations total 12.0 billion, with 4.1 billion within twelve months; that description includes capital assets as well as supplier, content and rights arrangements (Apple Inc., 2024, printed p. 25). A model confined to Note 12 would miss those reported aggregates. Conversely, subtracting 11.226 from 12.0 and labelling the difference short-term contracts is unsupported: the category and term populations have not been reconciled, and the precision differs.

At 28 December 2024 manufacturing obligations are 39.8 billion, all payable within twelve months. The interim report says contractual cash requirements have not changed materially since the annual filing except for commercial paper, manufacturing obligations and the State Aid tax payable (Apple Inc., 2025, printed p. 16). That qualitative statement is not an exact December lease measurement or evidence that every lease remained unchanged. The two manufacturing totals are dated observations over changing orders and payment horizons. Their

13.2 billion difference does not identify purchases or cash paid during the quarter.

To see what is missing, consider an explicitly simplified model in billions of dollars. Let O_0 and O_1 denote a comparable population of undelivered fixed-price orders at the two dates. Let N be new orders and F the value removed when goods are delivered. Assume no cancellation, price or currency change. Treat the displayed 53.0 and 39.8 as exact inputs for this illustration. These assumptions define the example; the disclosure does not establish this roll-forward convention for every reported obligation. The balance relation is

$$O_1 = O_0 + N - F, \qquad F - N = 53.0 - 39.8 = 13.2. \tag{2.5}$$

With no new orders, delivery is 13.2. With 5.0 of new orders, delivery is 18.2. Both leave the same closing undelivered amount of 39.8. The source balances alone cannot choose between these conditional movements; actual contract changes and delivery evidence could do so.

Payment is a further choice. Suppose the first history's 13.2 of delivered goods all remain unsold inventory, with no opening payables, prepayments or tax. Paying on delivery debits inventory and credits cash by 13.2: total assets and profit do not change within this subsystem. Delivery on credit instead debits inventory and credits trade payables by 13.2: assets and liabilities rise together, with no immediate cash or profit effect. Settling that payable later debits the payable and credits cash; consumption, sale or impairment would require further entries. These examples isolate recognition and settlement. They are not estimates of Apple's actual purchases or payments, or a fit to all its financial statements. No probability follows from writing two histories.

Delivery assumptions also need the narrative that surrounds the purchase table. Apple describes single or limited sources for some components, industry-wide shortages and price fluctuations, and capacity or yield constraints for new technologies. Suppliers may favour common components over custom ones. Its named manufacturing locations describe outsourced production, not a product-by-revenue-region allocation or ownership map (Apple Inc., 2024, Note 12, printed p. 46). A scenario in which input availability changes may alter output, variable lease payments, inventory and receipts together. The passage motivates those dependencies without identifying their size or probabilities.

The same note, and the interim contingency note, preserve management's dated assessment of unresolved legal claims. At each date, management states that there was not at least a reasonable possibility that Apple had incurred a material loss, or a material loss greater than a recorded accrual (Apple Inc., 2024, Note 12, printed p. 46) (Apple Inc., 2025, Note 9, printed p. 11). It does not make litigation exposure zero. Contract terms, supply constraints, commencement, delivery and settlement must therefore remain in the historical record alongside reported balances. They determine which future journal paths can be supported when a funding question is asked.

2.9 Apple: resources available for payment

The procurement and settlement choices just described determine when Apple needs money. Funding those payments requires a second account: what resources can be used, how securities become cash, and when customers pay. Annual and interim Note 4 distinguish these questions even though a single total might appear to answer them. The following table combines the notes' measurement and balance-sheet classifications. Amounts are in millions of US dollars at the displayed precision; the December date is the end of the first fiscal quarter, not another annual endpoint (Apple Inc., 2024, Note 4, printed pp. 36–37) (Apple Inc., 2025, Note 4, printed pp. 7–8).

Resource / measurement	30 Sep 2023	28 Sep 2024	28 Dec 2024
Cash and cash equivalents	29,965	29,943	30,299
Current marketable securities	31,590	35,228	23,476
Non-current marketable securities	100,544	91,479	87,593
Total fair value	162,099	156,650	141,368
Total adjusted cost	173,752	160,600	147,088
Net unrealised gains / losses	−11,653	−3,950	−5,720

Cash plus the two securities rows equals total fair value. Adjusted cost plus net unrealised gains and losses provides a second reconciliation of that same total; it is not an additional resource to add to it. The source calls the column *adjusted cost*. Relabelling every investment as amortised-cost debt would erase the different instruments represented in the table.

2.9.1 Restriction changes what can be spent

At September 2024, cash includes 2.6 billion dollars held in escrow, while marketable securities include a further 13.2 billion. Both are restricted from general use and designated for the State Aid tax obligation discussed earlier. Subtracting their displayed centres from the total gives 156,650 − 2,600 − 13,200 = 140,850 million. MD&A separately reports unrestricted cash, equivalents and securities of 140.8 billion (Apple Inc., 2024, printed pp. 25–26 and Note 4). The apparent difference does not establish conflicting disclosures: the restrictions and the MD&A total are rounded to tenths of a billion, whereas the table is in millions. If each amount is rounded to its nearest displayed quantum, the subtraction has possible bounds of 140,749.5 and 140,950.5 million. Those bounds overlap the 140,750 to 140,850 million rounding range for the MD&A total. This is a conditional compatibility check, not an issuer-stated rounding convention or a confidence interval.

The December footnote makes the distinction between maturity and restriction especially clear: *current* marketable securities include 3.3 billion dollars of restricted escrow funds, and cash includes 126 million. At the displayed centres, total resources less this disclosed escrow are 141,368 − 3,300 − 126 = 137,942 million. Cash plus current securities is 53,775 million; removing the same escrow gives 50,349 million. These are indicative subtractions, not exact freely spendable balances. Under the same nearest-rounding hypothesis the first subtraction ranges from 137,891 to 137,993 million. Interim MD&A supplies no numerical unrestricted total (Apple Inc., 2025, Note 4 and printed p. 16). A current classification therefore cannot serve as permission to spend an asset, and excluding escrow does not establish immediate liquidation at the reported price.

Maturity supplies a different kind of timing information. Note 1 tests the three-month cash-equivalent maturity at *purchase*, while annual Note 4 classifies marketable debt as available-for-sale and assigns current or non-current status using each instrument's underlying maturity (Apple Inc., 2024, printed pp. 34 and 37). Among non-current debt excluding mortgage- and asset-backed securities, the September maturity shares are 86%, 10% and 4% for 1–5, 5–10 and more than 10 years. The separate mortgage- and asset-backed population has shares of 14%, 9% and 77%. At December these become 85/11/4% and 13/10/77%, respectively (Apple Inc., 2025, Note 4, printed p. 8). Each triple divides its own population; the two triples cannot be added. Nor do contractual maturity bands establish a sale date, sale price or cash receipt schedule. Management's stated belief in sufficient resources also relies on ongoing operating cash and continued access to debt markets. It is a dated funding assessment, rather than a solvency guarantee derived from the table.

2.9.2 Security turnover differs from valuation

Securities can fund payments through sales or maturities, but a fall in their reported balance need not equal cash released. Fiscal 2024 cash proceeds from maturities and sales are 51,211 and 11,135 million, while purchases use 48,656. Net cash released is therefore $51{,}211 + 11{,}135 - 48{,}656 = 13{,}690$ million. The marketable-security balance falls by only $126{,}707 - 132{,}134 = -5{,}427$ million (Apple Inc., 2024, cash-flow statement, printed p. 33, and Note 4). To expose the missing reconciliation, let M_0 and M_1 be total current and non-current marketable securities at the period's opening and closing. Let C be maturity and sale proceeds less purchases, positive when cash is released. Define R as the unallocated difference between the stock movement and that cash measure, all in millions over the same period:

$$\begin{aligned} M_1 - M_0 &= -C + R, \\ \text{Fiscal 2024:} \quad -5{,}427 &= -13{,}690 + 8{,}263, \\ \text{First quarter 2025:} \quad -15{,}638 &= -13{,}335 - 2{,}303. \end{aligned}$$

For the quarter, the cash measure is $15{,}967 + 3{,}492 - 6{,}124 = 13{,}335$ million (Apple Inc., 2025, cash-flow statement and Note 4). The relation defines a residual; it does not explain it. Cash sale proceeds differ from the carrying amount removed on disposal, and changes in measurement or classification may also enter the stock. Apple's specific-identification policy for securities sold requires the disposed lots to recover their cost (Apple Inc., 2024, Note 1, printed p. 34). Neither the positive 8,263 nor the negative 2,303 identifies a fair-value, currency or reclassification journal.

The valuation columns help separate another tempting explanation. Across the annual dates, total adjusted cost falls by 13,152 million and net unrealised gains and losses rise by 7,703, producing a total fair-value decline of 5,449. Across the quarter, the corresponding changes are $-13{,}512$ and $-1{,}770$, producing $-15{,}282$. These totals include cash and equivalents, whereas M_0 and M_1 include only marketable securities. They also differ from the comprehensive-income statement's net-of-tax marketable-debt amounts: 6,054 million for the year and $-1{,}427$ for the quarter, including reclassification adjustments (Apple Inc., 2024, comprehensive-income statement, printed p. 30) (Apple Inc., 2025, comprehensive-income statement, printed p. 2). Endpoint unrealised balances and period OCI differ in population, tax and timing; their numerical differences cannot be assigned to a tax charge or realised gain without the missing movements.

Even reported cash needs matching populations. Annual cash-flow cash, cash equivalents and restricted cash opens at 30,737 million, exceeding opening balance-sheet cash and equivalents of 29,965 by 772. Both close at 29,943. Thus the cash-flow change is -794, while the balance-sheet line changes by -22; the cause and location of the earlier 772 remain unallocated. The quarter's endpoint populations agree, giving the same 356 million increase (Apple Inc., 2024, 2025, balance and cash-flow statements).

Valuation risk is likewise distinct from a payment. The notes describe Level 2 prices or models with significant observable inputs and counterparties that *generally* have high credit ratings. Annual Item 7A reports that a hypothetical 100-basis-point rate rise at all tenors would reduce investment fair value, including associated derivatives, by 2,755 million at September 2024 and 3,089 million at September 2023 (Apple Inc., 2024, printed p. 27). Those are dated issuer scenarios, not cash outflows or loss probabilities; the issuer's valuation model has not been replicated. Interim Item 3's no-material-change referral does not issue a new December sensitivity.

2.9.3 Future receipts need customers and collection dates

Receivables may turn into cash later, but their movement does not identify collections. Annual net trade receivables rise by 3,902 million, while the indirect cash-flow adjustment is −3,788. Their sum leaves an unallocated 114 million difference. In the quarter, the balance change of −3,771 and adjustment of +3,597 leave −174. Vendor non-trade changes match the opposite indirect adjustments exactly: +1,356/ − 1,356 annually and −3,166/ + 3,166 quarterly. Even that match supplies neither gross sales nor total collections (Apple Inc., 2024, 2025, balance and cash-flow statements). Sales, allowances, currency and noncash movements still need their own evidence before receipts can be dated.

Concentration describes who might pay, but overlapping descriptions cannot be summed as separate exposures. Carriers account for 41%, 38% and 33% of total trade receivables at the three successive dates. At December the note also identifies two customers individually accounting for at least 10%, with shares of 18% and 11%. It does not say whether they are carriers (Apple Inc., 2024, Note 4, printed p. 38) (Apple Inc., 2025, Note 4, printed p. 9). Let A be those two customers, B the carrier group, and $s(S)$ the share of total trade receivables owed by customers in a set S. Each customer belongs wholly to a set or does not. Treat the displayed percentages as exact centres for this illustration. Inclusion-exclusion gives

$$s(A \cup B) = s(A) + s(B) - s(A \cap B) = 29\% + 33\% - s(A \cap B).$$

If both large customers are carriers, their 29% is already inside the 33% carrier share, so the union is 33%. If neither is a carrier, the union is 62%. For a concrete construction, allocate receivables across eleven hypothetical customers with shares $18, 11, 9, 9, 8, 7, 4, 9, 9, 8, 8$ percent. These sum to 100 and only two reach 10%. Carriers can be the 18, 11 and 4 customers, totalling 33, or the first 9, 9, 8 and 7 customers, also totalling 33. The same marginal disclosures therefore permit both endpoint unions. If only one large customer is a carrier, the union is 44% or 51%; exact-centre membership does not permit an arbitrary continuously divisible overlap. These are conditional percentage examples, not recovered customer identities or probabilities of loss. The note's trade denominator is not automatically the balance sheet's *net* receivable line, so multiplying it by that line would add an unsupported measurement assumption. Required credit support or collateral applies to *certain* customers; it does not establish universal recovery.

Vendor non-trade receivables have a separate economic origin and denominator. Apple buys components from suppliers and sells them to manufacturing vendors that make subassemblies or final products for Apple. Those component sales are excluded from product net sales; any gain reduces product cost of sales when Apple sells the related final products. The two largest reported vendor shares are 44% and 23% at September 2024, and 43% and 24% at December, each pair totalling 67% on its own dated vendor population (Apple Inc., 2024, 2025, annual and interim Note 4). Equal combined shares do not establish unchanged counterparties, component cohorts or collection risk. A funding forecast consequently needs settlements and restrictions on assets alongside payment dates on obligations. Keeping reported assertions distinct from those missing facts is the starting point for representing the company's historical record.

2.10 Sales geography is not a map of costs or assets

The liquidity question leaves a related operating question. If demand weakens in one place, which part of Apple's sales and profit is exposed? Apple reports five geographic segments, but a segment is a management reporting division, not a list of incorporated companies, factories or product lines. Reading its two segment notes as though those were the same partitions would make a regional scenario appear more precise than the filing permits.

2.10.1 What the regional labels measure

Apple's fiscal 2024 Note 13 names Americas, Europe, Greater China, Japan and Rest of Asia Pacific. Americas covers North and South America. Europe also includes India, the Middle East and Africa; Greater China includes China mainland, Hong Kong and Taiwan. Apple says regional net sales are *generally* based on customer location and sales through retail stores in those geographic locations. Its segments sell similar products and services, but are managed separately around customers, distribution partners and market conditions (Apple Inc., 2024, Note 13, printed pp. 46–47). An India-customer sales shock therefore belongs with Apple's Europe segment under this disclosure's general attribution rule, even though India is in Asia. Neither this mapping nor the label "Greater China" locates the legal seller, the manufacturing partner or the cash-receiving entity.

The prior revenue reading established that product and regional net sales are two classifications of the same 391,035 million dollars in fiscal 2024 and 124,300 million in the first quarter of fiscal 2025; the totals cannot be added. The full Note 13 and interim Note 10 tables confirm five reported regional sales and operating-income pairs in each dated column. For example, Greater China has fiscal 2024 sales of 66,952 million and segment operating income of 27,082 million; in the quarter ended 28 December 2024 the amounts are 18,513 and 8,160 million (Apple Inc., 2024, Note 13, printed p. 47) (Apple Inc., 2025, Note 10, printed p. 11). The quarter is a distinct three-month period, not a fourth of the fiscal-year observation. Apple's fiscal calendar can contain 53 weeks, as it did in fiscal 2023, and its interim MD&A describes historically higher first-quarter sales associated in part with the holidays (Apple Inc., 2024, 2025, Note 1 and MD&A).

Management attributes the annual Greater China sales decline mainly to lower iPhone and iPad sales and reports an unfavourable renminbi effect. For the first quarter of fiscal 2025 it attributes the decline mainly to iPhone, while describing a favourable currency effect (Apple Inc., 2024, 2025, annual and interim segment operating performance). These are dated issuer explanations, not measured contributions of each driver or an estimated response equation. A future region-specific demand or exchange-rate path needs product, currency and contract information that the regional totals alone do not supply.

2.10.2 Why segment income cannot be treated as fully allocated profit

Apple defines each segment's operating income as third-party sales less the related cost of sales and directly attributable operating expenses. The annual total of segment operating income is 162,044 million, but the consolidated operating statement reports 123,216 million. Note 13 supplies the bridge: 31,370 million of research and development and 7,458 million of other corporate expenses, net, are subtracted. The latter category includes general and administrative compensation, various nonrecurring charges and other separately managed costs. The reported relation is

$$162{,}044 - 31{,}370 - 7{,}458 = 123{,}216 \quad \text{million dollars, fiscal 2024.} \tag{2.6}$$

The first-quarter fiscal 2025 bridge has the same form: $53{,}174 - 8{,}268 - 2{,}074 = 42{,}832$ million. Both endpoints agree with the consolidated statements of operations (Apple Inc., 2024, statements of operations and Note 13, printed pp. 29 and 47) (Apple Inc., 2025, statement of operations and Note 10, printed pp. 1 and 11). The annual "other corporate" amount is not identical to the 26,097 million selling, general and administrative expense line in the consolidated statement. Equating the two would change the reported classification.

Equation (2.6) identifies the company-wide cost pool, not how it would be attributed to regions. To see the difference, take a hypothetical allocation of the 38,828 million total corporate bridge

across Americas, Europe, Greater China, Japan and Rest of Asia Pacific of respectively 14,000, 10,000, 5,000, 4,000 and 5,828 million. Moving 1,000 from Americas to Greater China produces a second allocation of 13,000, 10,000, 6,000, 4,000 and 5,828 million. Both are nonnegative, sum to the same corporate total and reproduce the consolidated 123,216 million. Yet Greater China's hypothetical profit after these costs is either $27{,}082 - 5{,}000 = 22{,}082$ or $27{,}082 - 6{,}000 = 21{,}082$ million. Apple does not report either fully allocated figure, and the filing supplies no rule preferring one allocation. These constructions establish numerical nonidentification within the stated nonnegative illustration; they do not assert that all real cost allocations are nonnegative or that either construction describes Apple's internal books. Forecasting regional profit also requires the behaviour of directly attributable and corporate costs under the scenario.

2.10.3 Country sales and long-lived assets answer different questions

The annual note separately presents country sales for fiscal 2024, 2023 and 2022 and country long-lived-asset stocks at the ends of fiscal 2024 and 2023. It separately reports countries individually accounting for at least 10% of the respective totals and groups the rest; that selection rule does not make "Other countries" a single location. The 2024 displayed amounts illustrate the different populations:

Country class	2024 net sales	Long-lived assets
	USD millions	
U.S.	142,196	35,664
China, including Hong Kong and Taiwan	66,952	4,797
Other countries	181,887	5,219
Total	391,035	45,680

The China country-sales row equals Greater China segment sales in all three annual columns. The note's China footnote includes Hong Kong and Taiwan, so the equality is a useful check on these displayed populations. It does not turn that row into a legal-entity or production-location ledger. The U.S. country-sales row is below Americas segment sales by 24,849 million in 2024; assigning that difference to particular countries requires the precise country-to-region attribution and customer facts. Europe's segment label cannot be substituted for a European-country total because it includes India, the Middle East and Africa (Apple Inc., 2024, Note 13, printed pp. 46–47).

The 45,680 million long-lived-asset total equals net property, plant and equipment at the 2024 balance date, and the 2023 totals both equal 43,715 million at their displayed precision (Apple Inc., 2024, Notes 5 and 13, printed pp. 39 and 47). The equality is a checked comparison, not a definition of every asset's nature, owner or productive capacity. Country asset stocks are measured at an instant, while sales cover a year. Further, Apple says the information supplied to its chief operating decision maker for segment assessment excludes asset information (Apple Inc., 2024, Note 13, printed p. 46). Thus neither a segment return on assets nor an allocation of production capacity to the five sales regions is identified by the disclosed tables. The statement that hardware is made substantially by outsourcing partners located primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam concerns another population, and does not repair that missing asset/segment mapping (Apple Inc., 2024, Note 12, printed p. 46).

2.10.4 The same totals permit different product exposure

Note 2 reports 201,183 million of fiscal 2024 iPhone sales. Its four other product and service rows sum to 189,852 million. Along with Greater China's 66,952 million regional total, these are *margins* of an unreported product-by-region table. Let z be iPhone sales assigned to Greater China, with all amounts in millions of dollars at the displayed centres. The four cells for Greater China versus the other four regions must be

	iPhone	Other products and services
Greater China	z	$66{,}952 - z$
Other four regions	$201{,}183 - z$	$122{,}900 + z$

(2.7)

The row and column sums in (2.7) hold for any nonnegative cells; they do not select z. For example, $z = 36{,}000$ and $z = 38{,}000$ both meet the displayed numeric margins. For an explicit extension, let S_r be each other region's reported sales; those four regions sum to $391{,}035 - 66{,}952 = 324{,}083$. Assign each $(201{,}183 - z)S_r/324{,}083$ of iPhone sales; these amounts sum to the remaining iPhone total and are below S_r. Then split every region's remaining sales across the four non-iPhone categories in their disclosed company-wide proportions. This produces all five product/service and five region margins exactly. These proportional splits are mathematical witness constructions, not issuer allocations. The source adds that the product mix was *generally* consistent across segments, with a *moderately* higher iPhone share in Greater China. Both numerical examples put Greater China's iPhone share above the combined other-region share. The qualitative words have no disclosed tolerance, so neither matrix is certified as a complete interpretation of that statement (Apple Inc., 2024, Note 2, printed pp. 35–36). The interim note repeats the qualitative comparison for its own dated quarter (Apple Inc., 2025, Note 2, printed p. 6); it does not supply the missing cross-classified cells.

If a scenario *assumes* a 10% decline in Greater China iPhone sales alone, the two illustrative fiscal 2024 matrices imply 3,600 and 3,800 million less sales, respectively. The scenario percentage is a teaching assumption, not an estimated shock; neither sales change determines profit, cash flow, currency translation or future behaviour. The gap between those two conditional answers comes from the unobserved joint allocation, even before cost responses are considered. A reproducible model should retain the reported marginals, the qualitative mix statement, any admitted cross-classification and its source, and the unresolved allocation when the evidence does not determine it.

Finally, the annual and first-quarter MD&A state Apple's then-planned fourth-quarter fiscal 2025 adoption of ASU 2023-07 with retrospective transition (Apple Inc., 2024, 2025, Recent Accounting Pronouncements). This is a dated issuer statement. It neither supplies an already restated 2024 observation nor establishes the standard's operative provisions without inspection of that authority. A historical query must keep the older disclosure view separate from any later restated one when such a filing is actually acquired. The regional calculations here are checked source-bound comparisons and conditional witnesses, not a reconstructed Apple segment ledger or a forecast-ready regional model.

2.11 From geographic sales to legal entities

A regional sales forecast still leaves open which company sells the goods, owns the inventory or owes the supplier. Those identities matter when a scenario changes a particular entity's tax, funding or supply arrangements. Apple's geographic segments cannot answer that question: their sales are generally attributed by customer location and retail-store geography. The consolidation policy and subsidiary exhibit supply a different kind of information, about the companies represented in the group statements.

Annual Note 1 says that the consolidated financial statements include the accounts of Apple Inc. and its wholly owned subsidiaries. The filing also defines "Company" and "Apple" collectively in that way, unless otherwise stated. Interim Note 1 repeats the wholly-owned scope for its condensed statements and requires them to be read with the fiscal 2024 annual statements and notes (Apple Inc., 2024, definition of Company and Note 1, printed pp. 1 and 34) (Apple Inc., 2025, Note 1, printed p. 5). This is positive evidence about the reported consolidation perimeter. It is not a set of separate parent-company accounts, nor does it identify every legal entity or ownership link within that perimeter.

Exhibit 21.1 names the following subsidiaries and their jurisdictions of incorporation. The labels below are transcribed from the exhibit, including the separate U.S. state jurisdictions (Apple Inc., 2024, Exhibit 21.1, retained PDF p. 114).

Subsidiary	Jurisdiction of incorporation
Apple Asia Limited	Hong Kong
Apple Asia LLC	Delaware, U.S.
Apple Canada Inc.	Canada
Apple Computer Trading (Shanghai) Co., Ltd.	China
Apple Distribution International Limited	Ireland
Apple India Private Limited	India
Apple Insurance Company, Inc.	Arizona, U.S.
Apple Japan, Inc.	Japan
Apple Korea Limited	South Korea
Apple Operations International Limited	Ireland
Apple Operations Limited	Ireland
Apple Operations Mexico, S.A. de C.V.	Mexico
Apple Pty Limited	Australia
Apple Sales International Limited	Ireland
Apple South Asia (Thailand) Limited	Thailand
Apple Vietnam Limited Liability Company	Vietnam
Braeburn Capital, Inc.	Nevada, U.S.
iTunes K.K.	Japan

The exhibit expressly limits the list. Its footnote states: "Pursuant to Item 601(b)(21)(ii) of Regulation S-K, the names of other subsidiaries of Apple Inc. are omitted because, considered in the aggregate, they would not constitute a significant subsidiary as of the end of the year covered by this report." This preserves Apple's stated reason and date for the omission; it does not establish a numerical significance test for this model. The eighteen rows cannot be treated as an exhaustive group roster or as proof that an omitted subsidiary is irrelevant to every entity-specific question.

The same filing makes that incompleteness concrete. Its tax-risk discussion mentions Ireland and Singapore, "where a number of the Company's subsidiaries are organized," yet no Exhibit 21.1 row has Singapore as its jurisdiction (Apple Inc., 2024, Risk Factors, printed p. 16). The evidence supports subsidiaries organized there while leaving their names and ownership paths unresolved. Creating a company named "Apple Singapore" would turn a geographic reference into an invented identity.

2.11.1 Membership does not determine the ownership path

Wholly-owned membership and immediate ownership answer different questions. For a simple hypothetical example, a subsidiary could be owned directly by the parent, or through another subsidiary that the parent wholly owns. If every link is wholly owned, both arrangements preserve ultimate wholly-owned membership while producing different immediate parents. This

logical example does not select either arrangement for any company in the table. Neither the similar names of Apple Operations Limited and Apple Operations International Limited nor their common incorporation jurisdiction establishes an ownership link between them.

A model can therefore retain each named subsidiary, its reported jurisdiction and the general consolidation assertion without drawing unsupported edges. It should record an immediate parent, share class, ownership interest and effective interval only when the evidence identifies them. The September 2024 exhibit is not a control history for earlier years or an updated list at 28 December 2024. Repetition of the general policy in the interim note does not show that every subsidiary and ownership path remained unchanged. Determining the dated group requires incorporation, acquisition, disposal, reorganisation and control evidence as relevant; it cannot be recovered from the fiscal calendar alone.

2.11.2 Selling, manufacturing and owning are separate relationships

Apple India Private Limited illustrates the distinction between incorporation and sales attribution. The exhibit places its incorporation in India, while Apple's segment definition places India in Europe. Under the general customer-location rule, an India-customer sale belongs in that sales region; neither statement identifies its legal seller or assigns all India sales to that subsidiary. Even a complete subsidiary list would not supply the missing seller-by-customer-region relationship.

The Business discussion also separates distribution channels: direct sales through Apple's stores and sales force accounted for 38% of fiscal 2024 net sales, and indirect channels for 62%. The latter include third-party cellular carriers, wholesalers, retailers and resellers (Apple Inc., 2024, Markets and Distribution, printed p. 2). These are sales channel shares, not ownership shares or counts of companies. A trading relationship does not by itself establish that a counterparty belongs in the consolidated group.

Production adds a further distinction. Apple says substantially all its manufacturing is performed in whole or in part by outsourcing partners, primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam. It has also invested in manufacturing process equipment, much of which is held at certain outsourcing partners, and made prepayments under long-term supply agreements (Apple Inc., 2024, Risk Factors, printed p. 8). Its MD&A describes partners manufacturing subassemblies and performing final assembly and testing, alongside purchases from component suppliers (Apple Inc., 2024, Manufacturing Purchase Obligations, printed p. 25). Equipment custody, a supply contract and corporate control are thus separate relationships. A disruption at a partner may affect product supply, equipment recoverability and prepayments without supplying evidence that the partner should be consolidated. The affected assets, contracts and counterparties must be identified before amounts can be assigned to that scenario.

2.11.3 Tax and currency queries need facts beyond incorporation

The jurisdiction column also leaves entity tax residence and functional currency unresolved. Functional currency is the currency in which an entity measures its accounts; it is a different question from the dollars in which Apple presents its group statements or the currency of a particular invoice. The group foreign-exchange discussion describes Apple as a net receiver of currencies other than the U.S. dollar, with pricing and hedging responses to exchange-rate movements. It supplies no entity-by-entity functional-currency assignment (Apple Inc., 2024, Foreign Exchange Rate Risk, printed p. 27). An Ireland incorporation row therefore does not justify entering euros as the entity's functional currency, just as the India row does not establish rupees.

Tax evidence has its own dates and units of analysis. Annual Note 7 describes the historic tax allocation of profits of Irish branches of two subsidiaries without naming those subsidiaries in that passage. The MD&A describes the resulting group payment obligation and restricted escrow (Apple Inc., 2024, Note 7, printed pp. 39–40; State Aid Decision Tax Payable, printed p. 26). The four Ireland rows in Exhibit 21.1 do not identify which entities had those branches, their historic tax residence or the jurisdictional tax bases used in current accounts. A name match or country match would bypass the legal and temporal facts needed for the tax calculation.

These distinctions determine what a supported query can return. The retained sources answer questions about the stated consolidated scope and the named subsidiaries' incorporation jurisdictions. Entity-specific tax, currency and intercompany questions require additional records: legal counterparties, dated ownership and control, compatible entity books, functional currencies, tax bases and matched transactions. Group eliminations then require the applicable accounting methods as well as those facts. The later IFRS group examples explain their own assumptions; they do not establish the operative US GAAP treatment for Apple. Keeping those dependencies explicit preserves useful source knowledge while leaving the unanswered entity questions open.

2.12 Legal proceedings beyond the booked liability

Knowing the companies within a group does not identify all the events that could change their earnings. A legal proceeding can require a payment, but it can also change how a company sells its products or earns licensing revenue. Apple's reported App Store injunction illustrates the second channel: it concerns developers' external purchase links on the U.S. storefront, rather than a disclosed amount of damages. A forecast that represents legal exposure only as a liability balance would leave that operating consequence outside the model.

The annual Item 3 and interim Part II Item 1 each discuss Digital Markets Act investigations, the Department of Justice lawsuit, Epic Games and other legal proceedings. These are dated issuer reports, with different kinds of statements inside each matter: allegations, reported decisions, pending motions, management's position and possible consequences. Keeping those distinctions allows a model to use what is reported without deciding the case itself (Apple Inc., 2024, Item 3, printed p. 18) (Apple Inc., 2025, Part II, Item 1, printed p. 18).

2.12.1 A loss assessment answers a narrower question than exposure

Annual Note 12 and interim Note 9 give the same qualified assessment: "In the opinion of management, there was not at least a reasonable possibility the Company may have incurred a material loss, or a material loss greater than a recorded accrual, concerning loss contingencies for asserted legal and other claims" (Apple Inc., 2024, Note 12, printed p. 46) (Apple Inc., 2025, Note 9, printed p. 11). The sentence identifies whose assessment it is, concerns an incurred material loss, and allows for a recorded accrual. It does not disclose a total legal accrual, allocate an accrual to one of the named matters, or assert that all exposure is zero. Nor does it supply a numerical probability or materiality threshold.

The annual MD&A separately reports Apple's recognition policy: the company records a liability when it is probable that a loss has been incurred and the amount is reasonably estimable, a determination requiring significant judgment (Apple Inc., 2024, Legal and Other Contingencies, printed p. 26). That policy explains why a pending proceeding and a recognised liability are different facts. It is issuer-policy evidence; applying the operative US GAAP requirements to a particular case still needs the relevant authority and case facts. An allegation cannot supply

those facts, and a possible change in future sales is not itself the incurred loss described by the policy.

Settlement adds a further distinction. Annual Item 3 says certain matters settled during the fourth quarter of fiscal 2024 without an individually or aggregate material impact on financial condition or operating results. The interim passage makes that statement about the first quarter of fiscal 2025. These are two period-specific assertions, not the same settlement observation copied forward. Neither passage identifies the matters, settlement amounts, payment dates or earlier accruals. Both retain the warning that adverse resolutions above management's expectations could materially affect the reporting period. A payment schedule or an expense roll-forward therefore remains unresolved even though the settlement statements themselves are known.

2.12.2 The remedy, jurisdiction and procedural stage matter

The Digital Markets Act discussion reports two investigations opened by the European Commission on 25 March 2024. Article 5(4) concerns developers' communication and promotion of offers and conclusion of contracts with end users; Article 6(3) concerns default settings, app uninstallation and a browser choice screen on iOS. The Commission's 24 June announcement is described as preliminary findings alleging an Article 5(4) breach and the opening of a third investigation into new contractual requirements for third-party developers and app marketplaces. Apple states that it believes it complies and continues to change its compliance plan in response to Commission engagement. The preliminary allegation and Apple's position remain separate assertions.

Both filings say that, if the Commission makes a final violation determination, it can issue a cease and desist order and may impose fines up to 10% of Apple's annual worldwide net sales. They also say an order would apply immediately during an appeal unless a stay is granted. This preserves a conditional consequence, its stated sales base and the stay qualification. The percentage is not a fine already imposed, an expected loss or a booked liability. Even a conditional fine calculation would need the applicable operative law, relevant sales period and base, and assumptions about the determination and remedy. None is supplied by multiplying a convenient annual sales figure by the reported cap.

The Department of Justice discussion reports the civil antitrust lawsuit filed on 21 March 2024 in the District of New Jersey by the DOJ and state and district attorneys general. It alleges monopolization or attempted monopolization in the markets for "performance smartphones" and "smartphones." The DOJ seeks equitable relief; related state and federal civil litigation seeks monetary damages and other nonmonetary relief. Apple states that it has substantial defenses and intends to defend itself. Treating all these proceedings as one quantified damages claim would erase both the different requested relief and the absence of disclosed amounts.

Epic's case contains a reported court finding as well as unresolved procedure. Apple says the Northern District of California court found certain App Store Review Guidelines provisions to violate California's unfair competition law and issued an injunction against prohibiting developers' external purchase links. The stated scope is the U.S. storefront of the iOS and iPadOS App Store. Apple reports implementing a compliance plan and filing its statement on 16 January 2024. Epic's motion disputing that plan and seeking enforcement is reported as pending, opposed by Apple; Apple also reports filing a motion to narrow or vacate the injunction on 30 September 2024. These passages do not establish that the disputed plan was judicially accepted or quantify its sales effect. An EU-wide scenario cannot inherit the U.S. injunction merely because both matters concern the App Store.

The interim bodies repeat the substantive DMA, DOJ and Epic account, with acronym or

trademark presentation differences. Repetition is evidence of what Apple reported in each filing, not proof that every intervening legal fact was unchanged. The September 30 motion also falls after the annual September 28 period-end. Event date, reporting period and the date on which the report became available must therefore remain distinct in a historical information query.

2.12.3 A partner's remedy can change Apple's revenue

The risk discussion connects these matters to operations. Both filings report changes to U.S. developer communications and EU changes to distribution, payment processing, fee structures and developer tools. They also describe licensing revenue from Google and other search providers, whose arrangements are subject to investigations and proceedings (Apple Inc., 2024, Risk Factors, printed pp. 14–15) (Apple Inc., 2025, Part II, Item 1A, printed pp. 19–20). The interim adds that Google was found to have violated U.S. antitrust laws on 5 August 2024 and that the DOJ proposed remedies ranging from restrictions on licensing agreements to prohibiting commercial search-distribution terms with Apple for up to ten years. Apple says that, if implemented, those remedies could materially affect its ability to earn that licensing revenue. The finding is an earlier event newly described in this retained comparison; the remedies are proposed, not reported here as imposed.

That channel differs from an Apple fine. The affected economic state would include the search agreement, its revenue terms, duration, geography and substitution possibilities. Likewise, an App Store remedy requires the affected storefront, developer terms, fees, customer purchases and behavioural response. The filings identify these channels without supplying the contract economics needed to calculate them. They do not justify applying one loss percentage to all Services revenue or treating every customer as affected.

A conditional calculation should consequently specify which matter and remedy it assumes, when that remedy takes effect, which contracts it changes and how the parties respond. A monetary-loss calculation additionally needs recognition facts, any opening accrual and the settlement and payment schedule; expense recognition and cash payment can occur in different periods. The State Aid tax charge and escrow settlement, developed in Section 9.7, retain their own tax records and must not be counted again as a DMA loss. Until the missing facts and applicable treatments are supplied, the model can retain the reported proceedings and affected queries as unresolved, without assigning probabilities or producing numerical journals. This preserves useful knowledge about future exposure while keeping the distinction between a known assertion and a supported financial consequence.

2.13 When later disclosures change the historical query

A later filing can reveal more about an earlier period without identifying a different economic history. For example, a new segment-expense disclosure may make a previously hidden cost visible, while the transactions that generated that cost remain the same. To use that information correctly, a model must retain the original report, the later presentation and the date at which the additional detail became available. Apple's accounting-pronouncement discussion shows why those distinctions are needed before the later figures even exist.

Annual and interim MD&A describe planned changes to tax and segment disclosures; the interim adds expense disaggregation and a subsequent effective-date clarification (Apple Inc., 2024, Recent Accounting Pronouncements, printed p. 26) (Apple Inc., 2025, Recent Accounting Pronouncements, printed p. 17). The following table preserves Apple's stated adoption plans. The quarters are fiscal, and the descriptions are issuer reports of the updates, rather than an

independent determination of their operative requirements.

Update	Reported disclosure change	Apple's stated plan
ASU 2023-07	Segment expenses and information about the chief operating decision maker	Fiscal 2025 Q4, retrospective
ASU 2023-09	Income-tax reconciliation and taxes-paid detail	Fiscal 2026 Q4, prospective
ASU 2024-03, clarified by 2025-01	Expense components and selling-expense amount and definition	Fiscal 2028 Q4, prospective

The tax discussion says the rate reconciliation will include additional information and detail for items meeting a quantitative threshold. Taxes paid will be disaggregated into federal, state and foreign taxes, with further detail for significant individual jurisdictions. It gives neither the threshold value nor the future jurisdictional amounts. Those cash-tax populations also remain distinct from tax expense, payable and deferred balances; their reconciliation is developed in Section 9.7. An announced disclosure change cannot be used to allocate an old tax total among jurisdictions by assumption.

The segment discussion concerns expenses that are significant and regularly provided to the chief operating decision maker, together with that person's title, position and use of segment profit or loss in assessing performance and allocating resources. The future disclosure may answer questions about the information used by management. It does not identify the regional allocation of the existing corporate expense pool or change the current absence of segment asset information. The previously examined retrospective-adoption sentence remains a dated plan, not a later recast observation.

The interim expense passage reports ASU 2024-03's issuance in November 2024 and ASU 2025-01's clarification in January 2025. It describes purchases of inventory, employee compensation, depreciation and intangible-asset amortization, as applicable, within certain expense captions, a qualitative description of the remaining amounts, and both the amount and the company's definition of selling expenses. This describes disclosure populations; it does not provide their values or make selling expenses identical to the entire selling, general and administrative line. Purchases included in an expense disclosure likewise cannot be identified with cash paid to suppliers without the relevant timing and population facts.

These plans require separate historical query views. If a later filing presents an earlier segment period retrospectively, an original-information query retains the earlier disclosure, while a later-information query may use the checked recast presentation. The original must not be overwritten. A comparison also needs to establish the affected periods, definitions, units and basis before matching the two versions. The words "retrospective transition" alone supply neither a recast amount nor proof that a particular historical component will be available. For the prospective tax and expense plans, these passages supply no backward allocation of missing historical components. Exact transition requirements need the operative updates and any applicable amendments.

The January clarification illustrates the timing problem directly. It occurs after the interim period ended on 28 December 2024 but is described in the interim report. An event can therefore belong to the information in a report without occurring inside its reporting period. A historical forecast cutoff needs evidence of availability and the exact retained version, not merely the period-end or the update's issuance month. Similarly, an announced fiscal adoption quarter is not an observed adoption event; the later filing must be checked before the model records the plan as realised.

2.13.1 Unchanged critical estimates do not mean constant amounts

Annual Critical Accounting Estimates says management makes judgments, assumptions and estimates affecting reported amounts, using historical experience and other assumptions it considers reasonable to judge carrying values. Its uncertain-tax discussion says reserves can be adjusted as facts and circumstances change, and that final outcomes may differ from those reserves. Its legal-contingency policy is the incurred-loss and estimability condition examined above (Apple Inc., 2024, Critical Accounting Estimates, printed p. 26). These are descriptions of judgment and policy, not a disclosed numerical estimation model or an assurance that the estimates will prove accurate.

The interim references both annual and interim Note 1 and reports no material changes to critical accounting estimates since the annual filing (Apple Inc., 2025, Critical Accounting Estimates, printed p. 17). That statement preserves the annual dependency and its materiality qualifier. It cannot make every carrying amount, reserve or underlying assumption exactly constant, and it supplies no numerical tolerance for a comparison. A model should retain the reported policy and qualified continuity assertion alongside the actual dated balances, then request the missing estimate or transition facts when the query needs them. The present evidence supports those separate records; it does not establish current compliance, realised adoption or the historical expense components that future disclosures may reveal.

2.14 Claims against another party can still affect Apple

The estimate policy leaves a further question: whose claim can give rise to a cost for Apple? A company can face exposure through an agreement even when the claim is against someone else. Apple's annual risk discussion says its agreements sometimes include indemnification provisions that can subject it to costs and damages in the event of a claim against an indemnified third party (Apple Inc., 2024, Risk Factors, printed p. 13). The direction matters: the passage describes a possible obligation of Apple arising from another party's claim, rather than compensation owed to Apple. A model limited to proceedings naming Apple as defendant would miss that contractual channel.

The statement supplies neither an indemnified party nor a particular agreement. It also leaves the covered conduct, trigger, duration and limit unknown. To connect a third-party claim to a dated obligation, the model would need those terms and the facts that activate them. The existence of indemnification language somewhere in Apple's agreements cannot establish that a named case is covered, that a payment is due, or that a liability has been recognised. Those facts remain distinct from the management loss assessment examined in Section 2.12.

The full annual passage also describes unresolved ordinary-course claims and investigations, an increase over time in their number and alleged magnitude, and patent claims concerning cellular-enabled products. Apple reports defending infringement actions in U.S. and international courts, whose plaintiffs frequently seek broad injunctions and substantial damages. These are issuer statements about claims and requested relief. They do not identify a complete matter population, decide the infringement allegations or quantify the payments and product restrictions that will result.

Defense is a separate source of cost. The filing states, "Regardless of the merit of particular claims, defending against litigation or responding to government investigations can be expensive, time-consuming and disruptive to the Company's operations." Thus a scenario in which Apple successfully resists a claim need not be a scenario with no defense effort or disruption. The model therefore needs the response services, their amounts and dates, and related payment obligations. The damages outcome cannot identify those expenses or payments.

Settlement can alter recurring operations as well as resolve a dispute. Apple says it may enter agreements or other arrangements to settle litigation, without assurance of acceptable terms or that litigation will not occur. Those agreements can increase cost of sales and operating expenses, change business practices and limit products and services. A conditional settlement therefore needs its payment terms and its continuing effects on royalties, costs or offerings. An adverse resolution can additionally involve monetary damages, disgorgement, corporate measures or an injunction. The annual passage explicitly includes a resolution against an indemnified third party above management's expectations. Neither that possibility nor the separately reported quarterly settlements supplies a contract, amount or probability for a forecast.

2.14.1 Insurance describes a possible recovery, not a net loss

Apple says it maintains insurance for certain types of claims, while warning that coverage may be insufficient for all losses or all claim types. The qualification leaves both the covered population and the amount recoverable unresolved. A model cannot deduct an assumed insurance payment from damages or an accrued liability merely because insurance exists. It needs the policy period, insured party and beneficiary, covered event, exclusions, deductible, limit, insurer and recovery terms, together with facts about the claim. The applicable recognition and presentation treatment must also be established before a recovery amount becomes an accounting entry. The passages identify none of those amounts or terms.

Other annual insurance statements concern different exposures. The business-interruption discussion warns of recovery expenditures and lost sales, especially when critical components have single or limited sources, while saying insurance for certain types of losses may be insufficient (Apple Inc., 2024, Risk Factors, printed p. 6). The data-security discussion says coverage is intended for certain aspects of those risks and may not cover all losses or claim types (Apple Inc., 2024, Risk Factors, printed p. 12). The trade receivable discussion instead says a substantial majority of outstanding trade receivables lack collateral, third-party bank support or financing arrangements, or credit insurance (Apple Inc., 2024, Risk Factors, printed p. 16). These statements cannot be combined into one insured share of Apple's exposures. The receivable qualifier is qualitative; it identifies neither every uncovered customer nor a percentage. Insurance bought by Apple also does not make these exposures an insurer's issued-contract measurement problem.

The interim risk preamble incorporates the annual risks and states that there have been no material changes except those set out below (Apple Inc., 2025, Part II, Item 1A, printed p. 19). It preserves the annual dependency and an express exception; it does not assert identical policies, claims, coverage or loss amounts at the later date. Keeping the original risk, the later qualified statement and the separately reported updates allows a historical query to retain what was actually disclosed. The resulting account connects direct proceedings, contract indemnities, defense, settlement, operating restrictions and possible recovery without collapsing them into a single loss balance. Calculating their consequences still requires the dated contracts, claim facts and accounting treatment for each channel.

2.15 A regulation can change the feasible operating plan

A legal exposure need not begin with a claim or end with a payment. An import restriction can prevent a product from reaching a customer, leaving the company to change its supply route, offering or production plan. Apple's annual and interim general-regulation risk both say that its products could be banned, delayed or prohibited from importation despite procedures intended

to satisfy regulatory requirements. They also warn that measures may arrive with little or no advance notice and that Apple may be unable to mitigate all adverse impacts (Apple Inc., 2024, Risk Factors, printed pp. 13–14) (Apple Inc., 2025, Part II, Item 1A, printed p. 19). This describes a possible constraint on operations, separate from the damages, defense and insurance channels examined above.

The two reports preserve a broad list of regulated subjects: among them antitrust, privacy, consumer protection, intellectual property, financial services, trade, foreign exchange controls, cash repatriation, tax and employment. The interim adds national security to that list. It also expands the discussion of new and changing laws and regulations to name executive orders, directives and enforcement priorities. These additions appear in the cost-and-business-change, interpretation and specialized-application passages. They identify the scope of Apple's later risk statement without identifying a particular new instrument, effective date, jurisdiction or violation. A model cannot create a dated legal event merely because a subject is newly named in this comparison.

Both reports describe how changing requirements can increase costs, restrict a product or feature, alter its design, affect customer demand and change the business or supply chain. They describe further risks as products enter specialized applications such as health and financial services or expand the use of machine learning and artificial intelligence. The annual combines these consequences and the compliance-policy discussion in one paragraph; the interim splits and expands it. Paragraph count therefore cannot measure additional economic exposure. Conversely, the notice-and-import paragraph repeats the same substantive wording. That repetition establishes what Apple disclosed in each report, rather than identical underlying laws, products or supply routes.

The policy qualification is essential to the mechanism. Apple says it has implemented policies and procedures designed to ensure compliance, while giving no assurance that employees, contractors or agents will not violate laws, regulations or company procedures. Its separate import passage gives no assurance that programs will prevent a violation or a claim of one. Policy existence, compliance in a particular case, a claim of violation and effective mitigation are thus different facts. Treating the reported procedures as proof of compliance would remove precisely the uncertainty the source preserves.

A conditional operating scenario should start with the actual or assumed measure and its effective date, legal scope, affected entity, product and jurisdiction. It then needs the available supply routes, inventory, lead times, customer demand and management response. For an assumed import delay, those facts determine whether delivery is postponed, rerouted or lost, and which storage, production or sourcing costs change. Recognition, payment and any later claim must follow their own facts and treatments. The filings identify these possible channels but supply neither a tariff rate nor a universal sales reduction. Their reporting dates also do not establish when a particular rule arose or when these exact report versions became available. Keeping disclosed risk, operative authority, assumed restriction and response separate lets the model retain a useful operating question without inventing its answer.

2.16 Demand, disruption and recovery follow different paths

An open supply route is useful only if a customer still wants the product. Conversely, a customer may be ready to buy while a supplier cannot deliver a critical component. The operating scenario introduced above must therefore describe both demand and the ability to serve it. Apple's annual risk discussion connects global and regional economic conditions to consumer spending, the financial condition of its business partners and its own access to finance.

Its separate interruption discussion describes obstacles to manufacture, delivery and service provision (Apple Inc., 2024, Risk Factors, printed pp. 5–6). These passages give reasons to retain several possible operating histories behind the same reported financial totals.

Apple says that sales outside the United States account for a majority of net sales and that a majority of its supplier facilities are outside the United States. The two statements concern different populations. A sales region does not locate a factory, and a factory's location does not identify the customers whose orders depend on it. The segment definitions examined earlier generally attribute sales by customer and retail-store location, while manufacturing purchase obligations describe purchases from outsourcing partners and component suppliers (Apple Inc., 2024, Note 13, printed p. 46; Liquidity and Capital Resources, printed p. 25). Linking the two requires the actual product, component, supplier and route relationships. Geographic labels alone cannot create them.

The macroeconomic passage describes several possible effects of the same economic conditions. Recession, unemployment, inflation, tighter credit, higher interest rates and currency fluctuations can weaken consumer confidence and spending. Fiscal and monetary policy, market volatility and changes in income or asset values can also affect demand. Suppliers, manufacturers, logistics providers, distributors, carriers, other channel partners and developers may instead face financial instability, difficulty obtaining credit or insolvency. Apple further identifies trade-receivable collectibility, derivative-counterparty and financial-institution failure, limited access to new debt, reduced liquidity and declining financial-instrument fair values. The disclosure permits a scenario to connect these channels through common conditions; it supplies neither a joint probability law nor a requirement that every channel deteriorate together.

An interruption raises a different operating question: which activity becomes unavailable, and for how long? Apple's discussion includes political and trade disputes, geopolitical conflict, terrorism, natural disasters, public-health issues and industrial accidents. It says substantially all manufacturing is performed wholly or partly by outsourcing partners located primarily in China mainland, India, Japan, South Korea, Taiwan and Vietnam. Trade restrictions can raise input costs, require different suppliers or business relationships, or lead the company to cease offering affected products, services or third-party applications. Changing the supply chain may itself be costly and disruptive, and restrictions may arrive with little notice. Apple reports that past U.S.–China tensions led to tariffs and other restrictions affecting its business. This historical assertion is distinct from its conditional warning about further escalation; a future scenario still needs a specified measure, effective date and affected trade.

The same passage describes facilities in disaster-prone locations and asserts that climate change is increasing the frequency or intensity of certain natural disasters and extreme weather. It also identifies fire, power shortages, nuclear and other industrial accidents, hostile acts, ransomware, labor disputes and health emergencies. These are Apple's stated exposures, without facility-specific hazard estimates. Its requirement that suppliers maintain safe workplaces is separate from whether an accident occurs. The company reports that major public-health issues, including COVID-19, have adversely affected demand, operations and distribution, and warns of possible future effects. Travel restrictions, freight limitations and interruptions to production ramps supply concrete mechanisms; the paragraph does not quantify an historical loss or a future duration.

Consider two explicitly assumed histories for a particular product. In one, the supply route remains available but customers reduce orders. In another, orders remain but a critical component cannot arrive. The first requires a demand response; the second requires an inventory and delivery response. They can also occur together. To compare the histories, the model needs product-level orders, available components and finished goods, lead times, contractual commitments and feasible management actions. An assumed alternate supplier

must have capacity and suitable terms before it becomes an available response. A sales delay or lost order follows from those facts and assumptions; it cannot be obtained by applying one percentage to every reported balance.

Recovery adds another date. Apple warns that an interruption can require substantial time and expenditure to resume operations, with lost sales and greater consequences where critical components have single or limited sources. Note 12 separately explains that even components available from several suppliers can face industry-wide shortages and commodity-price fluctuations (Apple Inc., 2024, Note 12, printed p. 46). Ending an assumed interruption therefore need not restore production or sales immediately. Recovery spending can precede or follow resumed deliveries. The model must retain the dates of the interruption, restored capacity, deliveries, expenditure and settlement separately; the applicable accounting treatment then determines the journals. The insurance qualification already examined in the litigation discussion does not collapse those dates into a net loss or guarantee reimbursement.

The reported receivables, investments, debt and liquidity resources constrain particular dated histories, but they do not determine these operating responses. The investment sensitivity discussed above measures a specified interest-rate effect. Chapter 10 examines the distinct debt and currency measures; none identifies the broader disruption response. The interim's incorporation of annual risk factors, except for the changes it sets out, likewise does not establish unchanged facilities, orders or funding conditions (Apple Inc., 2025, Part II, Item 1A, printed p. 19). For the historical record, the useful distinction is now precise: retain Apple's assertions, the relationships needed to interpret them, and the analyst's scenario assumptions as separate information. Alternative demand, supply and recovery histories can then remain visible until evidence narrows them.

2.17 Product transitions change the relevant history

Keeping a supply route open does not ensure that it supplies the product customers now want. A new offering can replace demand for an earlier one while the company still holds its components, finished goods or purchase commitments. Apple's product-transition risk explicitly says that new products, services and technologies may replace existing offerings and produce lower revenues and lower profit margins (Apple Inc., 2024, Risk Factors, printed p. 7). A model that equates a successful release with an improvement in financial performance would remove a possibility the source asks it to preserve.

The preceding competitive-market discussion explains why a transition is not merely a production date. Apple describes price competition, short product life cycles, changing standards, rapid adoption of technology and price-sensitive customers. It says that developing almost the whole product solution requires substantial research and development, without assurance of the expected returns or successful marketing. Its reported intellectual-property portfolio also does not assure protection in every jurisdiction. The discussion of competitors imitating products or infringing rights remains an issuer assertion; it does not decide a particular infringement dispute. Apple's description of minority share in three global device markets and of competitors able to sell at little or no profit supplies competitive context, rather than a numerical market share or a forecast of Apple's margins. Services face a further set of alternatives, including free content. The company's expectation of more intense competition is a dated expectation, not an observed future result.

The transition passage identifies several separate conditions for a new offering: development must succeed on time, customers must accept it, production must reach suitable quantities and costs, and application software or other third-party support must be available. New

technology can create ramp-up or quality problems. Purchase commitments and inventory must also be managed in line with anticipated demand. Note 12 makes the supply condition concrete: new products often use custom components available from only one source; capacity can be constrained while yields mature or manufacturing expands, and a supplier may instead concentrate on common components (Apple Inc., 2024, Note 12, printed p. 46). A forecast of customer acceptance cannot establish that those components will be available.

Suppose, as an illustration, that customers switch from an earlier product to its replacement. If existing components can be used in the replacement and the supplier can expand output, the transition presents one set of feasible production choices. If the components are incompatible or the new component has constrained capacity, the same assumed customer response presents another. Neither history is identified by aggregate revenue or inventory. The comparison requires the old and new product definitions, component compatibility, inventory cohorts, committed purchases, cancellation terms, lead times and available capacity. Prices, quantities and costs would then be separate scenario inputs or model consequences. The risk language gives no substitution fraction, write-down amount or probability with which to combine the histories.

Even a delivered replacement has more than one financial date. Apple's revenue policy generally connects product revenue to transfer of control, while allocated revenue for bundled services is deferred over the estimated service period; collection follows its own terms (Apple Inc., 2024, Note 2, printed p. 35). Development, announcement, production, shipment, service provision and cash receipt therefore cannot share a single "launch" timestamp. The reported policy must be applied to the relevant contract and performance obligations. Any inventory disposition, commitment change or development-cost treatment needs its own facts and applicable accounting method. Preserving these distinctions lets the model ask whether an assumed transition is feasible and how its financial consequences would arise, while leaving actual product histories unresolved.

2.18 A supply agreement does not establish available output

Once a transition identifies the components a product needs, the next question is whether they can arrive on the required date. An agreement to purchase them does not answer that operating question. Apple's component-availability risk says that supply agreements cover many components but gives no assurance that they can be extended or renewed on similar terms, or at all. It also identifies supplier financial difficulties, business failure and industry consolidation as possible limits on supply (Apple Inc., 2024, Risk Factors, printed p. 8). The historical record must therefore distinguish the agreement, its effective term and the supplier's ability to perform. Renewal and delivery need their own evidence before a scenario can rely on them.

More suppliers do not eliminate every shortage. Apple describes significant supply and pricing risks for single or limited sources, while warning that even components available from several sources can face industry-wide shortages and commodity-price fluctuations. Its example of past semiconductor demand and supply shortages reports an adverse effect on obtaining components and products on commercially reasonable terms. The warning that such disruptions could recur preserves a future possibility; the passage supplies no duration, quantity shortfall or loss with which to calibrate it. A scenario may specify a common industry constraint, but it must state that assumption rather than derive independence or a diversification benefit from the number of suppliers.

Custom components present a different constraint. A sole source may initially have insufficient capacity while yields mature or manufacturing expands. Alternatively, the supplier may give priority to common components. A yield problem and a production-allocation decision can both

delay the same product, yet the possible responses differ. Apple separately reports past adverse effects from delayed or constrained component supply and delayed shipments of completed products by outsourcing partners. Component production, final assembly and delivery remain separate activities; an improvement in one need not establish completion of the others. The Note 12 supply discussion and the outsourcing passage retain the same distinction between sources, capacity, partner performance and control (Apple Inc., 2024, Note 12, printed p. 46; Risk Factors, printed p. 8).

For an explicitly assumed delay, compare restoring the current source with switching to another. The disclosure says that consequences can depend on the time needed to obtain sufficient quantities from either route. A named alternative is therefore only a proposed response until component compatibility, available quantities, lead times, price and contractual terms are established. Existing purchase obligations do not identify those facts or the release of the old commitment. The model should retain both responses with their separate prerequisites and delivery dates, leaving an infeasible or unresolved response visible. Only after the relevant events and accounting facts are specified can the resulting purchases, settlements or inventory movements generate journals. The filing does not select a route or assign probabilities to the alternatives.

2.19 Available output can still create losses and obligations

Obtaining components and completing delivery do not establish that a product can safely serve its intended use. Apple's defects discussion covers its hardware and software, third-party components and products, and service outages, slowdowns and errors. It reports that services have sometimes failed to perform as anticipated. It also warns that defects can create personal-injury, property or environmental risks, that specialized uses such as health can increase those risks, and that complex technologies such as artificial intelligence can expose users to harmful or inaccurate content. Detection and repair of every defect are not assured (Apple Inc., 2024, Risk Factors, printed p. 9). For the model, a delivered unit, an acceptable unit and a repaired unit are different states. The source identifies possible transitions between them, not a defect rate or a particular incident.

The financial consequences can extend well beyond inventory. Apple names product liability, recalls, replacement or modification, write-offs of inventory, property, plant and equipment or intangible assets, warranty and other expenses, litigation costs and regulatory fines. Quality can also affect reputation, acceptance, demand and the timing of new offerings. These effects must not all become one inventory expense. A customer obligation and a possible supplier recovery are also separate. The outsourcing passage says partner arrangements may include defect-expense reimbursement while Apple generally remains responsible to the consumer for warranty and out-of-warranty service (Apple Inc., 2024, Risk Factors, printed p. 8). Identifying a supplier does not establish a reimbursement right, amount or receipt date.

Excess stock raises a different question even when its quality is adequate. The inventory-risk passage says Apple records write-downs for product and component inventories that are obsolete, exceed anticipated demand, or have cost above net realizable value. It separately describes necessary cancellation fee reserves for orders of excess products and components. Inventory already held and an order not yet received are different exposures; cancellation depends on the actual order terms. Note 1's first-in, first-out policy identifies the reported cost-flow method, rather than a lot's realizable value or a cancellation charge (Apple Inc., 2024, Risk Factors, printed p. 9; Note 1, printed p. 34).

Apple also describes impairment reviews when circumstances indicate that long-lived assets,

including capital assets held at suppliers and inventory prepayments, may not be recoverable. It reports a write-down for excess carrying value over fair value if impairment is determined. This is the issuer's description of its assessment; an executable calculation still needs the actual asset classification, assessment basis and applicable accounting method. The company's belief that the named assets and commitments are currently recoverable is accompanied by no assurance against future charges. It neither proves a future impairment nor permits the model to erase that possibility. An inventory cost-flow calculation, a cancellation obligation and an asset impairment assessment cannot be substituted for one another.

The link to product transitions is the forecast used before delivery. Apple orders components and builds inventory ahead of announcements and shipments; manufacturing purchase obligations cover forecast requirements, typically for periods up to 150 days. It warns of excess or insufficient output, or failure to use firm commitments fully, when forecasts are wrong. The qualifier "typically" prevents that period from becoming a universal contract limit. Nor does the description make the separately reported Note 12 obligations with remaining terms above one year the same population as manufacturing obligations (Apple Inc., 2024, Risk Factors, printed p. 9; Note 12, printed p. 45).

Consider an explicitly assumed reduction in expected demand. Two companies could hold the same reported inventory amount while one holds saleable components for the new offering and the other holds incompatible earlier components, defective finished goods or commitments it cannot cancel cheaply. Their feasible responses and possible losses differ. Lot identity and cost, quality, compatibility, forecast dates, anticipated demand, realizable value, order terms and customer or supplier obligations distinguish the histories. Recognition and measurement then require their own evidence. The disclosure does not supply those facts or probabilities, so the model preserves the conditional histories without inventing a reserve or treating the forecast as observed customer demand.

2.19.1 Service provision requires continuing access

Adequate components and saleable inventory still leave a service question: can the company continue offering the software and content its customers want? Apple says its products and services require third-party intellectual property, while giving no assurance that the necessary licences will be available on commercially reasonable terms or at all. Its digital-content arrangements can be short and need not renew on similar terms. Content providers can also require new rights-management or security technology before access can continue (Apple Inc., 2024, Risk Factors, printed pp. 9–10). The relevant resource is therefore partly a dated right to offer particular content or functionality. Physical capacity alone cannot establish that right.

Developer support has a separate economic cause. Apple believes customers' hardware purchases depend partly on available applications, whose continued development and maintenance are not assured. It describes developers comparing platforms by market position, anticipated revenue, expected growth and the cost of developing and updating software (Apple Inc., 2024, Risk Factors, printed p. 10). An application can consequently become unavailable or unsuitable even while the hardware remains usable. A scenario needs the affected applications and developers' response; an estimate of component capacity cannot supply either.

Access also depends on how a service uses data and takes payment. Apple reports restrictions on collecting, using and transferring personal information, including transfers within its international group. It separately warns that failure to follow payment-card security standards could increase transaction costs or remove the ability to process cards (Apple Inc., 2024, Risk Factors, printed p. 15). These issuer descriptions identify possible constraints. Which constraint applies depends on the data, entity, jurisdiction, agreement and effective rule.

Suppose a content agreement permits a specified offering only until its expiry date. Under one explicitly assumed history it renews; under another it expires without replacement. Identical physical resources then permit different offerings. The comparison needs the service, customer population, territory, contract term and allowed substitutions, together with any new cost and demand response. A financial calculation must apply the relevant revenue, payment and remedy terms to the service actually provided. Apple's risk statements supply neither those contract facts nor probabilities for the alternatives. The historical record must preserve the continuing relationships as well as the reported balances if it is to support these different futures.

Chapter 3

Representing a company's historical record

Northstar's corrected receivable, Apple's debt measurements and its deferred revenue show why the source record must retain more than a label and a number. Section 2.7 distinguishes a reported balance, a derived movement, a dated expectation and an unquantified obligation. Section 2.8 adds the contract and delivery facts needed to turn a commitment into a future payment. The procedures below determine which assertions enter a model and which remain unresolved.

3.1 The record is a collection of assertions

The original record includes financial statements, notes, accounting policies, auditor opinions, management discussion, segment tables, security terms, regulatory returns and subsequent announcements. A complete ingestion process accounts for every page and every extracted block. A complete financial model is a stronger claim: it would require economically relevant information that public reports may never disclose. AccDSL therefore reports document coverage, semantic coverage and simulation coverage separately.

Each source span receives a disposition: structured observation, narrative assertion, relationship, policy evidence, contextual material, duplicate, or unresolved extraction. "Unresolved" remains visible. A commitment buried in a note is not discarded because it has no current balance-sheet line. An auditor's qualification is retained as evidence about the record; it does not become an automatic numerical haircut. Management's forecast is a dated assertion with an author and modality, not an observed future outcome.

An observation is an immutable record with the fields below. Values inferred from observations are separate records, linked by a derivation. This avoids overwriting a company's statement with an analyst's preferred reconstruction.

Field group	Minimum contents
Identity and source	Observation ID; document hash; filing identifier; PDF page and bounding box or HTML/XBRL locator; literal text; extraction version.

Field group	Minimum contents
Reporting context	Legal entity; reported group perimeter; statement or note; instant or interval; standard and taxonomy versions; dimensions such as segment, product, geography, seniority or measurement category.
Amount	Original string; parsed number; sign interpretation; unit; scale; currency; precision or rounding interval; missing-value category.
Meaning	Source label; taxonomy QName when present; one or more proposed concept mappings; whether the assertion is reported, estimated or conditional.
Timing	Economic validity when known; reporting period; publication and availability timestamps; ingestion timestamp; correction or supersession links.
Reliability and review	Extraction confidence, reviewer decision, source status and known contradictions, all distinct from a probability over histories.

A narrative assertion adds actor, predicate, object, scope, quantifier and modality. For example, "substantially all debt is fixed rate" has a source span and a qualitative quantifier. It does not assert $\alpha = 1$. A reviewed interpretation can introduce a numerical restriction such as $\alpha \geq a$, but the threshold a remains an analyst assumption unless the source defines it. Boilerplate policy language also does not prove that a company holds a particular instrument.

3.2 Different names, different measurements

An accounting concept is more than a row label. "Debt" can refer to legal principal, amortised carrying amount, undiscounted contractual cash flows or a fair-value disclosure. These amounts can all be correct at the same date. Likewise, "revenue" may be reported gross or net, a segment measure may differ from the consolidated measure, and an alternative performance measure may exclude selected expenses. A mapping must state its relationship: equivalent meaning, component of, broader than, presentation reclassification, or measurement transformation.

The common vocabulary will preserve source concepts as first-class identifiers. Company extensions may inherit a broad economic category without being declared identical to a standardised concept. For instance, an extension can be a component of borrowings while retaining an unresolved split between lease liabilities and other financing. Many-to-many mappings are allowed; each mapping has applicable dates, dimensions and evidence. An aggregation mapping does not silently become a measurement conversion.

Standards differ in recognition, measurement, presentation and disclosure. Consequently, conversion between IFRS, US GAAP and Chinese Accounting Standards for Business Enterprises (ASBE) is not generally a dictionary lookup. The original basis remains preserved, and any comparison basis is a separate derived book. The same principle applies to tax and prudential capital. A tax loss carryforward, accounting deferred tax asset and regulatory capital deduction can concern the same economic loss while belonging to different measurement systems.

3.3 Time, revisions and information availability

At least four times matter. A machine may enter service in November; its cost appears in the accounts for the year ending December; the annual report is published in March; and AccDSL ingests the report in April. A later report may revise the December comparative. These times answer different questions and must remain distinct.

The storage model uses validity intervals for economic assertions and transaction intervals for database versions, following the distinction in Snodgrass and Ahn (1986). Publication and availability timestamps are additional domain fields. Database transaction time records what AccDSL contained; it cannot substitute for what investors could have known. An "as known on" query applies an explicit availability rule and does not include a later restatement simply because its economic period is earlier.

AccDSL supports three named views: the original filing view; the latest restated view, with a stated information cutoff; and a harmonised research view, with explicit transformations. No view changes the stored source facts. Corrections have links to the replaced assertion and a reason. Where two sources conflict without a clear supersession relationship, both survive with a conflict record.

Instant values and interval flows have different types. A balance at 30 June is not a six-month flow. Calendar intervals use half-open boundaries $[t_0, t_1)$ internally; a filing period ending 30 June maps to the next boundary at 1 July in its reporting calendar. Quarter-only earnings may be derived from year-to-date values only if the periods, basis, perimeter, units and revision vintage are compatible. The derivation retains both source observations and their correlated rounding uncertainty. Fifty-two/fifty-three-week calendars, stub periods and reporting-date changes require explicit calendar objects.

Accounting-policy changes, estimate changes and error corrections also have different temporal effects. The applicable rule package decides whether a change is prospective, requires restatement, or adjusts opening balances. The engine never infers that effect from a generic "revision" flag.

3.4 The company boundary changes

Legal entity, reporting group and economic exposure are separate dimensions. Acquisitions, disposals, changes in control, joint arrangements, special-purpose vehicles and non-controlling interests can change the reporting perimeter. A group's increase in property need not be capital expenditure; it may reflect an acquisition, a currency translation or a revaluation. The bridge between opening and closing property must identify these movements before a scenario model can extrapolate investment.

The company graph therefore stores dated ownership, control and counterparty relationships. Control is an accounting conclusion with supporting evidence, not a rule that blindly consolidates every majority-owned entity. Group reporting combines entity books with a separate adjustment and elimination book. Intragroup receivables and payables require matching counterparties, currencies and dates. Unmatched eliminations remain unresolved; a balancing "plug" is not a recovered transaction.

Acquired opening balances, goodwill, purchase-price allocation, divested operations and non-controlling interests each have their own event types. Analyses can preserve reported scope or define a constant perimeter, but the latter is a derived modelling assumption, not the original record.

3.5 Further problems that the representation must handle

Several difficulties extend beyond differences in standards, industries and company labels. Recognition and settlement occur on different dates: a sale, invoice, cash receipt and impairment can belong to four periods. Net balances hide gross flows: repeated borrowing and repayment can leave the same closing debt. Netting can conceal offsetting exposures and liquidity needs. Different contractual maturities and repricing dates can share the same current carrying amount. Guarantees, undrawn credit facilities, litigation and covenants can be economically important before an asset or liability is recognised.

Foreign currency introduces transaction currency, functional currency and presentation currency, together with different translation conventions for stocks and flows. Currency translation effects must not be mistaken for cash receipts. Inventory and depreciation depend on vintages or lots; an aggregate stock and average age may be insufficient under a shock. Tax bases can evolve differently from carrying amounts. Deferred tax recognition requires assumptions about recoverability, rather than multiplication by a tax rate alone.

Public information is selectively disclosed and rounded. Duplicate XBRL facts may agree, overlap within rounding bounds or contradict one another. A dash can mean zero, immaterial, unavailable or inapplicable. Tables can mix units, span pages and embed signs in parentheses. Text and table references can be wrong in the source itself. AccDSL retains these distinctions and gives an analyst a local repair path without pretending that extraction confidence settles economic interpretation.

Finally, strategic behaviour matters. A firm may refinance, cut investment, raise equity, change prices, breach a covenant or fail. A bank may preserve a capital ratio by shrinking assets; an insurer may change reinsurance or asset allocation. Such responses cannot be deduced from the balance-sheet identity. They require explicitly calibrated or assumed policies, and alternative policies must remain separate when the available record does not distinguish them.

3.6 Account for a disclosure before translating it

The importer first freezes a filing package: documents, byte hashes, reporting dates, availability evidence and incorporated references. It compares the document's contents page and statement index with detected section and note headings. A missing heading, table continuation or referenced exhibit creates an obligation to resolve; successful text extraction cannot discharge it. Each PDF page and each HTML content unit then has a stable locator and a disposition. Child units such as a table or note belong to a parent page/section; the importer records overlap instead of counting the same content twice.

Three denominators remain separate. Acquired-document coverage counts source units with a disposition, including unresolved units. Semantic coverage counts identified assertions whose meaning and context have been reviewed. Simulation coverage counts the operations and exposures required by a specified query that have supported state and rules. The first can be complete while the other two are partial. An assertion inventory can itself be incomplete; until the page/section reconciliation is reviewed, semantic coverage is a lower-bound inventory rather than a percentage of all financial meaning.

Cross-reference traversal is a work-list algorithm. Start with the filing's statement and note references, resolve each target against the acquired manifest, add newly discovered targets, and stop when no unvisited target remains or a declared resource limit is reached. Document hash plus target locator prevents cycles from expanding forever. A target has one of four results: resolved, duplicate of a resolved source, unavailable, or ambiguous. The last two

remain obligations. Apple's proxy and prior-filed agreements show why closure over acquired files is weaker than closure over incorporated disclosures. A manifest may declare a bounded acquisition scope, but must show these outstanding references beside its completeness status.

A narrative translator records the actor, predicate, object, quantifier, modality, applicable entities/dates and qualifications before proposing any constraint. "All notes are fixed-rate" constrains contractual coupon type; it does not constrain the combined note-and-swap exposure. "May repurchase" creates an authorised action with limits, not an obligatory payment. "Substantially all" remains qualitative unless a separately reviewed assumption gives it a threshold. A conditional guarantee becomes an obligation with a trigger and exposure, not a present liability merely because its maximum amount is known. Unsupported predicates retain their literal text and links to affected balances. The reviewer accepts the translation separately from accepting that the text was extracted correctly.

3.7 Mappings must establish a financial relationship

A mapping record identifies source concepts, target concept, relationship, applicable entity/basis/period/dimensions, evidence, reviewer and any required transformation. Equivalence permits substitution only when all these contexts match. A component relationship permits inclusion in a sum only through a declared disjoint partition. A broader concept supplies containment, not an equality. Reclassification changes presentation while preserving the stated population and total; measurement transformation must supply an actual formula and inputs. These relationships cannot compose indiscriminately: component followed by equivalence remains component, while two measurement transforms compose only if their intermediate bases and information dates agree.

For example, Northstar's product revenue might split into 180 domestic and 80 overseas, while its customer channel split is 200 wholesale and 60 retail. Both classify the same 260. Adding all four numbers gives 520 and is invalid. The partition records therefore share a population ID but have different axes. To remove an axis, `sumOver` requires evidence that members are disjoint and exhaustive for that context. Where the source supplies only named segments, an explicit unallocated member carries the residual; it is not assumed zero. Geography-by-product cells cannot be recovered from the two marginals without further constraints or assumptions.

Apple supplies another counterexample: its vendor non-trade receivables arise from component sales to manufacturing vendors, whose gains reduce product cost of sales when the final products are sold (annual Note 4, p. 38). They cannot be relabelled as ordinary revenue-generating trade receivables without changing meaning. Likewise gross derivative assets and liabilities cannot be silently netted. Company extensions can define these distinctions and dated presentation changes; they cannot override mandatory recognition rules in a selected standards package.

3.8 Selecting original and revised assertions

The query procedure first filters assertions by the declared availability cutoff, then by an optional database-time cutoff. A day-only filing timestamp is conservatively available at the next local midnight in its named timezone; any more precise rule needs evidence. Within the eligible set, explicit supersession removes only the targeted assertions in matching contexts. The superseding assertion must itself be eligible. Follow the acyclic graph transitively; retain all maximal assertions if two eligible corrections conflict. File order and ingestion recency never decide which is true.

The original view selects the initial assertion chain in a named filing; the latest-available view selects eligible maximal versions; the harmonised view adds reviewed transformations to

those selections. A revision to revenue does not automatically revise every note or change the company's accounting basis. An estimate change takes effect in future measurements from its specified date. An error correction may change a comparative and an opening balance bridge, each as a new assertion with its own lineage. Northstar's explicit query fixture in Chapter 2 checks all these clocks without assuming that the most recently stored row is always the appropriate history.

3.9 What an accepted reading permits

Apple's inventory policy illustrates the distance between reading a sentence and reconstructing a business. Its reported use of FIFO supports a proposed cost-formula reading for the inventory and reporting contexts covered by that policy. It does not reveal the quantities and ages of the actual lots, establish that no write-down occurred, or supply the authority for every US accounting treatment. Those are separate inputs to a cost and impairment calculation. Similarly, a straight-line depreciation policy leaves useful lives, residual values, components and available-for-use dates to be established. Section 9.5 tests those limits against the reported asset balances, depreciation and cash payments, then constructs two different asset books compatible with the selected amounts.

The source preparation retains six located Apple policy passages: consolidation of Apple and wholly owned subsidiaries; comparative reclassification; FIFO; straight-line depreciation; the interim statement's estimates and annual-report reference; and its 52/53-week fiscal calendar. Each proposed reading carries the original document hash, extraction hash and exact character slice. These are company assertions. The consolidation sentence, for example, cannot exclude every partly owned entity without a control analysis. The calendar sentence does not make every future fiscal year 52 weeks or make a contractual interest year 365 days. The exact quotations and proposed predicates are retained in `../methods/source-translations-v1.json`; the meaning explained here is not an invented human acceptance of them.

A reading progresses through separate decisions. First, extraction review asks whether the located words and numbers were transcribed faithfully. Translation review asks whether the proposed predicate preserves their scope, quantifier, exceptions and reporting context. A model then admits that predicate, or an explicitly labelled assumption, for a particular purpose and information view. Accounting eligibility subsequently asks whether a method applies to the actual event. A faithful transcription of an erroneous company assertion is still faithful transcription. The four decisions therefore cannot share a single "approved" flag.

An explicit presumption in a standard also differs from filling an unknown with a convenient answer. Article 6 of the retained official ASBE 21 text requires both practical ability throughout use and economic benefit for a substantive substitution right. It specifies a presumption when that right is difficult to determine. Applying that conditional rule requires its stated circumstances; it does not transform missing lessor information into a reported fact. Nor may a translator assign arbitrary numerical thresholds to "substantive", "probable" or "highly probable".

3.10 Three meanings of disclosure coverage

The Apple annual/interim preparation inventories 125 quantities, 13 annual notes, 10 interim notes, 70 exhibit references and seven further package obligations. All 23 notes retain unresolved semantic content; none is recorded as fully formalised. Downloading their pages answers a useful but narrower question than constructing a complete company model.

Three populations make that distinction inspectable. Retained-unit coverage partitions acquired documents into nonoverlapping pages or smaller leaf units, including duplicates and unreadable material. Assertion-review coverage counts the separately enumerated material assertions and their review decisions. Query-exposure coverage identifies the sources, legal facts, methods and authority that a particular result requires. A fully downloaded annual report can have incomplete assertion review, and a reviewed debt total can leave a refinancing query unresolved because an incorporated agreement is missing. No weighted average of these populations establishes completeness.

Preparation begins with declared annual and interim roots and a policy for following incorporated references. Each edge is recorded as satisfied, duplicate or cyclic, outside the declared scope with a reason, inaccessible, or still unexpanded. A bounded traversal saves its unexplored frontier and budget so it can resume. An unknown external extent remains unknown. Acquired proxy and registration documents improve the Apple record, but do not prove that every executed amendment, issuance term and exhibit dependency has been closed. The present real package remains incomplete for that reason.

Container choice can itself create a false missing-source diagnosis. Apple's retained annual PDF contains securities and award exhibits absent from the retained HTML container; other incorporated award forms are present in earlier reports in the retained corpus. Locating those bodies resolves a source-access question. It still leaves their qualifications and outgoing references to be interpreted. Conversely, the cash restriction in Section 2.6 shows why a located note can change the meaning of an already extracted figure: correct arithmetic alone would not detect an erroneous "unrestricted" label.

3.11 Displayed precision does not determine a rounding rule

Suppose a statement displays 100 in millions. The scale establishes how the literal is read; it does not establish how the issuer rounded its underlying amount. Under an explicitly admitted nearest-rounding assumption, an integer display m at scale s permits the conservative compatibility enclosure

$$s(m - \tfrac{1}{2}) \le y \le s(m + \tfrac{1}{2}).$$

The closed endpoints accommodate an unknown tie rule within that nearest-rounding model. A specified tie rule can exclude an endpoint. Truncation of a positive amount instead permits $sm \le y < s(m + 1)$. Thus the illustrative display 100 corresponds to 99.5–100.5 million under the first assumption and 100–101 million, excluding 101, under the second. These are different restrictions on a history.

The retained Apple arithmetic uses the first enclosure as a conditional compatibility assumption; the issuer's rounding convention has not been established. Without that assumption or source support, the record states known display precision and unknown rounding convention and supplies no hard half-unit constraint. A one-unit difference between a displayed total and the sum of displayed components is consequently not enough to establish a contradiction. The source value is scaled once, while its literal and precision metadata remain available for inspection.

3.12 Public availability and system possession are different views

Northstar's correction explains why a result needs a reporting vintage. A historical forecast also needs to establish when the exact document bytes were available. A report downloaded today may carry an old date on its current web page; that does not prove that the same bytes

were public on that date. Public availability evidence and an ingestion receipt answer different questions.

An investor-information query supplies a cutoff K. Each admitted source must have supported public availability no later than K and evidence linking the selected bytes to that availability. Later ingestion is allowed. Later extraction review can read eligible old bytes, but an interpretation that uses later information fails the cutoff on its own dependencies. This view establishes public availability, not actual investor awareness or the ability to trade.

A system-as-of query instead supplies an instant S and requires an immutable receipt showing possession of those bytes by S. Publication cannot replace that receipt. An accepted-layer query also requires review decisions visible by S; a raw-layer query may return a then-unreviewed document. A query asking both what was public and what the system possessed must pass both filters. Unknown publication alone need not block a pure possession question.

Archival exploration permits inspection of retained sources with incomplete timing. With a cutoff, unknown publication yields an unresolved candidate, not a cutoff-qualified selection. Without a cutoff, the archive may display the retained lineage, explicitly without a historical-knowledge claim. A known later version is excluded from an earlier view and cannot suppress an eligible old assertion. An unresolved potentially relevant version keeps the affected answer unresolved even when an older assertion is displayed.

Date-only evidence requires particular care. For publication date D in a named timezone, let $[a_D, b_D)$ be the interval from local midnight to the next local midnight. A cutoff before a_D excludes the source; a cutoff inside the interval leaves availability unresolved; a cutoff at or after b_D supports admission on this date evidence. No arbitrary noon or fixed 24-hour duration is inserted. Missing timezone leaves the instant comparison unresolved.

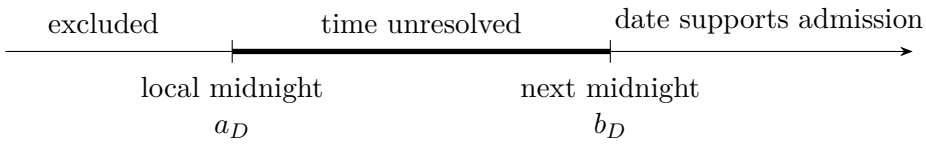


Figure 3.1: A dated publication supports historical admission only after its possible publication interval has ended. The line marks uncertainty about an instant, not a delay imposed on an already known timestamp.

Revision selection follows admission. Original selection returns eligible roots; latest-available selection returns eligible assertions without an eligible superseding descendant for the same context. Unrelated roots and multiple live leaves remain competing assertions. A harmonised view adds explicit transformations with their own admitted sources, method and opening bridges. File order never selects a winner.

The synthetic interim example has original first-half revenue 120, revised first-half revenue 110 and revised annual revenue 260. Compatible subtraction gives second-half revenue $260 - 110 = 150$. The arithmetic $260 - 120 = 140$ is easy but mixes incompatible vintages. An original-view display may show both source amounts while refusing that derived flow. A change in calendar or perimeter likewise requires a specific bridge.

The investigated real sources have not yet supplied a qualifying unresolved same-context contradiction. Known revisions, different reporting bases and synthetic conflicts remain useful cases of their own kinds. A genuine conflict case must first match entity, perimeter, concept, dimensions, unit, basis, period, versions and precision, and investigate extraction and supersession. The absence of a qualifying example in a bounded search is not evidence that the corpus contains no conflicts.

Chapter 4

What the literature contributes

The research problem crosses accounting, databases, programming languages, econometrics and document understanding. Each field resolves part of it. The review below distinguishes results that can be adopted directly, engineering ideas that need adaptation, and evidence that does not establish the intended product capability. Source versions and technical anchors are recorded in the accompanying research ledgers. The search includes work outside the original collection, with public-index searches and backward and forward citation searches through 2 October 2026. It is a scoped design survey, rather than a claim to enumerate every paper in these fields.

4.1 Accounting structure and company dynamics

Nehmer and Robinson (1997), especially Sections 2–5, supplies a useful algebraic starting point: balances occupy a structured state space, transactions change that state, and aggregation relates detailed accounts to reports. This supports a distinction between conservation constraints and permitted transactions. An entry can conserve the accounting total and still be an inadmissible transaction. AccDSL adopts that distinction in its posting validator. Its particular event vocabulary, evidence model and inverse-history semantics are new design decisions, not results established by the algebraic paper.

Doshi (2024) develops an accounting DSL with time, transaction logs and correction mechanisms. The relevant material is in Chapters 4 and 7–9 of the 2024 thesis; the original collection's filename misleadingly suggests 2026. It demonstrates why corrections and dates belong in the language rather than in informal comments. The thesis does not establish recovery of an entire company history from public reports or inference over economically distinct histories. Babcinețchi et al. (2022), pp. 342–345, offers a much smaller accounting language and grammar proposal. It is useful as a syntax precedent, but it provides insufficient semantics or evaluation to govern AccDSL by itself.

Shahnazarian (2004) connects accounting dynamics with behavioural and econometric equations. Sections 2.3–2.8 describe the accounting relationships; the optimisation and econometric discussion in Chapter 3 and the data and evaluation material in Chapters 4 and 8 show how the model is used with standardised Swedish corporate data. This is an important precedent for simulating companies under changed conditions. Its standardised tax-data starting point differs from heterogeneous public-report extraction, and its working-capital funds formulation should not be imported as a general modern cash-flow statement implementation. The transferable idea is the explicit separation of accounting evolution and behavioural assumptions.

Pierson (2020), Sections 3–4 and the reporting structures, makes that separation operational in system dynamics. Receivables, payables and wages accumulate differences between accrual and payment flows; reporting-period accumulators turn instantaneous flows into statements. AccDSL can use these structures as examples of coherent movement schedules. The model's payment delays, inventory approximations and simplified equity treatment remain modelling choices. They are not universal defaults for companies or evidence that an aggregate state is sufficient for all shocks.

Bank models demonstrate why journal generation needs economic state. Halaj (2016), Section 3, combines funding shocks, fire sales, loans, securities and risk constraints. The homogeneous portfolios and specified funding process make the model tractable, but limit instrument-level interpretation. Behn et al. (2019), Section 2, makes managers' choices over asset growth, debt, risk and payouts depend on capital and liquidity constraints. These models motivate interfaces for management policy and regulatory calculations. Their parameterisations must be calibrated and evaluated for the target application; passing their accounting identities would not validate an AccDSL forecast.

Two important neighbouring traditions remain explicit coverage qualifications. McCarthy's 1982 REA model is a foundational candidate for the relation between resources, events and agents; Caiani and co-authors' stock-flow-consistent agent-based benchmark is a candidate for simulation across interacting sectors. Verified primary PDFs could not be obtained in this retrieval campaign after institutional and publisher attempts. Their records are retained in the omission register, and no technical result from either is used to justify the design. AccDSL's event and counterparty records are specified and justified in Chapters 7 and 8. A later multi-company equilibrium engine would require expansion of the stock-flow-consistent literature before implementation.

4.2 Incomplete information and provenance

Financial disclosures are incomplete databases in a substantive sense. Multiple underlying records can yield the same published observations. Libkin (2014), Sections 2–5, explains the semantics of incomplete databases, labelled nulls, conditional tables and certain answers. Repeated occurrences of the same unknown must share a value. Conditional tables can represent dependence between alternatives that independent missing cells cannot express. This is directly relevant to a debt note and a cash-flow table that refer to the same unknown repayment.

AccDSL adopts shared symbolic unknowns and conditions. It extends the database idea with accounting constraints, continuous amounts, temporal transitions and economic event models. A "certain" answer means a statement that holds for every history in the declared feasible family. The family itself may omit unknown mechanisms; certainty is therefore always relative to a model scope. Imieliński and Lipski's original 1984 paper is a foundational backward lead whose publisher copy was inaccessible in this campaign. The inspected Libkin paper supplies the technical discussion actually used here.

Green et al. (2007), Definitions 3.1–3.2 and Section 4, treats provenance as annotations that compose through relational operations. Combining sources and alternative derivations can therefore have formal semantics, rather than being a list of unrelated citations. This is useful for report aggregation and explanation queries. Positive relational operations covered by provenance semirings do not by themselves specify every operation AccDSL needs: negation, nonlinear valuation, optimisation and probabilistic conditioning require their own derivation records. Accordingly, the implementation starts with a typed dependency graph and may use semiring annotations for supported relational subqueries. A dependency edge is not a posterior

probability.

Snodgrass and Ahn (1986) distinguishes historical validity from the time facts are stored. The archived paper is *Temporal Databases*, by Snodgrass and Ahn, not the later Jensen–Snodgrass paper initially returned by a search. Its snapshot, rollback, historical and temporal examples support the need to recover both old business facts and old database beliefs. AccDSL adds public availability time because a database insertion date cannot establish whether a historical investor could have seen a filing.

4.3 Inverse problems, identification and probability

Stuart (2010), Section 2.2, starts from an observation equation $y = G(u)$ and distinguishes inversion, regularisation and Bayesian inference. The discussion makes explicit that a regularising norm or prior is an additional assumption. AccDSL has a similar observation problem, although its unknowns include discrete accounting interpretations and event structures as well as continuous parameters. The infinite-dimensional well-posedness theorems in that paper are not automatically applicable to this mixed, potentially discontinuous model.

The relevant issue is often identification rather than computation alone. Giacomini and Kitagawa (2021), Sections 2.1–2.6 and Theorem 1 of the archived 2020 working version, shows how a class of priors can preserve uncertainty within an identified set. A likelihood that is flat over the unresolved part does not learn the conditional prior over that part. Posterior lower and upper probabilities summarise a class of posterior beliefs. The paper's regularity conditions and application to structural vector autoregressions do not prove that an arbitrary accounting program admits the same algorithms or guarantees. The design borrows the distinction between learnable quantities and assumption-dependent distributions; the simple accounting bounds in Chapter 6 are derived directly.

Hoeting et al. (1999), equations (1)–(3), gives Bayesian model averaging as a weighted sum of model-conditional distributions, with weights determined by model priors and marginal likelihoods. Sections 3 and 5 discuss computation and prior specification. This supports an optional weighted mode, provided AccDSL specifies the likelihood, priors and candidate model space. It does not justify giving equal mass to every syntactically different journal reconstruction. Two encodings of the same economic history must not gain extra weight by duplication. Model averaging also remains conditional on the considered model family, rather than repairing missing mechanisms.

Scenario semantics need similar care. Pearl (2009), Section 3.2.1 and equations (5)–(6), distinguishes an intervention that replaces a structural equation from ordinary conditioning. AccDSL will distinguish a supplied exogenous path, an observational conditional forecast and an intervention. A rise in rates imposed by a stress test is not automatically distributed as the collection of historical periods in which rates happened to be high. Identifying a causal effect requires assumptions beyond reproducing accounts.

4.4 Languages for rules and contracts

van Deursen et al. (2000) surveys DSL benefits, risks and implementation options. Mernik et al. (2005), Sections 1.1 and 2.2–2.5 of the archived CWI report, treats application libraries as serious competitors and distinguishes deciding, analysing, designing and implementing a DSL. Its implementation decision tree explicitly includes embedding in a host language. The literature therefore supports a conditional design decision: a DSL is attractive when domain

constraints and repeated notation justify it, but an ordinary typed library may be cheaper and equally expressive. AccDSL will implement one shared representation accessible from both a Python API and a textual language, so syntax does not become a prerequisite for testing the accounting model.

Merigoux et al. (2021), Sections 3–4 and Figures 8–9, provides Catala's scopes, exceptions, default calculus and translation into a conventional functional target. Its explicit empty and conflict outcomes are valuable precedents for accounting rules with exceptions. They are different from probabilistic uncertainty. AccDSL will preserve this distinction: conflicting applicable rules produce a diagnostic, while an evidence-supported alternative reading produces a conditional model branch. The paper's compiler correctness results do not establish correctness of our accounting rules or of an automatic translation from disclosures.

The inspected LegalMath code offers reusable ideas in its shared typed interpretation, partial observations, source references and validity/knowledge times. Its ordinary frontend execution selects a reading. AccDSL needs an additional execution mode that retains a family of readings, continuous unknowns and economic state. A bounded candidate generator is useful for search, but its candidate limit does not prove completeness.

Peyton Jones et al. (2000), Sections 3–5, builds complex financial contracts from combinators and separates their compositional meaning from a chosen valuation model. The ACTUS technical specification similarly separates contract states, scheduled events, state transitions and payoffs, with risk factors supplied through an interface (ACTUS Financial Research Foundation, 2026). These are strong reasons to represent contractual obligations independently of accounting classifications. Neither contract cash flows nor a derivative price determine the recognition, measurement and presentation rules of a complete IFRS or US GAAP book.

4.5 Extraction and financial reasoning

The document literature supports a staged ingestion process. PubTables-1M (Smock et al., 2022), in its canonicalisation procedure and evaluation section, distinguishes table detection, structure recognition and functional analysis. Its GriTS evaluation compares reconstructed table structure. TableBank (Li et al., 2020), Sections 3–4, uses document structure to obtain weak supervision for table detection and recognition. GTE and FinTabNet (Zheng et al., 2021), Sections 3–5, provide a financial-document setting and company-level dataset splits. These sources help select extraction components and evaluation units. Their table scores do not establish that a debt note has been interpreted correctly across years or that every off-balance-sheet exposure has been found.

The Docling report (Auer et al., 2024), Section 3, describes PDF parsing, page layout analysis and table structure recognition within a document-conversion pipeline. The locally archived version lists Christoph Auer first; this version difference is recorded in the catalogue. Docling is an ingestion candidate whose output must retain source locations and undergo accounting validation. Its code and model licences must be considered separately when packaging a deployment.

FinQA (Chen et al., 2021), Sections 3–6, provides questions with executable reasoning programs and supporting evidence. TAT-QA (Zhu et al., 2021), Sections 2–4, combines table cells and text spans with aggregation operations, and explicitly treats numerical signs in evaluation. DocFinQA (Reddy et al., 2024) extends financial questions to long documents and evaluates retrieval/context choices. FinBen (Xie et al., 2024) broadens the collection of financial tasks and model evaluations. These benchmarks can test retrieval and local numerical reasoning. Their question-answer format does not require a complete multi-period company representation,

joint alternative histories or scenario-dependent journal generation.

A directly relevant recent competitor is the auditable RAG architecture of Cosentino et al. (2026). Sections 3.1–3.2 describe persistent extraction, enrichment, retrieval and provenance records; Section 4.6 checks citation resolvability and failure recovery. Section 5 limits the interpretation of those results to the evaluated document-intelligence tasks. AccDSL should reuse the idea of persistent evidence and recoverable processing, while adding executable accounting and economic semantics. No performance comparison with that architecture has been run here.

4.6 Solvers and probabilistic execution

de Moura and Bjørner (2008), Sections 2–3, describes Z3's interfaces and theory-solving architecture. It supports choosing an SMT solver for logical choices and arithmetic constraints, but does not make arbitrary nonlinear simulation tractable. The first solver fragment is deliberately bounded: finite choices, exact linear arithmetic and selected calendar relations. Unsupported formulas remain in the representation with a solver diagnostic.

van de Meent et al. (2018) develops probabilistic programs through sampling, observations and inference; the inspected material includes the first-order language and weighted sampling discussion. Stan (Carpenter et al., 2017), its model language and sampler-output discussion, and Pyro (Bingham et al., 2019), Sections 2–3, show different ways to separate a probabilistic model from inference machinery. They are candidates for optional inference over an already-defined economic model. Exact accounting identities and discrete rule selection should not be turned into nearly satisfied likelihood penalties merely to fit an inference library. Nor should successful sampler execution be described as evidence that all interpretations were explored.

4.7 The remaining design problem

The combined literature provides algebra for journals, semantics for missing information, temporal storage, provenance, contract languages, rule languages, inference methods and extraction tools. The unresolved integration problem is to retain their distinctions while connecting them: a disclosure constrains economic histories; a history supplies contract and behavioural state; a scenario drives events; accounting rules produce journals; reporting rules render statements; and every step remains explainable.

The design is therefore an integration proposal with several local mathematical constructions. Its originality and empirical advantage have not been established by this survey. Its practical merit will depend on the acceptance cases, accounting review and comparative implementation work in Chapter 14.

Chapter 5

Architecture and reuse

5.1 One representation, several authoring methods

The architectural centre is a typed intermediate representation (IR), not a text parser. A document extractor, a human editor, a Python library and the AccDSL compiler all create the same records. This allows a modeller to begin with a reviewed structured file and later add convenient notation. It also allows source extraction to improve without changing accounting semantics.

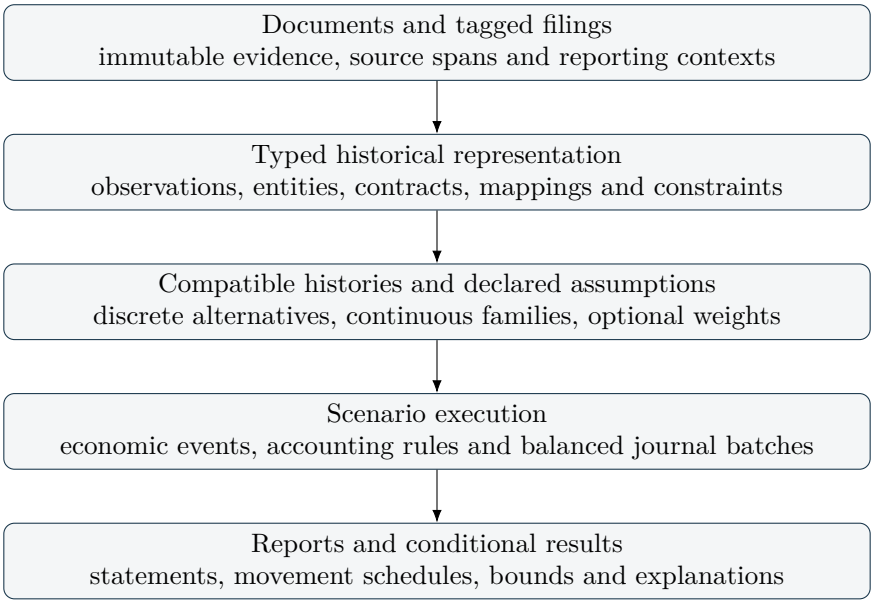


Figure 5.1: The simulation path. Every transformation records its inputs and assumptions; the source record remains available throughout.

Evidence storage and economic modelling must remain separate. If a company reports 100 of borrowing, a model can assume two loans of 60 and 40 without creating a new reported fact that the company disclosed those loans. The contract library consumes the assumed terms, the accounting module consumes the resulting events, and the explanation graph records the assumption at each affected result.

5.2 Why a DSL, and when another approach is better

The strongest baseline is a typed schema, SQL storage and ordinary Python code. That combination can implement the whole product. Its disadvantage is that repeated accounting requirements can become conventions scattered across functions: implicit currency conversion, missing period checks, hidden default values and duplicated rule selection. A bounded language makes these requirements part of typing and compilation. Its costs are language maintenance, diagnostics, migration and training.

Approach	Useful role	Limitation for this product
Typed schema and Python	Reference executor, flexible economic models, initial authoring API	Domain restrictions require disciplined APIs or a validator; arbitrary code complicates static analysis and provenance.
XBRL and OIM	Tagged evidence, contexts, dimensions, taxonomies and exchange	Reported facts and reporting relationships do not recover hidden economic histories or specify scenario behaviour.
Ledger languages	Recording known transactions, balance checks, export and differential testing	Alternatives, source interpretation and contract exposure need additional semantics.
Knowledge graph / SQL	Relationships, lineage, temporal queries and evidence navigation	A graph or schema alone does not define event execution or the probability meaning of alternatives.
Rules / Catala	Dated policy logic, defaults, exceptions, traceable rule evaluation	Ledger state, event scheduling and ensembles require an enclosing runtime.
Probabilistic program	Priors, likelihoods and inference over model parameters	Needs declared probability measures; cannot turn unspecified ambiguity into probability or supply accounting rules.
Spreadsheets	Familiar review and small transparent financial models	Versioned evidence, shared unknowns and branching histories are difficult to govern as the model grows.
AccDSL over shared IR	Common domain checks, reusable templates, explicit alternatives and deterministic posting semantics	Requires implementation and evaluation; broad notation without stable semantics would add cost.

The decision is to implement the IR, validators and Python reference API first, then the textual frontend. The same conformance cases must pass through both frontends. If the DSL fails to reduce ambiguity in representative authoring tasks, the IR and library remain useful. This is a reversible product decision, not a claim that a new language is inherently better than a library.

5.3 Open-source components

The following survey uses official repository descriptions, licence files and selected source interfaces archived on 2 October 2026. It is a source inspection, not an executed integration or performance evaluation. The machine-readable tool ledger records retrieval paths and available version metadata. Dependencies must be pinned when implementation begins; a branch name observed during this survey is not a release requirement.

Component	Proposed use and checked boundary	Licence / decision
Arelle	Parse and validate XBRL/iXBRL. Inspected `ModelFact`, context and unit accessors. Preserve taxonomy extensions and full contexts when importing.	Apache-2.0 core; first-choice adapter.
Docling	PDF layout, text and table conversion. Inspected converter interface; retain page coordinates and extraction settings.	MIT code; assess model licences separately. Candidate.
Table Transformer	Table extraction baseline associated with PubTables-1M; check finance-domain performance before selection.	MIT; repository marked archived in inspected metadata. Baseline, not a maintenance default.
Beancount	Posting and transaction data types; export selected concrete branches and compare balances.	GPL-2.0 project metadata; integration and redistribution choice needed. Optional adapter.
Ledger	Known-journal reference and export. Inspected transaction balancing code and manual date/assertion behaviour.	BSD-style licence; optional reference tool.
hledger	Alternative ledger import/export and accounting checks. Repository documentation inspected.	GPL-3.0 project metadata; optional process adapter.
Catala	Potential backend for deterministic policy modules; defaults and exceptions have useful semantics.	Apache-2.0; prototype only after core execution semantics stabilise.
ACTUS	Contract schedules, payoffs and transitions; external risk-factor interface. Verify event schedules against ACTUS tests.	Service Apache-2.0; technical specification CC-BY-SA-4.0. Candidate contract adapter.
QuantLib	Calendars, cash flows and valuation. Floating coupon source separates fixing dates and pricing. Accounting treatment remains external.	Modified BSD-style licence; candidate valuation adapter.
OpenFisca	Versioned tax-benefit formulas as a rule-system reference; inspected variable/formula metadata. Corporate-tax coverage must be supplied.	AGPL-3.0; optional, not the core runtime.

Component	Proposed use and checked boundary	Licence / decision
Z3	Finite alternatives, logical compatibility, linear constraints and conflict explanations.	MIT licence text; first solver candidate.
DuckDB	Local analytical queries over extracted facts and results.	MIT; candidate alongside SQLite transactional storage.
Stan	Continuous probabilistic models and diagnostics with an explicit target. Inspected model interface and paper.	BSD-3-Clause; optional inference backend.
Pyro / PyMC	Alternative probabilistic backends for model-specific inference. Pyro design paper and project descriptions inspected.	Apache-2.0 / Apache-2.0 licence texts; optional, no default selection.

The licence entries identify the inspected projects, not a legal conclusion about every combination of code, models and datasets. The default product architecture can communicate with optional tools through explicit adapters. This does not, by itself, settle redistribution obligations. That question is resolved when a concrete dependency and distribution method are selected.

XBRL's Open Information Model supplies standardised facts with concepts, entities, periods, units and dimensions (XBRL International, 2021). Those facts are an excellent ingestion source. AccDSL retains their original contexts and decimals rather than flattening them immediately into a company-by-year table. Calculation relationships can help validate reporting totals, but are not a general description of the transactions that generated those totals.

5.4 Responsibilities that AccDSL must implement

AccDSL owns the evidence and temporal model, typed accounting concepts, shared unknowns and alternatives, rule-package selection, journal validation, consolidation provenance and report explanations. It also owns the distinction between feasible sets and distributions. These responsibilities cannot be delegated implicitly to whichever parser or solver is convenient.

External components receive narrow requests and return typed responses. A contract adapter returns scheduled events, cash-flow quantities, state changes and source/model references; it does not directly mutate the ledger. A valuation adapter returns a measure, currency, date, method and numerical status; the accounting package decides whether and where to recognise the change. A solver returns feasibility or bounds with scope and completeness status. A language model proposes source-linked mappings or interpretations; it cannot commit an unsupported assertion as fact.

The architecture consequently permits deterministic reference execution, alternative solvers and later acceleration without changing the meaning of a historical observation or a journal entry. Performance work begins only after the same semantic cases pass through these interfaces.

Chapter 6

Historical compatibility and future simulation

6.1 Define what is observed before inverting it

Let D_k be the disclosures available under information cutoff k. Each numeric observation d_i has a context: entity or group perimeter, concept, measurement basis, dimensions, currency, period and precision. These contexts are part of the observation operator, not additional columns that can be discarded after extraction.

A candidate history h contains an opening economic and accounting state, events through the last historical date T, contract terms, accounting policies, consolidation relationships and latent quantities. Let M denote the declared model scope: its event templates, admissible rule versions and allowed unknowns. It excludes arbitrary events that the chosen model has no way to represent. The reporting operator $O_i(h)$ produces the quantity to which observation i refers, including its basis and perimeter.

Reported numbers are often rounded. If a number v_i is rounded to the nearest unit q_i, an illustrative admissible interval is $[v_i - q_i/2, v_i + q_i/2)$ under the corresponding tie convention. This half-open endpoint choice is used only when that tie rule is declared; Section 3.11 gives the conservative closed enclosure when nearest rounding is assumed but the tie rule is unknown. The actual rounding convention, scale and XBRL accuracy metadata govern the imported constraint. A disclosure with unspecified precision does not acquire a convenient tolerance silently. Write $\mathcal{I}_i$ for the justified observation interval, and $C_M(h)$ for the joint accounting, contractual and semantic constraints. Then the historical compatibility set is

$$\mathcal{H}_M(D_k) = \{h \in \mathcal{H}_M : C_M(h) \ \wedge \ O_i(h) \in \mathcal{I}_i \text{ for every applicable } i\}. \tag{6.1}$$

Qualitative assertions enter as reviewed logical conditions, rather than numeric intervals. Incompatible source contexts are never forced into the same equation.

Equation (6.1) distinguishes three outcomes. An empty set means that the observations and declared model are inconsistent. A singleton means that this model identifies a history at its chosen resolution. A larger set preserves unresolved alternatives. An empty set does not prove that the company's accounts are false: a wrong extraction, missing event type or incorrect perimeter can cause the inconsistency. A singleton does not prove that the model contains the real history.

6.2 Why published statements cannot identify all transactions

The simplest derivation uses fixed account mappings and a fixed basis. Let $z \in \mathbb{R}^n$ contain unknown historical movement amounts. For m disclosed aggregates, let $A \in \mathbb{R}^{m \times n}$ be the aggregation matrix and $y = Az \in \mathbb{R}^m$. If there is a nonzero vector v with $Av = 0$, then

$$A(z + v) = Az + Av = y.$$

Both movement vectors give the same reports whenever both satisfy the remaining constraints. Thus the feasible family is the affine set $z + \ker A$ intersected with those constraints. Accounting requirements can shrink this family, but do not generally eliminate it. For example, cash collections in a receivables bridge are constrained by

$$AR_T = AR_0 + \text{credit sales} - \text{collections} - \text{write-offs} + \text{other classified movements}.$$

If the note does not distinguish acquisitions, currency effects and write-offs, several collections paths may fit the same closing balance.

There is a second ambiguity beyond aggregation. In Equation (1.1), identical historical movements arise from different contractual reset terms. Even recovering z exactly would leave these terms unidentified. AccDSL therefore infers economically relevant states and contract terms alongside journal movements. A linear inverse solver over statement lines alone cannot solve this second problem.

6.3 The accounting state and its invariants

For one entity book with n_a accounts in one functional currency, let $b_t \in \mathbb{Q}^{n_a}$ be signed account balances, with debit amounts positive and credit amounts negative. Exact decimal amounts can be represented as rationals with a power of ten denominator. A journal batch j_t contains the signed functional-currency amounts posted by one atomic event or explicitly grouped event batch. Its balance condition is

$$\mathbf{1}^{\mathsf{T}} j_t = 0, \qquad b_t = b_{t^-} + j_t. \tag{6.2}$$

If $\mathbf{1}^{\mathsf{T}} b_0 = 0$, substitution gives $\mathbf{1}^{\mathsf{T}} b_t = \mathbf{1}^{\mathsf{T}} b_{t^-} + \mathbf{1}^{\mathsf{T}} j_t = 0$. Induction establishes conservation through every committed batch. Opening balances must therefore be supplied as a balanced, explained entry or a balanced imported state with unresolved composition explicitly identified.

This proof establishes a bookkeeping invariant. It does not establish that an amount belongs in the chosen account, that a contract exists, or that an accounting standard permits a treatment. Posting validation additionally checks account roles, ownership, currency conversions, applicability of rules and event preconditions.

The balance vector includes temporary revenue and expense accounts during the period. Period closing transfers their net balance to equity through balanced entries. A reporting function can include current profit in equity before closing, or use the closed book afterwards, but must not include it twice. Income and cash-flow reports use classified movements and exclude mechanical closing transfers from operating flows. Consequently statements are functions of balances, journals, contract state and reporting rules:

$$Y_{[u,v)} = R(b_{v^-}, J_{[u,v)}, X_{v^-}, \rho, [u, v)). \tag{6.3}$$

Here ρ specifies the reporting basis and perimeter, and X contains economic state beyond ledger balances. Cash-flow classification is part of R, not recoverable from the closing cash balance alone.

Each book has its own functional currency. Quantities such as shares, tonnes and foreign-currency units are typed separately and never added to this balance equation. A foreign-currency posting stores its original units and the functional-currency valuation with its rate source. Translation into a group's presentation currency and consolidation adjustments occur in a separate book under explicit rules.

6.4 Simulate events from economic state

For a chosen history h, the state at T is $X_T(h)$. It contains balances, open contracts, lots or cohorts, unpaid accruals, commitments and other quantities needed for future evolution. Let s identify a scenario definition, U_t^s its supplied economic drivers, a_t management's actions, and ε_t future innovations. A behavioural policy π chooses a_t using the permitted information $\mathcal{F}_t$:

$$a_t = \pi(X_t, U_{\le t}^s, \mathcal{F}_t; \theta), \tag{6.4}$$

$$e_{t+1} = \mathcal{G}(X_t, a_t, U_{t:t+1}^s, \varepsilon_{t+1}; \theta), \tag{6.5}$$

$$(j_{t+1}, m_{t+1}) = \mathcal{A}(e_{t+1}, X_t, r_{t+1}), \tag{6.6}$$

$$X_{t+1} = \mathcal{T}(X_t, e_{t+1}, j_{t+1}, m_{t+1}). \tag{6.7}$$

The generator $\mathcal{G}$ emits events, such as a sale, rate reset, payment or default. The accounting function $\mathcal{A}$ applies rule package r and returns journal postings plus measurement annotations m. The transition $\mathcal{T}$ updates economic and ledger state atomically. A default can therefore change the loan's legal status, expected recoveries, recognised allowance and cash collections through related, but distinct, records.

This ordering is a reference abstraction. Within a date, event dependencies and phases specify the actual order. Expectations used to measure an allowance or provision must be calculated from the information permitted at that date. They cannot use the realised future path of the simulation. An inner expectation model may use quadrature, analytic formulas or nested simulation; its approximation and seed belong in the measurement annotation.

Some scenarios specify a deterministic macroeconomic path while leaving idiosyncratic defaults random. Others supply a joint stochastic driver model. A risk-neutral measure used to value a derivative is distinct from the real-world measure used to simulate business outcomes. Every stochastic adapter declares its measure and conditioning information. The accounting function receives the resulting valuation with that declaration.

6.5 Preserve sets and distributions separately

Let $K_{h,s}$ be the probability law of future reports produced by the forward model conditional on history h, scenario s and its declared behavioural assumptions. A deterministic simulation is the special case of a point mass. When no probabilities over historical explanations are justified, the natural answer is a family of laws,

$$\mathcal{K}_{M,s}(D_k) = \{K_{h,s} : h \in \mathcal{H}_M(D_k)\}. \tag{6.8}$$

The system can report each explanation, bounds on a deterministic outcome, or lower and upper event probabilities across this family. Those are different result types. A range of conditional means is not a confidence interval, and an envelope of conditional quantiles is not itself a single predictive distribution.

If a defensible probability measure $\mu(dh \mid D_k, M)$ supported on $\mathcal{H}_M(D_k)$ is supplied, histories and report outcomes must have specified measurable spaces. Assume that $K_{h,s}$ is a probability

kernel: it is a probability measure on report outcomes for each h, and $h \mapsto K_{h,s}(B)$ is measurable for each measurable event B. The predictive law is then

$$\mathbb{P}(Y^s \in B \mid D_k, M) = \int K_{h,s}(B)\, \mu(dh \mid D_k, M). \tag{6.9}$$

For discrete explanations this is the usual weighted sum. For continuous unknowns, integration uses the declared measure over those unknowns, not a count of solver solutions. Bayesian weights require a prior and a likelihood; expert weights remain labelled as elicited. A likelihood must not treat a statement total and all its components as independent repeated evidence. Deterministic equality constraints require a reduced parameterisation or proper conditional-measure treatment, rather than an undocumented Gaussian penalty with near-zero variance.

Suppose the fixed-loan explanation receives weight w and the floating-loan explanation receives $1 - w$, solely as an illustration. Under the rate shock,

$$\mathbb{P}(I = 5) = w, \qquad \mathbb{P}(I = 7) = 1 - w, \qquad \mathbb{E}[I] = 7 - 2w.$$

The expected expense is useful but hides the two different outcomes. For a general scalar output with finite second moment, put $m(h) = \mathbb{E}[Y \mid h]$. Write $Y - \mathbb{E}[Y] = (Y - m(h)) + (m(h) - \mathbb{E}[Y])$. The cross term vanishes because $\mathbb{E}[Y - m(h) \mid h] = 0$. Taking expectations of the squared expression gives

$$\mathrm{Var}(Y) = \mathbb{E}_h[\mathrm{Var}(Y \mid h)] + \mathrm{Var}_h(\mathbb{E}[Y \mid h]). \tag{6.10}$$

The first term is uncertainty within a chosen explanation; the second captures differences between explanations. This decomposition requires the measure μ. Without it, the system reports the family in Equation (6.8), rather than manufacturing a variance.

Weights themselves can be uncertain. For an admissible class $\mathcal{W}$ of measures, AccDSL can compute

$$\underline{p}(B) = \inf_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h), \qquad \overline{p}(B) = \sup_{\mu \in \mathcal{W}} \int K_{h,s}(B)\, d\mu(h).$$

If every measure supported on the feasible histories is allowed and point masses are included, these bounds reduce to the infimum and supremum of $K_{h,s}(B)$ over histories: averages cannot lie outside the range of their integrands, and point masses attain, or approximate, the extremes. Narrower weight restrictions require an explicit justification and a separate optimisation problem.

6.6 Resolution and computational limits

For a requested horizon H and scenario family $\mathcal{S}$, histories may be grouped only when they have the same conditional law for all requested outputs:

$$h_1 \sim_{H,\mathcal{S}} h_2 \quad \Longleftrightarrow \quad K_{h_1,s} = K_{h_2,s} \text{ on those outputs for every } s \in \mathcal{S}.$$

This definition explains why individual sales can sometimes be replaced by cohorts, while fixed and floating loans usually cannot be merged for an interest-rate stress. Exact equivalence can be difficult to prove. A grouping justified only by approximate numerical agreement carries an error criterion and is not labelled exact. Source provenance remains available even when execution shares a common state.

Bounds require comparable honesty. A linear program can provide certified extrema for a supported linear model. A set of feasible sampled histories provides observed extrema: its minimum is no smaller than the true infimum, and its maximum is no larger than the true supremum. It therefore supplies an inner exploration of possibilities, not a certified outer bound.

Interval methods may provide outer bounds that no single history attains. Every result records which of these meanings applies.

The initial implementation offers exact bounded choices and linear constraints, symbolic continuous families such as $\alpha \in [0, 1]$, and explicit scenario execution at selected feasible states. General nonlinear partial identification, automatic model discovery and exhaustive enumeration of all economic histories are research extensions, not promises made by the core language.

6.7 A solver formulation for ordinary rounded disclosures

The debt example used an exact principal. Real disclosures often do not. Suppose debt is shown as 100 and interest as 5, each rounded to the nearest unit with the declared half-open intervals $D \in [99.5, 100.5)$ and $I_0 \in [4.5, 5.5)$. Using an unknown fraction α and unknown total D would introduce the product αD. That expression is bilinear and is outside the promised linear fragment.

Instead use principal amounts F and V. The compatible family satisfies

$$F, V \geq 0, \quad 99.5 \leq F + V < 100.5, \quad 4.5 \leq 0.05(F + V) < 5.5. \tag{6.11}$$

Future interest is $I_1 = 0.05F + 0.07V$. Its infimum is $0.05 \times 99.5 = 4.975$, attained with $F = 99.5, V = 0$. Its supremum is $0.07 \times 100.5 = 7.035$, approached as $F = 0$ and V tends to 100.5 from below, but not attained. A closed interval is a valid enclosure, but its upper endpoint must not be reported as attainable. The exact solver check separately tests that no better value exists and whether each endpoint is attainable.

For $D > 0$, the amount representation is equivalent to the fraction representation through $D = F + V$ and $\alpha = F/D$. At zero principal the fraction is undefined and must be omitted or treated as irrelevant, not recovered. A known fraction restriction $\alpha \geq a$ becomes $F \geq aD$ when $D > 0$ and a is fixed. If both a and D are unknown, that product remains nonlinear. Reformulation has a domain of validity; it is not permission to send arbitrary formulas to a linear solver.

The supported exact fragment contains rational linear equalities, weak and strict inequalities, and finite Boolean or enumeration choices. Guard partitions become finite disjunctions. Posting in integer cents introduces a different integer problem if performed before bounding. Consequently the request specifies whether bounds apply before quantisation, after a monotone declared rounding map, or to a bounded integer model. Approximate numerical valuations carry error metadata outside the exact rational kernel. Solver timeout, unsupported theory and unbounded objective have distinct responses; none is relabelled as infeasibility.

6.8 Building the family with explicit templates

The reconstruction input is a model-scope record, not an invitation to invent unlimited journals. It lists event templates, slots per entity and period, allowed contract classes, source mappings, opening-state domains, calendars, resolution and excluded operations. A template gives an incidence vector for postings and a transition for economic state. Northstar's annual bridges arise by substituting its collection, purchase and payment templates into opening-plus-movement identities.

For each allowed slot, create an activity flag and nonnegative amount; an inactive slot has zero amount. Instantiate the template's posting and state constraints, unify variables referring to the same observation or economic contract, and match report expressions to source domains

through validated context mappings. Add cross-note restrictions and choice-group dependencies to this same formula. No separate account-by-account reconstruction is allowed. Mark every constraint with its evidence, rule or assumption origin.

Enumerate the finite structural choices in a deterministic order. For each cell, ask for feasibility and retain the continuous formula, rather than enumerating real-valued solutions. An infeasible cell preserves a sufficient conflicting subset; an undecided cell preserves its formula and reason. Search is complete within the declared scope only when every finite cell has been decided, every applicable observation has been matched or explicitly qualified, and no resource truncation remains. A witness establishes existence in one cell. It does not establish complete search, and complete search does not establish that the scope includes the company's real history.

In Northstar the declared templates identify annual collections, purchases and supplier payments, but retain every $F \in [0, 100]$. Adding a verified note that $F = 60$ narrows this family by conjunction. Adding a proposed "all floating" reading to that revised view produces a rejected cell with the conflicting note attached. Adding a rule that expenses a capital contribution would instead fail treatment admission: inconsistent evidence and unsupported accounting have different diagnoses.

6.9 Which details may safely be combined

A resolution record names the outputs, horizon, scenario family, grouping key, required state, transformation and exact or approximate status. Safe exact transformations are deliberately few. Cash balances in the same book and currency can be added. Homogeneous inventory lots may be combined under a weighted-average policy if ownership, cost basis, impairment state and future costing treatment coincide. Debt principal can be added only within a group sharing currency, coupon/reset function, fixing and payment calendars, maturity, options, guarantees and measurement method. The transition and reporting maps must be additive on that group.

The last condition supplies a short proof obligation. If a transition obeys $T(x_1 + x_2, u) = T(x_1, u) + T(x_2, u)$ and reporting is additive for every admitted driver u, induction preserves equality after grouping. This does not apply to covenant thresholds, counterparty netting, nonlinear guarantees or lot-specific impairment. Two fixed loans with the same dates and rate can share an execution state under this test. The fixed/floating pair cannot; neither can two policies with different guarantees merely because their current reserves match.

A changed horizon, scenario, rule version or requested output invalidates the resolution approval and reruns the sufficiency check. Extending beyond a debt maturity requires schedule detail that an earlier short-horizon view could omit. Approximate cohorts retain the detailed source references and a stated error criterion tied to outputs. Unless that error is bounded or evaluated, the grouping remains an approximation awaiting evidence, not an exact quotient of the history family.

6.10 A finite probability model that can actually be evaluated

Optional probability weighting begins with disjoint economic hypotheses and a declared joint measure. As an illustration, assign fixed and floating hypotheses prior probabilities 0.6 and 0.4. The stress itself leaves these weights unchanged and gives mean interest $0.6(5) + 0.4(7) = 5.8$. Suppose genuinely new observed evidence has likelihood 0.2 under fixed and 0.8 under floating. The unnormalised posterior masses are 0.12 and 0.32; normalisation by 0.44 gives 3/11 and 8/11. These likelihoods are illustrative inputs, not extraction-confidence scores.

The finite reference algorithm enumerates joint hypothesis and innovation states, evaluates their declared joint mass and any observation likelihood once, checks the nonzero normalising constant, and propagates the entire statement vector on each state. It sums masses only across disjoint events. Two overlapping labels such as “fixed share at least one half” and “all fixed” are query views and cannot become two independent mixture components. Totals and their components share an observation block so their evidence is not counted twice. Continuous inference needs an additional parameterisation, density or conditional measure and an assessed numerical method; the finite algorithm supplies no guarantee for those extensions.

Under the prior, before the new observation is admitted, if the floating branch has an unsupported valuation operation, its mass 0.4 remains unresolved. The run may report the fixed result with successful mass 0.6, but cannot report 5 as the unconditional mean by renormalising away the failure. A separately requested mean conditional on successful execution must show that conditioning event and its mass. For unweighted families, an unresolved branch blocks a complete envelope over the family. Economic default is retained as a model outcome with remaining obligations, rather than being treated as a computational error to remove.

6.11 Numerical precision is part of a financial method

The affine family is useful for Northstar because the scenario fixes prices, rates and operating coefficients while the debt share remains unknown. Other contracts require roots or discount factors. Their numerical treatment must state what is exact, what is enclosed and when a posted amount is determined. An accounting name on a library call does not make its result exact.

Consider positive payments C_i at positive times t_i and a positive price P. The effective annual rate solves

$$f(r) = \sum_i \frac{C_i}{(1+r)^{t_i}} - P = 0, \qquad r > -1.$$

Each term discounts a payment to the valuation date. In this positive-payment case,

$$f'(r) = -\sum_i \frac{t_i C_i}{(1+r)^{t_i+1}} < 0.$$

Consequently a certified sign-changing bracket contains exactly one root. The method uses stated bracket bounds and an expansion cap, evaluates signs with outward error bounds, and bisects until the requested rate or value enclosure is reached. An uncertified sign, unavailable bracket or exhausted accuracy budget yields an unresolved numerical result. Mixed-sign cash flows can have several internal rates of return; the preceding monotonicity argument does not apply and cannot justify returning the first Newton iterate that converges.

Net proceeds 100 and a payment 121 in two years give exactly 10%. An immediate fee of 2 changes proceeds to 98 and changes that root. Annual-effective 21% for half a year has the exact accumulation factor $\sqrt{1.21} = 1.1$, so interest on 100 is 10. Simple proration would give 10.5 and computes a different contract. At annual-effective 10%, the half-year factor $\sqrt{1.1}$ is irrational. For rational time p/q and positive $1 + r$, exact rational power comparisons can enclose the positive qth root of $(1 + r)^p$. A negative p first requires the reciprocal. General exponential discounting needs a separately specified error-controlled method.

A posting quantum of 0.01 does not eliminate this uncertainty. The whole certified monetary enclosure must lie in one rounding cell under the admitted tie rule. An enclosure from 1.0049 to 1.0051 crosses a half-cent boundary and does not establish one posted amount. Refine within a declared budget or retain rounding unresolved. This rule separates evidence for a numerical answer from the accounting authority to use that answer.

6.12 Updating a law without counting evidence twice

Two financial histories can be compatible with the same reports without being equally probable. Once an explicitly justified law is supplied, genuinely new evidence may change it. For finite histories h, prior mass $p(h)$ and an admitted observation likelihood $L(D \mid h)$, Bayes' rule gives

$$p(h \mid D) = \frac{p(h)L(D \mid h)}{\sum_g p(g)L(D \mid g)}.$$

The denominator is the probability of that evidence under the declared model. If it is zero, the condition is impossible in that model; dropping incompatible histories and renormalising an empty collection cannot repair it.

For two histories with assumed prior masses 1/3 and 2/3 and likelihoods 3/4 and 1, the unnormalised masses are 1/4 and 2/3. Their sum is 11/12, so the posterior masses are 3/11 and 8/11. If the second history's cash calculation fails, the first still has mass 3/11 and unresolved cash retains mass 8/11. A report giving the successful history probability one would answer a different, success-conditioned question.

An observation has a semantic information identity as well as a file locator. Reimporting the same observation through another file does not create a second measurement. Repeating an identical admitted condition is idempotent: it keeps the law and information lineage unchanged. Submitting an already consumed observation as a fresh likelihood block is rejected as duplicate information. Equal selected member sets alone do not prove that two conditions have the same meaning. An admitted conditional kernel may divide a region's existing mass when it is refined; the number of new representatives never determines that division.

Continuous inference adds a measure choice. One must specify coordinates z, domain, base measure, prior, observation mechanism and any transformation Jacobian. Stuart (2010, Section 2.2, equations (2.5)–(2.8)) supplies the inverse-problem formulation, but does not establish identification or easy sampling for a mixed accounting model. In particular, an exact accounting equality can confine a model to a lower-dimensional set. Multiplying an ordinary Lebesgue density by its zero-volume indicator is not a posterior construction. One must parameterise the constrained model with its declared measure or supply an actual measurement-error model. Adding convenient small noise changes the target.

6.13 A continuous example whose posterior can be checked

A linear Gaussian model supplies a tractable comparison for later numerical inference. Assume the unknown economic quantity has a Gaussian prior $z \sim \mathcal{N}(m_0, V_0)$ with positive definite covariance V_0. Let the observation be $y = Hz + \epsilon$, where ϵ is independent Gaussian noise with positive definite covariance R. The measurement-error assumption is substantive: it describes this example, not rounding in every financial statement.

Multiplying prior and likelihood adds their quadratic forms. Terms that depend on z are

$$\begin{aligned}&(z - m_0)^\top V_0^{-1}(z - m_0) + (y - Hz)^\top R^{-1}(y - Hz)\\ &\quad = z^\top (V_0^{-1} + H^\top R^{-1} H)z - 2z^\top (V_0^{-1} m_0 + H^\top R^{-1} y) + \text{constant}.\end{aligned}$$

Completing the square therefore gives posterior covariance V and mean m through

$$V^{-1} = V_0^{-1} + H^\top R^{-1} H, \qquad V^{-1} m = V_0^{-1} m_0 + H^\top R^{-1} y.$$

The prior and observation contribute precision and precision-weighted information. An implementation should solve factored linear systems rather than explicitly form these inverse matrices, and reject an invalid covariance or unsupported rank condition.

For $m_0 = 0$, $V_0 = 4$, $H = 1$, $R = 1$ and $y = 2$, posterior precision is $1/4 + 1 = 5/4$, variance is $4/5$ and mean is $(4/5)2 = 8/5$. A conditional cash quantity $100 + 10z$ has mean 116 and variance 80. A second *independent* measurement of 2 would give precision $9/4$, variance $4/9$ and mean $16/9$. Reusing the first measurement would produce those same numbers but would be wrong for the stated information. The analytic example can distinguish numerical error from information double counting.

For a bounded non-Gaussian scalar model, quadrature requires a stated domain or tail truncation and sufficient integrand error bounds to enclose the normalizer and requested moments. Without a positive lower normalizer bound, the ratio remains unresolved. Mixed models can enumerate finite choices and integrate each conditional continuous model, but combining branches requires their evidence factors, not draw counts. Optional MCMC needs target-specific parameterisation, tuning, independent chains, tractable correctness comparisons, convergence and Monte Carlo error checks. This proposal supplies no generally validated sampler or universal tolerance.

Nor can a more elaborate inference engine identify a quantity that the observation model never sees. If reports constrain only total debt 100, the likelihood may be constant in its floating share x. Its conditional prior then remains unchanged. A uniform prior is a chosen assumption, not a discovery from those reports. The inspected working version of Giacomini and Kitagawa (2021, Sections 2.1–2.3, equations (2.2)–(2.4)) motivates retaining posterior classes where a conditional prior cannot be revised. For Northstar there is a simple direct deduction: under all probability measures on $[0, 1]$, the expectation of $5 + 2x$ lies in $[5, 7]$, with endpoints attained by point masses. This deduction imports no asymptotic theorem from that paper's structural-vector-autoregression setting.

6.14 A stress is not an observed condition or an intervention

The interest-rate stress in Northstar fixes a future driver path. It supplies no new evidence about the historical debt share. An observational forecast instead conditions a joint law on information. A causal intervention needs structural equations and an exogenous disturbance law, because it replaces the mechanism determining a chosen variable while leaving other mechanisms in place (Pearl, 2009, Section 3.2, equations (5)–(9) and (15)–(17)).

An elementary example makes the difference numerical. Let U equal zero or one with probability $1/2$ each, and assume

$$X = U, \qquad Y = 2X + U.$$

Observing $X = 1$ establishes $U = 1$ and hence $Y = 3$. Intervening to set $X = 1$ replaces the first equation. It leaves the original law of U, so Y is 2 or 3 with equal mass and mean 2.5. Selection of a high-investment company and an imposed investment policy can therefore answer different questions even under one fully specified model. Financial filings alone do not establish these causal equations or identify their effects.

An intervention request must name its target, replacement equation, structural parents and disturbance law and satisfy acyclicity or a separately supported equilibrium rule. Correlated disturbances remain joint. The initial company profile supports imposed exogenous stress; the continuous and causal methods described here require separately admitted extension schemas and signatures. They do not silently expand the initial finite-law interface.

At every decision date, behavior and measurement receive the admitted historical view and

permitted simulated past. The scheduler's knowledge of a complete future stress path is not permission for a manager to foresee it. A perfect- foresight policy needs an explicit additional assumption. Common random numbers can improve a paired numerical comparison while establishing no causal effect in actual company data.

Chapter 7

The accounting language

7.1 Scope of the first language

The historical AccDSL Core 0 proposal describes entities and books, accounts, observations, shared unknowns, constraints, alternative interpretations, contract instances, accounting rules and reporting mappings. It has pure expressions and a restricted posting effect. Economic forecasting algorithms can live in typed plugins; unrestricted numerical code does not need to be reimplemented in the DSL. The historical grammar in Appendix B fixes the preserved teaching examples. The proposed portable records and first company profile below define the new authoring boundary; the first matched text frontend supports the declared debt fragment.

The historical Core 0 compiler has six ordered stages: parsing with source locations; name and module resolution; type and temporal-context checking; lowering to the IR; constraint and rule-applicability analysis; and execution-plan construction. The compiler never silently supplies an economic assumption to make a program compile. A program can be valid as a historical representation while requiring additional assumptions before a particular simulation can run. The proposed portable compilation sequence is specified separately in Section 7.3.

The language's meaning is defined by its IR and evaluator, rather than by the surface spelling alone. Formatting changes cannot change a model's semantic hash. A changed source span or reviewer decision changes evidence provenance even if the numerical model remains the same. The system therefore maintains separate evidence, semantic-model and execution identifiers.

For the preserved Core 0 examples, the complete grammar is supplied as `spec/core-0.lark`; the original design directory retains `spec/core-ir.schema.json` and `spec/core-contract.md`. They define reference resolution, expression arity, type/context checks, operations, failures and identity. Structural schema validation alone does not establish those semantic properties. ASCII identifiers and UTF-8 strings are used in this revision; currency suffixes, dates and decimal literals have explicit lexical rules. The grammar is checked by parsing the complete demonstration file, not by claiming that its parser also implements a compiler or solver.

7.2 Types that protect meaning

The following table uses the preserved Core 0 type vocabulary. Its distinctions motivate the more precise portable types introduced below.

Type	Meaning and checks
`Money<C>`	Exact amount in currency C. Addition requires identical currency; conversion requires a dated rate and policy.
`Quantity<U>`	Units of shares, goods or other commodities. Quantities cannot be added to money or used as a functional-currency posting.
`Rate<B>` and `YearFraction`	Rate with basis B, plus a separately computed accrual fraction. Simple and compound accrual functions are distinct.
`Instant`, `Period`, `Calendar`	Instant observations and flows over intervals; interval operations preserve calendar conventions.
`Account<K,Book>`	Account role K and owning book. Cross-book effects require separate balanced batches linked by an economic event.
`Observation<T>`	Source assertion with contextual and precision metadata; accessing its value retains its constraint and provenance.
`Symbol<T,D>`	Shared unknown with domain D; repeated references denote the same value in a history.
`Choice<E>`	A finite structural choice over enumeration E. No probability is implied.
`Distribution<T>`	Explicit probability measure with parameter, conditioning and measure metadata. It is not a Symbol or a Choice.
`SourceRef`, `RuleRef`, `ModelRef`	Versioned evidence or implementation references; unresolved references prevent concrete execution.
`Measurement<T>`	Value plus valuation date, basis, method, assumptions and numerical status. Recognition remains a rule decision.

The portable profile spells the rate type `Rate<convention>` and makes both calendar and accrual convention explicit in `YearFraction<calendar, convention>`. `InstantValue<T>` and `Flow<T, [start,end)>` distinguish dated stocks from interval flows; their context carries the calendar. An approximate numerical component of a historical `Measurement<T>` uses `Approximation` with its method, error description and environment references. An exact measurement remains exact; valuation context and recognition authority are separate in either case.

Core expressions use exact integers and rationals, Boolean logic, enumerations, dates, records, lists, comparisons and registered pure functions. A decimal literal is parsed exactly, not first converted through binary floating point. Numerical plugins can return approximate measurements, but must declare their error or numerical status and the rounding used when posting money.

Missingness is a tagged type with distinct values: `notDisclosed`, `notApplicable`, `unreadable`, `withheld`, and `unresolvedMeaning`. Zero is an ordinary known number. A missing datum can be replaced by a declared Symbol or assumption in a model branch; the source observation stays missing. An unknown is not necessarily random, and an estimate is not necessarily missing. On portable import, `notDisclosed`, `unreadable`, `withheld` and `unresolvedMeaning` become `Unknown` with the corresponding reason and obligation references. `notApplicable` becomes the explicit `NotApplicable` variant, never an unknown or zero.

Addition of monetary observations also requires compatible entity, perimeter, time, dimensions and basis. These checks operate on contextual expressions in addition to the currency type. An analyst can deliberately aggregate dimensions or transform a basis using an explicit function.

For example, `sumOver(segment, ...)` records the dimension being removed and requires the segment partition to be declared; it does not assume every disclosed segment total is additive.

7.3 A portable record has an exact meaning

A constructor and a textual frontend can describe the same financial model only if they exchange values whose identity is independent of a Python object or an incidental JSON writer. The proposed portable record therefore has six fields: `kind`, `schema`, `key`, `revision`, `body` and `dependencies`. Kind and schema identify a registered version; key is its stable, schema-scoped logical name; revision is a nonnegative integer. The body obeys that schema's required, optional and nullable fields. Unknown fields reject.

A typed reference has exactly kind, schema and digest. The digest covers the complete target record. Every body reference is also listed in dependencies; additional dependencies need a schema-declared role. The submitted closure is finite and acyclic. A receipt can refer to its request and old state, but the old state cannot refer forward to that receipt. Stable business names can relate counterparties without creating a cycle of content references.

Financial quantities use a reduced rational with signed numerator text n and positive denominator text d, with $\gcd(|n|, d) = 1$ and zero represented by $0/1$. The decimal token 2.50 therefore denotes $5/2$ exactly. Revisions, ordinals and integer counts remain integers; a Boolean is never an integer. Approximate measurements instead retain decimal text, method, error description and environment. A rounded decimal cannot silently acquire exact status.

An absent required field is invalid. Unknown financial content is an explicit value with a reason and outstanding evidence references. It can be retained, but cannot satisfy a method that requires the missing fact. Null is permitted only where the particular schema gives it a meaning; not applicable, unknown and zero are distinct. A candidate also differs from a value admitted in a specific immutable view. Admission checks the actual decision and its ancestors; wrapping a candidate in an "admitted" type is insufficient.

Portable bytes use UTF-8 JSON with ASCII object keys sorted by their byte order, no insignificant whitespace, no floats, duplicate keys, nonfinite numbers or invalid Unicode scalar values. Decoded strings retain their code points. An escaped Unicode character and the same literal character encode equally; composed and decomposed spellings remain different unless an explicit named transformation relates them. Global NFC normalisation would silently change some source text and is therefore not part of this encoding.

The writer emits literal non-ASCII scalars, escapes quotation marks and reverse solidus, and escapes control characters U+0000–U+001F. It uses the five named JSON escapes where defined and lowercase `\u00xx` for the remaining controls; solidus is not escaped. Integers use minimal base-ten spelling without plus signs, leading zeroes or negative zero. Dependencies, module declarations by key, dimensions by axis and explicit all-of/any-of operand sets are sorted after rejecting duplicates. Other arrays, including journal lines, event schedules and diagnostic priority, retain order.

With these rules, the record digest is

$$\mathrm{SHA256}\big(\mathrm{UTF8}(\text{“accdsl.portable.v1/”} + \mathrm{schema}) \parallel \texttt{0x00} \parallel \mathrm{canonical}(\mathrm{record})\big).$$

This recognises representational identity, not economic equivalence or source truth. Original documents retain a separate hash of their raw bytes. A blob reference names that byte hash, algorithm and length; it is not a typed record reference. A submission record links those bytes to media type, transport and decoder, preserving a malformed request without pretending it was valid.

Historical Core 0 and replay examples keep their original encodings and IDs. A declared legacy import retains original bytes, decoder, old identity, portable output and field-by-field comparison. If one historical record maps to several portable records, an import mapping names every output and conversion. Import does not grant financial admission in a new view. This separation permits old examples to remain reproducible while the proposed portable representation becomes precise enough for two implementations to exchange.

7.4 The first company profile and its authoring boundary

The proposed `CorporateJointHistory/1` profile connects one legal entity and book per branch to annual/interim sources, revisions, working capital, payroll, tangible assets, simple financing, an explicitly declared tax model and complete statements. Its types preserve entity, perimeter and currency even where an operation is outside this initial scope. A real company cannot become eligible for the profile by dropping its subsidiaries, leases or contingencies.

Typed constructors must express every included declaration in the portable IR without opaque Python functions. The initially matched text frontend covers the bounded debt and constraint fragment. Whole-company textual authoring remains a subsequent frontend obligation; the whole-company typed model is part of the initial domain. Thus domain scope and authoring syntax are separate decisions. The following Core 0 listing teaches the shared debt family under its preserved historical grammar. It is not a promise that the first new parser supports every historical declaration. The matched portable debt example uses floating share x, while the following listing uses fixed share α. The substitution $x = 1 - \alpha$ preserves the entire family, including its endpoint results.

Backend admission examines expanded expressions. Multiplication of an unknown quantity by an unknown price is bilinear even when hidden inside a function called revenue. The initial exact reconstruction fragment admits rational affine constraints, finite choices and bounded integers under explicit templates. Unsupported functions return a capability diagnostic. A type-correct historical representation may therefore be retained even when a requested forecast still requires an unavailable method or an additional admitted assumption.

The portable `CompileRequest` supplies source kind, source bytes or a typed module, schema and profile references, the full package lock, evidence/time view and requested capability target. Its stages are strict decoding and source locations; name/import resolution; dependency and cycle checks; static type and context checks; admission-layer checks; expanded-operation capability analysis; and canonical record production. The earlier six-stage workflow describes the historical Core 0 frontend; it does not add execution planning to this portable compiler or permit it to call a solver.

The result contains candidate IR, source map, unresolved inputs, capability analysis and a validation receipt bound to every input. Invalid structure or types yield no executable IR. Valid structure with an unresolved financial input can yield a retained candidate whose affected target has execution admission false. Source maps retain decoded Unicode code-point half-open ranges for text or constructor paths for typed input. Formatting can change source-map identity without changing financial-model identity.

For a matched example, both authoring routes must describe debt 100, floating share $x \in [0, 1]$ and conditional interest $100((1 - x)/20 + 7x/100) = 5 + 2x$. Endpoints are 5 and 7; $x = 1/2$ gives 6. Changing the floating assumption from 7% to 9% changes the upper endpoint to 9. Independent arithmetic is needed because two frontends could share the same wrong constant. Agreement between them alone does not establish correctness.

7.5 A small model with a continuous interpretation family

The following source uses a synthetic observation whose full context is stored in the evidence registry. The example imports the demonstration debt package, which uses one-year simple accrual, no fees and immediate rate resets. It is not an IFRS rule package.

All dependencies are instantiated in `fixtures/demo-debt-package.json`: the source reading, balanced opening state, currency precision, contract state and dates, reset and interest operations, closing and report mappings, request and expected outputs. `fixtures/borrowing.acc` is the complete source, including the retained earnings account and an `opening` declaration. The shorter listings below explain its declarations in stages. They do not replace that full source.

```
module BorrowingExample version "1";
use DemoDebt version "1";
source note7 = external("synthetic-note-7");
entity Acme functional USD;
book Local entity Acme basis DemoDebt version "1";
account Cash : Asset book Local;
account Debt : Liability book Local;
account InterestExpense : Expense book Local;

observation debt : Money<USD> instant @2025-12-31
  = 100 USD from note7;
symbol alpha : Real in [0, 1]
  because "The fixed-rate proportion is not disclosed";
```

The remaining declarations use this same unknown fraction in both loan components. A change to the fraction therefore changes the whole debt mix consistently.

```
let fixedPrincipal = alpha * value(debt);
let floatingPrincipal = (1 - alpha) * value(debt);
constraint split:
  fixedPrincipal + floatingPrincipal == value(debt);

contract loans : DebtMix book Local {
  principal = value(debt);
  fixed_fraction = alpha;
  fixed_rate = 0.05 per year;
  initial_floating_rate = 0.05 per year;
}
alternative allFixed { assume alpha == 1; }
alternative allFloating { assume alpha == 0; }
scenario rateShock {
  set market_rate = constant(0.07 per year);
}
```

The parent model retains every $\alpha \in [0, 1]$. The two named alternatives are useful conditional views of that family, not a declaration that the endpoints exhaust it. Requesting the parent family's exact bounds produces $[5, 7]$ for the next year's expense under `rateShock`. Requesting a conditional run of `allFixed` supplies $\alpha = 1$. Requesting a weighted mixture requires a separate weight model; the two labels do not create equal weights.

An observation declaration is a reference to evidence plus a proposed typed reading. The compiler checks that the registry context agrees with the declared period and type. If the evidence is rounded, `value(debt)` denotes a shared constrained value rather than a silently exact 100. This synthetic registry explicitly declares an exact value, which keeps the illustration deterministic.

7.6 Posting rules and explicit exceptions

An event type declares the information needed by a rule. The example below assumes an event representing interest accrued and paid on the same date. Separate accrual and settlement events are needed when those dates differ.

```
event InterestPaid(principal: Money<USD>,
                   rate: Rate<annual_simple>,
                   fraction: YearFraction) book Local;

rule recogniseInterest for InterestPaid as e
when e.principal >= 0 USD {
  let charge = roundMoney(simpleInterest(e.principal, e.rate, e.fraction),
                          currencyScale(USD), "half_even");
  require charge >= 0 USD;
  post {
    debit InterestExpense charge;
    credit Cash charge;
  }
}
```

The rule cannot perform file I/O, mutate another entity or sample a random number. It returns a proposed batch. The runtime verifies balance and all preconditions before committing state. Negative interest requires an explicit supported rule; it is not mishandled by this intentionally restricted example. Rounding is a declared demonstration convention, not a universal accounting default.

Rules can declare `overrides ruleName`. The resulting override graph must be acyclic. For a concrete event, the evaluator determines all applicable rules and removes any that are overridden by another applicable rule. Exactly one maximal rule must remain for a required recognition operation. An empty set of applicable rules produces `UnsupportedTreatment`; more than one maximal rule produces `RuleConflict`. File order and a hidden numeric priority never break ties.

If a guard depends on a symbolic value, the solver partitions the current feasible family into the guard and its negation, subject to the existing constraints. This is a logical partition, not a random draw. An incomplete solver returns `UnknownApplicability`, not a guessed rule. Statically provable disjointness or total coverage can be cached; otherwise the runtime performs the same checks on every concrete event.

7.7 Alternative readings, assumptions and updates

An alternative adds constraints and possibly a supported semantic mapping to its parent model. Its record contains a rationale, supporting spans, opposing evidence, author, creation time and scope. Child alternatives inherit all parent constraints; they cannot quietly discard an inconvenient disclosure. A deliberate source rejection is a separate review decision with an explicit reason and a new evidence view.

Two independent alternatives may coexist, while two mutually exclusive readings of the same text require an explicit choice group. The representation stores these relationships in a constraint graph. It does not take a Cartesian product blindly. For example, interpreting an instrument as a liability affects both its balance-sheet account and the treatment of its periodic payments; those choices must be linked through one shared treatment variable.

New evidence narrows an existing family by conjunction when it concerns the same definitions and reporting basis. If it changes the basis or corrects an earlier assertion, the system creates a

revised evidence view and records the relationship. Posterior reweighting, when enabled, is a separate operation with likelihood provenance. A branch that becomes inconsistent is retained as historically considered and rejected, with the evidence that rejected it.

7.8 Modules and stable intermediate representation

A package identifies its namespace, semantic version, dependencies, effective dates, jurisdiction, supported entity classes, recognition operations and measurement policies. Standard packages define allowable treatments; industry packages add economic objects and specialised rules; company packages map source concepts and declare evidenced elections. A company extension cannot override a mandatory standard rule merely because its package is loaded last. An override is valid only at a declared extension point and within the selected standard's policy domain.

The IR has separate node classes for evidence, contextual observations, expressions, domains, constraints, alternatives, contracts, economic events, measurement results, rules, journal batches, report mappings and runs. Each node has a stable identifier and explicit input references. No expression stores executable host-language text. Plugins are referenced by versioned identifiers with typed signatures. This permits a second implementation to execute the same supported program without evaluating arbitrary embedded code.

The IR serialisation version is independent of the DSL version and of any accounting-standard edition. Migrations preserve the original payload and record changed meanings. A syntax migration can be automatic when semantics are unchanged. A changed recognition policy requires a new model version, even if the source file still parses.

7.9 Errors are part of the interface

The compiler and runtime return typed diagnostics with a source span, affected entity and period, dependency chain and suggested local action. The principal diagnostic categories have distinct recovery paths:

Diagnostic	Meaning and next action
`TypeMismatch`, `ContextMismatch`	Resolve incompatible units, periods, entities or measurement bases; do not coerce implicitly.
`MissingEvidence`	Supply a source or label a model input as an assumption.
`InfeasibleHistory`	Inspect the conflicting evidence and model constraints; preserve the rejected candidate.
`RuleConflict`, `UnknownApplicability`	Resolve rule scope or preserve a justified conditional partition.
`UnsupportedTreatment`	Add a reviewed capability or leave the affected output incomplete.
`UnbalancedBatch`	Reject the atomic update and diagnose the posting rule.
`SolverIncomplete`, `NumericalFailure`	Preserve the target and inputs; repair or change the computational method.
`InsufficientSimulationState`	Add declared conditional assumptions or request a family of results.

Warnings can describe low extraction confidence or an approximate economic model, but they do not replace these errors. A required unknown input prevents a concrete unconditional result. The analyst can add a declared conditional assumption, request bounds, or leave that output incomplete. Every path has a different, explicit result type; there is no generic fallback to zero.

Chapter 8

Reconstruction and execution procedures

8.1 Ingest, reconcile and construct histories

The historical procedure starts with a filing manifest. It records documents, reporting dates, publication dates, revisions and expected statements and notes. Tagged filings are parsed through an XBRL adapter; PDF and HTML sources are segmented into text, tables, captions and references. The original bytes and extraction output remain immutable. Re-running an improved extractor creates a new extraction version with links to the same source.

For each table, the importer first resolves headers, units, signs, periods and dimensions. It then proposes concept mappings. A deterministic validator checks subtotal arithmetic, duplicate facts, comparative consistency and cross-note references. Every proposed repair creates a new reading. The system may suggest that an OCR value is 10 rather than 70, but only a reviewed correction or an explicit alternative allows that reading into a validated model.

Reconciliation proceeds from local totals to movement schedules. A debt bridge may contain opening debt, issuance, repayment, acquired debt, interest accretion, currency effects, fair-value changes and closing debt. Every component receives a movement type. An unexplained residual is stored as a residual with a source and period; it is not automatically classified as repayment or issuance. A model can introduce an unknown decomposition of that residual, subject to evidence and economic constraints.

The reconstruction engine then performs the following finite procedure within a declared model scope:

1. Build shared variables for observations, opening states, contract terms and supported movement amounts. Attach original rounding domains.
2. Add identities, roll-forwards, account restrictions, contractual relationships and cross-period links. Add each semantic reading as a condition on the same joint model.
3. Propagate types, units, intervals and simple equalities. Reject context mismatches before invoking a numerical solver.
4. Ask the solver for feasibility. On infeasibility, return a sufficient conflicting subset of constraints, its source spans and model assumptions. A solver's conflict core need not be the smallest possible explanation.
5. Retain finite alternatives and continuous solution families symbolically. Materialise

example histories only when needed for review or execution.

6. Check whether each requested simulation has sufficient economic state. Missing maturities, reset dates or cohort composition become declared assumptions or explicit unresolved inputs.

Unrestricted journal enumeration would be unbounded: arbitrary offsetting transactions can be inserted without changing reports. The model therefore declares allowed event types, time resolution, relevant accounts, lot/cohort resolution and limits on latent contract instances. These limits describe the represented family; they are not evidence that all real possibilities were searched. The compiler canonicalises syntactic duplicates and equivalences established by its supported rewrite rules. It does not decide arbitrary economic equivalence. Probability measures attach to declared economic hypotheses or parameters, never to a count of encodings; adding a duplicate representation therefore cannot create extra probability mass or coverage.

The first implementation uses symbolic linear bridges and finite structural choices. Nonlinear economic models can be evaluated on feasible states through plugins. A solver timeout preserves the unresolved family and returns `SolverIncomplete`; it does not prune the family as impossible. Search budgets belong to run configuration and are recorded with the results.

8.2 Condition a run without rewriting history

A run selects an evidence view and cutoff, a historical family, a forecast starting date, scenario definitions, accounting packages, management policies, output queries and numerical settings. A run may use a concrete history, an exact symbolic family, a finite set of conditional histories, or weighted draws from a declared distribution. The output records which mode was used.

Suppose no debt maturity is disclosed. An analyst may run a conditional case that assumes all debt matures after the forecast horizon. The run is valid under that assumption even though the historical representation remains incomplete. The interface must display the assumption next to the result and provide sensitivity cases when the assumption is material. It must not label the assumed maturity as a discovered historical fact.

Management behaviour normally uses only information available at the decision date. A perfect-foresight stress case is a separate, explicitly labelled policy. Scenarios should share the same underlying innovations when estimating paired differences where appropriate, with stable stream identifiers so branch order does not change the draws. Common random numbers improve the comparability of paired simulations; they do not imply that every branch has the same economic event counts or shock distribution.

8.3 Schedule and commit events

Each event carries an economic occurrence time, accrual interval when relevant, settlement time, entity, contract or cohort identifier, currency, inputs and provenance. Its ordering key consists of date, declared phase, dependency edges and a deterministic tie identifier. The reference phases are: opening and scheduled contractual changes; market observations and rate resets; business events and decisions; accrual and measurement; settlement; period closing and reporting. A contract can require a more specific dependency. These phases are an engineering starting point, not a universal description of market timing.

Two events may share a tie order only if their order is either specified or irrelevant to the result. If payment before remeasurement and payment after remeasurement differ, an unresolved

ordering is a model ambiguity or an error. The engine does not conceal it behind alphabetical event IDs.

```
execute_event(operation_request):
    validate exact input closure, bindings and historical admission
    validate separate posting authorization, journal and delta
    lookup stable entity/book/branch/business-event key
    if exact committed fingerprint: return original receipt
    if different committed fingerprint: return event conflict
    compare expected opening state and head generation
    atomically recheck key/head and commit journal, delta, state, receipt
    return receipt, emitted dependent events and diagnostics
```

The atomic unit includes the economic state and its accounting effects. If posting fails, neither state changes. A repeated event ID with identical inputs returns the existing result; the same ID with different inputs is a conflict. Here the inputs include the request identifier, expected head and execution settings, not only the financial payload. Section 8.5 defines the stable key and full fingerprint. Consolidation runs after entity books reach the required date and stores its adjustments separately. A failure in one scenario branch cannot mutate another branch's state.

8.4 From an economic event to permission to post

A delivery can change revenue, inventory and receivables at once. Before any book changes, the method needs to know which contract was performed, which opening state it consumes, which evidence supports that reading and which accounting profile applies. Balance alone cannot supply those facts. AccDSL therefore separates the economic basis, permission to use a treatment, proposed entries and the transaction that makes them durable.

The event basis binds an immutable event to opening state, context, branch, time view, chart, authority, method, evidence, admissions, assumptions and view. Its business identity is the four-tuple of entity, book, branch and business event. The branch declares whether it is synthetic or real. A synthetic teaching profile can justify a teaching posting; it cannot authorize a real book merely because the resulting arithmetic is identical.

The method first evaluates reporting framework, jurisdiction and dates; then entity and transaction scope; contract and recognition facts; elections; numerical capability; and account/context bindings. Every material finding survives. A false required fact is ineligible, an unknown required fact is unresolved, and an event outside the method is outside scope. A known false finding must not disappear behind a later unknown one. Missing operative authority never becomes source-supported permission.

The native operation `accounting.assess@1` remains read-only assessment. Its `assess_only` request and false posting permission cannot be promoted by editing a flag. The separate `accounting.authorize@1` operation produces a `PostingAuthorization` from the complete basis and a supported posting profile. It permits posting only when the treatment is eligible, the method is fully specified for the event and the branch mode matches the profile's evidence scope. An unresolved or unsupported assessment is still a useful completed assessment, but it contains no permission to post.

An authorized proposal contains both a journal and an independent economic delta, each bound to the same basis and authorization. Journal lines name account role, debit or credit, positive exact quantity, currency/context, entity, book, event and method. Zero lines are omitted; a negative effect reverses sides. The journal balances within each currency and book. The delta separately checks quantities and contractual memory: inventory movements consume available

lots, cash follows settlement, share counts follow ownership events and debt face amount remains distinct from carrying value.

The commit arguments have exactly four references:

```
AccountingCommitArguments/v1 {
  basis_ref, authorization_ref, journal_ref, delta_ref
}
```

The expected opening state and branch generation appear once, in the outer operation request. A reference to a native assessment in the authorization field fails its reference type. False posting permission fails authorization. An unknown posting schema returns unsupported operation with no journal commit. All bindings are checked before looking up a prior receipt.

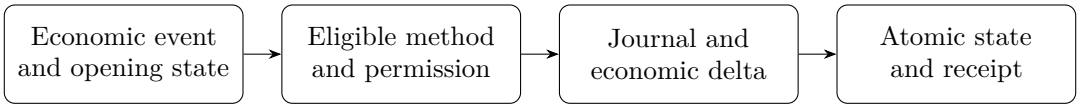


Figure 8.1: Each step answers a different question. Accounting assessment may inform the permission decision, but only the final transaction changes the book.

For a delivery, the basis identifies the exact performance obligation, quantity, allocation and control finding. The opening state contains cumulative revenue, billing, collection and inventory. The selected method derives the recognition and lot movements; the report later consumes that same committed path. Combining a journal from one method version with a delta or opening state from another is invalid even if the totals happen to balance.

8.5 A lost acknowledgement must not create a second sale

Suppose a sale commits but the response is lost. The client knows the request, but cannot tell whether the server committed it. Retrying under a new event ID could duplicate revenue. Retrying the original request must instead return the original immutable receipt.

The lookup key is

$$(\text{entity},\ \text{book},\ \text{branch},\ \text{business event}).$$

Payload, event version, opening state and request settings belong to the complete fingerprint, not this key. A renamed run or bumped event version cannot evade a conflict. A separately commissioned hypothetical projection uses a new branch namespace; reopening the same projection uses its existing one.

A correction of an already committed event is a new business event linked to its predecessor and an admitted correction method, such as reversal and replacement or a dated adjustment. It preserves the original receipt. The accounting treatment must apply to the actual correction; absent treatment authority leaves it unresolved rather than licensing a generic reversal.

Commit first validates input shape, closure, bindings and authorization in the requested historical view. It then looks up the stable event key. An equal fingerprint returns the old receipt with no additional commit; a different fingerprint returns an event-identity conflict. Only an absent key proceeds to the expected-state and generation comparison. This order is essential: after a successful sale the current head has changed, so checking freshness before replay would reject the very retry needed to resolve the lost response.

A new candidate is assessed and constructed without mutation. One transaction then rechecks key absence and the expected head, inserts journal, delta, closing state, dependencies and

receipt, and advances the head. It commits all or none. If another client has committed the identical event, return that receipt. A different event that advanced the head makes this request stale. A changed payload under the same key conflicts. Diagnostic or scratch records never count as economic transitions.

A timeout consequently has three possible observations: committed, not committed or unknown. Unknown retains the original key, fingerprint and expected head for receipt lookup or exact retry. Refreshing the head changes the request and cannot be presented as the same retry. Missing committed dependencies are an integrity failure, not permission to reconstruct an approximately similar receipt.

8.6 Funding, tax and other simultaneous relations

Some financial models have circular dependencies. Interest affects taxable profit; tax and investment affect funding needs; funding affects interest. The reference executor first tries to resolve these through explicit timing. For example, beginning-of-period debt can determine current interest, with new borrowing affecting the next accrual period. That is acceptable only when the model's timing assumption is stated and appropriate.

Where the economic model genuinely requires simultaneous decisions, the dependency graph identifies the strongly connected component and delegates it to a declared equation or optimisation solver. The solver must return a solution, residuals, admissibility checks and uniqueness or multiplicity status. A contraction iteration requires a justified convergence condition; an arbitrary fixed iteration count is not a mathematical solution. Multiple admissible equilibria become alternatives when material. No solution produces a model outcome such as financing failure, or an explicit numerical failure, depending on the cause.

Funding policy must include available facilities, limits, covenants, collateral and allowed capital actions. The program cannot issue unlimited debt to keep cash positive. An inability to meet payments may be a legitimate default outcome. Default ends or changes contractual and accounting processes under the relevant event rules; it is not quietly discarded as a bad simulation draw.

8.7 Worked example: identical history, different future journals

Return to the company in Chapter 1. Immediately after the historical interest payment and closing, cash is 95, debt is 100 and equity is −5. There are no other assets, liabilities or transactions. Next year's expense is 5 in the all-fixed branch and 7 in the all-floating branch under the rate shock. The generated entries and closing statements are:

Output	All fixed	All floating
Debit interest expense / credit cash	5	7
Closing cash	90	88
Closing debt	100	100
Current-year profit	−5	−7
Closing equity after profit transfer	−10	−12
Cash change	−5	−7

The balance-sheet identity holds in each branch: $90 = 100 - 10$ and $88 = 100 - 12$. The demonstration reporting profile classifies interest paid as operating cash flow. A different

permitted reporting profile changes that classification, while preserving the cash movement and source event. This example proves the intended distinction and its arithmetic; it does not estimate the probability that either debt structure describes a real company.

8.8 Worked example: a bond has several relevant amounts

A two-period zero-coupon bond raises cash of 100 and promises 110.25 at maturity. Assume a 5% effective rate per period, no transaction costs, no default, and amortised-cost treatment under a demonstration rule. The effective-interest schedule follows directly from $100(1.05)^2 = 110.25$:

Date	Opening liability	Interest accretion	Closing liability
Issue	0	0	100
End of period 1	100	5	105
End of period 2, before payment	105	5.25	110.25
Redemption	110.25	0	0

Issuance debits cash and credits the liability by 100. Each accretion debits interest expense and credits the liability. Redemption debits the liability and credits cash by 110.25. Legal face amount, carrying amount and market fair value remain separate fields. A market-rate shock may change fair value without changing this contractual payment schedule or this demonstration amortised-cost path.

Cash-flow reporting uses the actual payment event and its economic components; the annual accretion is non-cash. The applicable reporting package decides how to classify the maturity payment, rather than inferring it solely from the liability debit. Fees, modifications, early redemption, embedded options, hedges and foreign currency require additional event and measurement rules.

Ordinary equity issuance similarly separates cash, share quantity and legal capital. Ten new shares at 12, with par value 1, produce cash of 120, share capital of 10 and premium of 110 under a simple par-value profile. Shares outstanding increase by ten units. Direct issuance costs of 3 reduce the specified equity account and cash under the chosen illustrative treatment. The accounting rule and the cost's eligibility must be explicit; this example does not prescribe every jurisdiction's treatment.

8.9 Reporting and explanations

The reporting engine produces the balance sheet, income statement, comprehensive income, cash-flow statement, changes in equity and note schedules when the necessary modules are present. Each output carries its basis, perimeter, period, scenario and interpretation. A comparative table cannot combine an original prior-year figure and a restated current-year comparative without displaying that choice.

An explanation is a dependency graph with typed edges: extracted from, interpreted as, assumed by, constrained by, generated by, measured by, posted by, aggregated into and supersedes. A cash-flow line must lead back to payment events; a fair-value movement must lead back to valuation inputs; a consolidated amount must show eliminations. A provenance path establishes how a result was obtained. It does not by itself establish that the underlying economic model is correct.

8.10 The corporate reference package as operations

The Northstar example defines a closed, reproducible corporate package rather than a universal standard. Its events separate performance, invoicing and settlement even where the example makes the first two simultaneous. Every payload includes entity/book, currency, economic date, period, source or assumption IDs and an information-view ID. Amounts are exact before the declared posting quantisation. Quantities and money are checked separately.

Event	Preconditions and state change	Posting
Deliver/invoice	Goods owned and available; control transferred; quantity removed at selected cost; unconditional amount due added	Debit receivable, credit revenue; debit cost of sales, credit inventory
Collect	Receipt no greater than selected unpaid receivable; reduce that balance; excess must be separately classified	Debit cash, credit receivable
Receive goods	Accepted owned quantity at supported unit cost; create supplier obligation and inventory lot	Debit inventory, credit payable
Pay supplier	Identified obligation and sufficient available cash; reduce both	Debit payable, credit cash
Payroll	Service period and cash obligation evidenced or assumed	Debit payroll expense, credit cash; accrual variant uses a payable
Acquire plant	Control obtained, cost known, in-service date/life/residual recorded; add asset cohort	Debit plant cost, credit cash or payable
Depreciate	In-service asset with remaining depreciable amount; apply dated schedule, capped by that amount	Debit depreciation, credit accumulated depreciation
Interest	Contract, reset state and accrual fraction known; create interest obligation, then settle if cash available	Debit interest, credit interest payable; debit payable, credit cash
Current tax	Reference tax base positive; 20% teaching rule; apply prepaid tax before further payment	Debit tax expense; credit prepaid tax or payable; settlement separately affects cash
Dividend	Explicit authorised distribution, sufficient funds and permitted capital action	Debit retained earnings, credit dividend payable; then settle
Close	Required period operations completed; preserve original movements	Transfer temporary income/expense balances to retained earnings

The current example combines accrual and settlement when they share a date, but the separate obligation remains the general operation. Customer advances are debit cash/credit contract liability and become revenue only on the supported performance event. Inventory costing requires an admissible policy; negative inventory and collection beyond an identified claim are

rejected. Non-cash acquisitions, write-offs, impairment and currency movements require their own classified templates. An unknown residual never enters one of these operations merely to force a reconciliation.

Reporting maps asset and contra-asset balances into the balance sheet and liability/equity balances with their opposite sign. Before closing, current profit is included once in equity; after closing it is already in retained earnings. Income uses period movements in income/expense accounts, and OCI uses its own designated accounts. Cash flow sums cash legs by economic operation: Northstar's collections, suppliers, payroll, interest and taxes are operating, plant payments investing, and debt/equity proceeds or repayments and dividends financing. Non-cash depreciation is excluded from direct cash flow and added back only in the indirect reconciliation. Note schedules use the same contracts, lots and classified movements as the statements.

8.11 Scenario meaning and information access

An `ExogenousStress` supplies a path or joint driver law while holding the historical evidence model fixed. `ObservationalConditioning` additionally supplies actual observations and a joint probability model; the finite method in Chapter 6 evaluates the conditional distribution when its normalising mass is positive. `StructuralIntervention` requires a named structural model and replaces designated equations, retaining its exogenous noise. An absent structural model gives `UnsupportedScenario`, not an intervention inferred from an accounting identity (Pearl, 2009, Section 3.2.1). The textual Core 0 `scenario` declaration is an exogenous stress only; the other modes require the explicit typed API records.

At each decision date the runtime constructs an information view containing eligible disclosures, committed state through that date, currently observed drivers and the declared forecasting model. Future realised innovations and future outcomes are inaccessible. A stress path may be announced in advance, but this is an explicit visibility policy. Otherwise management sees only its current and past values. A bank's allowance at the opening date may use a forecast distribution of defaults; it must not inspect the outer simulation's future realised default. Inner valuation randomness has a separate stream domain from outer realised outcomes.

Funding follows a declared waterfall: available cash, committed eligible facilities up to their limits, and separately authorised capital actions. Each action checks collateral, covenants and settlement timing. If an amount remains due, record an economic funding failure and the unpaid obligation. If required valuation or accounting is unsupported, stop the affected dependencies as incomplete. The branch manifest preserves the last committed state, completed outputs, unavailable outputs and reason. These distinctions let an analyst see a meaningful bad outcome without confusing it with broken computation.

Chapter 9

Accounting methods for an operating company

9.1 A common core with distinct accounting books

The common core governs evidence, time, money, events, journals and provenance. Reporting standards govern recognition, measurement and presentation within that core. The design preserves three levels of meaning: economic terms such as a promise to pay cash; an accounting treatment under a particular standard; and a company's reported label or extension. These levels are connected by explicit rules and mappings, rather than collapsed into one universal chart of accounts.

A rule package includes a standards register with jurisdiction, issuer class, effective dates, transition provisions, permitted elections, paragraph references, implementation version and tests. The package selector uses the entity, reporting basis, period and adopted elections. It must distinguish a standard's mandatory effective date from permitted early adoption. For example, the IFRS Foundation's IFRS 18 overview states an effective date of 1 January 2027 with earlier application permitted (IFRS Foundation, 2026c). A 2026 historical report cannot be assigned that presentation regime solely because the latest installed package includes it.

IFRS, US GAAP and ASBE packages share economic contracts and events where appropriate, but produce separate accounting books. A reconciliation between books consists of explained adjustments with origins in differing treatments, not a renaming of accounts. The first implementation can preserve all three source bases even while only a selected subset has executable rules. Such a source is represented as observed information with unsupported simulation treatments, rather than being silently converted to the supported basis.

The official IFRS overviews for IFRS 9, IAS 32 and IFRS 17 establish the relevant scope of financial-instrument measurement, presentation and insurance contracts (IFRS Foundation, 2026d,a,b). They are orientation sources, not substitutes for reviewed paragraph-level implementations. The retained official Ministry of Finance ASBE 21 page now supports selected lease provisions. The official 2017 ASBE 22 text also supports the selected credit-loss, interest and write-off distinctions worked through in Chapter 11 (Ministry of Finance of the People's Republic of China, 2017). Entity-specific issuance and amendment chains and broader ASBE authority remain incomplete. Chapter 12 compares the dated ASBE 25 premium-allocation and incurred-claim provisions with the retained AASB 17 text (Ministry of Finance of the People's Republic of China, 2020). The Ministry's 2025 explanation supports the stated entity-class timetable, while Ping An's reports supply an actual adoption and comparative-restatement example (Section 12.7). Direct verification of the operative issuance and implementation notices

remains open, including the detailed conditions of the later simplifications. Current ASC access is also unresolved. Historical amendments and issuer policies inform proposed comparisons without substituting for operative authority. These source limits are addressed explicitly in the following dated-method discussion.

9.2 Non-financial companies

The initial industry package covers sales, collections, purchases, payments, inventory movements, payroll, capital expenditure, depreciation and basic financing. Revenue recognition is a separate rule from invoicing and cash collection. A service contract may generate a performance event before an invoice, creating a different asset from an unconditional receivable. Customer advances create obligations before revenue. Returns, rebates and variable consideration need separate assumptions and subsequent adjustments.

Inventory state includes quantity, cost, location, ownership and lot or cohort. A costing method is selected within the reporting package's permissible domain. An aggregate stock can support a simple model only if its approximation is declared. A commodity-price shock can expose differences between old and new inventory that an average value suppresses. Fixed-asset state likewise requires acquisition dates, components, useful lives, residual values, depreciation method, tax basis and impairment history when those affect the requested forecast.

Industry extensions add economic mechanisms without bypassing the accounting core. Construction requires performance and completion states; extractive industries require reserves, depletion and restoration obligations; utilities may require rate-regulation treatment; software businesses may require contract liabilities and development-cost decisions. The module manifest states what is represented, what is executable, and what remains aggregated.

9.3 A matched inventory case across reporting profiles

Standards packages need operation-level decisions. For interchangeable goods, the inspected AASB 102 text incorporating IAS 2 permits FIFO or weighted average in paragraph 25 and explains both in paragraph 27 (Australian Accounting Standards Board, 2015). The reproduced ASBE 1 text specifies FIFO, weighted average or specific identification in Article 14; its official Ministry version could not be retrieved in this review, so that reproduction has provisional source status (Ministry of Finance of the People's Republic of China, 2006a). Caterpillar's company-authored 2024 report states that it principally uses LIFO and supplies a FIFO comparison (Caterpillar Inc., 2024, Note 1.C). That is evidence of a disclosed US-GAAP company policy; full ASC 330 paragraph authority remains an acquisition task. It is not a rule that every US company must use LIFO.

Take the same purchases in three separate books: ten units at 5, followed by ten at 7, then delivery of twelve units. FIFO charges $10(5) + 2(7) = 64$ to cost of sales and leaves inventory 56. LIFO charges $10(7) + 2(5) = 80$ and leaves inventory 40. Periodic weighted average uses cost $120/20 = 6$, charges 72 and leaves 48. The physical deliveries and supplier cash flows agree; profit and carrying inventory differ by the selected costing rule. A concept mapping cannot remove that difference. The FIFO/average selector in an IAS 2 profile rejects LIFO as an unsupported election under the inspected paragraph. The US company profile admits its evidenced LIFO policy while retaining its narrower authority. An ASBE package remains source-provisional until the attributed text and applicable amendments are officially verified.

Inventory impairment illustrates another required memory. Under inspected AASB 102 para-

graphs 28–34, an item costing 56 written down to net realisable value 50 records expense 6. If the conditions reverse and net realisable value becomes 54, reversal is 4, capped by original cost; the previous write-down must therefore remain in state. The reproduced ASBE 1 Article 19 also addresses reversal within the earlier provision. This paragraph is not used to assert the US treatment, whose primary authority was unavailable. Source limitations constrain the treatment claim, while the separate-book method and exact costing calculations remain fully specified.

9.4 Operating events

The operating methods use the selected revenue, inventory and asset provisions of Australian Accounting Standards Board (2022c, 2015, 2022a). The lease and gross-tax case uses the dated lease, income-tax and cash-flow compilations (Australian Accounting Standards Board, 2023c,b,a). Each rule still needs the entity, period, transaction facts and elections described below.

Every event uses the event-basis, authorization and commit contracts described in Section 8.4. The event inputs described here are required in addition to identity and date. Each object identifier resolves to its contract, context and admitted source or assumption. A purported cash collection does not establish a sale; a purchase order does not establish receipt.

Revenue, billing and collection are separate transitions. A delivery event names the contract, performance obligation, quantity, allocated price and control evidence. Recognition first releases a related contract liability; earned consideration not yet unconditional creates a contract asset. Both credit revenue. A missing control finding or unresolved price constraint leaves the recognition unresolved.

An invoice or right-to-payment event names the contract, bill and unconditional amount. It debits the receivable and credits an existing contract asset, with any excess recorded as a contract liability. A collection names that receivable, settlement and amount and debits cash while crediting the receivable. An excess payment needs a separate customer-advance method; it cannot be credited to revenue merely because cash arrived. These transitions retain cumulative performance, billing and collection independently.

Purchasing starts when the company receives owned goods, not when it places an order. The event records lot, quantity, unit cost and ownership evidence, debits inventory and credits the payable. Consigned or unreceived goods cannot pass that ownership condition. Consumption or sale identifies an available lot allocation, quantity and admitted cost formula; it debits cost of goods sold or an eligible production asset and credits inventory. Overselling does not create stock. Supplier payment separately debits the payable and credits actual cash. Insufficient cash blocks that settlement and can trigger a separately modelled default; it never creates a hidden loan.

A tangible-asset event records component cost, residual value, useful life, depreciation method and the date it becomes available for use. Initial debit to PPE has a corresponding cash or payable credit; depreciation later debits expense or an eligible asset and credits accumulated depreciation. Employee service records earned pay, deductions and employer obligations for the service period, separately from settlement. Unknown labour or tax law cannot become a zero payable.

Closing derives revenue, expenses, OCI and owner movements from those events using admitted statement mappings. It checks cash, inventory, financing and tax roll-forwards rather than introducing a balancing entry. These event fields form the method's required inputs; each identifier resolves to the same contract, context and evidence used by its authorization.

AASB 15 paragraphs 9, 22, 31, 35–43, 47–59, 73–86 and 105–108 provide the selected revenue conditions; the retained compilation dated 31 December 2022 applies to annual periods beginning on or after 1 January 2023 and before 1 July 2026, covering the stated for-profit FY 2024 example. Variable consideration is estimated under a selected method and admitted only subject to the significant-reversal constraint, not recognized solely because its unconstrained expectation is positive. Principal versus agent, returns, warranties, contract modifications and distinct bundles require their own scoped decision records. If their facts are unresolved, retain alternative eligible methods; do not borrow the simple delivery rule by label.

The following schedules make the distinction between performance and billing explicit. For one performance obligation with price 100, cumulative revenue 60, unconditional bills 50 and collection 30, AR=20, CA=10, CL=0 and cash=30. Revenue 60 is represented by those three assets. Billing before performance changes intermediate balances but not the final result. For bills 80 instead, AR=50, CA=0, CL=20; net assets remain 60. Cumulative performed-service cost 40 then 70 posts 40 then 30, not 110. A downward revision follows the correction method and keeps the old result tied to its original view.

For this simple contract, let R be cumulative recognised revenue, B unconditional billings and C collections, with $0 \le C \le B$. The unpaid bills determine accounts receivable (AR); performance in excess of billing is a contract asset (CA), while billing in excess of performance is a contract liability (CL). After each event,

$$\text{AR} = B - C, \quad \text{CA} = \max(R - B, 0), \quad \text{CL} = \max(B - R, 0), \quad \text{cash received} = C.$$

This recurrence applies per contract, with no impairment, variable consideration or unbilled prepayment; it is not permission to offset unrelated contracts. For billing 80 before delivery, debit AR 80/credit CL 80. Recognizing service 60 later debits CL 60 and credits revenue 60. Collecting 30 debits cash 30/credits AR 30, leaving AR 50 and CL 20, without a second receivable or an artificial contract asset. If service comes first, recognize CA 60/revenue 60, then bill AR 80 against CA 60 and CL 20. Both orders end at the same state. Cash received before an unconditional bill needs an explicit advance event that debits cash and credits CL and records the advance applied to later billing; it cannot pass through collect(AR).

Inventory uses identified cost for non-interchangeable items or the admitted cost formula for interchangeable items. AASB 102.9, 23, 25 and 28–34 support lower of cost/net realizable value, relevant cost formulas and write-down/reversal conditions. A method never chooses LIFO merely to fit a reported balance. In a second lot example, purchases are 10 units at 2 and 10 at 3, followed by sale of 12 units. FIFO gives cost 26 and stock 8 units/cost 24. Weighted average gives cost 30 and stock 20. Under the same given selling price 48, margin differs 22 versus 18. Both alternatives require their own permitted policy, consistent application and evidence; matching total cash does not remove the difference. NRV 16 for the remaining FIFO lot records expense 8, inventory allowance 8; recovery to NRV 22 reverses 6, leaving allowance 2. Recovery to 30 is capped at original cost 24. US LIFO remains a separately sourced question.

The retained AASB 116.7/15, 29–30 and 48–62 support cost and depreciation. Asset 100, residual 10, three-year even use gives 30 per full year. A year-two estimate change after first-year depreciation to residual 10 and three remaining years gives (70−10)/3=20 prospectively; an error in original cost is a correction, not an estimate change. Component replacements derecognize the replaced component's remaining amount before recognizing its replacement; no perpetual double asset. Borrowing-cost capitalization, impairment, provisions, employee benefits and share-based payment are specified as additional scoped methods below, not silently assigned zero in real-company cases.

9.5 Apple: asset balances do not identify asset movements

The preceding methods need receipts, lots, useful lives and settlement dates. Apple's disclosures supply some of the balances those events must explain, but they do not supply the events themselves. Its FIFO and straight-line policies narrow the permitted cost calculations without identifying the actual inventory lots or asset cohorts (Apple Inc., 2024, Note 1, printed p. 34). Comparing the notes with the cash-flow statement shows why filling those missing inputs by subtracting balances can produce a wrong history.

9.5.1 An inventory adjustment is not purchases or payments

Between 28 September and 28 December 2024, components fall from 3,627 to 2,792 million dollars while finished goods rise from 3,659 to 4,119 million. Total inventory falls from 7,286 to 6,911: the component changes, −835 and +460, sum to −375 million (Apple Inc., 2025, Note 5, printed p. 9). Those two classes do not reveal quantities, unit costs or the transfers and sales that produced the balances.

The quarter's cash-flow statement reports +215 million for inventories within its reconciliation of net income to operating cash (Apple Inc., 2025, printed p. 5). That is an indirect reconciliation adjustment, not total cash paid to suppliers. Even replacing it with minus the stock change would give +375, a different quantity. The difference $215 - 375 = -160$ million remains unexplained by these selected disclosures. In fiscal 2024, inventory rose from 6,331 to 7,286, while the cash-flow adjustment was −1,046; relative to the −955 stock-change shortcut the difference is −91 million (Apple Inc., 2024, printed pp. 31 and 33). Neither residual is an identified currency effect, write-down or other journal entry. Production, consumption, ownership, valuation and payment histories must be established before they can explain the differences.

9.5.2 Testing a proposed fixed-asset history

The fixed-asset notes distinguish three measurements. The table shows gross property, plant and equipment (PP&E), its accumulated depreciation and the remaining carrying amount, all in millions of dollars (Apple Inc., 2024, Note 5, printed p. 39) (Apple Inc., 2025, Note 5, printed p. 9).

Amount	30 Sep 2023	28 Sep 2024	28 Dec 2024
Gross PP&E	114,599	119,128	120,615
Accumulated depreciation	70,884	73,448	74,546
Net PP&E	43,715	45,680	46,069

Accumulated depreciation is shown here as a positive deduction: gross less that deduction equals net. During fiscal 2024, gross cost increases 4,529, accumulated depreciation increases 2,564 and net PP&E increases 1,965. Yet the note reports PP&E depreciation expense of 8.2 billion, and the cash-flow statement reports capital payments of 9,447 million. Its broader depreciation-and-amortisation adjustment is 11,445 million (Apple Inc., 2024, printed pp. 33 and 39). These amounts differ in meaning; the 8.2 billion also has coarser displayed precision. The change in accumulated depreciation cannot be substituted for the year's depreciation charge.

Consider a deliberately limited asset book to test a tempting reconstruction. Let G_0, G_1 be opening and closing gross cost and B_0, B_1 accumulated depreciation, recorded as positive amounts. Let A be gross additions, R the original cost of retired assets, d the depreciation

charge and S the accumulated depreciation removed with those assets. Assume there are no other movements and all depreciation is expensed. For this illustration, use the displayed amounts as exact, including $d = 8{,}200$ million. The two balances must then satisfy

$$G_1 = G_0 + A - R, \qquad B_1 = B_0 + d - S, \qquad 0 \le S \le R. \tag{9.1}$$

The inequality requires a nonnegative carrying amount for the retired assets in this cost model. If additions are equated with the 9,447 of cash payments, the gross balance requires $R = 9{,}447 - 4{,}529 = 4{,}918$. The depreciation balance requires $S = 8{,}200 - 2{,}564 = 5{,}636$. The proposed book removes more accumulated depreciation than original cost, by 718 million. It is infeasible under these combined assumptions. That finding rejects the proposed history; it does not establish an error in Apple's filing or identify which assumption must change.

The result is not sensitive to small rounding differences under a specified rounding model. If the million-dollar balances and payment are rounded to the nearest million, and 8.2 billion to the nearest 0.1 billion, the excess $d - 9{,}447 + (45{,}680 - 43{,}715)$ lies in the closed outer envelope $[666.5, 769.5]$ million. Its lower endpoint is $8{,}150 - 9{,}447.5 + 45{,}679.5 - 43{,}715.5 = 666.5$. This is a conditional precision check, not a verified issuer rounding rule; the exact tie convention remains unknown.

Allowing additions to be partly unpaid provides two simple repairs. Retain the same selected opening and closing balances, 8,200 depreciation and 9,447 cash payments. Assume retirement at the start of the year with zero proceeds, depreciation of surviving assets, and new acquisitions at year end. There are no opening capital payables, taxes or other asset movements. The figures below are constructed histories, not estimates of actual Apple transactions.

Annual movement (USD million)	History 1	History 2
Gross additions	10,529	11,529
Gross cost retired	6,000	7,000
Accumulated depreciation removed	5,636	5,636
Retired carrying amount, expensed	364	1,364
Cash paid for additions	9,447	9,447
Additions remaining payable	1,082	2,082

Both histories satisfy equation (9.1); their gross additions can also be allocated to match the three reported asset classes. For History 1, acquisition debits PP&E cost 10,529 and credits cash 9,447 and a capital payable 1,082. Depreciation debits expense and credits accumulated depreciation by 8,200. Retirement debits accumulated depreciation 5,636 and loss 364, while crediting gross PP&E cost 6,000. Net PP&E rises 1,965 and cash falls 9,447, so total assets fall 7,482. Liabilities rise 1,082 and equity falls 8,564 through the two expenses: $1{,}082 - 8{,}564 = -7{,}482$. History 2 has the same asset change but liabilities rise 2,082 and equity falls 9,564. These are coherent subsystem consequences, not a fit to all of Apple's statements or an exhaustive family of real asset histories. Neither history receives a probability or preference from this arithmetic.

The following quarter supplies a new snapshot, not the missing annual movements. Its gross, accumulated and net changes are 1,487, 1,098 and 389 million, whereas capital payments are 2,940 and broad depreciation/amortisation is 3,080 (Apple Inc., 2025, printed pp. 5 and 9). The inspected interim note does not separately identify PP&E depreciation expense. The lease note adds a related distinction: finance-lease right-of-use assets are included in PP&E, while operating-lease right-of-use assets are in other noncurrent assets. The 1.0 billion of noncash right-of-use acquisitions covers the disclosed lease population (Apple Inc., 2024, Note 8, printed p. 42); adding it all to PP&E or to cash capital expenditure would erase that distinction.

The method therefore preserves reported balances, cash payments, expense and unresolved movement components separately. A future cost or funding query still needs lot and asset ages, useful lives, residual values, impairment and settlement facts. The real filing has supplied constraints and ruled out one simple history, while leaving several different explanations open.

9.6 Lease and tax integration

A lease connects asset use, a financing obligation and income tax. The following schedule extends the operating-company example by separating these three measurements. The retained lease and tax dossiers supply the dated provisions, modification case and assumptions. The official lease compilation is dated December 2023 despite the older filename. The right-of-use (ROU) asset represents the admitted use of the leased asset. Its cost, the lease liability, accumulated depreciation, remaining payments, rate version and option judgments are separate state fields. Tax retains carrying amounts, legal tax bases, reversal dates, enacted/substantively-enacted rate evidence, recovery support, and original profit/OCI/equity allocation. The deferred tax asset (DTA) and deferred tax liability (DTL) are retained gross before any permitted offset presentation. A liability schedule is not a tax-base schedule.

For the two annual 121 payments at 10%, initial liability and ROU are 210. Year-one interest is 21, liability after payment 110, depreciation 105 and ROU 105. Year two interest is 11, and both balances finish at zero. A same-asset modification just after year one creates two future payments 144 at revised 20%, PV 220, so liability increases 110 and ROU increases 110 to 215. A payment index reset has different rate rules; it cannot reuse that modification blindly.

To demonstrate connection to company statements, assume a synthetic entity with opening cash/equity 500, no other opening balances, and annual cash service revenue 400 less cash operating costs 100, all recognized in their years. There are no dividends, other assets, debt, losses or permanent differences. Add the base lease above. Assumed tax law deducts cash lease payments against taxable income, attributes those deductions to the lease liability, gives zero tax bases to ROU/liability, and taxes profits at enacted 25%. Sufficient usable profits and all recognition conditions for DTA are assumed. Tax is paid the following year. No current-period tax payment is invented from tax expense.

At commencement debit DTA 52.5 and credit DTL 52.5 in addition to the lease recognition. At year-one close DTA is 27.5 and DTL 26.25. The DTA reduction 25 is deferred expense; DTL reduction 26.25 is deferred benefit, net benefit 1.25. Taxable profit is 300−121=179; current tax/payable 44.75. Pretax book profit is 300−105−21=174, total tax 43.5 and profit 130.5. At year two, deferred expense 1.25 combines with current tax 44.75 to give total tax 46 on pretax profit 184.

Closing balance	2024	2025
Cash	679	813.25
ROU asset	105	0
Deferred tax asset	27.5	0
Lease liability	110	0
Deferred tax liability	26.25	0
Current tax payable	44.75	44.75
Equity	630.5	768.5

2024 assets 811.5 equal liabilities 181 plus equity 630.5; 2025 assets 813.25 equal liabilities 44.75 plus equity 768.5. For this case, elect financing classification for lease interest under the

retained AASB 107 regime. The operating and financing cash totals below depend on that presentation election. Direct operating cash is 300 then 255.25 after the first tax payment; financing lease cash is −121 each year; no investing cash arises. Indirect operating cash in year one is profit 130.5 plus depreciation 105 plus interest 21 minus deferred benefit 1.25 plus unpaid current tax 44.75 = 300. Year two gives profit 138 +105 +11 +1.25 =255.25. The second tax payable settles next year. Zero OCI and no owner transactions mean the equity change equals cumulative profit. This case tests gross tax, cash timing and financial-statement coherence, not a real entity's tax law.

9.7 Apple: recognising tax and settling the obligation

The lease example separates a tax charge from the payment that follows it. Apple's fiscal 2024 disclosures show why that separation matters when several tax jurisdictions, deferred balances and a large disputed obligation enter the same accounts. Its annual income-tax expense was 29,749 million dollars; net cash paid for income taxes was 26,102 million (Apple Inc., 2024, printed pp. 29 and 33). The 3,647 difference is not, by itself, an increase in tax payable. Deferred tax and other balance movements must first be distinguished from current tax and settlement.

9.7.1 The expense and balance reconciliations answer different questions

Note 7 divides the 2024 expense between current and deferred amounts for federal, state and foreign taxes. In millions of dollars, the reported components are as follows (Apple Inc., 2024, Note 7, printed p. 40). Parentheses in the source represent negative expense, or a tax benefit.

Tax category	Current	Deferred	Total
Federal	5,571	−3,080	2,491
State	1,726	−298	1,428
Foreign	25,483	347	25,830
Sum	32,780	−3,031	29,749

The current charge is therefore 32,780 and the deferred benefit 3,031. The tax payable has its own scope. Note 6 reports current income taxes payable of 26,601 and noncurrent income taxes payable of 9,254 at 28 September 2024, against 8,819 and 15,457 a year earlier. Combining both maturity classes gives 35,855 against 24,276, an increase of 11,579 (Apple Inc., 2024, Note 6, printed p. 39). Current expense less net cash taxes is only 32,780 − 26,102 = 6,678. Reclassifying an amount between current and noncurrent cannot explain the difference in their combined total.

To expose the missing movements, let T denote that combined tax payable and N the net deferred tax asset reported in Note 7. Let C be current tax expense, D deferred tax expense (negative for a benefit), and P net cash taxes paid, all for the same annual period and in millions of dollars. Define U and V as the residuals needed to connect the reported amounts:

$$\Delta T = C - P + U, \qquad \Delta N = -D + V. \tag{9.2}$$

These are diagnostic definitions, not assertions that every tax balance has the same recognition or settlement scope. The first relation gives $U = 11{,}579 - 6{,}678 = 4{,}901$. Setting that residual to zero would produce the wrong closing payable. Assigning it to currency translation, uncertain tax positions or another journal would require the missing transaction and classification evidence.

The deferred balance illustrates the second relation. At September 2024, gross deferred tax assets of 34,873 less a valuation allowance of 8,866 leave 26,007. Deducting deferred tax liabilities of 6,805 gives net deferred tax assets of 19,202, compared with 17,251 a year earlier (Apple Inc., 2024, Note 7, printed p. 41). The increase is 1,951, so $V = 1{,}951 - 3{,}031 = -1{,}080$. The deferred benefit in profit does not explain the whole net-balance movement. Its allocation among currency, equity, other comprehensive income and other causes remains unresolved here.

Note 6's 19,499 deferred-tax-asset presentation is also different from Note 7's 19,202 net position. Their difference of 297 is an arithmetic consequence, not a separately disclosed liability line or authority to offset jurisdictions. The reported temporary differences—including research expenditure, deferred revenue, leases and depreciation—need their own tax bases and recovery conditions. The note reports indefinitely available Irish foreign-tax and California R&D credit carryforwards, but also a valuation allowance for those credits and some other differences. An indefinite term does not establish that an asset is recoverable (Apple Inc., 2024, Notes 6–7, printed pp. 39–41).

9.7.2 A recognised charge, restricted assets and a later payment

Apple attributes its large 2024 State Aid charge to the September 2024 court decision confirming the European Commission's recovery decision. This is the issuer's account of the dispute and its accounting consequence. Note 7 describes a 15.8 billion dollar payable to Ireland, partly offset by a 4.8 billion US foreign tax credit and an 823 million decrease in unrecognised tax benefits (Apple Inc., 2024, Note 7, printed pp. 39–40).

The narrative calls the net charge 10.2 billion; the statutory-rate reconciliation gives the more precise amount 10,246 million. Subtracting the displayed narrative centres gives $15{,}800 - 4{,}800 - 823 = 10{,}177$, which is 69 million lower. That is compatible with the coarser inputs: if the two billion-dollar amounts are rounded to the nearest 0.1 billion and 823 to the nearest million, the component calculation lies in the closed outer interval $[10{,}076.5, 10{,}277.5]$ million, containing 10,246. The 10.2 billion charge is compatible with it as well. This is a conditional rounding check, with the issuer's precise rule and tie convention unverified; the centre difference does not establish a conflicting source assertion.

The tax charge does not make the settlement funds disappear at recognition. At 28 September 2024 the escrow comprised 2.6 billion dollars of cash and cash equivalents and 13.2 billion of current marketable securities, restricted from general use (Apple Inc., 2024, Notes 4 and 7, printed pp. 37 and 39). The annual liquidity discussion separately identifies the obligation and the funds held to settle it. The following quarter reports both a payment and the remaining obligation (Apple Inc., 2024, printed p. 26) (Apple Inc., 2025, printed p. 16).

State Aid obligation and settlement	EUR billion	USD billion
Obligation at 28 Sep 2024	14.2	15.8
Payment during the following quarter	11.0	11.9
Obligation at 28 Dec 2024	3.3	3.4

The three rows are different dated quantities. In dollars, a movement equation with opening obligation 15.8, payment 11.9 and closing obligation 3.4 requires an additional net movement of $3.4 + 11.9 - 15.8 = -0.5$ billion. Under nearest rounding to 0.1 billion, its outer interval is $[-650, -350]$ million, excluding zero. The same calculation in euros gives $3.3 + 11.0 - 14.2 = 0.1$ billion, with outer interval $[-50, 250]$ million, which includes zero. Neither calculation supplies a common exchange rate or an exact exchange-gain entry. The interim tax discussion mentions

foreign-currency revaluation effects related to the decision, but does not allocate this 500 million dollar residual (Apple Inc., 2025, printed p. 16).

The remaining funds also retain their asset classifications. At December 2024, restricted cash and cash equivalents were 126 million dollars and restricted current marketable securities 3.3 billion (Apple Inc., 2025, Note 4, printed p. 8). Their displayed sum, 3,426 million, is compatible with the rounded 3.4 billion obligation. It is not all cash available for ordinary operations. The opening 2.6 billion of restricted cash alone could not have funded the reported 11.9 billion payment; securities sales, maturities, transfers and valuations need their own histories.

For an isolated settlement illustration, suppose 11,900 million of restricted cash is available and the amount paid equals the liability's carrying value at settlement. Debit tax payable and credit restricted cash by 11,900. Assets and liabilities each fall by 11,900; equity and current-period tax expense do not change in this posting. This assumes the required cash conversion has already occurred and omits any associated gain, loss or fee. It explains settlement of an existing obligation without pretending to reconstruct Apple's funding and currency entries. The quarter separately reports tax expense of 6,254 and net cash taxes of 18,651 million (Apple Inc., 2025, printed pp. 1 and 5); neither should be replaced with the Ireland payment or treated as another copy of the preceding year's charge.

9.7.3 Uncertain tax positions do not supply a payment distribution

The last part of Note 7 preserves another population: gross unrecognised tax benefits. Its 2024 movement is $19{,}454 + 1{,}727 - 386 + 2{,}542 - 1{,}070 - 229 = 22{,}038$ million. The increases concern prior- and current-year positions; the decreases concern changed prior-year positions, settlements and expired limitation periods. Interest and penalties are excluded from this rollforward (Apple Inc., 2024, Note 7, printed p. 41). The 1,070 settlement-related decrease is a movement in that gross balance, not an independently identified total of cash paid to tax authorities.

Of the rounded 22.0 billion gross amount, the issuer says that 10.8 billion, if recognised, would affect its effective tax rate. It also describes a reasonably possible decrease of approximately 5–13 billion over the next twelve months, primarily concerning transfer pricing and deemed repatriation tax, while noting uncertainty about examination timing. These are different conditional statements. The range is neither a probability law nor a promised cash payment or expense release. A forecast still needs jurisdictional outcomes, recognition, interest, penalties and settlement dates. Retaining those unknowns keeps the tax note connected to the company's funding needs without turning management's dated assessment into an invented future transaction.

9.8 Returning to the company forecast

The Northstar case remains the broader ordinary-company teaching baseline: collections, purchases/payments, inventory, payroll, PPE, financing, tax, statements, annual/interim and revision views are already specified. Its 2024 collection/purchase/payment solution is 245/155/145. The 2025 forecast in the Northstar reference fixture opens with cash 74.6 and equity 90.6, spends 20 on capital expenditure at period end, pays dividend 8 and uses a base floating rate 7%. Its simplified tax is settled in the year earned. The running case uses fixed-rate share α; here floating-rate share $x = 1 - \alpha$ describes the same uncertainty. For 2025 the same unknown

floating fraction $x \in [0,1]$ determines every statement amount:

$$\begin{aligned} I(x) &= 5 + 2x, & \text{pretax profit}(x) &= 82 - 2x, \\ \text{tax}(x) &= 16.4 - 0.4x, & \text{profit}(x) &= 65.6 - 1.6x, \\ \text{operating cash}(x) &= 48.6 - 1.6x, & \text{closing cash}(x) &= 95.2 - 1.6x, \\ \text{equity}(x) &= 148.2 - 1.6x, & \text{assets}(x) &= 288.2 - 1.6x. \end{aligned}$$

Receivables 55, inventory 40, net PPE 98, payables 40 and debt 100 complete the balance sheet. Liabilities are 140 for every x, so the common cash/equity slope preserves the accounting identity. There is no default distribution on x. At a 9% floating rate the all-floating history has interest 9, profit 62.4, cash 92, equity 145 and assets 285. The annual cash bounds still cannot establish daily payment solvency without event-date cash flows.

The lease/tax example extends the interaction evidence without changing the original Northstar fixture or pretending that its teaching 20% tax formula is an IAS 12 method. In a real-company package, each required method is admitted separately and unsupported operations remain visible. The extended obligations for pensions, impairment, provisions, valuation and share-based payment are in the additional corporate methods that follow; omission from a finite company case never proves the actual company has none.

9.9 Employee benefits

The additional methods below use the selected IAS 19, IAS 36, IAS 37, IFRS 2, IFRS 13 and IAS 23 provisions in the original 2023 EU incorporating regulation (European Commission, 2023). This is historical mechanism support; later amendments and actual entity facts remain separate applicability inputs.

DefinedContributionState retains service periods, promised contribution formula, contributions due/paid and applicable legal terms. For earned service requiring contribution 10, debit employee expense 10/credit payable 10; payment settles it. Calling a plan defined contribution needs the limited-obligation facts; a guaranteed benefit or additional funding obligation cannot disappear by name.

The defined-benefit obligation (DBO) measures accrued employee benefits under the stated actuarial assumptions. DefinedBenefitState retains employee/cohort service, benefit formula, projected salary, retirement/turnover/mortality assumptions, discount-curve currency/term, obligation, plan-asset fair value, asset ceiling, contributions, benefits and remeasurements. IAS 19.67–74 requires projected unit credit and the stated service attribution; .83 supplies discount-rate requirements. Assumptions are separately admitted measurements, not constants provided by the accounting rule. A one-employee toy plan promises 100 for each completed service year, paid after year two, with certain completion, zero salary growth and supported 10% discount. At year-one end accrued benefit 100 due in one year has PV 1000/11. In year two that opening obligation accretes 100/11 and the second earned benefit adds 100; obligation 200 then settles 200. Merely using current salary times years without timing would give the wrong year-one obligation.

A separate funded-plan roll-forward supplies an independent statement example: opening DBO 100, plan assets 80 and net liability 20; current service 12; interest 5 on DBO, interest 4 on assets at 5%; contributions 10 at year end; benefits 8 paid from plan at year end; actuarial obligation loss 7; actual asset return 6. DBO closes 100+12+5−8+7=116, assets 80+6+10−8=88, net liability 28. Expense is 12+1=13, OCI loss is 7−(6−4)=5. Debit service expense 12/credit net pension liability 12; debit interest expense 1/credit liability 1; debit OCI 5/credit liability 5; debit liability 10/credit cash 10. End liability 20+13+5−10=28. Plan-paid benefits reduce

both DBO and plan assets and do not cause a second employer cash payment. IAS 19.113, .120–127 supports these components; remeasurement OCI is not recycled to profit under 122. A surplus requires the asset ceiling and applicable minimum funding analysis, not recognition of any calculated negative liability.

Unknown actuarial inputs give conditional obligations or measurement_unresolved. Curtailment, settlement, amendment and multi-employer/insured arrangements need their own fact conditions and measurement-date remeasurement. A buyout is not settlement if obligations remain. Sensitivities vary one assumption only when that conditional meaning is explicit; correlated risks require joint scenarios.

9.10 Impairment and provisions

ImpairmentState retains asset/CGU boundaries, carrying amounts, allocated goodwill, corporate assets, recoverable-amount inputs and previous impairments/depreciation. IAS 36.9–10 determines indicators and mandatory annual tests. Recoverable amount is the higher of value in use and fair value less disposal costs (.18); it is not the lower of two pessimistic scenarios. VIU needs the supported cash flows, forecast horizon/growth, discount and exclusion of inappropriate financing/tax or uncommitted restructuring effects under 33–57. Compute conditional values if those estimates are unknown; do not infer them from historical profitability.

For a cash-generating unit (CGU), assets excluding goodwill 100 plus goodwill 20, recoverable amount 90: loss 30 first removes goodwill 20 and then reduces other assets 10 subject to their floors under 104–105. Debit impairment 30/credit goodwill 20/credit asset allowance 10. If the asset floor is 95 the requested reduction to 90 is impermissible; apply the remaining allocation/recognition conditions, not a forced plug. For a standalone cost-model asset with carrying 80, recoverable 60, recognize 20 loss. At a later measurement before further depreciation, supported recovery to 90 permits reversal only 20 to the no-impairment 80 ceiling (.117–119). Goodwill impairment is not reversed (.124). Different dates require the hypothetical unimpaired depreciation path, not the old nominal cost as cap.

ProvisionState retains obligating past event, present legal/constructive duty, outflow assessment, estimate distribution, settlement dates, risk/discount basis, reimbursements and evidence. IAS 37.14 requires present obligation, probable outflow and reliable estimate; .36–47 governs best estimate/risk/discount. An intended future operating expenditure without that duty is not a provision. Assume an established obligation to pay 121 in two years, deterministic amount, appropriate 10% rate and no double-counted risk: recognize expense/provision 100, accrete interest 10 then 11, settle 121. The passage-of-time increase is borrowing cost under 60. If the obligation is an eligible asset restoration cost, initial debit follows that asset's method rather than always expense; later changes need the applicable restoration interpretation. Contingent liabilities/disclosures remain visible when recognition conditions are not met; no generic 0.5 threshold may substitute for the relevant standard's required analysis.

9.11 Share-based compensation

AwardState distinguishes equity versus cash settlement, grant/vesting/service dates, granted units, grant-date fair value, service/nonmarket/market conditions, expected units vesting, cumulative expense and changes/cancellations. IFRS 2.7–11, 15–23 supports the employee equity-settled method. With 100 units, grant value 3, two-year service condition and all expected to vest, year-one cumulative expense 150 and equity 150. At vesting only 80 satisfy the service condition: cumulative 240, second-year expense 90. Debit employee expense and

credit equity. Failure of a market condition is different under 21; if the employee satisfies other vesting conditions, it does not automatically reverse the grant-date measured service expense. Unknown conditions cannot be represented by a single forfeiture percentage without their type and evidence.

Cash-settled awards remeasure the liability under 30–33. Same 100 expected units, half service and current unit fair value 4 gives liability/expense 200 at year one; 80 actual units and value 5 at settlement gives total 400, second expense 200, then debit liability 400/credit cash 400. Reusing the equity grant value 3 would be wrong. Actual award valuation requires the specified model/market inputs, option rights, volatility, dividends and calibration; the fair values 3/4/5 are case inputs. Modifications and settlement-choice arrangements require their own conditions, with the original minimum expense and increment analyzed under the applicable provisions rather than overwriting the original grant.

9.12 Fair value and capitalization boundaries

A ValuationRequest names the unit of account, market/date, asset/liability characteristics, principal/accessible market, technique, input hierarchy, counterparty/own non-performance risk where relevant, calibration and sensitivity. IFRS 13.24, .42–43 and .67–80 guide the selected measurement. Quoted identical asset price 7 for 10 units gives 70 in a qualifying Level 1 case; a blockage discount based only on holding size cannot reduce it to 63 under 69/80. A present-value illustration 111 due next year at an independently supported 11% rate gives 100; that equality does not establish the rate, market comparability or accounting classification. Historical EIR, an impairment discount, risk-neutral pricing and a physical expected return answer different questions.

Borrowing-cost capitalization requires a qualifying asset, eligible costs, commencement/suspension/cessation dates and the specific/general-borrowing calculation, before moving interest into cost. IAS 23.8–25 supplies these rules in the retained historical EU text: for specific borrowing, eligible actual costs less temporary investment income; for general borrowing, the applicable weighted rate and cost ceiling. Commencement requires expenditures, borrowing costs and necessary preparation activities; extended inactive periods can suspend it, and substantial completion ends it. Assume specific borrowing 100 outstanding during a full eligible construction year at 5%, with an admitted expenditure schedule and temporary investment income 1 on temporarily unused funds during that period. Net capitalized cost is 4. Real admission still needs the dated standard, actual schedule and eligibility evidence. It is not a default 5% add-on to all assets. Internally developed intangibles likewise need research/development separation and all applicable recognition criteria; a forecast of future sales does not establish an asset.

The extension interface returns the stated measured components, proposed journal, economic delta, disclosures and unresolved inputs under the event-basis, authorization and commit contracts. Calculation success does not permit another operation to fill a missing authority/fact with a default. Real-company acquisition/tax/employee/valuation evidence and the full exceptions remain explicit internal preparation, with proposed answers and competing treatments retained for any later authorized domain review.

9.13 Rule selection

The proposed selection procedure distinguishes operative authority from historical source support. The retained sources include official AASB compilations, the original EU incorporating regulation and official ASBE lease and income-tax texts (Australian Accounting Standards

Board, 2022e,c, 2023c,b; European Commission, 2023; Ministry of Finance of the People's Republic of China, 2019, 2018, 2006b, 2022). Historical US lease additions are separately identified in the retained amendment source (Financial Accounting Standards Board, 2016b).

AuthorityRequirement contains issuer/framework, jurisdiction, entity class, annual-period/application interval, paragraph and local source Ref, incorporating instrument, amendment chain, transition, elections and unresolved dependencies. RuleSelection binds those requirements to the accounting event and evidence view. The algorithm evaluates chronological applicability first, then scope and facts, then available elections/transition, then measurement/presentation. If two supported rules can apply, retain the explicit alternative and required election; if one is contradicted by facts, mark ineligible. Missing current authority gives unresolved, never eligibility through a familiar rule name.

An unchanged historical report remains an assertion under its own basis and vintage. A harmonized comparison creates an adjustment book and reconciliation, with changed recognition/measurement explained. Label mapping alone cannot convert US GAAP to IFRS or ASBE. The four clocks and source version remain part of each rule dependency; later knowledge cannot leak into an original as-of view. Transition entries reconcile opening assets, liabilities, tax and equity, rather than changing old source bytes or plugging an unexplained amount.

9.14 IFRS and incorporating jurisdictions

The retained official AASB texts provide explicit IAS/IFRS incorporation for their stated entity/version scope. AASB 9/15 compilation dates are 31 December 2022 with the source covers' application window; AASB 16 compilation 8 is 31 December 2023, AASB 112 compilation 7 is 29 September 2023. Their filenames are not date evidence. AASB 102's older retained edition is selected historical support for inventory, not current 2024 applicability certification. The retained EU 2023/1803 original provides official EU-adopted IFRS 3/10/IAS 19/21/36/37/IFRS 2/13 passages inspected in this method investigation; its historical adoption scope and later amendments remain separate. The retained amendment investigation supplies additional source evidence, not a blanket acceptance of all rules in these standards.

Initial real-method admission is therefore claim-specific: the selected revenue, lease/tax, financing, credit, group/FX and insurance cases must resolve each named applicability prerequisite. For synthetic demonstrations, declare the chosen dated provisions and assumptions and use the synthetic book mode. No assertion of whole-IFRS compliance follows from those entries.

The base lease in the saved dossiers (two annual 121 payments, 10% rate, no qualifying exemptions or extra costs) has liability 210 initially, 110 after year one, 0 after year two. IFRS-equivalent cost-model ROU is 210/105/0; expenses 126 then 116. Income tax needs its own legal deductions and gross temporary differences, as the connected corporate case demonstrates. Revenue timing, inventory cost formula, financial instrument classification, control and contract boundaries are required rule predicates, not accounting options chosen after seeing which outcome fits best.

9.15 US GAAP proposed methods and authority limitation

The saved official FASB ASU 2016-02 Section A and later 2020-05/2021-09 amendments contain inspected historical ASC 842 additions. Their covers explicitly say an ASU is not authoritative. Current Codification access returned 403 again in this source investigation; the proposed methods cannot be marked current-authority-admitted. ASU amendments, historical FAS 109,

issuer accounting policy and a model's memory are all insufficient substitutes for the operative ASC text and full applicable amendment chain.

The same lease facts include ten-year remaining equipment life, fair value 1000, alternative use to lessor, no ownership transfer, no purchase option, two-year term and PV 210. Under the inspected historical 842-10-25-2/25-3 provisions, the stated facts support the proposed operating classification. Historical 842-20-25-6 and 35-3 yield single lease expense 121 per year, liability 210/110/0 and ROU 210/ 110/0. First year debit lease expense 121/credit liability 21/credit ROU 100; payment debits liability 121/credits cash 121. Next expense 121 decomposes into liability interest 11 and ROU reduction 110. The finance comparison uses the separate depreciation 105 plus interest 21/11 path. Changing classification facts is not a free election. The 75%/90% heuristics in the historical guidance cannot become universal requirements. This complete numerical contrast is useful despite the remaining current-authority limitation.

Proposed ASC 740 temporary-difference treatment retains tax base, reversal dates, enacted rates, valuation allowance and source-specific allocation/presentation. The saved historical FAS 109 and ASU 2019-12 provide inspected enactment support, not a completed current ASC 740 implementation. Under the matched PPE case, book 800/tax 600 at assumed 25% yields DTL 50 and current tax 75 on taxable 300. A future 30% rate that is merely substantively enacted but not legally enacted cannot be imported from the AASB 112 rule into this proposed US treatment; the declared 2025 closing temporary difference is 300, giving DTL 75 at the enacted 25% rate versus DTL 90 at 30%. This deferred-tax comparison is separate from current tax 75 on taxable profit 300. Actual legal enactment remains a separate jurisdictional fact.

The proposed CECL method for in-scope amortised-cost assets estimates lifetime expected cash shortfalls from origination using past/current/forecast evidence, with a documented reversion method beyond a reasonable/supportable forecast horizon and applicable instrument exceptions. It retains allowance roll-forwards, write-offs, recoveries, purchased-credit-deteriorated handling and off-balance commitments separately. It must not copy IFRS 9's 12-month/SICR staging to the same record. A candidate contrast: lifetime discounted loss 6 with 2 attributable to default events in the next year gives IFRS general Stage 1 candidate 2 versus CECL lifetime candidate 6, conditional on scope and method support. No current ASC 326 text was retrieved, and the attempted official amendment path returned 403; this contrast remains an explicitly unadmitted proposed reading, not a source-verified US requirement or posting example. Full current ASC 326, insurance 944, consolidation 810, FX 830 and relevant later amendments are specific internal source tasks before corresponding real admission.

9.16 ASBE lease adoption and deferred tax

For an ASBE lease, selecting the lease standard and selecting the related income-tax treatment require different application dates. Article 68 of the revised ASBE 21 states a general date of 1 January 2019, but the Ministry of Finance's 2018 issuance notice specifies the enterprise groups. Paragraph I requires application from that date for enterprises listed both domestically and overseas, and for overseas-listed enterprises preparing financial statements under IFRS or ASBE. Other enterprises applying ASBE must apply the revision from 1 January 2021 (Ministry of Finance of the People's Republic of China, 2018, issuance notice, para. I).

Paragraph II permits earlier application for enterprises whose parent or subsidiary is listed overseas and prepares its overseas financial statements under IFRS or ASBE. Their early date cannot precede simultaneous application of the specified 2017 financial-instruments and revenue standards. Paragraph III discontinues the 2006 lease standard and its application guidance when the enterprise applies the revision. Thus a 2020 reporting date alone does not establish

use of the revised lease standard for an enterprise in the second group: its qualifying early adoption would need evidence. The retained official Chinese texts include the 2018 notice and attachment and the Ministry's later ASBE 21 page, published on 28 October 2019 (Ministry of Finance of the People's Republic of China, 2018, 2019). They replace the earlier secondary reproduction as support for the inspected wording.

Articles 4–8 require control of identified use, economic benefits and direction; article 6 includes both conditions for substantive substitution and the express presumption when difficult to determine. Articles 14–18 specify commencement, ROU cost and PV, using implicit rate or incremental borrowing rate if the former cannot be determined. Articles 20–23 separate cost-model depreciation and periodic interest; articles 25–29 distinguish reassessment/index changes/modifications and their discount-rate rules. Articles 30–33 define the exemptions; a two-year term is not short-term. Articles 60–67 specify transition choices and disclosures; the chosen transition cannot be inferred from a present closing liability (Ministry of Finance of the People's Republic of China, 2018, attached standard, arts. 4–8, 14–33, 60–67).

For the declared base lease, these inspected recognition/measurement provisions support proposed ROU 210/105/0 and liability 210/110/0 with expense 126/116, under the same stated facts, depreciation assumption and fixed rule version. A same-asset modification is tested against 28–29, rather than labeled a separate lease without increased scope and commensurate consideration. A 2019 first-adopter using article 61(2) records cumulative opening changes without adjusting comparatives, while 61(1) is retrospective; comparison datasets must retain that distinction and the entity's actual election.

The tax question is why equal opening lease balances can still require separate deferred-tax assets and liabilities. The Ministry's original ASBE 18 defines tax bases and temporary differences in articles 4–9 and contains initial-recognition exceptions in articles 11 and 13. Article 13 also limits a deferred-tax asset to the probable taxable profit available to use the deduction. Articles 16–23 govern measurement, recovery or settlement assumptions, subsequent review, allocation and noncurrent classification (Ministry of Finance of the People's Republic of China, 2006b). The original text must be read with later interpretations.

Interpretation 16 removes the initial-recognition exemption for a single transaction that is not a business combination, initially affects neither accounting profit nor taxable income or loss, and creates equal taxable and deductible temporary differences. Section I(a) expressly includes qualifying initial lease liabilities and right-of-use assets; the corresponding deferred-tax liabilities and assets are recognized separately under ASBE 18. Equal carrying amounts alone do not establish equal temporary differences, and the asset's recoverability condition still applies. Section IV makes this treatment effective from 1 January 2023; Section I(b) permits early application from the year of issuance, 2022, with disclosure (Ministry of Finance of the People's Republic of China, 2022, secs. I(a)–(b), IV). This tax date is distinct from the enterprise's lease-standard adoption date. The operative text now supports the treatment previously illustrated by the retained official 2023 tax application case.

For the synthetic two-year lease, retain the assumed payments of 121, 10% discount rate and straight-line depreciation. All amounts use the same illustrative monetary unit. Assume that tax deductions are allocated to the liability, both tax bases are zero, the applicable tax rate is 25%, and sufficient taxable profit satisfies the deferred-tax asset recognition condition. Also assume that the transaction meets Section I(a)'s conditions and falls within its application period. At commencement the ROU asset and liability are each 210. The taxable ROU difference gives a deferred-tax liability (DTL) of 52.5; the deductible liability difference gives a deferred-tax asset (DTA) of 52.5. The entry debits DTA and credits DTL by those amounts. At the first annual close, liability 110 and ROU 105 give DTA 27.5 and DTL 26.25. Reducing the DTA by 25 creates deferred-tax expense; reducing the DTL by 26.25 creates a benefit, leaving net

benefit 1.25. Both balances reverse by expiry. Applying the old exemption to omit both opening deferred balances is wrong for these qualifying facts and dates: it conceals the required gross balances and the later tax effect.

Transition also needs more than an opening journal entry. Section I(b) requires adjustments for qualifying transactions between the earliest comparative opening and application. For qualifying existing balances at that opening, cumulative effects adjust opening retained earnings and the relevant financial-statement items. Consider a separate transition example with cash 500, ROU 105, lease liability 110 and equity 495, with no deferred tax yet recognized. Under the same assumed tax bases, rate and recoverability, recognition debits DTA 27.5, credits DTL 26.25 and credits opening retained earnings 1.25. The resulting balance sheet is

$$\underbrace{500 + 105 + 27.5}_{\text{assets } 632.5} = \underbrace{110 + 26.25}_{\text{liabilities } 136.25} + \underbrace{495 + 1.25}_{\text{equity } 496.25}.$$

The equity adjustment records the cumulative effect at the comparative opening. If the remaining lease year falls within the comparative year, the DTA of 27.5 and DTL of 26.25 reverse during that year, giving comparative deferred-tax expense 1.25. The opening adjustment alone therefore does not complete the transition. Neither ASBE 21's article 61(2) lease-adoption election nor Interpretation 16 Section III's no-comparative-adjustment rule for its separate share-payment topic may be substituted for Section I(b)'s tax transition.

These are gross recognition and measurement amounts. Separate recognition and noncurrent classification do not determine whether an actual balance sheet may offset DTA and DTL. That decision needs the applicable presentation provisions and entity/tax-authority facts. Actual tax law, tax bases and rates, recoverable profit, the enterprise's adoption and elections, and later amendments remain to be established before applying the example to an issuer. Full ASBE 22 credit, consolidated statements, group combinations and insurance revisions also retain specific source dependencies; similar amounts do not establish full IFRS equivalence.

9.17 Matched-case outcomes and decision

Input changed or withheld	Required consequence
Framework changes with same lease facts	Preserve different expense/ROU paths and source strength, not just rename accounts
Method source present but target entity's adoption unknown	Unresolved applicability, no real posting authorization
ROU/liability initially equal	Still evaluate gross tax differences and later divergent balances
Historical rule supports method but a later reporting year is requested	Recheck operative amendments/transition; no automatic extension
Issuer policy discloses LIFO, regulator text unavailable	Retain issuer assertion; do not upgrade it into universal US authority
Foreign basis label replaced by local label	Reject as a standards conversion without recognition/measurement bridge

These decisions establish a common procedure and specific proposed treatments. The recovered ASBE lease notice and tax interpretation support the stated application schedules and conditional example. Broader authority coverage remains incomplete: operative US sources, ASBE sector methods, actual tax law and presentation conditions, and the applicable amendment and entity-election evidence still require internal investigation.

Chapter 10

Financing, ownership and the group boundary

A company can preserve the same operating assets while changing who supplies its funding and who shares its results. The following cases move from debt cash flows to ownership, then to the reporting boundary of a group. Each numerical example states its own contract; the coupon-bearing convertible below and the later zero-coupon compound example are different instruments.

10.1 Debt and hybrid financing

The contract layer for a debt instrument stores issuer and holder perspective, currency, issue and settlement dates, issue proceeds and legal face amount. Its payment terms include coupon/index/spread, reset and fixing dates, day-count convention, payment calendar, amortisation schedule and maturity. Its rights and protections include seniority, collateral, covenants, options, conversion terms, guarantees and linked hedges. Missing terms are explicit unknowns. A single carrying amount cannot stand in for this record.

The accounting layer stores the classification decision and its evidence, initial measurement, carrying amount, accrued interest, effective-rate method, unamortised fees or discounts, fair-value disclosures, impairment or credit adjustments where applicable, and presentation mappings. Contractual interest, effective-interest expense, realised cash payments and fair-value changes are separate quantities. The zero-coupon example in Chapter 8 provides a minimal conformance case for this separation.

The lifecycle must support issuance, recognition, accrual, reset, coupon, principal repayment, modification, derecognition, default, recovery and conversion. A modification event first determines which accounting operation the selected standard requires; it does not always mean issuing a new loan. An embedded option can affect valuation and classification without being exercised. A derivative linked as a hedge requires both the economic contract and any applicable accounting designation; economic risk reduction alone does not establish hedge-accounting eligibility.

Hybrid instruments make interpretation particularly consequential. IAS 32's overview discusses liability/equity classification and the separation of a compound instrument (IFRS Foundation, 2026a). The executable classification must consider the full terms and the selected standard version. US GAAP or ASBE treatment must be implemented under its own rules rather than inferred from an IFRS classification. Where the terms are not disclosed sufficiently, alternative

classifications can coexist only if each is actually compatible with the known terms and applicable rules.

10.1.1 Apple: debt balances and cash repayment

Apple's borrowing shows why the distinction between a contract and its carrying amount matters for a funding forecast. A fall in reported debt might suggest how much cash the company repaid. Its annual disclosure lets us test that interpretation by separating principal from the adjustments included in the carrying amount. The following amounts are in millions of US dollars at the displayed precision; changes run from 30 September 2023 to 28 September 2024 (Apple Inc., 2024, Note 9, printed pp. 43–44).

Term-debt component	2023	2024	Change
Principal	106,572	97,341	$-9,231$
Net premium, discount and issuance costs	-356	-321	$+35$
Hedge accounting fair-value adjustment	$-1,113$	-358	$+755$
Carrying amount	105,103	96,662	$-8,441$

The cash-flow statement reports no net term-debt issuance proceeds and cash repayments of 9,958 million for fiscal 2024 (Apple Inc., 2024, printed p. 33). To connect that cash movement to the table, first calculate the difference between the change in reported principal and net term-debt cash flow. It is $-9{,}231 - (-9{,}958) = 727$ million. Adding the reported measurement-component changes then gives the carrying-amount movement. With Δ denoting closing less opening balance, the two displayed-amount identities are

$$\Delta\text{principal} = -9{,}958 + 727 = -9{,}231,$$
$$\Delta\text{carrying amount} = -9{,}958 + 727 + 35 + 755 = -8{,}441.$$

The 727 is an unallocated balance-to-cash difference. It does not identify an exchange movement or a journal. Likewise, the 35 is the change in a net unamortised balance, rather than an identified issuance fee or interest charge; the 755 is a change in the debt's hedge adjustment, rather than a cash receipt. Together they explain 790 of the 1,517 million difference between the carrying-amount movement and net cash repayment. A forecast that used the 8,441 carrying decline as repayment would therefore compute the wrong cash amount for this observed year. The cash statement's proceeds measure is net; it must not automatically become legal face issued in another period.

The maturity schedule answers a different funding question. Its six future principal buckets sum to 97,341 million, with 10,930 in fiscal 2025. The current carrying amount is 10,912. Their 18 million difference does not allocate a discount or hedge adjustment to the current portion. The measurements differ, and their time boundaries must also be retained: Note 1 supplies the fiscal-year default, the issuance table explicitly labels its 2024–2062 range as calendar years, and MD&A separately reports amounts payable within twelve months. MD&A also reports future term-debt interest payments of 38.5 billion dollars, including 2.6 billion within twelve months (Apple Inc., 2024, printed pp. 25, 34 and 43). Those cash requirements cannot replace period interest expense or a present value. The dated Level 2 fair-value disclosures likewise measure something different: the annual note reports 88.4 billion dollars at September 2024, while the interim note reports 82.7 billion at December 2024 alongside a rounded carrying amount of 94.8 billion (Apple Inc., 2025, Note 6, printed p. 10).

10.1.2 Net borrowing can conceal refinancing

Commercial paper exposes a second limitation of aggregate financing data. Apple describes unsecured short-term promissory notes used for general corporate purposes, including dividends and repurchases. The annual note says their maturities are generally less than nine months; that description does not identify each note or trace individual borrowed dollars to a use. At the annual statement amounts, commercial paper increases from 5,985 to 9,967 million, a rise of 3,982, while net cash proceeds are 3,960. The resulting 22 million difference remains unallocated. During the following quarter the balance falls from 9,967 to 1,995, while net cash repayments are 7,944; the corresponding unallocated difference is −28 million (Apple Inc., 2024, Note 9 and balance and cash-flow statements) (Apple Inc., 2025, Note 6 and balance and cash-flow statements).

Even a complete explanation of those differences would leave refinancing activity unresolved. Annual Note 9 reports net cash for paper with maturities of at most 90 days, but separate proceeds and repayments for longer maturities. In fiscal 2022, for example, the longer-maturity proceeds of 5,948 and repayments of 7,257 give net repayments of 1,309 million. Adding short-maturity net proceeds of 5,264 gives total net proceeds of 3,955. For fiscal 2024 the longer-maturity cash rows are zero and the short-maturity net inflow is 3,960. Neither that net amount nor a closing balance reveals how many issues and redemptions occurred between reporting dates.

Consider a hypothetical incremental borrowing of 1,000 million US dollars, issued at par and repaid after 60 days. Stipulate simple interest of 5% using actual days over a 360-day year, with no fees, taxes, hedges or early settlement. The rate is an assumption of this example; Apple's disclosed 5.00% weighted rate at September 2024 does not establish the rate on such a transaction. Issuing and repaying the principal adds $1{,}000 - 1{,}000 = 0$ to net principal cash flow and leaves zero principal outstanding, yet supplies 1,000 million between those dates. The cost over the 60 days is

$$\text{interest} = 1{,}000 \times 0.05 \times \frac{60}{360} = \frac{25}{3} \approx 8.3333 \quad \text{million US dollars.}$$

Issue debits cash and credits principal 1,000. Accrual debits interest expense and credits interest payable 25/3. Settlement debits principal 1,000 and interest payable 25/3, and credits cash 3,025/3. Each entry balances; after settlement the incremental liability is zero, while cash and equity are each lower by 25/3 relative to the otherwise funded baseline.

This example isolates what a net principal-flow observation fails to reveal. It is not a reconstructed Apple history: interest expense, interest cash and other disclosures could distinguish or exclude the added borrowing. Within that scope, it shows why a financing model needs dated issuance, accrual and redemption events. Matching a net financing row does not establish either intraperiod liquidity or funding cost.

10.1.3 A rate scenario needs an exposure definition

The contract's coupon is also insufficient to describe exposure after hedging. Apple's annual debt note states that interest-rate swaps effectively convert some fixed-rate US-dollar notes to floating rates, while cross-currency swaps effectively convert some foreign-currency notes to US-dollar notes. The effective-rate ranges include interest, discount or premium amortisation and applicable hedge adjustments. These are relationships between separate contracts and measurements, not a single rate that can be applied to total principal (Apple Inc., 2024, Note 9, printed p. 43).

The filing supplies a useful, more specific constraint. Annual Item 7A reports that a hypothetical increase of 100 basis points at all tenors, including associated derivatives, would increase annual term-debt interest expense by 139 million dollars at September 2024 and 194 million at September 2023 (Apple Inc., 2024, printed p. 27). Multiplying Note 4's 2024 interest-rate contract notional of 14,575 million by 1% instead gives 145.75 million. It does not reproduce the reported sensitivity, even allowing for nearest rounding of the displayed amounts. The difference cannot be assigned to a particular reset date or swap feature from these aggregates. For 2023 the same shortcut gives $19{,}375 \times 0.01 = 193.75$, consistent with rounded 194; that numerical match does not validate the shortcut or recover the issuer's model.

The scenario's measure matters as much as its shock. In the same table, the 2024 investment-portfolio response is a 2,755 million decline in fair value. It cannot be added to an annual interest-expense increase as though both were the same loss measure. At December 2024 the reported interest-rate notional is still 14,575 million, while the carrying amount of term debt subject to fair-value hedges is a rounded 13.3 billion, compared with 13.5 billion at September. These aggregates describe different populations or measurements. The interim statement that market risk has not changed materially refers the reader to annual Item 7A; it does not publish an updated 139 million sensitivity (Apple Inc., 2025, Note 4 and Part I, Item 3).

Hedge accounting and cash presentation also remain distinct. Annual Note 1 presents derivative assets and liabilities at gross fair values. Note 4 classifies derivative cash flows with the items hedged, which are *generally* operating activities; that qualifier does not classify every derivative settlement as operating. The maximum disclosed horizon for hedging debt-related foreign-currency cash-flow variability is 18 years, and Apple states that some exposures may be left unhedged and that offsets may be partial (Apple Inc., 2024, Notes 1 and 4, printed pp. 34 and 37–38).

Finally, Item 7A's currency-risk statistic has its own population. Apple reports a Monte Carlo estimate, at 95% confidence over one day, of 538 million dollars at September 2024 and 669 million at September 2023 for adverse fair-value changes in its foreign-currency derivatives. The assessment excludes underlying currency exposures, whose changes generally offset derivative changes. This derivative VaR is neither a hard maximum loss nor a measure of the company's net currency risk; the issuer's simulation has not been replicated here. A financing simulation can retain these dated scenarios as disclosed constraints, but still needs actual holdings, currencies, payment calendars, reset terms and hedge relationships to calculate future cash and accounting consequences. The next question is how financing choices change the quantities and claims of the company's owners.

10.2 Equity and ownership

Equity has both monetary accounts and security quantities. The security register records share class, issued and outstanding quantities, treasury holdings, par or stated value where applicable, voting rights, dividend rights, redemption features, options and conversion relationships. Monetary equity accounts include contributed capital, reserves, retained earnings and other components required by the selected basis. These records evolve together through issuance, repurchase, cancellation, split, dividend and conversion events.

A stock split changes quantities and per-share measures without necessarily changing aggregate monetary equity. A cash dividend affects distributions and cash rather than current operating expense under the corresponding ordinary equity treatment. Repurchases and employee share awards require their own rules. Earnings per share uses a dated quantity schedule and rules for potential ordinary shares; simply dividing profit by the closing share count is insufficient when

shares change during the period.

Funding simulations must connect the quantity register to market assumptions. An equity issue at a scenario-dependent price changes cash raised, shares and dilution. The model must state whether it targets a fixed cash amount, a fixed share count or a capital ratio. These are different management decisions even when the historical statement would report the same aggregate proceeds.

10.2.1 Apple: shares through time

Apple's disclosures show why the ownership record needs dates as well as quantities. In fiscal 2024, its opening 15,550,061 thousand common shares, less 499,372 thousand repurchased and plus 66,097 thousand issued net of employee-tax withholding, give 15,116,786 thousand closing shares. That bridge accounts for the change in the reported stock. It does not say how long each share participated in the year's earnings. The reported basic weighted average is 15,343,783 thousand shares (Apple Inc., 2024, Notes 3 and 10, printed pp. 36 and 44).

The unit conversion matters before any timing inference. Net income is 93,736 million dollars, whereas the denominator is in thousands of shares. Thus $93{,}736 \times 1{,}000/15{,}343{,}783 \approx 6.109$ dollars per share, consistent with reported basic EPS of \$6.11. Using closing shares instead gives about \$6.201: it computes a different ratio. Adding the reported 64,312 thousand incremental shares from dilutive awards gives the diluted denominator of 15,408,095 thousand and EPS of about \$6.084, reported as \$6.08. These are checks using displayed amounts. Conditional nearest-rounding intervals also overlap the reported two-decimal EPS; the issuer's tie convention and undisclosed exact amounts remain unverified.

The first quarter of fiscal 2025 makes the distinction again. Income of 36,330 million divided by 15,081,724 thousand basic shares, with the same factor of 1,000, gives about \$2.409. The diluted denominator adds 69,141 thousand shares and gives about \$2.398, consistent with the reported \$2.41 and \$2.40 (Apple Inc., 2025, Note 3, printed p. 7). Those denominators describe three months; adding them to other quarterly averages would not produce an annual share count. Period lengths and consistent measurement rules are needed to combine averages.

Even a weighted average leaves dates unresolved. To see exactly what it constrains, consider a deliberately simplified common-share model for the 364 days of fiscal 2024. Apple reports a 52-week year ending 28 September 2024 (Apple Inc., 2024, Note 1, printed p. 34). Let S_0 be the opening number of ordinary shares, and let ΔS_j be a signed change after d_j whole days have elapsed since 1 October 2023. Positive changes issue shares; negative changes repurchase them. An event then affects the remaining $364 - d_j$ days. Assume no splits or other quantity changes, and aggregate all net issuance into one event. The average is the opening holding plus each change weighted by its remaining time:

$$\overline{S} = S_0 + \sum_j \Delta S_j \frac{364 - d_j}{364}, \qquad 0 \le d_j < 364. \tag{10.1}$$

All quantities in this equation are ordinary shares and event days are integers. An immediate change receives full weight; a change after 363 days receives only one day's weight. Closing shares depend on the sum of changes, while equation (10.1) also constrains their timing. Neither relation separately identifies every change and date.

The construction must also respect information outside the annual EPS note. Apple reports fourth-quarter purchases of 35,697, 42,910 and 33,653 thousand shares in the intervals 30 June–3 August, 4–31 August and 1–28 September, respectively (Apple Inc., 2024, Item 5, printed p. 19). A history putting all repurchases in the middle of the year fails that disclosure even if its annual

average is correct. The later interim EPS note also reports a first-quarter fiscal 2024 basic average of 15,509,763 thousand shares. Both constraints are included in the following two authored quantity histories.

Event	History 1	History 2
First-quarter purchase	−40,298 (45)	−40,298 (45)
First-quarter purchase	−40,298 (46)	−40,298 (46)
Net issuance	+66,097 (182)	+66,097 (91)
Further purchase	−131,136 (179)	−15,643 (159)
Further purchase	−175,380 (180)	−290,873 (160)
June–August interval purchase	−35,697 (290)	−35,697 (290)
August interval purchase	−42,910 (322)	−42,910 (322)
September interval purchase	−33,653 (350)	−33,653 (350)

The table expresses changes in thousands of shares; parentheses give elapsed days, not negative signs. Each quantity is multiplied by 1,000 when used in equation (10.1). Treat the displayed source centres as exact solely for this illustration. Both histories start at 15,550,061,000 shares, repurchase 499,372,000, issue a net 66,097,000 and end at 15,116,786,000. Their first 91 days average 15,509,763,000 shares: the two equal early purchases reduce the average by $40{,}298{,}000(46 + 45)/91 = 40{,}298{,}000$. For the full year, the signed time-weighted changes sum to −75,085,192,000 share-days in each history. Dividing by 364 reduces the opening count by 206,278,000, giving the reported 15,343,783,000 average. An independent sum of daily balances gives the same result.

After 100 elapsed days, however, the first history has 15,469,465,000 shares and the second 15,535,562,000. The earlier issuance makes the difference. The fourth-quarter purchase dates lie inside their reported intervals, but none of these illustrative dates is an estimate of Apple's actual transactions. This establishes non-uniqueness for the stated quantity constraints. Prices, award-level settlement and monetary journals impose further conditions; the two examples are not a complete reconstruction of the filing or an exhaustive family, and no probability attaches to either. A query for shares on an intermediate date must therefore retain the timing alternatives or obtain additional evidence.

10.2.2 Awards, repurchases and cash measure different movements

The next quarter shows how another disclosure can narrow an unknown. Note 7 describes about 100 million repurchased shares, while Item 2's interval table gives 41,627, 32,784 and 25,379 thousand, totalling 99,790 thousand. The balance sheet reports 15,040,731 thousand closing shares against 15,116,786 thousand at the preceding year end. At the displayed centres, the quantity bridge requires a net other increase of $15{,}040{,}731 - 15{,}116{,}786 + 99{,}790 = 23{,}735$ thousand shares (Apple Inc., 2025, printed pp. 3, 10 and 20). This narrows the total without identifying the events composing it. Repurchase authority is different again: both the note and programme footnote explicitly say that the programme does not oblige Apple to acquire a minimum amount of shares.

Restricted stock units have their own movement record. In thousands of RSUs, the fiscal 2024 bridge is $180{,}247 + 80{,}456 - 87{,}633 - 9{,}744 = 163{,}326$: opening awards plus grants, less vesting and cancellations. The next quarter gives $163{,}326 + 61{,}391 - 35{,}683 - 2{,}278 = 186{,}756$ (Apple Inc., 2024, Note 11, printed pp. 44–45) (Apple Inc., 2025, Note 8, printed p. 10). These gross award movements are different from net common shares issued or incremental shares in the diluted EPS calculation. Apple reports that most vested RSUs were net share settled and that approximately 31 million shares were withheld for employee taxes in 2024. Subtracting

that rounded amount from 87.633 million vested RSUs does not explain the 66.097 million net issuance line in full.

The 2022 employee plan permits several award types. Its RSUs generally vest over four years subject to continued employment, settle one-for-one in common shares and carry dividend-equivalent rights. "Generally" cannot determine the vesting dates of every grant. The prior plan ceased authorising new grants in the third quarter of 2022; that does not say its outstanding awards vanished. Likewise, the approximately 24 million RSUs excluded as antidilutive in fiscal 2023 are a period-specific disclosure, not a universal exclusion rule (Apple Inc., 2024, Notes 3 and 11, printed pp. 36 and 44–45). A future dilution calculation needs the applicable award terms and EPS rules as well as quantities; the arithmetic above does not validate that methodology.

Monetary statements impose another set of distinctions. The following amounts are millions of dollars, with reductions and payments shown as positive magnitudes (Apple Inc., 2024, printed pp. 32–33 and 45) (Apple Inc., 2025, printed pp. 4–5 and 10).

Reported measure	Fiscal 2024	Fiscal 2025 Q1
Repurchase reduction in equity	95,846	23,476
Cash-flow repurchases	94,949	23,606
Employee-tax withholding reduction in equity	5,605	2,993
Cash-flow employee-tax payments	5,441	2,921
Share-based compensation in equity	12,034	3,383
Share-based compensation expense	11,688	3,286

The withholding reductions combine the contributed-capital and retained-earnings rows: $3{,}993 + 1{,}612 = 5{,}605$ annually and $1{,}891 + 1{,}102 = 2{,}993$ in the quarter. These differences remain unallocated here; assigning them to excise tax, transaction fees or unsettled trades would require further evidence. The narrative's $95.0 billion and $23.3 billion purchase amounts also retain their own precision and transaction scope. The separate employee-tax settlement narrative cannot silently replace the cash-flow line.

Finally, grant-date value, vesting-date value, compensation expense and future unrecognised cost answer different questions. Apple's stated policy recognises expense on a straight-line basis for its estimate of awards that ultimately vest (Apple Inc., 2024, Note 1, printed p. 34). Its unrecognised RSU cost increases from $19.4 billion over a weighted-average 2.4 years to $29.2 billion over 2.9 years at the quarter end. Dividing either amount by its weighted period does not disclose a constant annual expense schedule or a cash liability: different cohorts have different remaining recognition periods, and future vesting estimates can matter even under the stated straight-line policy. The ownership model therefore needs linked award cohorts, recognition periods, share events and cash settlements. Keeping those dates and measurements distinct allows an actual later grant or settlement disclosure to narrow the histories without overwriting the reported EPS or inventing a missing event.

10.3 A compound bond exposes classification dependencies

Consider an issuer receiving 100 for a two-year bond paying annual coupons of 5 and principal 100 at maturity, with a holder option to convert that principal into ten ordinary shares. Assume its currency is the issuer's functional currency, no other options, fees or contingent settlements, and that the conversion feature qualifies as equity under the selected rule. A comparable liability without conversion yields 10%. The inspected IAS 32-equivalent text requires separate liability and equity components and measures equity as the residual (Australian Accounting Standards Board, 2022b, paragraphs 28–32, AG30–AG32).

The liability's present value is

$$L_0 = \frac{5}{1.1} + \frac{105}{1.1^2} = \frac{11050}{121} \approx 91.3223, \qquad E_0 = 100 - L_0 \approx 8.6777.$$

Issue debits cash 100 and credits these two components. Under the stated effective-interest schedule, the first coupon leaves liability $1.1L_0 - 5 = 1050/11 \approx 95.4545$; after the second coupon it is 100. Redemption debits liability and credits cash 100. Alternatively, conversion at maturity debits liability and credits equity 100 while issuing ten shares; the original equity component remains in equity. AG32 supports the absence of a gain or loss on that maturity conversion. Exact rational valuations are kept before the declared monetary rounding.

A variable-number share settlement, cash alternative or other clause can change classification. The contract record must therefore retain these terms; it cannot select liability/equity accounts independently of the coupon, conversion and settlement rules. Early repurchase and induced conversion invoke separate operations: the inspected AG33–AG35 text describes allocation and the induced-conversion consideration difference. They are not implemented by changing the maturity date of the simple example. US-GAAP and ASBE classification require their own dated authority; the IFRS-oriented split is never silently reused under another basis.

10.4 Schedules and initial classification

The following liability and ownership methods use selected dated AASB 9 and AASB 132 provisions (Australian Accounting Standards Board, 2022e,b). Issuer, holder, contractual cash flow and accounting measurement remain separate throughout the schedule.

DebtContract contains parties, currency, principal, issue/settlement dates, payment calendar, fixed/index/spread/reset terms, compounding/day-count, business-day adjustment, principal changes, fees, options, security, priority and legal default/remedy terms. Each term has evidence/admission provenance. The economic schedule emits due cash independently of the recognition method. A market-value adapter may value that schedule but cannot choose amortised cost, FVOCI, FVTPL or liability/equity classification.

An investor's classification evaluates business model and contractual cash-flow features; a debt label alone cannot establish the AASB 9.4.1.2/4.1.2A conditions. At initial recognition the selected 5.1.1 rule measures fair value and qualifying costs as appropriate. An unknown day-one observable-value condition yields unresolved under 5.1.1A/B5.1.2A. The retained acquisition cases distinguish FVTPL expensed fees, amortised-cost fees, original EIR and POCI credit-adjusted EIR. No rate method inferred from market value can overwrite the contract rate.

For an issuer with a plain amortised-cost debt, net proceeds 100 and one legally due payment 121 at the end of two annual periods, the exact EIR solves 100=121/(1+r)^2, hence r=10%. Issue debit cash 100/credit debt 100; year-one debit interest 10/credit debt 10; year-two debit interest 11/credit debt 11; redemption debit debt 121/credit cash 121. Outstanding contractual payment stays 121 until settled even while carrying value is 100 or 110. Total interest 21 equals cash outflow 121 minus proceeds 100. An immediate transaction fee 2 changes net proceeds to 98 and the EIR root; retaining 10% is wrong. The numerical root method is in the numerical-method discussion; an irrational rate is not silently an exact rational.

A fractional period uses the declared convention. Annual-effective 21% over half a year produces factor 1.1 and interest 10 on 100; simple proration gives 10.5 and is wrong for that contract. A floating coupon resets from the admitted fixing available at the reset date, not a later observed rate. Actual day-count and holiday-calendar versions are immutable inputs. Unknown fixings preserve conditional paths, rather than defaulting to the latest value.

10.5 Modification, extinguishment and default

At a modification, retain original and replacement terms, counterparty, fees, effective date, legal-release evidence and original EIR. Determine derecognition before measurement. AASB 9.3.3.1–3.3.3 and B3.3.1–B3.3.6 govern the selected issuer-liability cases. The quantitative ten-percent test compares the PV of new terms, including specified borrower/lender fees, at original EIR against remaining original PV. It is not a universal asset test, nor proof that every change below ten percent is immaterial; changed currency, rights and other qualitative terms need the applicable analysis. Insufficient evidence remains unresolved. Third-party payment without legal release does not extinguish debt.

On the first anniversary above, old remaining 121 has PV 110. Replace with 133.1 due at the original maturity date, one year after the modification, with no fees and an explicit supported substantial-terms conclusion: PV 121 differs by exactly 10%, satisfying the inclusive test. Assume separately supported new fair value 118; debit old debt 110, debit extinguishment loss 8, credit new debt 118. The new EIR is $133.1/118-1$, not automatically the old rate. If the replacement is instead a stated non-substantial same-currency reduction to 115.5, no fees or other changes, PV 105 leads to debit debt 5/credit modification gain 5 under the B5.4.6 recalculation mechanism, then interest 10.5 and settlement 115.5. The actual change below ten percent still needs the qualitative finding. Repurchase of carrying amount 110 for 108 extinguishes the repurchased obligation: debit debt 110/credit cash 108/credit gain 2. Unamortised fees belong in the carrying amount before that comparison, not a second gain adjustment.

If cash available is 80 against debt 121 due, record the due obligation and an unpaid/default economic state; payment 80 leaves contractual unpaid 41 only when that partial payment actually occurs. Do not cancel the remaining liability, recognize a waiver or issue a loan unless a separately admitted event authorizes it. Covenant acceleration, cross-default, cure periods, collateral realization and restructuring require the actual legal terms and priority waterfall. A scenario may explore specified alternatives with explicit assumptions; it may not silently select the cheapest settlement as an accounting fact.

10.6 Equity and compound instruments

EquityInstrument retains class, voting/distribution/liquidation rights, issued, treasury and outstanding counts, par/legal-capital requirements and conversion terms. Share counts are integer quantities separate from currency balances. Issue 100 ordinary shares at 2 with no fees: debit cash 200/credit equity 200, issued and outstanding 100. Repurchase 10 at 3 for treasury: debit treasury-equity 30/ credit cash 30; outstanding 90, issued 100, treasury 10. Reissue 5 at 4: debit cash 20/ credit treasury-equity 15/credit other equity 5; outstanding 95, treasury 5. There is no trading profit under the selected AASB 132.33 rule; a statutory par split requires that jurisdiction's facts. Dividend declaration creates a liability only at the admitted point at which the obligation exists; proposal of a dividend is not itself payment or automatically an expense.

For a qualifying fixed-for-fixed issuer conversion option in functional currency, AASB 132.28–32 separates liability fair value and residual equity. Assume total issue proceeds 100, standalone debt fair value 90, zero coupon, maturity two years and cash redemption 100 or fixed 10 ordinary shares; no extra derivative, tax or fee. Issue debit cash 100/credit debt 90/credit conversion-equity 10. Effective interest accretes debt 90 to 100. At maturity conversion under original terms debits debt 100 and credits issued-share equity 100; the original equity 10 stays in equity (a permitted internal transfer is not income). Shares increase 10 and no redemption cash leaves. Redemption instead debits debt 100/credit cash 100; unexercised equity is not remeasured into

profit. AG 32 supports no conversion gain/loss in that stated case. Variable share count, cash alternatives, foreign currency and changed conversion terms require classification/modification analysis; they cannot be forced into this example.

Financial statement mappings separate interest, modification/extinguishment results, owner transactions and cash settlement. A financing-liability note reconciles opening carrying amount, cash proceeds/repayments, effective interest, fees, conversions, modifications and FX. Total shares and earnings-per-share denominators need dates and dilution conditions; outstanding shares at year end are not a substitute for a weighted period denominator. Full option valuation, hedge accounting, EPS and legal-capital regimes remain explicitly additional methods; the proposed records preserve their required inputs.

10.7 Control and an elimination book

The consolidation procedure selects the control graph at each date, aligns supported accounting policies and presentation currencies, combines controlled entity books, and posts acquisition, elimination and non-controlling-interest adjustments in a separate book. A control conclusion needs evidence and a rule version. Ownership percentage alone is insufficient. Unknown purchase allocation, currency translation or unmatched intragroup differences remain named unresolved adjustments rather than unexplained balancing entries.

For a simplified acquisition of a different subsidiary from the goodwill case that follows, assume control is established, all net assets are already at the relevant acquisition measurements, no tax or goodwill arises, and non-controlling interest is measured proportionately. A parent has cash 60, investment 40 and equity 100; an 80%-owned subsidiary has cash 30, plant 20 and equity 50. Consolidation debits subsidiary equity 50, credits investment 40 and credits non-controlling interest 10. Group assets are cash 90 plus plant 20, and group equity is 100 attributable to the parent plus 10 to other owners.

A subsequent intragroup loan of 10 shifts cash from parent to subsidiary, creates a receivable and payable, and is eliminated by debiting payable and crediting receivable 10. Group totals remain 110. An unmatched loan currency or settlement date would block that exact elimination. If the parent instead sells its entire interest for 44 immediately after the initial state and loses control, the group removes subsidiary cash 30, plant 20 and non-controlling interest 10, recognises cash consideration 44 and gain 4; remaining cash/equity is 104. More complex acquisitions and retained interests require additional measurement operations; this example fixes the structure and arithmetic without certifying a complete consolidation standard.

10.8 Perimeter and consolidation state

The acquisition, control and currency methods below use the inspected IFRS 3, IFRS 10 and IAS 21 provisions in the retained 2023 EU text (European Commission, 2023). The acquisition with goodwill is a new example; it does not change the preceding zero-goodwill case.

ControlAssessment records investor/investee, relevant activities, rights and their substantive/protective character, variable returns, ability to use power, principal/agent analysis, effective interval and evidence. IFRS 10.5–9 requires power, returns and their connection, with reassessment on changed facts. An 80% share count is evidence to analyze, not an automatic substitute for control. Unknown control yields separate conditional perimeters and unresolved group results; no entity disappears merely to make reconciliation easy.

PerimeterVersion binds control intervals, ownership classes, report cutoffs, functional/pre-

sentation currency, policy-alignment adjustments and the exact source entity books. ConsolidationAdjustment is a distinct group book with matched counterparty/event references; legal-entity journals remain intact. The procedure is: select time-compatible entity books; apply supported policy alignment; translate them; add like balances/movements; eliminate investment against acquisition equity; eliminate intragroup balances, income/cost and unrealised gains; allocate profit/OCI to owner/NCI; reconcile all statements and movement notes. IFRS 10.19–23/B86–B96 support these steps. Reporting-date differences need B92–B93's conditions and intervening adjustments; a generic nearest-date join is invalid.

10.9 Acquisition and ownership case

Assume a business combination, no common-control transaction, parent obtaining control and 80% of ordinary residual interests for cash 96. Acquiree identifiable assets fair value 160 and liabilities 60 give net 100. There is no prior interest, contingent consideration, tax difference, separately unrecognized intangible, transaction cost or impairment; those absences are synthetic facts. Select proportionate-share NCI under IFRS 3.19: NCI 20 and goodwill 96+20−100=16 under .32. In parent books debit investment 96/credit cash 96. In the consolidation overlay, debit acquiree acquisition equity 100 and goodwill 16, credit investment 96 and NCI 20. Only post-acquisition income enters group performance. Asset measurement exceptions, deferred tax and a bargain gain reassessment under .36 would change the calculation; goodwill cannot absorb an unexplained missing balance.

If post-acquisition net profit is 20, all accrued in retained earnings with no dividend/OCI, NCI increases 4 to 24 and parent's attributable profit is 16. Parent buys another 10% for 15 while retaining control: transfer 12 from NCI (10% of current net 120), credit cash 15 and debit parent equity 3. Remaining NCI 12; goodwill 16 unchanged; no new profit. IFRS 10.23/B96 makes this an equity transaction. This example has equal ordinary rights; preference rights need their actual allocation and cannot use a simple ownership fraction.

In a separate alternative starting before that additional purchase, sell all 80% for 130 when net assets 120, goodwill 16, NCI 24 and OCI zero. On loss of control, group gain=130+24−(120+16)=18. Derecognize individual assets/liabilities and goodwill, remove NCI, recognize consideration and gain. The separate parent book's investment gain is not added again. If 20% is retained and 60% sold for 100 with retained fair value 35, group gain=100+35+24−136=23; recognize the retained investment 35. Its subsequent associate/financial-asset method needs separately established significant-influence/classification facts. IFRS 10.B98–B99 also requires the appropriate prior-OCI treatment; this case assumes none.

10.10 Intercompany inventory and tax

Assume parent's external inventory cost 60 is sold to its controlled subsidiary for 100 on credit, all still held at report date, tax ignored in the base case. Parent records AR 100/revenue 100 and COGS 60/inventory 60; subsidiary inventory 100/ AP 100. Consolidation debits AP 100/credits AR 100 and debits revenue 100/credits COGS 60 and inventory 40. Group inventory 60, revenue and COGS zero, and no group profit arises. When subsidiary later sells externally for 120, group COGS is 60 and profit 60, not 20; release the previously eliminated 40 in the continuing adjustment roll-forward. Reversing a closing overlay and recreating a new one must not recognize the profit twice.

If the purchaser's independently supported tax base is 100, group carrying amount 60 creates

deductible difference 40. At assumed applicable 25% and with all IAS 12 DTA conditions satisfied, recognize group DTA 10/deferred-tax benefit 10; reverse on outside sale. The original tax expense and cash obligations in the seller's tax jurisdiction remain. Different rates, recoverability or tax-law transfer rules change the answer; tax is not simply eliminated with group profit. For an upstream subsidiary sale, attribute elimination and related tax effects to the appropriate subsidiary/NCI performance; downstream parent's profit elimination does not mechanically reduce NCI by the same fraction.

10.11 Functional currency versus group presentation

EntityCurrencyState names the functional currency and its evidenced rationale, foreign-currency nominal balances, monetary/nonmonetary classification, historical or fair-value measurement dates, rates and sources. IAS 21.21–23 and .28 require transaction-date translation and closing-rate remeasurement of monetary items; historical-cost nonmonetary assets retain transaction-date rates. A functional currency change is prospective under .35–37. Hyperinflation, lack of exchangeability and hedge accounting require separate applicable methods; do not import the historical .26 text into a later amended period without checking it.

Example: buy goods for foreign 100 at functional 1.2 per foreign unit; debit inventory 120/credit payable 120. Closing rate 1.3 gives payable 130 and FX expense 10, while inventory remains 120 before any NRV test. Pay at 1.25: debit payable 130/ credit cash 125/credit FX gain 5. Cumulative expense 5 matches 125 cash less 120 asset cost. Translating the inventory again to 130 would be wrong for this declared basis.

Group presentation is a different operation under IAS 21.39–41. A subsidiary has opening net assets 100, equity 100, no debt or distributions, annual profit 20 entirely earned at rate 1.25, and closing assets 120. Opening historical rate 1.2 and closing 1.3 give translated assets 156, contributed/opening equity 120 and profit 25; translation OCI 11 reconciles the difference. It is explained as $100\times(1.3-1.2)+20\times(1.3-1.25)=11$, not a free balancing plug. If ownership is 80%, parent/NCI shares of profit are 20/5 and OCI 8.8/2.2 under the stated equal rights. Use transaction rates; an average is acceptable only as a justified approximation, not when significant fluctuation invalidates it. Goodwill and fair-value adjustments follow IAS 21.47's foreign-operation treatment, rather than freezing them automatically in the parent's currency.

Group reports retain the translation reserve by entity, ownership and event. Loss of control requires the appropriate reserve reclassification under IAS 21.48 and IFRS 10.B99; a no-control-loss change transfers the relevant attribution in equity under applicable conditions. Cash translation effects reconcile cash separately from operating/investing/financing movements. An intercompany monetary balance's exchange result cannot simply be eliminated as if no currency exposure existed; IAS 21.45 and net-investment .32 require the substantive classification.

Chapter 11

Credit losses and the bank balance sheet

The funding and default distinctions become central in a bank. A loan creates an asset, customer payments affect settlement liquidity, and a forecast of credit loss affects an allowance before cash is lost. The first case connects funding to a simple allowance; the later dated loan schedule separates credit stages, interest recognition and eventual recovery.

11.1 Banks: accounting capital is not regulatory capital

A bank package needs instrument and cohort states for loans, securities, deposits, wholesale funding, repos, derivatives, collateral and commitments. For interest-rate analysis, currency and repricing buckets matter alongside remaining maturity. For credit analysis, exposure, default state, loss severity, recoveries, guarantees and their dependencies matter. For liquidity analysis, deposit behaviour, collateral eligibility, encumbrance and payment timing matter. These states can be represented at an explicitly chosen aggregation level; the public record rarely identifies every contract.

11.1.1 A reported group boundary

JPMorgan Chase's filings make the boundary between a bank, its parent and its securities issuers concrete. In the 2022–2025 reports, Item 1 identifies JPMorgan Chase Bank, N.A. as the principal bank subsidiary and describes an intermediate holding company, JPMorgan Chase Holdings LLC (the IHC). The reports also describe J.P. Morgan Securities LLC as a principal non-bank subsidiary and place J.P. Morgan Securities plc and J.P. Morgan SE beneath the bank. These are reported relationships for the stated dates, not a complete legal-entity graph (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

The attached Exhibit 21 tables qualify their scope: they list significant legal-entity subsidiaries, some parents of significant subsidiaries and certain entities designated for resolution planning, while excluding subsidiaries that would not be significant in aggregate under the stated SEC definition. Their indentation shows how the issuer presents the rows, but it does not supply ownership percentages, tax residence, functional currency or unrestricted transfer rights. A model therefore keeps a reported display relationship separate from a confirmed control edge.

Exhibit 22.2 adds a different edge. It names JPMorgan Chase Financial Company LLC as issuer of Global Medium-Term Notes, Series A, and JPMorgan Chase & Co. as guarantor. Financial

Company LLC does not appear in those qualified Exhibit 21 lists. The 2022–2023 text describes Series A notes as currently offered and sold under one cited registration; the 2024–2025 text also says that the issuer may issue debt securities and warrants under a later cited registration. The descriptions say that the relevant securities are fully and unconditionally guaranteed, but they give no outstanding principal, guarantee valuation or accounting recognition. They also do not establish that a warrant was issued or that an earlier note was redeemed.

The same reports state that the parent depends on subsidiary cash flows to pay its securities. The bank's dividends are subject to regulatory restrictions, and the IHC cannot pay dividends or extend credit to the parent when stated capital or liquidity thresholds, or management and board limits, apply. This means that consolidated cash, a subsidiary's assets and the parent's distributable funding are different state variables. The example gives the simulator a place to record the legal entity, guarantee and conditional funding path before it assigns a cash movement; it does not infer a balance or a probability of default.

Note 33 makes the funding distinction measurable for 2025. Its parent-only income statement reports net income of US$57,048m. The parent-only cash-flow statement subtracts US$71,163m of subsidiary and affiliate net income, reports a US$14,115m parent company net loss in that reconciliation, then adds US$50,000m of cash dividends from subsidiaries and US$8,583m of other operating adjustments to reach US$44,468m of operating cash. Neither the income figure nor the dividend alone is the period's ending cash.

At 31 December 2025, the parent reports US$39m of cash and due from banks and US$9,751m of deposits with banking subsidiaries, together US$9,790m. The cash-flow footnote says this measure includes US$8.0bn of intercompany time deposits. The same parent-only balance sheet reports US$669,449m of equity-method investments in bank and bank-holding subsidiaries; that asset is not parent cash. Operating cash of US$44,468m and investing cash of US$34m, less US$44,512m used in financing, leave closing cash and deposits US$10m below their opening balance. These dated amounts distinguish accounting income, existing cash and conditional subsidiary transfers. A future debt-service calculation still needs payment dates, permissible transfers, deposit terms and funding actions (JPMorgan Chase & Co., 2026, pp. 19, 310–311).

Debt amounts add a second distinction. Note 20 puts US$253,039m of long-term-debt carrying value under Parent company for 2025, while Note 33's parent-only balance sheet shows US$254,633m. The US$1,594m difference equals Note 20's separately displayed junior-subordinated-debt subtotal. The same comparison gives US$1,603m in 2024. This exact two-year arithmetic match does not establish how that row relates to the parent-only balance, or which legal entity owes each instrument. Its accounting and contractual relation still needs source support.

Timing requires a different measure. Note 20 places US$14,298m of parent carrying value in its under-one-year bucket; Note 33 reports US$14.3bn of parent long-term debt contractually maturing in 2026. They align at the footnote's displayed precision, but carrying value includes premiums, discounts, costs and valuation adjustments rather than simply principal payable. For the separately reported consolidated zero-coupon population, Note 20 gives US$18.1bn of carrying value against US$57.6bn of aggregate principal at the notes' respective maturities. The firm's US$435,206m consolidated long-term-debt total and US$47.6bn of subsidiary debt and structured notes guaranteed by the parent are also separate from the parent's own debt balance; the latter guarantee amount supplies neither a liability valuation nor a payment forecast. A dated parent debt-service path still needs instrument terms, cash dates and permissible funding transfers (JPMorgan Chase & Co., 2026, pp. 281–282, 310–311).

The complete Form 10-Ks describe Series A as the issuer's direct, unsecured general obligations, ranking equally with its other unsecured and unsubordinated debt. Claims under the parent guarantee generally rank behind those of subsidiary creditors. Their summary of the 2016 Indenture requires the parent, on demand after issuer nonpayment, to make full and punctual payment of principal, interest and other amounts due: a guarantee of payment, not collection (Sections 14.01–14.02). It excludes amounts the issuer is excused from paying or beyond those actually due. Issuer defenses and counterclaims carry over, except those arising solely from insolvency or bankruptcy status. These descriptions refer to the Indenture and applicable prospectus supplements; they specify reported terms but do not supply a current balance, valuation or accounting recognition.

The complete filings also describe one particular Series A population: the Callable Fixed Rate Notes due 10 June 2032. They report US$5,000,000 of aggregate principal outstanding at 31 December 2022 in the 2022 filing, at 31 December 2023 in the 2023 filing, and at 31 December 2024 in both the 2024 and 2025 filings. The last report repeats a 2024 as-of date; this passage supplies no 2025 year-end balance. The notes were issued on 10 June 2021 and bear 2.60% annual interest payable semiannually on 10 June and 10 December. They mature on 10 June 2032 unless an earlier contractual outcome applies. The issuer may redeem all, but not part, on 10 June 2031 at principal plus accrued and unpaid interest. Upon acceleration after a continuing event of default, the stated amount is US$1,000 per US$1,000 principal plus accrued and unpaid interest. The descriptions report NYSE listing under JPM/32, DTC global book-entry issuance, and the possibility that a reopening increases aggregate principal outstanding (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

These dated amounts are reported principal, not carrying or fair value. Nor do they establish the accounting classification or value of the parent guarantee, the entire Series A population, or an unchanged balance between the stated dates. Each filing's Exhibit 4.4 identifies the 19 February 2016 Indenture and incorporates it by reference to Exhibit 4(a)(7) of Form S-3 file 333-209682, filed 24 February 2016. The complete filings summarize that instrument; its text and applicable supplements or amendments require their own inspection before being used as complete legal terms.

The filings then describe general Series A terms that may apply to securities including the Callable Notes. In their summary, unpaid interest must continue for 30 days to become an event of default; principal and other amounts default when due. An issuer covenant breach has a 90-day period after the specified written notice. For listed continuing payment, issuer-covenant or guarantee events, the trustee or holders of at least 25% of the aggregate principal then outstanding for the relevant series may, by written notice to the issuer and guarantor, declare principal and accrued interest due. Specified continuing issuer insolvency events accelerate the debt automatically without a declaration. Guarantor insolvency alone is excluded and does not authorize acceleration under this description, while a payment default by the guarantor remains a separate possibility (JPMorgan Chase & Co., 2023, 2024, 2025, 2026). A history must therefore keep the failure, notice, elapsed time, actor, declaration and changed due schedule distinct, rather than record one group-wide default flag.

Collective remedy control also differs from an individual's right to payment. The summaries condition an ordinary holder remedy action on prior trustee notice, a request by holders meeting the stated 25% threshold, 60 days of trustee inaction, indemnity and no inconsistent majority direction. They separately protect each holder's right to receive and sue for payment when due against impairment without that holder's consent. The summary permits certain amendments without holder consent under Section 8.01. For consent-based modifications under Section 8.02, it specifies a majority in aggregate principal amount of each affected outstanding series, whereas listed changes to maturity, amounts, currency, payment method, redemption, payment suit or specified holder repurchase rights, voting threshold or adverse guarantee terms require each affected holder's consent. These thresholds attach to different actions and voting populations, not a generic creditor vote. The 2022 default-and-waiver subsection omits the trustee-conflict paragraph that the 2023–2025 versions place there. The 2022 filing discusses a potential default, trustee conflict and possible resignation later under its trustee heading. This change in placement does not establish a contract amendment. The original Indenture and series-specific supplements remain uninspected (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

Reported succession terms give another reason to keep the obligor separate from the debt claim. Under the general Series A summary, a successor to the issuer in a covered merger, consolidation or transfer of all or substantially all assets must be a U.S. corporation or limited liability company that expressly assumes due payment and the issuer's covenants. A successor to the guarantor must instead be a U.S. corporation that assumes the full guarantee and the guarantor's covenants. The continuing issuer or guarantor is the other stated route. Each party's route also requires that no default or event that would become a default with notice or time has occurred or would be continuing immediately after the transaction. The descriptions deem certain resolution-related transfers of the guarantor's material assets to meet the all-or-substantially-all category; they do not define a numeric percentage (JPMorgan Chase & Co., 2023, 2024, 2025, 2026).

The same summaries say a recapitalization, change of control or highly leveraged transaction alone receives no additional holder protection from these merger provisions; a particular supplement may provide a put or higher interest. The model therefore needs a dated successor, assumed payment and covenant duties, separate issuer/guarantor conditions and any evidenced series-specific protection. Succession itself does not prove repayment, accounting derecognition or unrestricted access to the group's cash. These are reported terms that may apply, pending inspection of the original Indenture and applicable supplements.

Loan origination can debit loans and credit a customer deposit in the bank's book. Subsequent transfer to another bank produces settlement flows. Treating all loan growth as an immediate cash outflow would therefore encode a specific funding assumption, not a universal banking identity. A system-wide model must also link counterparties and settlement assets; single-company simulation records external counterparties without claiming to close the whole economy.

An allowance calculation is a dated measurement based on forecast information. For a single horizon, let EAD be a fixed exposure at default, PD the probability of default, and LGD the mean loss fraction conditional on default. Ignoring discounting, the illustrative expected loss in currency units is

$$L = EAD \times PD \times LGD.$$

With exposure 100, default probability 0.02 and loss given default 0.4, it produces 0.8. Changing those inputs to 0.04 and 0.5 produces 2, so the allowance increase is 1.2 if there are no other movements. This calculation is a demonstration model. A standard-specific expected-credit-loss implementation must determine horizon, staging or other classification, discounting, forward-looking information, write-offs and purchased-credit-impaired or other special cases. The simple formula is not an IFRS 9 or US GAAP compliance implementation.

The exposure entering an allowance calculation also needs a defined population. JPMorgan's 2025 report gives four different amounts for lending-related commitments at 31 December 2025:

Reported quantity	US$ millions
Contractual lending-related amount	1,817,307
Commitments in the allowance-methodology table	603,947
Allowance for lending-related commitments	5,071
Lending-related carrying value	6,278

Notes 13 and 28 define these populations and measurements separately (JPMorgan Chase & Co., 2026, pp. 258–262, 295–297). The contractual total contains consumer and wholesale commitments; the allowance table excludes specified populations. The commitment allowance is reported in accounts payable and other liabilities. Carrying value can also contain guarantee liabilities, fair-value measurements and deferred fees or purchase adjustments, as applicable. For standby letters of credit and other financial guarantees, the report explicitly reconciles a US$175m allowance and a US$441m guarantee liability to US$616m of carrying value. Treating either the contractual total or carrying value as the allowance would compute the wrong quantity.

The report also distinguishes contractual capacity from expected funding. Customers may leave facilities unused, refinance them or let them expire; some commitments can be reduced or cancelled under their contracts or law. The reported policy records no allowance for future draws on unconditionally cancellable commitments. That exclusion concerns the loss measurement; a liquidity scenario still needs the conditions under which a customer can draw, the bank can cancel, and a payment requires settlement funds. Those conditions and their dependence on borrower distress cannot be recovered from a commitment total. Comparisons also need the same reporting basis: specified wholesale amounts are net of risk participations here but gross in Federal Reserve filings. Parent guarantees of subsidiary debt are expressly outside the Note 28 table (JPMorgan Chase & Co., 2026, pp. 258, 295–296, 300).

The historical FASB amendment ASU 2016-13 helps explain these distinctions. Its paragraph 326-20-30-11 measures off-balance expected credit losses over the contractual period of a present obligation to extend credit, except for an obligation the issuer may cancel unconditionally. For a covered commitment, the estimate considers both whether funding will occur and the expected loss on commitments that do fund. Example 10 keeps funded credit-card loans within loss measurement while excluding an allowance on unconditionally cancellable undrawn credit, even though the bank has sometimes advanced funds before detecting default. Paragraph 460-10-30-5 separately requires a contingent expected-loss liability and a noncontingent fair-value guarantee liability for covered financial guarantees. These are historical amendment provisions, not verified current Codification; applying them to the 2025 report requires the actual contracts and later rule history (Financial Accounting Standards Board, 2016a, PDF pp. 2, 108, 118, 144, 173–176).

A commitment to acquire loans makes the time distinction especially clear. The report dates the Apple Card forward purchase commitment to 30 December 2025, its announcement to 7 January 2026, and expected closing to approximately 24 months later. The commitment disclosure includes about US$104bn of estimated credit exposure at closing, including about US$23bn of estimated drawn loans, while the reported transaction allowance is US$2.2bn (JPMorgan Chase & Co., 2026, pp. 47, 260, 296). The expected drawn loans at that future closing are not loans already owned at the 2025 reporting date. A reconstruction therefore retains the forward commitment, the future portfolio estimate and the dated allowance as different facts. A forecast must supply closing and subsequent draw assumptions before generating loan and settlement events. These reported policies and amounts support the distinctions; their application to a new calculation still requires the operative accounting rules and actual contract facts.

Regulatory capital is computed in a separate prudential layer with its own perimeter, eligibility, deductions, risk weights and transition provisions. Accounting equity feeds a reconciliation to capital; it is not automatically Common Equity Tier 1. Liquidity and capital constraints influence the management policy, which in turn generates asset sales, funding or capital events. A breached ratio can be an economically meaningful scenario outcome. No hidden capital injection is permitted to repair it.

11.2 A bank's cash, allowance and capital in one path

The bank reference state contains settlement cash 30, gross loans 100, allowance 0.8, deposits 110 and equity 19.2. Originating a loan of 20 first debits loans and credits the borrower's deposit. A subsequent transfer of 15 to another bank debits deposits and credits settlement cash. Loan origination and external cash use are consequently distinct events. Receive loan interest 6 in cash and credit deposit interest 2 to customer balances. Remeasure the allowance to 2.4, posting an additional loss of 1.6. The resulting state is cash 21, gross loans 120, allowance 2.4, deposits 117 and equity 21.6. Net assets are $21 + 120 - 2.4 = 138.6 = 117 + 21.6$.

Now 25 of deposits runs off. The declared funding policy uses cash first, then a committed facility with limit 10. It draws 4 and settles 25, leaving cash zero, deposits 92 and wholesale debt 4. A runoff of 40 instead leaves 9 unpaid even after the maximum facility draw. The simulator records default and retains the unpaid deposit obligation. It does not assume unlimited central-bank or shareholder funding.

The prudential calculation is a separate, illustrative policy. Deduct 2 from accounting equity to obtain capital 19.6; apply a 75% risk weight to gross loans for risk-weighted assets 90. The resulting ratio is 19.6/90. Neither the deduction nor the weight is a Basel default. Actual prudential packages require jurisdiction, exposure class, eligibility, buffers and dated rules, with a reconciliation from the accounting book. The structure follows the need to model capital and liquidity jointly in Behn et al. (2019, Section 2), without transferring that paper's estimated preferences or shock parameters to this bank.

For a richer credit adapter, partition borrower cohorts by currency, vintage, credit state, remaining maturity, collateral and measurement category. At date t, supply scenario probabilities, conditional default-time probabilities, exposure paths, conditional loss severity and discount factors. Its expected loss calculation has the declared form

$$L_t = \sum_s p_s \sum_{k \in H} d_{t,k} \Pr(\tau = k \mid s, \mathcal{F}_t) \mathbb{E}[\mathrm{EAD}_k \mathrm{LGD}_k \mid \tau = k, s, \mathcal{F}_t].$$

The conditional product expectation retains dependence between exposure and severity. The horizon H concerns eligible default times; associated cash shortfalls may continue later. The rule package decides twelve-month versus lifetime default scope, credit-state transfers, interest basis

and write-off recognition. Changes in allowance must be attributed to origination, derecognition, transfers, revised expectations, currency and write-offs. Contract and prudential support can be specified now, while full IFRS 9/CECL/ ASBE classification and paragraph-level validation remain later obligations.

11.3 State and event order

The dated credit methods use the selected amortised-cost, interest and impairment provisions of AASB 9 (Australian Accounting Standards Board, 2022e). The following loan has different inputs from the earlier liquidity example, so its allowance and final equity are calculated separately.

BankState contains cash/reserves by settlement venue and currency; deposits by customer, access rights and maturity; assets by contract, gross carrying amount, allowance, original EIR, accrued cash flows, collateral and credit state; undrawn commitments; funding lines and covenants; accounting equity; regulatory mappings; and evidence/information views. LoanState separately retains origination credit risk and current risk, SICR finding, credit-impaired/POCI status, forbearance, modification, write-off and recovery history. Aggregating by today's loan balance alone loses both staging evidence and later response to shocks.

Each time step first admits information known then, forms permitted credit and liquidity forecasts, chooses behavior under that information, and schedules due events. Accrual, actual settlement, impairment measurement and regulatory queries have separate business identities. A deposit received from outside debits cash/reserves and credits deposit liability. A loan credited to the borrower's deposit debits loans and credits deposits without initially transferring reserves; a transfer to another bank then debits deposits and credits reserves. A loan paid directly outside instead debits loan/credits reserves. These two funding paths cannot be treated as identical cash events. Deposit interest debits interest expense/credits deposit or accrued payable; withdrawal requires settlement funds.

11.4 Expected credit loss

For each admitted scenario j and date t, calculate contractual due flows minus the cash expected to be received, including supported collateral/enforcement cash and timing; discount the shortfall under the applicable original EIR or specified exception. With mutually exclusive exhaustive scenario weights w_j, ECL is the sum of w_j times those present-value shortfalls. Default-time hazards must be converted to disjoint default-time probabilities; summing cumulative PDs over dates counts the same default repeatedly. Collateral recoveries cannot be credited both in LGD and as a second recovery cash flow. A shorthand PD×LGD×EAD is admissible only if its definitions and discounting reproduce the specified cash-shortfall calculation; it is not a universal replacement.

Expected credit loss (ECL) measures the discounted expected cash shortfall. A significant increase in credit risk (SICR) changes the general method's default horizon. Purchased or originated credit-impaired (POCI) assets follow a separate method. For the general approach, twelve-month ECL considers lifetime shortfalls arising from possible defaults during the next twelve months; it is not just unpaid cash in those twelve months. SICR uses the change in lifetime default risk since initial recognition, reasonable forward-looking information and the applicable rebuttable presumptions. Do not install a two-times-PD or 30-day rule as an unconditional universal trigger. Lifetime ECL follows a supported SICR finding; cure needs a supported reversal of that finding. Subsequently credit-impaired assets use original EIR on

amortised cost for interest in subsequent periods; POCI has a separate credit-adjusted-rate and cumulative-change method.

The 2017 ASBE 22 text makes the two horizons explicit. Article 47 defines credit loss as the present value of all contractual cash flows less expected receipts; a late receipt can therefore create a loss even when its nominal amount is paid in full. Article 48 limits twelve-month ECL by the date of the possible default event, while retaining the resulting cash shortfalls over the asset's life. Article 52 assesses changes in default risk, rather than changes in the allowance, and Article 58 requires probability weighting, time value and reasonable, supportable information (Ministry of Finance of the People's Republic of China, 2017, Articles 47–48, 52, 58). These provisions support the following dated comparison. Article 84 states 1 January 2018, but applying the standard to a particular company also requires its issuance schedule, subsequent amendments and adoption facts.

The general measurement interface requires scenario_cashflows, law_ref, information_view, horizon_admission, credit_assessment and discount_rule. An unweighted history set can produce conditional losses or loss bounds, but cannot produce an unbiased probability-weighted ECL by silently using equal weights. Return measurement_unresolved for the ECL claim. Trade receivables and certain other assets use the separately applicable simplified approach; the general loan staging rule must not overwrite AASB 9.5.5.15–16.

11.5 Full loan and allowance roll-forward

Assume a fixed-rate non-POCI amortised-cost loan held to collect its contractual cash flows, originated for 100, with one terminal payment 121 two years later, original EIR 10%, no fees, collateral, modifications, FX or taxes. Since no payment falls due before maturity, an earlier default in this example is a separately established qualitative credit event, such as financial distress under the admitted default definition; it is not a missed instalment. Article 52 of ASBE 22 requires a definition consistent with credit-risk management and consideration of qualitative indicators (Ministry of Finance of the People's Republic of China, 2017). At origination the assumed lifetime scenarios are: default in the first twelve months with probability 0.1, default during the following year with probability 0.1, and no default with probability 0.8. Either default gives terminal collection 60.5. The first twelve-month ECL includes lifetime loss from the first scenario only: $0.1\times(121-60.5)/1.1^2=5$; initial lifetime ECL would be 10. Debit loan 100/ credit cash 100; debit impairment 5/credit allowance 5. Gross 100, allowance 5, net 95.

At year-one accrual the loan has survived the first window and is not yet credit-impaired: debit gross loan 10/credit interest 10. Survival alone would give remaining default probability $0.1/0.9=1/9$. A subsequent, separately admitted credit forecast revises that conditional probability to 0.2; it is an authored forecast update, not a Bayes calculation from survival alone. Its lifetime weights are 0.8/0.2, with a separately established SICR finding; $\text{ECL}=0.2\times60.5/1.1=11$. Debit impairment 6/ credit allowance 6. Gross 110, allowance 11, net 99. The ratio change alone is not the defined SICR test. Only twelve months remain, so twelve-month and lifetime ECL both equal 11 under this revised forecast. This amount cannot distinguish the two classifications.

Immediately afterward, a distinct, ordered credit-impairment event changes expected maturity collections to 60.5. This is a forecast mean, not a guaranteed or maximum recovery. Required allowance is $110-60.5/1.1=55$; debit impairment 44/credit allowance 44. Net 55. ASBE 22 Article 59(5) measures this non-POCI credit-impaired asset by gross balance less discounted expected receipts. Articles 38 and 39(2) then make amortised cost, after the allowance, the basis for subsequent-period interest (Ministry of Finance of the People's Republic of China, 2017).

The bank is assumed still to have reasonable prospects of fuller recovery before maturity. The separate finding that some contractual cash will not reasonably be recovered first arises at settlement on this branch. If that finding arose earlier, ASBE 22 Article 43 would require the corresponding earlier write-off. Expected collections alone do not justify postponing an already-required write-off.

In the next full year interest revenue is 55×10%=5.5. The gross contractual accretion is 11. One consistent decomposed posting debits gross loan 11, credits interest revenue 5.5 and allowance 5.5: gross 121, allowance 60.5, net 60.5. The allowance increase of 5.5 is attributed to discount unwind in its roll-forward, with no additional impairment expense in this posting. This reconciles gross and net-interest bases without counting the same expected loss again. Collect 60.5: debit cash/credit gross 60.5. When no reasonable expectation of remaining recovery is established, debit allowance 60.5/credit gross 60.5. Both balances reach zero. A mere delinquency is not that write-off finding. Subsequent unexpected cash recovery is recorded with its own recovery-income method and evidence; it does not restore an old live receivable by default.

Moment	Cash, opening 200	Gross loan	Allowance	Deposit liability	Equity, opening 20
Initial loan and allowance	100	100	5	180	15
Year-one accrual and SICR	100	110	11	180	19
Credit impairment	100	110	55	180	−25
Maturity accrual	100	121	60.5	180	−19.5
Collection and write-off	160.5	0	0	180	−19.5

Total interest 15.5 less impairment 55 is loss 39.5, equal to 100 advanced minus 60.5 received. The negative equity state survives; no missing branch or capital plug restores solvency. These five dates jointly test allowance, recognition and bank balance-sheet arithmetic. The same loss cannot also be deducted directly from gross before the stated write-off.

For a non-derecognized loan modification, AASB 9.5.4.3 requires gross PV at original EIR and a modification result, then allowance and SICR reassessment under 5.5.12. For example gross 110 changed to a single future 110 at original 10% gives revised gross 100 and modification loss 10 before any allowance change. ASBE 22 Articles 42 and 56 similarly separate remeasurement of a non-derecognised asset from its credit-risk reassessment (Ministry of Finance of the People's Republic of China, 2017). This is separate from the financing method's issuer-liability ten-percent test. A derecognition decision requires its own asset conditions and cannot reuse that threshold.

11.6 Funding, capital and decision policies

Liquidity simulation settles the dated net payment queue by venue/currency, respecting encumbrance, available cash, committed line terms and admissible actions. Haircut/eligibility and transferability come from a versioned policy; market value alone is not available liquidity. With cash 20, due withdrawal 30 and an admitted usable facility 8, even a full draw leaves shortfall 2. Record the default/unresolved settlement according to the event's legal model. Raising exactly 2 of new equity without an investor event is an invented financing source. Interbank contagion requires explicit counterparties and settlement priorities; it cannot be inferred from aggregate deposits.

A RegulatoryAssessment reads an accounting view and an independent jurisdiction, date, perimeter, eligibility, deductions and exposure-weight policy. It outputs capital components, RWA/exposure measures, liquidity measures and reconciliations with their distinct denominators. Illustratively, equity 20 minus an assumed regulatory deduction 2 yields capital 18; assumed RWA 125 gives 14.4%. Those inputs are synthetic, not a Basel rule, national capital approval or default threshold. No binding capital-policy numbers have been inspected for a target bank here. Use the Basel/national source and actual bank facts before real admission.

Behavior under capital/liquidity constraints is a separate economic model. Specify feasible actions, objectives, information set and ties before solving; failure to find an action returns policy_failure or default under the stated contract. Halaj/Behn sources can motivate candidate behavior but cannot calibrate it for an individual bank. A forecast experiment must compare simple fixed-book, contractual runoff and explicitly estimated policy baselines with time-safe data, before claiming a predictive improvement. CECL and other bases retain separate authority/method branches in the dated-authority discussion.

Chapter 12

Insurance service, finance and claims

An insurer receives cash before much of its service is delivered. Cash premium, service revenue and profit therefore follow different schedules. A short coverage example first connects gross insurance to reinsurance. The subsequent two-year cases introduce unearned profit, loss allocation and finance one change at a time.

12.1 Insurance: cohorts and future service

An insurer requires policy cohorts or groups with issue dates, coverage, premiums, benefits, expenses, lapses, claims development, guarantees and reinsurance relationships. Asset portfolios need their own contracts and valuations. The liability state must distinguish expected future cash flows, discounting, risk adjustments or margins, unearned profit measures where applicable, incurred claims and movements between opening and closing states. The IFRS 17 overview's distinction between measuring future cash flows and recognising profit as service is provided motivates this separation (IFRS Foundation, 2026b).

The module API accepts policy/cohort state, economic assumptions, non-economic assumptions and information date. It returns a measurement vector, movement attribution and proposed recognition events. Experience differences, changes in assumptions, finance effects and service effects remain separately identifiable. Reinsurance held has its own record and measurement; it is not an automatic percentage offset to every insurance liability.

Scenarios can alter discount curves, mortality, morbidity, lapse rates, inflation, catastrophe losses and policyholder behaviour. Their joint dependence and cohort exposure must be declared. A portfolio total can match history while hiding very different guarantee and duration profiles. This makes alternative economic histories especially valuable, but also limits what public-data reconstruction can identify. A full insurance package needs actuarial and accounting review beyond the core release.

12.2 Insurance service and reinsurance in a complete cohort

The example uses one-year policies and a one-year proportionate treaty starting when those policies are recognised, with uniform service for each. Both groups qualify for premium allocation through the one-year conditions in paragraphs 53(b) and 69(b); paragraphs 62–62A determine treaty recognition. Assume no acquisition cash flows, investment components, significant financing or onerous remaining coverage. Claims and recoveries are expected within a year of the claims being incurred. The inspected AASB 17 text incorporating IFRS 17 specifies the movements in paragraphs 53–59, allocation in B126 and separate reinsurance requirements in 63–70A (Australian Accounting Standards Board, 2022d). Gross and ceded groups retain distinct identities.

Begin with capital and cash of 20. Receive premium 120 and recognise a remaining-coverage liability of 120. Pay reinsurance premium 24 and recognise a coverage asset of 24. After half the uniform service, recognise revenue 60 by releasing gross coverage and reinsurance expense 12 by releasing its coverage asset. Estimated incurred claims are 40, of which 30 is paid; the remaining-claim risk adjustment is assumed to be 2. Gross claims expense is therefore 42 and incurred-claims liability is 12. Assume the treaty reimburses half the gross expected claims, recovery cash of 15 has arrived, the risk transferred is valued at 1, and reinsurer non-performance has zero expected effect in this example. The reinsurance claims asset is $20 + 1 - 15 = 6$, and recovery income is 21.

Cash is $20 + 120 - 24 - 30 + 15 = 101$. Assets are cash 101, remaining reinsurance coverage 12 and claims recovery 6, totalling 119. Liabilities are remaining gross coverage 60 plus incurred claims 12, totalling 72. Profit is $60 - 42 - 12 + 21 = 27$ and equity is $20 + 27 = 47$. These balances reconcile without netting reinsurance against the insurance liability. If counterparty credit risk changes, the recovery measurement changes through its own inputs; it does not mechanically halve every gross liability or risk adjustment.

Longer-duration business adds a fulfilment-cash-flow vector, discount curve, risk adjustment, contractual service margin, loss component and coverage-unit schedule. Select portfolio and annual cohort, assess onerous grouping, project cash flows within contract boundaries, discount with the declared financial assumptions, measure non-financial risk separately, and attribute subsequent movements to new business, experience, assumptions, finance and service. Future-service changes adjust the supported service-margin operation subject to the loss rules; current/past-service effects enter their specified result components. Unsupported variable-fee, transition or loss-recovery operations must stop those dependencies explicitly. A PAA example does not implement these longer-duration methods by implication.

An actuarial reserve adapter also needs a concrete estimation contract. For cumulative claims $C_{i,k}$ by origin year i and development year k, the chain-ladder point method estimates each factor from matched observed cells, $\widehat{f}_k = \sum_i C_{i,k+1} / \sum_i C_{i,k}$, then projects the latest cumulative claim along the remaining factors. A triangular example with rows $(100, 150, 180)$, $(120, 180)$ and (140) gives factors 1.5 and 1.2, outstanding amounts 36 and 112 for the unfinished rows, and total 148. Mack (1993, Sections 2–3, equations (1)–(3)) makes explicit the conditional-mean and variance assumptions and independence across origin years; its aggregate-error discussion retains dependence from shared estimated factors. A single final-development observation cannot supply an empirical tail variance without an additional assumption. The adapter therefore returns the point projection and marks uncertainty estimation unavailable in this tiny example. It supplies neither a full predictive distribution nor an IFRS risk adjustment. Calendar effects, inflation, changing claims handling and tail selection require separate sensitivity and validation before real use.

12.3 Group state and general measurement

The following cases apply selected general-measurement and service-allocation provisions from the retained AASB 17 compilation (Australian Accounting Standards Board, 2022d). They extend the premium-allocation example without presuming actuarial calibration or a complete insurer model.

The general method distinguishes fulfilment cash flows (FCF), including an explicit risk adjustment (RA), from the contractual service margin (CSM), which holds unearned profit. The liability for remaining coverage (LRC) concerns future service; the liability for incurred claims (LIC) concerns service and insured events already experienced. Keeping these components separate makes a change in future estimates different from an unpaid claim.

Contract records retain insured risks, premiums/benefits, service and settlement dates, options/guarantees, cancellation/repricing rights and acquisition costs. Group admission evaluates portfolio/similar-risk management, onerous status and required issuance cohorts under AASB 17.14–24, recognition under 25–28 and boundary under 34. Cash outside the boundary belongs to future contracts, not this group's profit. GroupState retains expected dated cash flows, current discount curves, inception rates, explicit risk adjustment, CSM, loss component, LRC/LIC, coverage units, service experience and assumption vintages. Same premium totals are not sufficient state for merging different guarantees or claim tails.

General measurement follows 32–38: fulfilment cash flows are probability-weighted future net outflows, adjusted for timing/financial risk without double counting, plus explicit non-financial risk adjustment. CSM removes initial unearned gain under its conditions; an initial net loss is recognized under 47, not negative CSM. Paragraphs 40–44 separate remaining coverage and incurred claims. For a nonparticipating group, accrete CSM at the specified locked-in rate, apply future service changes at their required rate to CSM up to its available amount, record any excess loss and allocate current service over current plus future coverage units under B119. B96–B97 distinguish future service from incurred-claim changes, financial changes and current experience; not every revision is a CSM adjustment.

A MeasurementInput must resolve cash-flow/law, discount, risk-adjustment method, service grouping and coverage-unit evidence. Missing probabilities or RA support permit conditional sensitivity cases but block a source-supported measurement. Risk adjustment is the compensation the entity requires under 37; a standard deviation, confidence quantile or arbitrary percentage is only a candidate method until justified for that entity. Disclosures and the confidence-level equivalent, where applicable, are separate obligations. Market-consistent financial inputs and entity risk compensation must not be conflated.

12.4 Complete two-year general-measurement case

Take a two-year nonparticipating group with one premium 120 received at inception, claims expected/actually incurred and paid 40 each year, zero discount rate, RA 10 initially released 5 with each equal service year, no acquisition costs, investment component, lapses, taxes, finance changes or reinsurance. Coverage units are 1 in each year because equal benefits and coverage are assumed. Before premium payment, expected net cash outflow is $80-120=-40$, RA 10, FCF -30 and CSM 30, with net initial carrying amount zero. After the premium debit cash 120/ credit LRC 120. Its components are remaining expected claims 80, RA 10 and CSM 30.

Each year releases expected service claims 40, RA 5 and CSM 15: debit LRC 60/ credit insurance revenue 60. Incur claims 40: debit service expense 40/credit LIC 40; payment debits LIC 40/credits cash 40. End of year one: cash 80, LRC 60 (claims 40, RA 5, CSM 15), LIC 0

and equity/profit 20. End year two: cash 40, liabilities 0, cumulative equity/profit 40. Revenue 120 and claims 80 reconcile to premium 120 less claims cash 80. No investment component has been mistaken for revenue.

At year-one close, a separate pre-next-service change increases the remaining claim estimate from 40 to 60, RA unchanged. FCF rises 20; consume remaining CSM 15 and recognize service loss 5. LRC rises from 60 to 65, comprising FCF 65, CSM 0 and loss component 5 as a tracked portion, not an added liability on top. Debit service loss 5/credit LRC 5 with component transfer CSM→FCF 15. A decrease of 8 in the revised future claim estimate, before any additional service/finance movement, first reverses loss 5 and restores CSM 3. FCF 57 plus CSM 3 gives LRC 60. Credit loss reversal 5/debit LRC 5. This tests 48–52 ordering. Later systematic allocation of loss component to releases/finance must reduce it to zero by coverage end; it cannot be amortized as revenue a second time.

In a separate case, an already-incurred unpaid claim estimate rises 40 to 48. Debit service expense 8/credit LIC 8; do not consume future-service CSM. A discount-rate change is financial under B97; its profit/OCI route follows the selected permitted finance policy. The zero-rate base case does not remove the need for separate current and locked-in rate calculations in a real group.

For the unreversed onerous alternative, carry the loss component through the remaining service year. Opening LRC 65 comprises future claims 60 and RA 5, with CSM 0 and loss component 5 within that LRC. A declared proportional allocation assigns 5/65 of each expected claim/RA release to the loss component, consistent with paragraphs 49–52 and B124's revenue exclusions. The expected claim release allocates 60/13 to loss reversal and the RA release 5/13; the total is 5, leaving loss component 0. Debit LRC 65, credit revenue 60 and credit loss reversal 5. Separately incur and pay claims 60 through LIC. Period profit is 60−60+5=5; combined with first-year profit 20 and the earlier onerous loss 5, lifetime profit is 20. Closing cash 20 equals equity 20 and all liabilities are zero. Treating all 65 as revenue while also reversing loss 5 overstates profit by 5.

12.5 Nonzero discount and service allocation

A second complete case keeps premium 120, two equal coverage units and RA 10, but expected and realized claims are 44 after one year and 48.4 after two years. Both the inception and current discount curves remain a flat 10% throughout; there are no new estimates, options, fees, taxes, investments or OCI election. This RA schedule is an authored measurement input, not an actuarial calibration: it declines by 5 per year, with its entire change in the service result under 81. All other insurance finance effects go to profit or loss. The retained source supports the mechanics in 44, 80–88, B119 and B123–B124.

At inception the present value of the two expected claims is

$$\frac{44}{1.1} + \frac{48.4}{1.1^2} = 40 + 40 = 80.$$

FCF before premium is 80+10−120=−30, CSM 30, and after premium LRC 120. In year one the claim PV accretes 8; CSM accretes 3 at the locked-in rate. Allocate the resulting CSM 33 across the two current/future units, releasing 16.5. Revenue is expected claims 44 plus RA 5 plus CSM 16.5=65.5. Service expense 44 and finance expense 11 leave profit 10.5. After paying 44, cash 76 and LRC 65.5 (claim PV 44, RA 5, CSM 16.5) reconcile to equity 10.5.

In year two claim PV accretes 4.4 and CSM accretes 1.65. Release all remaining CSM 18.15, RA 5 and expected claims 48.4, giving revenue 71.55. Service expense 48.4 and finance expense

6.05 leave profit 17.1. Final cash/equity 27.6 equals premium 120 minus claims 92.4; all group liabilities are zero. Total revenue 137.05 less finance expense 17.05 equals premium 120. Setting revenue equal to cash premium each year or releasing 15 of CSM each year would ignore the specified financing. These are separate component calculations followed by debit finance expense/ credit LRC for accretion, debit LRC/credit revenue for service and the stated LIC recognition/settlement entries. Cash is deliberately assumed uninvested.

12.6 PAA, reinsurance and valuation methods

The preceding premium-allocation case states separate eligibility and recognition timing for gross insurance and reinsurance. PAA under 53–59 requires the one-year coverage or nonmaterial-difference condition; a short payment lag is not the eligibility test. LRC starts with received premium and applicable acquisition adjustments, then releases insurance revenue under the prescribed service pattern. LIC still measures incurred obligations using the applicable fulfilment-cash-flow rules; the allowed short-settlement discount simplification needs its own conditions. Emerging onerous facts require a loss assessment; a premium balance cannot be presumed sufficient.

A one-year policy can leave claims unpaid for several years. The distinction between the period of insurance service and the period until settlement is therefore essential even when PAA is available. The official ASBE 25 text separates three timing tests. Article 50 permits PAA when each contract's coverage is no longer than one year, or when the stated no-material-difference condition holds. Article 54 concerns a different interval: the time between each portion of service and the related premium due date. If that interval does not exceed a year, the remaining-coverage liability need not be adjusted for financing. Article 56 instead measures from claim incurrence to expected payment or receipt. Its option to omit the time-value and financial-risk adjustment requires settlement within a year of incurrence (Ministry of Finance of the People's Republic of China, 2020, Articles 50, 54, 56). The retained AASB 17 compilation makes the corresponding distinctions in paragraphs 53, 56 and 59(b).

Consider a separate group of issued policies with exactly one year of coverage. All amounts are illustrative currency units. Premium 120 is due and received when coverage starts, with no earlier onerous trigger; group recognition is therefore at that date under Article 16. Assume valid insurance grouping, uniform risk release, no acquisition costs, investment component, reinsurance, tax, other expenses or other movements, and no facts indicating onerous remaining coverage. Elect the permitted remaining-coverage financing exception. After half a year, recognise revenue 60 and retain remaining coverage of 60. Articles 53 and 57 support this roll-forward and time allocation; Article 57 requires a different allocation when risk release is not predominantly time-based (Ministry of Finance of the People's Republic of China, 2020). These assumptions isolate the claim measurement from other changes in the group's result.

At that reporting instant a claim is first incurred, with an expected payment of 60.5 exactly two years later. This is a mean estimate, not a certain settlement. Assume an appropriate flat annual effective discount rate of 10% and a separately estimated non-financial risk adjustment of 5 in current measurement units. The inputs are authored for the example; applying them to an insurer would require market and actuarial evidence. Articles 21, 23, 25 and 26 require current probability-weighted cash flows, appropriate discounting and a separate non-financial risk adjustment (Ministry of Finance of the People's Republic of China, 2020). The claim's cash-flow present value is $60.5/1.1^2 = 50$, so its liability is $50 + 5 = 55$. The risk adjustment is not discounted a second time. The claim has only just been incurred, so no intervening finance accrual enters this measurement.

Record debit cash 120/credit remaining coverage 120 on premium receipt, then debit remaining coverage 60/credit insurance revenue 60 for the service provided. Recognising the claim gives debit insurance service expense 55/credit incurred claims 55, consistent with Article 31. The group then has cash 120, remaining coverage 60, incurred claims 55 and a cumulative result of 5: $120 - 60 - 55 = 60 - 55 = 5$. These are balances attributable to the group at mid-year, not a complete insurer forecast. Using the undiscounted $60.5 + 5 = 65.5$ for this two-year claim would overstate the required liability by 10.5 and understate the result by the same amount. The shortcut is wrong under the stated rules and assumptions. A zero rate would remove that discounting difference while leaving the separate risk adjustment intact. Articles 86(3) and 97 also require disclosure of the distinct claim components and the PAA/financing choices (Ministry of Finance of the People's Republic of China, 2020).

This comparison uses identified editions. The Ministry page reproduces the 2020 ASBE 25 text; Article 120 states 1 January 2023, but that sentence does not settle every entity group's adoption schedule. Section 12.7 distinguishes the Ministry's later explanation of that timetable from a company's reported adoption. The retained December 2022 AASB 17 compilation applies to annual periods beginning on or after 1 January 2023 but before 1 July 2026. The shared result in this example establishes a scoped measurement comparison. Applying it to a particular entity still requires its applicable rules and contract facts; later amendments and broader equivalence require separate evidence.

Reinsurance has its own group, boundary, recognition time, counterparty risk, cash-flow, CSM/cost and possibly loss-recovery component under AASB 17 paragraphs 60–70A. Do not multiply every gross insurance liability by a quota share: excluded claims, limits, deductibles, timing, reinstatements and counterparty non-performance matter. Claim 100 with an eligible 50% recovery but expected 90% collectability gives conditional expected recovery 45, not 50; altering transferred insurance risk and reinsurer non-performance are different inputs. This simple calculation does not establish treaty recognition or RA by itself. Gross liability 100 remains; reinsurance recovery is a separate asset/measurement with its own authority.

For direct participation, test B101's conditions using the actual underlying items, substantial-share and variation judgments. Retain underlying fair values, entity fee share and guarantee/risk-mitigation choices; paragraphs 45 and B101–B118 specify the variable-fee adjustments to CSM, with onerous and mitigation exceptions. A local component illustration of a qualifying share: underlying-value increase 20 and entity's fixed 10% share gives candidate variable-fee change 2; eligibility, guarantees and other FCF changes must be evaluated before posting. The full option/guarantee valuation and transition calculations are further supported profiles, with source and numerical obligations, not an assumed linear fee model.

The claims-development example above supplies an estimation input, not a recognition rule. Its point projection is 148, but the single observed final factor leaves its variance unresolved. In the general method the factors are

$$\widehat{f}_k = \frac{\sum_{i \in O_k} C_{i,k+1}}{\sum_{i \in O_k} C_{i,k}},$$

where O_k contains origin years with both required observed cells. Each incomplete row is projected through the remaining factors. The same triangle therefore has ultimate claims 180, 216 and 252, with unpaid amounts 0, 36 and 112. Tail development, calendar inflation, zero or negative cells and dependence need explicit extensions. The conditional means, variance scaling and independent-origin-year assumptions behind the inspected Mack error formulas (Mack, 1993, equations (1)–(3), Theorems 1–3) cannot turn this tiny triangle into a predictive law or a justified insurance risk adjustment.

12.7 Adoption dates and the restatement of 2022

A standard's effective date and the economic date of a reported balance need not coincide. An insurer adopting new rules in 2023 may report a revised balance for December 2022 while retaining the earlier report as a record of what it originally disclosed. The system must preserve both versions and explain which one a calculation uses.

The Ministry of Finance's explanation dated 12 June 2025 describes the original ASBE 25 timetable. It specifies application from 1 January 2023 for enterprises listed both domestically and overseas, and for overseas-listed enterprises preparing financial statements under IFRS or ASBE. Other enterprises applying ASBE were scheduled to apply the revised standard from 1 January 2026. Its account of the 2025 notice says existing adopters, including enterprises that had adopted early under the 2020 notice, must continue applying it. Insurers facing genuine difficulties and seeking deferral must submit written reasons to the Ministry's Accounting Department and the relevant institutional supervision departments of the National Financial Regulatory Administration. That statement does not establish an automatic deferral merely because preparation is difficult (Ministry of Finance of the People's Republic of China, Accounting Department, 2025, answers on background and implementation arrangements).

The same explanation limits the new simplification choices to unlisted enterprises; matters outside those choices remain subject to ASBE 25. It names grouping, measurement, presentation and transition as affected areas, including comparative information and the transition date. These are source-supported descriptions of the Ministry's explanation. The operative 2020 and 2025 notices have not yet been directly verified in this collection, so the explanation does not supply the detailed conditions needed to implement those choices (Ministry of Finance of the People's Republic of China, Accounting Department, 2025, answer on the relationship between the notice and the standard).

Ping An provides a concrete adoption history. Its 2022 ASBE annual report states that the group would adopt the revised insurance standard for accounting years beginning on 1 January 2023. Its 2023 report confirms adoption on that date and restatement of comparative information (Ping An Insurance (Group) Company of China, Ltd., 2023, Note XIV.2, pp. 315–316) (Ping An Insurance (Group) Company of China, Ltd., 2024, Note IV.34, p. 192). The following balances all refer to the consolidated group at 31 December 2022, in RMB millions. The first column comes from the 2022 report; the second is the restated comparative in the 2023 report. Changes are calculated from those columns; the asset, liability and parent-equity changes also match the three rows of Note IV.34's reported bridge (Ping An Insurance (Group) Company of China, Ltd., 2023, pp. 146–148) (Ping An Insurance (Group) Company of China, Ltd., 2024, pp. 141–143 and 192).

Balance	Original	Restated	Change
Assets	11,137,168	11,009,940	$-127,228$
Liabilities	9,961,870	9,823,944	$-137,926$
Total equity	1,175,298	1,185,996	$10,698$
Parent shareholders' equity	858,675	869,191	$10,516$
Non-controlling interests	316,623	316,805	182

The distinction between total equity and equity attributable to the parent matters. Assets less liabilities rise by $-127,228 - (-137,926) = 10,698$. Of that increase, 10,516 belongs to parent shareholders and 182 to non-controlling interests. Treating the parent-equity row as total equity would leave an unexplained difference of 182. These are changes between reported accounting bases for the same date, not cash received in a later period.

The transition also changes how later growth is calculated. Ping An reports assets of 11,583,417 at the end of 2023. Subtracting its restated 2022 comparative gives a reported-balance increase of $11,583,417 - 11,009,940 = 573,477$. Subtracting the original 2022 amount instead gives 446,249, mixing the old and new bases and understating that comparable movement by 127,228 (Ping An Insurance (Group) Company of China, Ltd., 2024, p. 141). The first difference is a movement in reported asset balances; it is not a cash-flow measure or an explanation of every economic cause of growth.

An original-source query for assets at 31 December 2022, explicitly selecting the 2022 report, therefore returns 11,137,168. A query selecting the 2023 report's restated comparison for that date returns 11,009,940. Each assertion retains its report identity, consolidated scope, RMB-million scale, accounting basis and economic date. A link records that the later assertion restates the comparison; it does not delete the original. A request for a comparable annual movement must reject the mixed-basis pair unless an explicit reconciliation has first aligned it. Because the basis and disclosure version differ, these two amounts are not a same-context source conflict.

The information date remains a separate question. The 2023 report says its board authorised the statements on 21 March 2024 (Ping An Insurance (Group) Company of China, Ltd., 2024, Note I, p. 154). Neither that authorisation nor the 31 December reporting date establishes when the exact retained document became public. The corpus has archive-date leads but no verified publication date or historical byte vintage for either selected report. A query for information available to investors at 31 December 2023 must therefore retain an unresolved availability result for this source selection; it cannot silently import the later restated balance. The corpus's October 2026 revalidation timestamps describe local evidence handling, not investor knowledge in 2023.

Finally, Note IV.34 reports financial-asset reclassification, including use of the classification overlay for relevant assets derecognised during 2022. It also states that some insurance groups used modified-retrospective or fair-value transition because retrospective application was impracticable. The headline bridge consequently cannot be reconstructed by applying one percentage to every insurance liability. The example establishes how to retain and compare disclosed histories; reconstructing the transition itself still requires the affected groups, asset classifications and underlying measurement inputs (Ping An Insurance (Group) Company of China, Ltd., 2024, Note IV.34, p. 192).

12.8 Interfaces, exclusions and acceptance cases

Insurance events are group_recognize, premium_settle, service_release, claim_incur, claim_settle, assumption_change, onerous_change and treaty_change, each bound through the event-basis, authorization and commit contracts to the exact group/version, evidence and information view. Journals and component roll-forwards must reconcile independently; a component reallocation is not always a net liability change. Invalid grouping, missing rate/law, unsupported option valuation, ambiguous contract boundary or unknown service units produces the corresponding unresolved/outside-scope result. Failed numerical paths retain their coverage/mass, without successful-only normalization. Modified contracts need the applicable derecognition analysis; no new ID can erase unfavorable old-group obligations.

Initial transition requires an actual method choice and eligibility for full retrospective, modified-retrospective or fair-value transition under AASB 17 Appendix C, with acquisition, grouping, past service and historical-rate records. This collection does not claim an executed transition or a general VFA valuation. The nonzero-rate general case and onerous allocation through expiry above are finite specification examples. Required later acceptance still includes a priced guarantee, acquisition/reinsurance interaction, transition and actuarial backtesting. These retained obligations prevent the two-year illustration from being described as full insurance-domain readiness.

Chapter 13

Implementation contracts

13.1 Reference implementation and component boundaries

The proposed reference implementation uses Python for orchestration and a deterministic reference evaluator, exact decimal/rational arithmetic for booked amounts, SQLite for transactional metadata and versioned records, and columnar files for large scenario outputs. DuckDB is an optional analytical reader. These are convenience choices justified by inspectability and modest initial scale; they do not establish production performance. The IR and conformance tests allow a later compiled executor without changing accounting meaning. Committed journals, economic deltas, receipts and branch heads belong to the one transactional store, including for simulation branches. Columnar files hold derived analytical exports, never independently writable committed books.

```
accdsl/
  evidence/       documents, spans, filing contexts, extraction versions
  model/          typed IR, entities, books, symbols, contracts
  language/       grammar, parser, resolver, type checker, lowering
  constraints/    propagation, solver adapters, bounds and conflict cores
  accounting/     rule selection, posting validation, closing, consolidation
  economics/      contract and behavioural plugin interfaces
  execution/      scheduler, branch state, stochastic streams, run records
  reporting/      statements, notes, movement schedules, explanations
  storage/        versioned records, migrations, result readers
  interfaces/     Python API, CLI, structured import/export
  packages/       reviewed demonstration and accounting modules
```

The first service boundary is local and synchronous. A job abstraction can be added for extraction and simulation without requiring distributed execution. The model compiler returns an immutable execution plan; the executor receives that plan and run inputs. Neither should retrieve live market data implicitly. Input datasets and rule packages are selected and versioned before execution.

13.2 Persistent records

The table identifies logical storage relations and their obligations; exact versioned record schemas determine their payloads. Candidate calculations and diagnostics remain separate from the committed accounting relations.

Relation	Key information and constraints
`document`	Hash, original path/URI, filing ID, entity, publication and availability times, media type, revision links. Original bytes are retained.
`source_span`	Document hash, page/element locator, bounding box, text/table representation, extraction run ID.
`observation`	Context ID, value or missingness tag, precision domain, source-span IDs, semantic-reading ID, review status.
`context`	Entity/perimeter, concept, basis, currency/unit, instant or interval, dimensions and taxonomy version. Contexts are immutable.
`assertion_version`	Logical assertion ID, payload hash, economic validity, ingestion time, public availability, superseded version and reason.
`model_node`	IR node ID, type, serialised payload, dependency IDs, package/model version; no executable arbitrary text.
`constraint`	Expression, scope, source/assumption references, supported solver fragment, applicable alternative condition.
`alternative`	Parent model, added conditions, rationale, evidence, coverage declaration and rejection history.
`economic_event`	Stable entity/book/branch/business-event key; versioned event, contract/cohort, dates, quantities and dependencies.
`branch`	Branch key and immutable definition reference, including entity, book and required synthetic or real book mode.
`branch_head`	Branch owner, current state reference and generation; updated only by the expected-head transaction.
`posting_authorization`	Exact basis, posting profile, method, authority, findings, permission and historical view. A permission record alone does not establish a committed event.
`journal_batch`	Committed balanced postings with event key, book/basis, date, method and measurement references. Candidate journals are separate; membership here requires the complete accounting transaction.
`economic_delta`	Committed contractual and physical state change, bound to the same basis and authorization as its journal.
`committed_receipt`	Unique stable event key, full request fingerprint, original request, opening/closing state, journal and delta references. Published atomically with those records and the advanced head.
`review_view`	Immutable parent view, decisions and dependency selection revisions; preserves historical interpretation.
`dependency`	Typed source and derived-record edges, including candidate-set membership needed for transitive invalidation.
`search_checkpoint`	Semantic manifest, partition, explored/unresolved regions, frontier, used budget and result references, committed together.

Relation	Key information and constraints
`run`	Evidence and model hashes, cutoff, scenarios, branch selection, weight model, solver/numerical settings, seeds and environment.
`report_cell`	Run, branch/scenario, context, amount or result family, dependency references and completeness status.

Persistent identifiers are opaque; display labels can change without changing identity. References use foreign keys or validated content-addressed links. The full logical key of a reported fact includes its dimensions, not merely company, label and year. Duplicate assertions are stored as distinct source claims until a documented duplicate-resolution operation relates them.

Version records are append-only. A derived transaction-time interval can be closed by the next version event without deleting the original payload. Validity intervals can overlap when assertions conflict. Such overlap is diagnosed and resolved by the evidence view, not prohibited by a database constraint that would erase one assertion. Scenario states use structural sharing of immutable historical nodes and branch-local changes.

Large simulation output is indexed by run, explanation, parameter draw, scenario, innovation path, entity, book, date and account. Reducing this to means across all dimensions would destroy the product's central feature. Online summaries may compress draws within a declared distribution, while retaining conditional summaries and representative replay identifiers. A retention policy that removes individual traces must state which explanations can still be reconstructed.

13.3 Public API and result types

The following unversioned names describe the user workflow. They are conceptual shorthand, not the portable operation registry. The exact versioned requests are specified below; Python, command-line and any service binding must exchange the same typed values:

```
ingest(manifest, extractor_config) -> EvidenceVersion
compile(source_or_ir, evidence_view, package_lock) -> ModelPlan | Diagnostics
reconcile(plan, cutoff, solver_config) -> HistoryFamily | Diagnostics
condition(family, assumptions) -> ConditionalFamily | Infeasible
simulate(family, scenario_set, run_config) -> RunResult
render(run_id, report_request) -> ReportBundle
explain(node_id, evidence_view_or_run_id) -> ExplanationGraph
compare(run_a, run_b, alignment_spec) -> Comparison
```

`HistoryFamily` contains a model-scope declaration, shared variables, constraints, finite choices, feasibility status and solver-completeness status. It may contain witness histories without claiming that those witnesses enumerate the family. `RunResult` is a tagged union of concrete outputs, conditional samples, certified bounds, explored ranges and unresolved outputs. Probability-bearing results reference a probability-model record. Results without that reference cannot be presented as probabilities.

Comparison requires an alignment specification: entity/perimeter, dates, reporting basis, scenarios and shared innovations where applicable. Comparing IFRS and US GAAP books holds economic events fixed and changes accounting rules. Comparing management responses holds the specified scenario and initial information fixed and changes the behavioural model. These are different questions and must not be combined into an unexplained difference.

13.4 Versioned requests and independent results

The workflow names above become portable operations only when their arguments, input versions and effects are fixed. Each request names an exact operation, profile and signature; an immutable view; typed arguments; execution settings; and the expected head for a mutation. The body is closed:

```
OperationRequest {
  operation, request_id, workspace_key,
  profile_ref, signature_ref, view_ref, arguments,
  execution: {mode, budget, environment_ref},
  expected_head: null | {owner_key, state_ref, generation}
}
```

All fields are present. The mode distinguishes a specification example from a runtime request. Budget states operation-count and elapsed-time limits, or a reason either is inapplicable. The expected head appears only in this envelope. For accounting its owner is the entity/book/branch; for review it is the review namespace; for probability selection it is the family/evidence/time owner. Resolving a mutable "current" alias happens before constructing the request, and a mutation compares the resolved revision again.

The fingerprint covers the complete request, including its request identifier, expected generation and execution settings. In accounting the lookup namespace is still the business-event tuple in Section 8.5, not request ID. Changing the request ID for a committed event changes the fingerprint and conflicts. Source and review mutations use their own operation/workspace/request namespace. A transport correlation identifier and an economic event are not interchangeable identities.

The result preserves the submission even when decoding fails. Once exact-byte retention succeeds, it has the following fields:

```
OperationResult {
  submission_ref, request_ref, operation, input_view_ref,
  disposition, statuses, payload, diagnostics, coverage,
  unresolved, continuation_ref, commit_observation,
  receipt_ref, evidence_mode
}
```

The validated request reference remains null until all required input checks, including bindings and authorization, pass. Rejected input has null payload. If submission retention itself fails, the caller receives a transport failure with its original correlation, not a fabricated stored submission or a known commit outcome. A read can inspect an earlier committed receipt while its own commit observation says no new commit occurred.

Completed, partial, rejected, unsupported, cancelled, internal failure, unknown commit and replay are operation dispositions. They do not replace the separate structural, mathematical, computational, empirical, accounting, forecast, product and human statuses. Computing a zero normalizer can complete while conditioning on that event is undefined. A valid symbolic family can have only partly explored regions. A supported insolvency scenario can be fully evaluated while its company defaults. These distinctions prevent an infrastructure error from erasing an economically important path.

For example, let an admitted law assign mass 3/5 to a path with cash 120 and 2/5 to an unavailable cash calculation. The output retains both regions and their masses. If separate evidence bounds the unavailable cash by $[80, 110]$, the unconditional mean is bounded by

$$\tfrac{3}{5}(120) + \tfrac{2}{5}[80, 110] = [104, 116].$$

Without those bounds, even this mean interval is unsupported. Reporting a mean of 120 would silently condition on computational success.

Portable-only validation proceeds through decoding, field shape, reference integrity, cross-record binding and time context, capability and admission, calculation, transaction comparison and publication. Native interfaces preserve their own explicit stage order; the native accounting sequence checks recognizable dependencies before shape and source versions before bindings. At the earliest failed stage, return every independently decidable diagnostic in canonical field order. A malformed parent prevents its dependent checks, but does not hide unrelated sibling errors. Later stages are marked unevaluated.

Appendix A gives the exact operation names and argument meanings. Financial method names such as delivery or claim settlement are versioned members of a package; they are not invented new platform endpoints. In particular, continuous inference and causal methods need their own admitted extension signatures before the finite/exogenous initial profile can execute them.

13.5 Correction changes a view without rewriting its history

A correction can affect a result that never explicitly selected the corrected source. Suppose a query selected one available report because it was the only eligible candidate. Admitting a second report may create a conflict. The query therefore depends on candidate-set membership as well as on its selected record. Invalidation must follow both dependencies transitively.

Each correction creates an immutable new view with parent and changed decisions. Old frozen outputs retain their original views. A current query must check ancestor dependencies even for descendants created after an invalidation scan; otherwise delayed computation could revive a stale reading. Unrelated closures remain usable. Self-supersession, incompatible replacement scope and a missing required predecessor reject before mutation.

The mathematical correction and the audit of affected work are separate records. A `CorrectionBasis` depends on the unaffected base law, replaced reading, corrected source, target admission and replacement method. An `InvalidationReport` additionally depends on the inspected store snapshot and lists affected descendants. Computing another optional descendant changes the report, not the corrected law. This also keeps the content-reference graph acyclic.

Probability calculation, admission and selection are three effects. A selected law belongs to the owner consisting of family, target evidence view, target time and purpose. Correcting a source does not mutate the old owner's selected pointer or automatically select the new candidate posterior. In the new owner, an inherited stale selection returns selection superseded with an explicit readmission and reselection obligation. With no inherited selection it returns selection required. A historical query naming the old view remains valid in that frozen scope.

13.6 Durability and migration have observable obligations

One logical transactional store owns records, dependency edges, review views, branch heads, journals, economic deltas, receipts and search checkpoints. Original documents reside in immutable byte storage. Their exact bytes must be durable and verified before a metadata transaction refers to them. A crash between blob creation and metadata commit may leave an unreferenced blob; it must not leave an accepted reference to absent bytes. Analytical tables, caches and PDFs derive from a named snapshot and do not become separate writable versions of truth.

The proposed local store serializes commits even when several clients submit requests. A single

writer still needs expected-head comparisons. An event-log implementation is also possible if its commit marker, projection visibility and replay obey the same observable rules. Neither a document trace nor a choice of SQLite establishes physical-media durability or multiwriter throughput. Those require crash, reopen and fault-injection evidence in the selected environment.

A search checkpoint commits the semantic manifest, partition identity, explored and unresolved regions, frontier, used budget and result references together. A cell is not complete before its result and frontier transition are durable. Resume names that exact checkpoint and manifest. Increasing a scheduling budget can continue it; changing evidence, methods, scenario, family or traversal meaning requires a new manifest and an explicit reuse relation. Cancellation preserves the last committed frontier.

Export first stages a closed manifest, verifies every required record and blob, then publishes the manifest atomically. An incomplete export remains explicitly incomplete. Import verifies required bytes before committing metadata and grants no new financial admission. A failed report export cannot undo a journal or justify posting it again.

A representation-only migration retains old and new bytes and IDs plus a field/meaning mapping. Its input names the old and new schemas, migration method, source snapshot and declared semantic relation; the common request envelope supplies the expected head for its transaction. It must preserve typed quantities, context, uncertainty category, reference closure, admission, event identity and supported reference query results. Changing a rounding policy, calendar, recognition rule or probability law is a new model, not a format migration. A failed comparison leaves old heads and objects usable.

Exact reruns compare values on the same evaluated domain. An elapsed-time cap can visit different search cells on different machines, so equal partial coverage cannot be presumed. Reproducible prefix comparisons need a pinned traversal and operation-count budget; resume uses the saved frontier. Approximate methods use their declared financial error comparison, and random streams name economic owner, date, shock kind and draw index independently of branch order.

13.7 Plugin interfaces

Every plugin declares a version, supported types, input information dates, determinism, numerical measure and capabilities. A contract plugin implements `schedule`, `observe`, `transition` and `payoff`; it may return future conditional events. A behaviour plugin implements `decide` and declares which state and scenario information it uses. A measurement plugin implements `measure` and returns an amount or vector with a basis and numerical status. A reporting plugin maps classified journals and balances to output contexts.

The plugin response must include all inputs that materially affect results. An external calculator's global evaluation date, market-curve cache, calendar version or random seed cannot remain invisible. For example, QuantLib's floating coupon interface makes fixing-date conventions and pricing objects relevant inputs; the adapter records them rather than relying on process-global state. ACTUS risk-factor observations receive explicit dates and identifiers.

The constraint interface accepts a typed formula and a requested operation: feasibility, witness, enumeration of declared finite choices, or bound. It returns `sat`, `unsat` or `unknown`, plus evidence appropriate to the operation. Timeouts and unsupported theories return `unknown`. The history layer exposes that solver outcome as `SolverIncomplete`, retains the affected region and marks its coverage unresolved. These are layer-specific descriptions of incomplete computation, not infeasibility. A witness proves existence within the formula, not completeness of a recovered historical family.

13.8 Human review and model-assisted extraction

The review interface should put the source passage beside the proposed observation, context and reconciliation impact. An analyst can accept a mapping, add an alternative, reject an extraction, define a symbolic unknown or request more evidence. These actions create structured changes with reasons. The useful unit of review is a coherent disclosure and its links, not an isolated number detached from its note.

A language model can identify candidate tables, draft concept mappings, propose contract terms and suggest alternative readings. Its output must reference source spans and a schema. Deterministic checks reject fabricated source references, invalid units, inconsistent periods and unbalanced postings. A model-generated interpretation that passes those checks remains a proposal until its evidentiary status is established. Token probabilities and self-reported confidence are never automatically used as weights over economic histories.

For routine local research use, a lightweight interface and immutable source versions are sufficient. Multi-user approval hierarchies, cryptographic attestations and deployment security systems are outside this initial design. They can be added for a concrete operational requirement without changing the scientific representation.

13.9 Failure recovery and reproducibility

Extraction jobs and simulation jobs write explicit progress records and atomic results. Restarting a failed document conversion reuses its document hash and creates either the same idempotent extraction result or a new version under changed settings. Restarting a simulation resumes only from a committed state whose model and input hashes match the run. Numerical failures retain the inputs needed for a focused diagnostic.

A run manifest records code revision when available, package lock, source hashes, cutoff rule, model hash, scenario data, priors or weights, seed/stream scheme, solver settings, decimal and rounding conventions, calendars, environment and result locations. Absence of a usable Git revision is recorded as unavailable with content hashes, rather than inventing a commit.

13.10 Review decisions have consequences for later results

A proposed reading starts in `Proposed`. Acceptance records a reviewer, date, reason, verified locator and context, together with the reconciliation result. Acceptance means that the stated layer has been reviewed: accepting an extraction of a company's sentence does not accept its accounting treatment. Competing defensible readings enter `AcceptedAlternative` under a choice group. Rejection preserves the source and reason. A correction creates a new reading; replacing an accepted reading requires an explicit `Supersede` action. A prior reading therefore remains reproducible without entering the latest model.

Suppose OCR reads Northstar's original 120 as 720. The source span supports 120; correcting extraction creates a new reading linked to the erroneous one. The later revenue correction from 120 to 110 is different: it changes the company's reported assertion. The original extraction remains faithful to its document. Finally, a reviewer may accept that two notes conflict without accepting either as a unique model constraint. The view then contains a conflict set requiring alternatives or further evidence.

A changed reading or mapping invalidates descendant constraints, families, measurements, reports and cached runs for a new view. Unaffected dependencies can be reused. Old frozen

runs retain their original decisions and evidence IDs; they are marked stale for current use, not overwritten. A journal derived under the old view is never silently attached to a new source. The local workflow needs no multi-user approval hierarchy to enforce these rules.

13.11 Identity must survive harmless reordering

The portable serialisation in Section 7.3 preserves decoded Unicode code points without global NFC normalisation. It uses sorted ASCII keys and exact rational financial quantities, while integer metadata remains integer. Only schema-declared sets are sorted. Source identity hashes original bytes. The old Core 0 encoding survives only for its historical fixtures through an explicit legacy import; it does not define new portable digests. Neither encoding proves arbitrary economic equivalence.

SHA-256 with a domain/version prefix identifies three different closures. Evidence identity includes source bytes, locations, extraction and review versions. Model identity includes typed IR, assumptions, rules, calculator locks and resolution, while excluding formatting and source byte offsets. Run identity adds the evidence view, horizon, scenarios, outputs, calendars, numerical policy, random-stream scheme and environment. A corrected source can change evidence and run identity without changing the numerical model; a new rate assumption changes the model as well.

Each economic business event has a stable identity within its entity, book and branch. Its lookup key is the four-tuple in Section 8.5; event version, payload, expected head and request settings belong to the full fingerprint. Exact replay returns the original receipt before a new-event freshness check. Changed inputs under that key conflict. A new run label or version cannot make the event post again. Independent events may reorder only when read/write sets are disjoint or a package proves commutation, including funding constraints. Alphabetical ordering cannot settle an economic timing ambiguity.

Stochastic keys contain seed, coupling group, entity, contract/cohort, economic date, shock kind and draw index. Branch enumeration order is absent. Shared coupling groups may use the same uniform draw with different marginal transforms. The reference fixture maps the first 64 bits of a SHA-256 key digest, interpreted as unsigned big-endian integer v, to $(v + 1/2)/2^{64}$. This fixes example replay; it does not validate a production Monte Carlo generator. Research runs select an assessed counter-based generator and record transforms and numerical environment. Inner measurement draws have different shock domains from outer realised defaults.

Schema migration preserves old and new payloads and a field/meaning mapping. A representation-only migration must demonstrate equivalent supported semantics; a changed recognition rule creates a new model. Restart resumes only a committed state whose locks and identities match. SQLite owns source, review, accounting and run metadata transactions; content-addressed files hold immutable bytes and JSON, while DuckDB reads versioned analytical exports. Each mutable transition has one owner.

13.12 Adapters exchange typed records

The portable interfaces described in this chapter govern new exchanges. The original `spec/core-contract.md` and `spec/core-ir.schema.json` retain the historical Core 0 examples. A named decoder and comparison link those examples to the portable records; their older fields do not silently extend a new closed signature. Every method returns a success record or a diagnostic with code, source IDs, entity/period, dependency path and retry status. The principal sequence

is `ingest`, `review`, `compile`, `reconstruct`, `simulate`, `report`, with `bound`, `explain` and `replay` over frozen identities.

Arelle supplies candidate facts with QName, entity, context, period, dimensions, unit, accuracy, nil status and locator. Duplicates survive for review. Its archived `ModelInstanceObject` interface exposes these properties; validation success does not establish economic truth. Docling supplies page/table proposals, locations and conversion status. Missing headers and failed pages remain unresolved. Z3 receives named rational-linear assertions and finite choices; witnesses are checked against exact inputs and unknown results retain their reasons. The reference checks exercise Z3; the other selected adapters remain interface designs grounded in inspected official source snapshots.

ACTUS receives contract terms, calendars and dated risk factors and returns schedules, transitions and payoffs. AccDSL applies accounting separately. QuantLib receives an explicit evaluation date, instrument, curves, fixings, calendar/day-count and measure; it returns dated measurements with numerical status. No process-global valuation date may leak between branches. Beancount exports concrete journals for later differential checks; it does not own symbolic families. Import failures retain request ID and source hash so retries are idempotent. Licence/version evidence remains in the software ledger; integration acceptance requires pinned releases or source digests.

13.13 A successful extraction can still lose the fact

Consider an Inline XBRL amount whose literal is `1,234`, whose admitted numeric transform treats the comma as grouping, whose scale is -3, and whose context and unit identify a particular dollar observation. The resulting amount is $1234 \times 10^{-3} = 1.234$. Scaling that normalized value again gives 0.001234 and is wrong. An adapter must retain the literal, transform, scale, context, unit, nil status and original occurrence so that this error can be found.

The same principle applies to dates and dimensions. A lexical instant of 31 December remains distinct from the exclusive 1 January boundary used for an interval. Publication and ingestion are separate timestamps. Two equal numbers with different contexts remain separate occurrences. A nil fact is unknown, not zero. Dropping a typed dimension or context reference yields a preservation failure for that fact even if the resulting Markdown looks clear.

A PDF can preserve characters that the reader never sees. The retained HSBC 2023 annual report has a cover reading “Annual Report and Accounts 2023”, but both PyMuPDF and Poppler also return 2024 in the extracted cover text (HSBC Holdings plc, 2023, cover, physical PDF p. 1). Inspection of the drawing order explains the discrepancy: the earlier 2024 text is covered by an opaque dark rectangle, after which the 2023 text is painted. This is an observed difference between extracted characters and the visible title. It does not establish an accounting contradiction or explain why the file was constructed that way. Agreement between two text extractors would not resolve it.

The proposed exchange keeps the original PDF, the unchanged raw extraction, the page locator and a render from the same bytes. The annotation’s `literal` retains both years; separate `readings` record the raw extraction and the proposed visible title. A linked inspection record identifies the extractor versions and options, rendered image, relevant drawing order and reviewer. Thus a raw-text search still finds 2024, while the inspected cover supports a visible report-year reading of 2023. Rejecting 2024 as the visible report year must not delete its extracted occurrence. The current example has author inspection; independent adjudication remains pending.

If an earlier review had already accepted 2024 as the report year, the correction would require a new reading and an explicit `Supersede` decision. Results depending on that reading would need revalidation in the new evidence view; frozen earlier results would retain their original evidence. This follows the review rules already stated above. Even after the visible title is resolved, the cover year supplies no publication or availability timestamp for these particular bytes. An as-of query must therefore retain the unresolved availability condition. This single-page case motivates preservation and review behavior; it supplies no extraction-accuracy ranking or general visibility test for other pages.

The inspected Arelle interfaces support native XBRL contexts, dimensions, transformations and validation. Its internal object IDs do not replace stable source identity. The inspected Docling source offers useful layout and table proposals, but its examined XBRL graph path does not retain every required fact/context/unit/typed-dimension field. That path needs a preservation sidecar or must report unsupported conversion for affected records. These are source audit findings, not measured accuracy rankings of the tools.

A returned layout conversion can also conceal missing pages. At the pinned Docling implementation, the timeout branch assigns `PARTIAL_SUCCESS` when pages were expected, even if none finished. The public `convert_all` interface permits that status through `raises_on_error=True` if the remaining stages return normally. Thus absence of an exception cannot establish that a three-page request produced any extracted content. The assembly code can restore missing page numbers with empty page objects; counting those objects does not count successful processing (Docling contributors, 2026). These are source-derived failure cases, not observed failures on Apple's filing.

The proposed exchange therefore keeps the original page inventory, requested range, successfully processed pages, empty placeholders and structured errors separate. If only page two was requested and processed, that is success on the declared subset; pages one and three remain outside the conversion claim. If the three-page request times out before any page finishes, all three remain unprocessed despite the partial-success status. Retain the source and the unfinished pages for another attempt. A deadline is also distinct from stopped execution: the inspected shutdown code waits for worker threads for a bounded time and can report that a thread is still alive. Later execution must confirm termination within its isolated process before treating the attempt as stopped (Docling contributors, 2026).

A requested processing stage may also remain unperformed. In the same Docling source, automatic OCR selects among engines according to platform, installed packages, languages and runtime availability. If it selects no engine and has recorded no language-specific rejection, it logs a warning; the subsequent call returns the pages without OCR. Language-specific rejection can instead produce an exception. Thus `do_ocr=True` records a request, not evidence that text recognition occurred. The proposed adapter records the requested mode, selected engine and model identity, and processing outcome for each page. A page that requires OCR but did not receive it remains unprocessed, even if other stages produced layout regions (Docling contributors, 2026).

Identifying the engine still leaves its model to be fixed. The inspected default layout preset names Heron at revision `main`, while the TableFormer downloader names the model repository's `v2.3.0` tag. Neither name supplies the hashes of the weights and preprocessing files actually used. The proposed adapter binds those files, configurations and dependency versions before use. Disabling remote inference services also does not disable model acquisition: the layout helper downloads when no artifact directory is supplied. With a supplied directory it requires the model's subfolder, but finding that folder does not check its contents against the requested revision. Local file verification and later isolated execution must therefore establish the intended offline behavior; a source setting alone cannot do so (Docling contributors, 2026).

The retained Apple annual filing makes the dependency problem concrete. Its issuer archive includes the instance, the issuer's schema and its calculation, definition, label and presentation linkbases. Those files also refer to US GAAP and SEC Reporting Taxonomy definitions, SEC schemas and XBRL definitions held elsewhere. The issuer archive alone therefore cannot supply every definition needed to interpret its tagged facts. For this particular archive, a static inspection now resolves the recorded imports, link references and typed-domain references to retained files, checks target namespaces and fragment identifiers, and checks that instance facts have declared concepts, contexts and units. This establishes a source dependency map. XML Schema and XBRL validation, the issuer's interpretation of each concept, and the completeness of the wider disclosure package remain separate questions.

An offline parser needs more than a map of intended filenames. At the inspected Arelle source version, the cache lookup can return a local pathname when the file is absent; subsequent loading records an `IOerror`. The proposed adapter must check each mapped file and its hash before loading, retain the missing dependency and affected records, and return an unresolved-dependency result. It must not infer successful retrieval from a nonempty pathname or silently substitute a newer taxonomy. Original schemas remain unchanged; an external resolver or isolated cache associates their declared addresses with the retained bytes. Any built-in cache fallback must belong to the same checked source set (Arelle contributors, 2026).

Nor is a successful API return necessarily a validation verdict. In the same source version, the controller's return value takes accumulated validation errors into account only when `validationExitCode` is enabled. The proposed adapter sets it explicitly, requests validation, disables automatic formula execution, and examines the returned status, diagnostic messages and model errors together. It keeps the model open while copying the necessary facts, contexts, units and original occurrence references, then closes the session. One session runs at a time in its owning thread within a process; the inspected API uses shared package and plugin state. These are source-grounded interface requirements for a later implementation, not results from an executed Arelle adapter (Arelle contributors, 2026).

Each adapter therefore pins code, dependency, model and taxonomy versions, licenses, input and output schemas, transformations and field-loss behavior. Preparation preserves bytes and receipts before parsing, inventories the package, attaches occurrences and contexts, records normalization, checks required fields independently and submits candidates for review. Partial conversion retains both successful units and the exact failures or unexplored frontier. Conversion counts cannot accept a package or an accounting treatment.

The proposed extraction comparison uses the same retained files, roots, budgets and taxonomy resources. Manual located source units and transparent native extraction provide the reference; applicable pdfplumber/Arelle configurations, plain Docling and explicit context/footnote repairs form the candidate ladder. Transcription, context, table structure, mapping, source support, retention, historical admission and cost are separate outcomes. Lost hashes or locators, look-ahead, silent omissions and untracked downloads veto the affected exchange. The current 31 development annotations cannot establish a heldout accuracy ranking, and cell counts within one report are not independent observations.

Reuse also has distribution conditions. Arelle's Apache and Docling's MIT code licenses do not by themselves clear every model, taxonomy or dataset. ACTUS technical-specification reuse retains its share-alike obligations; QuantLib, Z3 and ledger tools retain their own code and dependency terms. The Beancount and hledger comparison needs an explicit distribution boundary for copyleft components. License admission records actual versions, notices, models and data, rather than treating a repository's headline license as universal.

EasyOCR makes the dependency question concrete. Version 1.7.2 requires `python-bidi` without a version constraint and imports its `get_display` function when the OCR module loads. Although the call reorders Arabic recognition results, selecting English does not remove the import or

package requirement. In the inspected `python-bidi` 0.6.11 source, this import selects a Rust-backed implementation; the older Python implementation has a different import path. Its own source states LGPLv3-or-later terms. The inspected release is a candidate for dependency analysis, and its source-level call signature alone does not establish compatibility with an installed OCR configuration (Jaided AI, 2024; Kriheli and contributors, 2026).

The same package shows why an accompanying notice still needs checking. Its third-party list names older PyO3 and `unicode-bidi` versions than its bundled Cargo lockfile. The exact locked sources retain MIT-or-Apache terms, so the version discrepancy does not demonstrate a licence conflict. Other locked packages add conditions: `target-lexicon` includes the LLVM exception to Apache 2.0, while generated `unicode-ident` tables require Unicode data-file terms alongside the chosen MIT or Apache licence (Kriheli and contributors, 2026; PyO3, Servo, Rust and dependency contributors, 2026). The adapter must therefore bind the selected build and retain each applicable notice. Verified source archives help prepare that decision; actual wheel contents, model rights and the intended distribution still need separate assessment.

The actual taxonomy notices show why that distinction changes the design. The inspected FASB notices permit specified financial reporting and unchanged incorporation in works that explain or assist implementation, subject to their notice and third-party conditions. They do not provide a general permission to edit taxonomy definitions. Keeping the original files and recording location mappings separately respects that distinction. The notices retrieved for this inspection are unversioned; their relationship to the terms attached to the 2024 files still needs to be established before a distribution decision. XBRL's public-document notice separately directs software and schemas to a Software Notice, so permission for an explanatory document cannot simply be applied to a schema bundle. Exact dependency licences, component-specific terms and the intended distribution remain explicit preparation tasks (Financial Accounting Foundation, 2026b,a; XBRL International, 2026).

Chapter 14

Acceptance and implementation sequence

The reference cases now make several of the acceptance obligations executable as design checks. Northstar provides full statements and a known compatible history; Apple supplies real source contexts and a bounded marginal stress; bank, insurance, consolidation and inventory fixtures test the core's broader state requirements. They do not constitute a production implementation.

14.1 Three different questions require different evidence

Engineering correctness asks whether the implementation follows the specified types, transitions and arithmetic. Accounting validity asks whether the selected rules faithfully implement the applicable standards and disclosed policies. Scientific validity asks whether the economic assumptions and uncertainty representation support the intended forecasts or comparisons. A balanced journal answers only part of the first question.

The implementation must therefore maintain separate validation results. Compilation and deterministic examples cannot establish forecasting quality. Agreement with a public report cannot identify an undisclosed contract term. A lower extraction error on a document benchmark cannot establish that the engine preserved every materially different scenario response.

14.2 Core conformance cases

Test	Required case	Pass condition
T01	Exact debit/credit batches, including reversals and period closing	Every committed book conserves signed balances; imbalance rejected atomically.
T02	Money, quantities, currencies, instants and interval flows	Invalid additions and context combinations rejected; explicit conversions traced.
T03	Original filing, later correction and as-known cutoff	Earlier query reproduces the original evidence view; later view shows revision.

Test	Required case	Pass condition
T04	Annual and interim cumulative flows with revised comparatives	Quarter derivation only across compatible contexts and vintages.
T05	Fixed/floating borrowing with identical historical postings	Both explanations survive; next-year expense is 5 and 7 under the stated shock.
T06	Continuous fixed share $\alpha \in [0, 1]$	Exact model returns $I = 7 - 2\alpha$ and extrema 5, 7 without assigning probabilities.
T07	Duplicated encoding of one history	Canonical duplicates preserve results and declared weights; no extra probability mass from an additional encoding.
T08	Linked interpretation across debt classification and payment treatment	Impossible combinations rejected through shared constraints.
T09	Rounded totals, duplicate facts and contradictory notes	Correct interval compatibility; no automatic zero or balancing plug.
T10	Infeasible or unsupported constraint problem	Conflicting sources exposed; timeout/unsupported case reported as unknown.
T11	Zero-coupon bond and ordinary equity examples	Carrying amounts, payments, share quantities and report mappings match derivations.
T12	Foreign-currency asset, translation and cash movement	No mixed-unit balance test; translation effects separated from cash flows.
T13	Acquisition and intragroup balance elimination	Perimeter and adjustment book reproduce group totals without inventing capex.
T14	Conflicting recognition rules and unknown guards	Explicit conflict or symbolic partition; file order has no effect.
T15	Insufficient cash with constrained funding	Funding failure/default retained as an outcome; no hidden borrowing.
T16	Expected-loss measurement before future default is realised	Measurement uses only information allowed at that date.
T17	Repeated/reordered independent events and interrupted execution	Idempotent replay; branch isolation; identical committed outputs.

Test	Required case	Pass condition
T18	Conditional distributions and optional weighted mixture	Branch distributions retained; mixture agrees with declared weights and total probability; unweighted family never labelled probabilistic.

For synthetic cases, expected histories, excluded histories and scenario consequences are known by construction. Tests should include counterexamples that balance but have the wrong economic effect, such as treating a non-cash acquisition movement as operating cash. Such tests discriminate between arithmetic validity and the intended semantics.

Differential tests against an existing ledger check only the shared supported subset. ACTUS schedule comparisons check contractual events, and QuantLib comparisons check selected valuation calculations under matched conventions. Disagreements must be classified as different conventions, unsupported cases or implementation errors before one tool is treated as an oracle.

14.3 Extraction and reconstruction evaluation

A company corpus must include different standards, industries, languages, filing formats, extensions, interim calendars and restatements. Company and time splits are required so repeated report templates or revised versions do not leak into evaluation. Public benchmark tables are useful development material, while acceptance uses complete disclosure packages that include notes and their relationships.

Reviewers annotate source facts, contexts, mappings, required relationships, known ambiguities and unsupported assertions. Relevant measures include correct source-linked values and contexts, unresolved extraction volume, coverage of note obligations, reconciliation failures and invented facts. Report each separately. A single overall score can hide a serious error in a small but consequential financing note.

Historical reconstruction should test whether the known synthetic history remains feasible, whether known incompatible histories are rejected, and whether economically distinct alternatives are preserved. On real companies, public data generally cannot establish the true journal. Review instead assesses evidential compatibility, unsupported commitments and conditional economic consequences. Later disclosures may provide discriminating evidence for previously ambiguous terms.

No universal numerical threshold for semantic coverage or extraction accuracy is fixed in this document. Before a release evaluation, an accounting-reviewed test corpus and task-specific acceptance thresholds must be recorded in an evidence contract. Hard conformance failures, fabricated citations, silent discarding of valid alternatives and undocumented accounting adjustments veto acceptance regardless of an aggregate score.

14.4 Evaluate the DSL against the strongest practical baseline

The comparison uses the same IR, accounting examples and tasks. One group of models is authored through the typed Python API; another through the DSL. Both receive equivalent documentation and examples. Tasks include encoding a movement schedule, adding an alternative

interpretation, changing an effective date and tracing a simulated amount. Measure correctness, material omissions, time to a reviewed result, ease of explaining a change and maintenance effort.

This comparison tests whether the notation and compiler help users; it is not a comparison between careful DSL code and unstructured spreadsheets. If the DSL offers little advantage, retain the shared IR and library and narrow the textual language. Small pilots nominate improvements. A claim of superiority requires a predeclared comparison and uncertainty evidence, not a few favourable demonstrations.

14.5 Forecasting evaluation comes later

The proposed experiment asks whether a conditional company method adds out-of-time predictive information while preserving statement coherence and unresolved exposure. The comparator ladder uses the same available sources and targets: last-observation and seasonal naive forecasts; contractual runoff and accounting-identity forecasts; tuned classical statistical and conventional financial models; the structural model with one selected interpretation; the same structural model preserving alternatives; and declared extensions. This separates changes in economics from changes in the representation of ambiguity. Tuning stays within training windows. Test issuers or time blocks remain untouched, and unavailable predictions stay in the reported denominator.

For probabilities, predeclare proper scores, calibration, interval coverage, sharpness and dependence-sensitive joint checks. Unweighted feasible sets have coverage and width under a stated set interpretation, not probability intervals. Promotion needs a declared score difference with uncertainty evidence on matched eligible targets and no accounting or information-time veto. Issuer and time dependence require a justified paired resampling or equivalent inference design; overlapping reports are not independent replications. Runtime, a few losses and small-seed tail differences remain descriptive without that support. No such forecast experiment has been run for this proposal.

Look-ahead, broken conservation, a changed target or missing scope can invalidate an experiment. A valid comparison in which the proposed predictor performs poorly instead rejects that candidate and motivates its stated repair. The two outcomes must not be conflated when choosing the next experiment.

Forecast evaluation uses historical information cutoffs, fixed data vintages, declared economic scenarios and proper separation of training and testing periods. Every level of the comparator ladder above uses that same information boundary, so extra information cannot silently account for an apparent benefit.

Construct the inexpensive comparators before choosing a complex forecasting method. A no-growth operating forecast with scheduled contractual financing tests whether the model adds value beyond existing obligations. A constant margin and turnover model tests whether detailed operations improve on familiar financial ratios. A trailing-average growth model tests whether fitted dynamics improve on recent experience. A fixed-exposure contractual stress model tests whether estimated management responses help beyond repricing the existing book. Each comparator must generate coherent accounts through the same reporting rules and use the same historical information. In synthetic cases, the known hidden state also supplies an oracle conditional on the stipulated scenario.

Evaluate these comparisons separately under stable conditions, large rate resets, refinancing pressure, credit deterioration and changes in reporting perimeter. A pooled average can conceal failure in the situation the product is intended to explain. Supported underperformance against

a cheap comparator in a declared critical situation vetoes promotion of the complex forecasting method and must lead the result discussion. These checks are falsification tests, not training targets or evidence of superiority when merely passed.

Evaluate point forecasts where they are a stated target, distributional calibration when probabilities are justified, and decision-relevant robustness when only feasible sets are available. A wide feasible set is not automatically a good prediction, and an unweighted set cannot be assessed as if it were a 95% predictive interval. In multi-scenario stress tests, scenario severity and probability remain separate inputs.

Any serious numerical comparison must follow the repository's research plan, assumption audit, baseline, veto, uncertainty and result-record policies. The present document runs only deterministic checks of its own examples. It neither selects a scientific forecasting default nor reports comparative performance.

14.6 Capability contracts for sector packages

Package	Required state and operations	Minimum acceptance evidence
Core corporate	Working-capital bridges, lots/assets, ordinary debt and equity, period closing and cash-flow mappings	Exact synthetic cases, reviewed public filing reconstruction, all statement reconciliations.
Financial instruments	Terms, calendars, resets, options, measurement categories, modifications and hedges	Contract schedule comparisons, valuation checks, paragraph-based posting cases and transition cases.
Banking	Credit cohorts, deposits, collateral, liquidity, regulatory capital, management actions	Allowance roll-forwards, repricing and runoff stresses, capital reconciliation and default/funding-failure cases.
Insurance	Policy groups, future cash flows, service and finance components, reinsurance and asset interaction	Cohort movement reconciliation, assumption change cases, service recognition and reviewed actuarial calculations.
Multi-basis reporting	Versioned IFRS/US GAAP/ASBE rules, entity elections and transition treatment	Matched economic-event cases with explained basis differences and independent accounting review.

Every package declares capabilities at operation level. "Supports bonds" is too vague; "supports fixed-rate, single-currency, non-callable amortised-cost issuance and redemption under this reviewed rule set" is implementable and testable. Unimplemented operations retain their source and contract representation while producing an explicit capability error when execution requires them.

14.7 Implementation milestones and dependency order

Stage	Concrete implementation deliverable	Exit evidence
P0	Portable records and contextual types; source/review/time selection; typed constructors and versioned IR; separate posting authorization and atomic journal/delta/state/receipt transactions for the running cases	T01–T04, T09 and T17; source inspection and exact receipt replay.
P1	Admitted history templates, finite choices, shared symbols, linear solver adapter and historical bridges; conditional simulation and report explanations	T05–T10, T14 and T18; synthetic family preserved.
P2	Matched debt text frontend with identical portable IR; connected corporate methods through the typed authoring API; document/XBRL import adapters	Parser/API equivalence; T11–T13; reviewed three-date public-company case.
P3	Contract and valuation adapters, dated standard packages, richer financing and tax/funding policies	Contract/valuation differential cases; accounting review; T15 and transition cases.
P4	Bank package, allowance models, regulatory reconciliation and management response alternatives	T16; banking-specific reviewed histories and stresses.
P5	Insurance cohort package, reinsurance and actuarial measurement adapters	Reviewed cohort roll-forwards, information-date and service/finance tests.
P6	Optional probabilistic inference, model-weight sensitivity and broader scenario research	Explicit targets, sampler/numerical validity, prior sensitivity and downstream evaluation.

These stages specify dependency order, not dates or staffing estimates. Exit checks are cumulative: a case can recur when a later stage changes the inputs or execution path that it tests. Banking and insurance representation requirements influence P0 types, although their measurement engines arrive later. Probability weighting is optional; P1 already produces useful conditional cases and bounds. A failure of a chosen extractor or inference algorithm triggers a local replacement or repair, rather than abandoning the product's representation requirements. The parser/API comparison in P2 covers the matched debt fragment. Complete textual authoring of the corporate profile is a further frontend task; its absence does not remove corporate declarations from the typed API.

14.8 Decisions and replaceable assumptions

The requirements fixed by this design are preservation of evidence and alternatives, explicit time and basis, balanced and admissible posting, economic state beyond journals, and distinct set/probability semantics. Replaceable choices include the storage engines, parser generator, solver, numerical libraries, aggregation resolution and user interface. Every scientifically material aggregation, behavioural parameter, prior or truncation rule must record provenance, purpose,

failure mode and a diagnostic before it becomes a default.

The immediate work is to finish and review the governing proposal, including its real-source, authority, domain and research obligations. Product implementation follows completion of that work and explicit implementation authorization. In the table above, P0 establishes portable records, review/time selection, durable receipts and typed authoring; P1 builds admitted histories; P2 connects the corporate methods and matched textual frontend. These stages are a design sequence, not an already running product campaign.

14.9 Concrete conformance and corpus protocol

The fixture directory contains independently entered inputs and expected outputs for Northstar, rounded borrowing, evidence views and sector cases. `check_repair_conformance.py` recomputes exact journals and statements, constructs the joint linear bridges in Z3, checks strict bounds, and exercises replay, atomic rejection, finite Bayes updating and funding failure. `check_core_contract.py` checks grammar/schema examples, canonical identities, stable streams, information restrictions and source inventory. These scripts are reference oracles for portions of the design. They do not test an unimplemented full DSL compiler, all adapters or standards compliance.

Every conformance case must specify input files, the operation requested, expected values or typed errors, comparison convention and supported scope. Exact money, finite probabilities, linear witnesses and example journals compare as rationals. Approximate valuations require an error bound or a justified tolerance tied to the valuation scale. A missing expected result cannot become a passing test simply because the operation returned something. An implementation release must pass all applicable positive and negative fixtures and preserve unresolved operations; the arithmetic oracle alone does not release it.

The retained panel contains all 192 target annual/interim report slots for 2018–2025 across nine issuers and three evidenced accounting families, with eight consecutive reporting years per issuer. Here "complete" applies to the source-acquisition panel: every declared slot has a retained report version. It does not mean that each selected document contains every related filing section. For example, the selected JPMorgan shareholder annual reports for 2022–2025 explicitly omit Form 10-K pages (respectively 1–42 and 304, 1–44 and 322, 1–48 and 334, and 1–42 and 328). The complete issuer-hosted 10-K candidates and their bounded semantic annotations are preserved as a separate source layer; they do not retroactively change the selected panel or establish full package closure.

Apple, Caterpillar and JPMorgan Chase each contribute one annual and three quarterly reports per year; the other six issuers each contribute one annual and one half-year report. Thus $3 \times 8 \times 4 + 6 \times 8 \times 2 = 192$. This is source acquisition, not semantic closure or a representative random sample. The 244 retained original versions include historical alternatives to the active panel. Apple remains a detailed development package, with Caterpillar supplying a contrasting inventory policy; bank, insurer, IFRS and ASBE reports are also retained.

Publication dates remain unverified for 136 of the 192 accepted reports; 56 have filing-date metadata. A historical forecasting population needs the actual byte-availability evidence described in Section 3.12, not imputed publication dates. The 31 provisional annotations across four reports are exposed development material, not heldout answers. Policies, note links, incorporated documents, changed comparatives and adjudicated meanings still require task-specific preparation. Synthetic revisions provide controlled cases without substituting for a required real conflict. The corpus protocol and active panel preserve their distinct populations and versions.

Annotators first record literal/context readings without seeing the model's predictions, then mark source conflicts, mapping relationships, narrative qualifications and admissible alternatives. A second accounting-aware reviewer adjudicates material disagreements while preserving both initial annotations. Reports from the same issuer family and duplicated tables cannot be split between development and heldout evaluation. Forecast tests freeze the actual publication cutoff; later revisions are separately labelled outcomes and may not enter predictor inputs. Extractor upgrades rerun the same frozen corpus.

For the initial supported release, loss of a source span, wrong entity/unit/ period/basis, invented evidence, hidden financing, dropped feasible branch, incorrect cash reconciliation or unreported unsupported treatment blocks acceptance. All selected critical facts and operations must have exact reviewed answers or the expected unresolved response. These zero-tolerance conditions follow from the product's meaning-preservation requirements. Non-critical extraction quality is reported by stratum and error category; no arbitrary overall accuracy threshold certifies whole-report understanding. An extraction benchmark needs its own sample size and uncertainty protocol before making statistical accuracy claims.

The DSL and typed-library comparison uses identical packages, evidence, schemas and runtime. Users perform the same tasks: correct a context, add a shared uncertainty, represent a restatement, add a supported event and explain a forecast difference. Task order is counterbalanced and experience recorded. Both frontends receive the same validation and error messages. Measure successful completion, material semantic errors, time and maintenance effort with uncertainty appropriate to the participants; report no winner from the current syntax checks. A textual language earns its maintenance cost only if that comparison supports its use.

14.10 Workload limits and release evidence

The first integration workload is one local company package, up to 10,000 observations, 100 declared aggregate event slots per period, 64 materialised finite-choice cells and a 30-second budget per solver operation. These are provisional engineering limits for bounded failure behaviour, not measured capacity or scientific defaults. They must be configurable and recorded. Reaching a limit yields partial coverage with the unvisited formula/cells, never an exhaustive-history claim. Peak memory and latency are measured only after an implementation exists; a slower but semantically correct result does not become a scientific failure.

Engineering release needs compiler/API equivalence, transactional replay and adapter subset checks. Accounting release additionally needs paragraph-level authority, dated applicability, exact fixtures and independent domain review for every promoted operation. Forecast promotion additionally needs preserved information sets, the stated baseline ladder, statistical uncertainty and critical-regime vetoes. Human comprehension review remains separate from all three. This version supplies a richer design candidate and reference checks while preserving these outstanding acceptance decisions.

Appendix A

Portable operation reference

These proposed versioned operations share the request and result envelope in Section 13.4. Required inputs are explicit even when a list is empty. An expected head mentioned below refers to the common envelope, never a second argument. The registry is a lookup reference after the financial and transaction meanings have been explained in the body. These are specified exchanges, not deployed endpoints.

A.1 source.prepare@1

package manifest, source snapshot, adapter/config lock, prior review, traversal budget and optional continuation → ExtractionPreparation

Acquires/preserves sources and constructs candidates under the source-preparation method; no semantic admission

A.2 review.propose@1

target Ref, layer, decision kind, evidence Refs, actor, rationale, replacement Ref or null → ReviewProposal

Candidate construction; accepts no layer

A.3 review.commit@1

ReviewProposal, expected review view/revision → immutable ReviewDecision and new view

Review transaction under the review and storage rules; admissible action/layer required

A.4 time.select@1

package/view, investor/system/exploratory mode, cutoffs, revision and unknown-time policy → selected assertions, conflicts and temporal obligations

Pure selection under the time-admission rules; no invented ingestion or publication time

A.5 model.compile@1

the versioned CompileRequest → candidate Module, source map, diagnostics and target admission/capability report

Candidate construction; no solver or posting

A.6 history.construct@1

admitted Module, opening, scope, template registry, query/horizon, resource budget → HistoryFamily, partition, witnesses and unresolved formulas

History-construction contract; witnesses are examples within the family, not its definition

A.7 history.refine@1

frozen family, additional admitted constraint or refinement, dependency view, budget → child family and inclusion relation

Candidate construction; record whether correction replaces constraints instead of intersecting them

A.8 history.query@1

family, typed joint observable, bound/feasibility request, capability manifest → symbolic/partial/exact answer, coverage and endpoint attainability

Pure; unsupported expressions remain explicit

A.9 uncertainty.evaluate@1

exact native uncertainty request, law/attachment/admission and operation-specific event or observation block → native UncertaintyResult and candidate derived bundle

Native admission/attachment/chaining rules; never selects its result automatically

A.10 uncertainty.admit@1

candidate bundle, target time/evidence/purpose, actor, rationale, requested assumption scope, expected selection revision → admitted attachment and decision

Admission transaction; author-stipulated measures keep that label

A.11 uncertainty.select@1

admitted attachment/receipt, rationale and expected selection revision → new selected-view revision

Selection transaction; no probability recomputation

A.12 scenario.project@1

family, exogenous scenario, policy/method lock, horizon/calendar, explicit information policy, resource budget → conditional event/journal/state proposals by history and scenario

Candidate construction; budgets preserve unresolved regions; no historical-book commit

A.13 accounting.assess@1

native assessment request, event, opening, context/time, authority/method, evidence/assumptions and admissions → TreatmentAssessment

Pure accounting assessment; inherited `assess_only` profile cannot post

A.14 accounting.authorize@1

AccountingEventBasis and posting-profile Ref → PostingAuthorization

Pure evaluation under a separately supported posting profile; an assess-only result cannot grant posting permission

A.15 accounting.propose@1

PostingAuthorization, AccountingEventBasis and posting/rounding method → JournalProposal and independent EconomicDelta

Candidate construction; revalidate bindings; unsupported method gives no journal

A.16 accounting.commit@1

AccountingCommitArguments/v1 and envelope expected branch head/generation → StateTransitionReceipt

Single atomic transaction under the event-commit contract; the separate commit schema, separate from native assess-only

A.17 report.build@1

frozen branch/family result, statement-map version, interval/view, typed requested statements and presentation policy → joint statement family, reconciliation and coverage

Pure; derive from retained state/journals, no balancing plug or hidden branch selection

A.18 record.explain@1

output Ref, its frozen view, requested lineage depth → source/method/assumption/decision paths and unresolved dependencies

Pure; depth truncation is explicit; cannot substitute today's view

A.19 work.resume@1

checkpoint Ref, original semantic manifest, scheduling budget → operation-specific resumed result

Resume the original operation; checkpoint ownership/atomicity under the durability contract; no model change hidden as resume

A.20 store.export@1

snapshot/root Refs, format/decoder, destination → closed manifest or explicit incomplete export

Publish a verified export manifest atomically under Section 13.6; no domain mutation

A.21 store.import@1

closed manifest, exact blobs/records, decoder and target namespace → import receipt and original-to-local mappings

Verify before metadata transaction; import alone grants no financial admission

A.22 store.migrate@1

migration request under Section 13.6 → preserved originals, new representation, comparison/mapping and receipt

Transaction only after required semantic comparisons; no changed-model-as-format claim

A.23 Closed accounting records

The following field lists are exact for the named versions. Every reference uses the portable kind/schema/digest type; plural references are schema-declared sets. Unknown fields reject. Source and admission sets can be empty only when the method's admitted assumption scope explicitly permits and explains that.

```
AccountingEventBasis/v1 {
  identity: {entity_id, book_id, branch_key, business_event_id},
  event_ref, opening_ref, context_ref, branch_ref, time_ref,
  chart_ref, authority_refs, method_ref, evidence_refs,
  admission_refs, assumptions_ref, view_ref
}
PostingAuthorization/v1 {
  basis_ref, posting_profile_ref, method_ref, authority_refs,
  status, findings, posting_permitted, evidence_level, view_ref
}
AccountingCommitArguments/v1 {
  basis_ref, authorization_ref, journal_ref, delta_ref
}
```

Identity fields are nonempty ASCII names. Each singular reference is nonnull. Authority references identify applicable support or a specifically typed synthetic authority for an explicitly

synthetic example. Authorization status is eligible, unresolved, ineligible or outside scope; posting permission is a Boolean and can be true only for eligible status under a supporting profile. A branch's immutable book mode is synthetic or real, and a profile names its supported modes. Synthetic evidence cannot authorize a real-book posting. Journal and delta both identify the same basis and authorization. Their domain payloads obey the package's pinned journal and state schemas. No earlier record refers forward to the final receipt.

A.24 Method and extension definitions

A versioned financial method supplies the following nonempty definitions:

```
MethodDefinition {
  method_id, version, event_kind,
  state_schema, event_schema, journal_schema, delta_schema,
  output_schema, required_contexts, authority_requirements,
  preconditions, algorithm, account_roles, report_mappings,
  numerical_capability, examples, unsupported_cases
}
ExtensionPackage {
  package_key, version, record_schemas, operation_signatures,
  supported_contexts, state_requirements, authority_refs,
  method_refs, capability_requirements, migration_refs,
  example_refs
}
```

Each precondition has an identity, typed predicate, required input/admission references and separate false/unknown consequences. Domain state and event fields are specified with the financial methods in the body. Schema names and package members resolve through immutable locks. An empty authority list makes no accounting-authority claim. Registering a schema does not establish the method's entity, instrument, reporting-period or probabilistic applicability. The full machine-oriented contracts and finite exchanges accompany the proposal in the architecture and methods directories; their exact versions are pinned by the manuscript's dependency index.

Appendix B

Core grammar, records and execution details

B.1 Preserved Core 0 teaching grammar

The grammar governing the preserved Core 0 teaching examples is reproduced from `spec/core-0.lark`. It does not enlarge the first matched text frontend beyond the portable profile described in the body. Names are case-sensitive ASCII identifiers; strings are UTF-8 with JSON escaping. All literal words in the grammar are reserved. The companion contract specifies valid dates, exact rational literals, argument arity, type/context checks and package resolution. A syntax parse is not semantic validation.

```
// Normative Core 0 syntax, Lark EBNF. UTF-8 input; ASCII identifiers.
// Longest token wins; keywords cannot be declaration identifiers.
start: "module" NAME "version" STRING ";" declaration*
?declaration: use | source | entity | book | account | observation | symbol
            | choice | binding | constraint | alternative | contract | event
            | rule | scenario | report | opening
use: "use" NAME "version" STRING ";"
source: "source" NAME "=" expr ";"
entity: "entity" NAME "functional" NAME ";"
book: "book" NAME "entity" NAME "basis" NAME "version" STRING ";"
account: "account" NAME ":" NAME "book" NAME ";"
observation: "observation" NAME ":" type time_spec "=" expr "from" NAME ";"
time_spec: "instant" DATE | "period" DATE "to" DATE
symbol: "symbol" NAME ":" type "in" domain "because" STRING ";"
domain: "[" expr "," expr "]" | "{" expr ("," expr)* "}"
choice: "choice" NAME "in" "{" NAME ("," NAME)* "}" "because" STRING ";"
binding: "let" NAME [":" type] "=" expr ";"
constraint: "constraint" NAME ":" expr ";"
alternative: "alternative" NAME "{" assumption* "}"
assumption: "assume" expr ";"
contract: "contract" NAME ":" NAME "book" NAME "{" field* "}"
field: NAME "=" expr ";"
event: "event" NAME "(" [parameters] ")" "book" NAME ";"
parameters: NAME ":" type ("," NAME ":" type)*
rule: "rule" NAME "for" NAME "as" NAME ["overrides" NAME ("," NAME)*] "when" expr "{" rule_statement* "}"
?rule_statement: binding | require | post
require: "require" expr ";"
post: "post" "{" posting+ "}"
posting: SIDE NAME expr ";"
scenario: "scenario" NAME "{" setting* "}"
setting: "set" NAME "=" expr ";"
report: "report" NAME "for" NAME "{" field* "}"
opening: "opening" NAME "for" NAME "from" expr ";"
type: NAME ["<" type ("," type)* ">"]
?expr: disjunction
?disjunction: conjunction (OR conjunction)*
?conjunction: comparison (AND comparison)*
```

```
?comparison: sum [COMPARE sum]
?sum: product (ADD product)*
?product: unary (MUL unary)*
?unary: UNARY unary | postfix
?postfix: atom ("." NAME)*
?atom: number | STRING | DATE | BOOL | call | NAME | "(" expr ")" | array | record
call: NAME "(" [expr ("," expr)*] ")"
number: NUMBER [NAME | "per" "year"]
array: "[" [expr ("," expr)*] "]"
record: "{" [record_field ("," record_field)*] "}"
record_field: NAME ":" expr
SIDE: "debit" | "credit"
OR: "or"
AND: "and"
UNARY: "-" | "not"
COMPARE: "==" | "!=" | "<=" | ">=" | "<" | ">"
ADD: "+" | "-"
MUL: "*" | "/"
BOOL.2: "true" | "false"
DATE: /@[0-9]{4}-[0-9]{2}-[0-9]{2}/
NAME: /[A-Za-z_][A-Za-z0-9_]*/
NUMBER: /(?:0|[1-9][0-9]*)(?:\.[0-9]+)?/
STRING: /"(?:[^"\\\x00-\x1f]|\\(?:["\\/bfnrt]|u[0-9a-fA-F]{4}))*"/
%import common.WS
%ignore WS
%ignore /\/\/[^\n]*/
```

B.2 Required core record definitions

The historical closed machine-readable definitions remain in the original design directory at `spec/core-ir.schema.json`. They define contexts, source spans, observations, narratives, coverage units, mappings, partitions, entity/control records, books, accounts, symbols, choices, expressions, constraints, assumptions, alternatives, contracts, events, rules, packages, measurements, states, journals, scenarios, information views, review decisions, evidence views, model scopes, historical families, probability models, report mappings, result cells, branches, runs and diagnostics. For those historical records the schema fixes field spelling and structural shape, and its companion contract fixes reference resolution and operations. New portable records instead obey the exact versioned contracts introduced in Chapters 7, 8 and 13. No historical field list grants new posting authority.

For example, these selected records reproduce the schema's field names. Each field is required; a nullable link is represented by explicit null.

```
EvidenceView {
  id, cutoff, system_as_of, mode,
  documents[], observations[], decisions[],
  conflict_sets[][], transforms[]
}
Context {
  id, entity, perimeter, concept, basis, measurement,
  time, unit, currency, taxonomy_version, dimensions[]
}
Symbol {
  id, type, context, domain, uncertainty_kind,
  scope, dependencies[], admission, provenance[]
}
Journal {
  id, event, book, date, accrual_period,
  postings[], rule, measurements[], assumptions[],
  precondition_hash, explanation
}
ResultCell {
  id, context, history_condition, scenario, path, kind,
  value, expression, lower, upper,
  lower_attained, upper_attained, sample_ref,
  certificate, reason, dependencies[]
}
```

A value is a reduced rational, or a typed symbolic/domain reference. A posting is nonnegative with its side carrying the sign, in the book's functional currency. Original units and the valuation reference survive conversion. Monetary contexts require currency; a period flow requires a calendar and exclusive end boundary. Historical zero postings may preserve traceability but establish no economic event. New portable proposals omit zero lines and retain their explanation in the event and findings instead. The amount may have finer precision before the declared posting-rounding operation.

Schema validation rejects unknown fields but cannot prove that a treatment is permitted, a reference points to the correct kind of record, or two periods are comparable. Those semantic checks precede solver or runtime dispatch. The structural fixture and complete demonstration source/package are supplied separately so that a future implementation can check both layers explicitly. `fixtures/core-ir-demo.json` supplies an authored expected lowering; the reference checker evaluates its selected debt expressions. A general source-to-IR compiler and complete external-registry validation remain to be implemented. Domain records retain disclosed display text and rounding rules alongside their exact bounds.

B.3 Typing and partial-information rules

For expressions x and y in the same currency C, the money addition rule is

$$\frac{x : \mathrm{Money}\langle C\rangle \quad y : \mathrm{Money}\langle C\rangle}{x + y : \mathrm{Money}\langle C\rangle}.$$

Context compatibility is a separate premise for observations. There is no analogous implicit rule for different currencies. Explicit conversion uses a dated measurement and records its rate and purpose. A rate-times-principal calculation is typed through a declared accrual convention; the example's annual simple rate and year fraction produce money in the principal's currency.

Known expressions evaluate normally. Shared symbols evaluate to symbolic expressions subject to a joint constraint store. Missing source values cannot enter arithmetic until the model supplies a symbol or a conditional assumption. Boolean guards over symbols produce formulas, not the value false. This avoids the common error in which "unknown whether impaired" silently means "not impaired." Division requires a proven nonzero denominator or a partitioned domain that excludes zero; otherwise the expression remains unsupported for concrete execution.

Logical queries first require a nonempty feasible family. An empty family returns `Infeasible`; it must not produce vacuous answers to substantive questions. For a nonempty family, constraint evaluation has three results. `entailed` means every state in the declared feasible family satisfies the proposition. `refuted` means none does. `undetermined` means both remain feasible or the solver has not decided; the response distinguishes those causes. Probabilistic truth is a separate query requiring a measure. A language-level Boolean value must never be inferred from an arbitrary probability threshold.

B.4 Operational rule selection

Let $V(e, h)$ be the applicable rules for event e in concrete history h, after package/date selection and guard evaluation. Let $r \prec r'$ mean that r' validly overrides r through the declared acyclic graph. Define

$$\mathrm{Max}(V) = \{r \in V : \nexists r' \in V \text{ with } r \prec r'\}.$$

For one required recognition operation, a singleton selects that rule; the empty set yields `UnsupportedTreatment`; a larger set yields `RuleConflict`. Multiple distinct operations, such as measurement and recognition, are evaluated through separate declared interfaces. A rule that intentionally produces no postings must belong to a supported non-recognition event operation, whose economic state effect is still recorded.

Given a selected rule, the evaluator computes all local bindings and preconditions, emits proposed postings, verifies Equation (6.2), validates account/context restrictions and commits the event transaction. Failure returns the original state plus a diagnostic. This defines the atomicity requirement independently of the storage technology.

B.5 Example run request and response

The following request executes two named conditions under one scenario. It does not request an aggregate probability distribution.

```
{
  "model": "BorrowingExample@1",
  "evidence_view": "synthetic-2025-exact",
  "starting_state": "after-2025-close",
  "horizon": {"start": "2026-01-01", "end_exclusive": "2027-01-01"},
  "conditions": ["allFixed", "allFloating"],
  "scenarios": ["rateShock"],
  "mode": "conditional",
  "weight_model": null,
  "reports": ["balance_sheet", "income", "cash_flow", "equity", "journals"],
  "package_lock": "demonstration-packages-v1"
}
```

The response contains two separately labelled results with interest expense 5 and 7, cash 90 and 88, debt 100 in both, and equity -10 and -12. A request for the parent symbolic family returns the formula $7 - 2\alpha$ and, under exact linear bounds, $[5, 7]$ before monetary quantisation. The

API rejects an unconditional mean request without a weight model. A user can instead request an arbitrary clearly labelled representative case, which does not stand in for the feasible family.

B.6 Requirement traceability

Requirement	Primary implementation area	Conformance cases
R1	Evidence registry and extraction adapters	T03, T09
R2	Context/version model and cutoff queries	T03, T04
R3	Types, mappings and package selection	T02, T12, T13
R4	Symbols, choices and constraint graph	T05–T10, T14
R5	Posting, closing and consolidation	T01, T11–T13
R6	Contract and behavioural interfaces	T05, T11, T15, T16
R7	Conditional output and probability types	T06, T07, T18
R8	Explanation graph and report dependencies	T09, T11–T13
R9	Capabilities and versioned modules	T02, T14–T16
R10	Run records, scheduler and replay	T03, T17, T18

Appendix C

Research basis and outstanding review

C.1 Source collection and reading scope

The companion library contains 32 verified paper PDFs: 25 retrieved from external sources and seven preserved copies from the original collection. Filenames use the title, first author's surname and year. Publication years are used where an archived working version underlies a later publication; the catalogue identifies the actual version read. The original collection has not been renamed or modified. Standards pages, contract specifications, software source snapshots and metadata are stored separately from papers.

The survey considered accounting algebra and DSLs, corporate and bank simulation, incomplete and temporal databases, provenance, inverse problems, partial identification, model averaging, causal scenarios, contract languages, probabilistic programming, document extraction and financial reasoning. OpenAlex and Crossref supplied dated metadata, while primary publisher, institutional and author copies supplied technical support. Public-index results were checked against the downloaded title pages; HTTP success alone was not accepted as a valid paper download.

Technical reading was targeted to the mechanisms used in this design. The full PDF is archived even where only specified sections support the argument. No claim is made that every theorem, appendix or empirical result in every archived paper was independently replicated. Representative support anchors are listed below; the source-support ledger contains the per-source scope and qualifications.

Source	Inspected technical support	Design use
Nehmer and Robinson	Sections 2–5: account states, transactions and aggregation	Conservation plus admissibility.
Doshi	Chapters 4, 7–9 of the 2024 thesis	Dates, logs, corrections and DSL scope.
Shahnazarian	Sections 2.3–2.8; relevant Chapter 3 equations; data and evaluation sections; Appendix B	Accounting/behaviour separation; limits of data transfer.
Pierson	Sections 3–4 and reporting structures; supplementary Vensim equations	Accrual/payment bridges and reporting accumulators.

Source	Inspected technical support	Design use
Libkin	Sections 2–5: incomplete states, conditions and certain answers	Shared unknowns and joint alternatives.
Green et al.	Definitions 3.1–3.2; Section 4	Composable relational provenance; limits outside positive operations.
Stuart	Section 2.2, observation equation and prior/regularisation discussion	Inverse-problem formulation, not imported convergence guarantees.
Giacomini and Kitagawa	Sections 2.1–2.6 and Theorem 1 in CWP12/20	Prior sensitivity and classes of posterior beliefs.
Hoeting et al.	Equations (1)–(3); prior and predictive-evaluation discussion	Optional model-weighted predictions.
Merigoux et al.	Sections 3–4, default calculus and translation figures	Scopes, exceptions, empty/conflict outcomes.
Peyton Jones et al.	Sections 3–5, combinators and valuation semantics	Contract composition separate from valuation assumptions.
FinQA / TAT-QA	Dataset operations, annotation, model and evaluation sections; FinQA evaluation source	Evidence-linked local arithmetic; benchmark boundaries.
DocFinQA / FinBen	Context/retrieval and financial-task evaluations	Long-document difficulty and task coverage.
PubTables / GTE	Canonicalisation, dataset construction, methods and evaluation	Table structure and financial-domain extraction.
Cosentino et al.	Sections 3.1–3.2, 4.6 and 5	Persistent RAG provenance; scope of auditability evidence.

C.2 Version and access qualifications

Several discrepancies mattered. Doshi's thesis is dated 2024 despite the original filename. The archived Mernik text is the 2003 CWI report underlying the 2005 article. Giacomini and Kitagawa's copy is the April 2020 working paper underlying the 2021 article. The Docling v5 copy lists Auer as first author. The probabilistic-programming tutorial is an arXiv version revised in 2021, first submitted in 2018; a merged index DOI was not used. The Libkin author PDF has a footer DOI that resolves to another paper; title/author metadata from the indexes resolves the cited work to DOI `10.1145/2594538.2594561`. These discrepancies remain documented rather than being hidden by a cleaned bibliography.

McCarthy (1982), Imieliński and Lipski (1984), Tamer (2010), and Caiani and co-authors (2016) are retained as important blocked candidates. Access failures and rejected wrong-content downloads are recorded. The design does not use their uninspected technical claims. More specialised work on accounting measurement, process mining, multi-company equilibrium, actuarial inference and learned forecasting is recorded as expansion work for the corresponding implementation stage. The survey supports the core architecture without pretending that those later modules have a complete literature or rule base.

OpenAlex retraction flags and available Crossref correction/version relations were inspected

alongside local versions. No reviewed source was excluded for a discovered retraction. Missing correction metadata is marked unavailable; absence of a discovered notice does not certify that no notice exists. Citation counts are dated index metadata, and venue rankings were not available in this campaign. Neither is used as evidence of correctness.

C.3 Companion records and status

The six research ledgers are `source-support.json`, `citation-venue.json`, `backward-snowball.json`, `forward-snowball.json`, `claim-support.json`, and `omissions.json`. They record technical anchors, metadata sources, considered leads, omitted or blocked candidates and the distinction between source claims and this document's derivations. Citation leads with checked seed references are distinguished from cross-survey additions and topic expansion leads. Forward searches retain the most-cited and most-recent results returned by the recorded bounded queries; they do not establish exhaustive citation coverage. The separate `tools.json` records software evidence. Raw retrievals and text extracts permit a later reviewer to revisit the decisions.

The research library includes Mack's reserving paper, with its conditional-mean/variance assumptions, proofs, aggregate-error dependence and numerical comparisons inspected. It also archives official AASB texts incorporating IAS 2, IAS 32 and IFRS 17, distinguishing their Australian additions and applicable editions. The corporate filings are evidence of reported amounts and policies, rather than substitutes for every underlying standard. The ASBE 1 reproduction has not been verified against an accessible official current text, and official FASB/ASC access remains incomplete. These limitations are recorded at the specific operations they affect.

Every one of the 179 unique candidates in the acquired forward-query files has an individual title/metadata relevance decision. This includes previously omitted discovery files associated with inaccessible seed papers. The generation script requires an explicit decision and stops on an unseen candidate. Screening is still distinct from technical reading: deferred provenance, temporal-query, bank-optimisation, extraction and actuarial follow-ups remain identified expansion risks. An omitted-paper reason does not make that paper technically reviewed or make the survey exhaustive.

This document specifies the product; it does not report a completed engine. The companion review records cover deterministic example arithmetic, bibliography/build checks, rendered-page inspection and author review. Human comprehension review, independent accounting review and implementation conformance remain separate future evidence. They are not replaced by a clean LaTeX build or by the quantity of literature collected.

Bibliography

ACTUS Financial Research Foundation. ACTUS technical specifications: contract states, events, payoffs and risk-factor interface, 2026. URL https://github.com/actusfrf/actus-techspecs. Official source, accessed 2 October 2026.

Apple Inc. Form 10-K for the fiscal year ended September 28, 2024, 2024. URL https://www.annualreports.com/HostedData/AnnualReportArchive/a/NASDAQ_AAPL_2024.pdf. Company-authored filing, archived mirror; financial statements and Notes 1, 4, 9; accessed 2 October 2026.

Apple Inc. Form 10-Q for the quarter ended December 28, 2024, 2025. URL https://www.sec.gov/Archives/edgar/data/320193/000032019325000008/aapl-20241228.htm. SEC EDGAR inline XBRL filing; balance sheet and Note 6; accessed 2 October 2026.

Arelle contributors. Arelle source snapshot: Python session, controller, cache and document loading, 2026. URL https://github.com/Arelle/Arelle/tree/01f6a4656c2c6551d2f05f31c9d74366d133ee7e. Pinned commit 01f6a4656c2c6551d2f05f31c9d74366d133ee7e. Session.run/get_models, CntlrCmdLine.run, RuntimeOptions.validationExitCode, WebCache.getfilename and ModelDocument.load inspected; local exact line/hash evidence in adapter-dossier.json, taxonomy-reuse-20261007. Source inspection only; no adapter runtime experiment.

Christoph Auer, Maksym Lysak, Ahmed Nassar, Michele Dolfi, et al. Docling Technical Report. Technical report, arXiv:2408.09869v5, 2024. URL https://arxiv.org/pdf/2408.09869.

Australian Accounting Standards Board. AASB 102 Inventories, 2015. URL https://www.aasb.gov.au/admin/file/content105/c9/AASB102_07-15.pdf. Official adopted IAS 2 text; comparison section and paragraphs 23–34; Australian additions distinguished.

Australian Accounting Standards Board. AASB 116 Property, Plant and Equipment, compilation 6, 31 December 2022, 2022a. URL https://www.aasb.gov.au/admin/file/content105/c9/AASB116_08-15_COMPdec22_01-23.pdf. Registration F2023C00192; paragraphs 7, 15, 29–30 and 48–62.

Australian Accounting Standards Board. AASB 132 Financial Instruments: Presentation, compiled December 2022, 2022b. URL https://www.aasb.gov.au/admin/file/content105/c9/AASB132_08-15_COMPdec22_01-23.pdf. Official adopted IAS 32 text; selected classification paragraphs, 28–32 and AG30–AG35.

Australian Accounting Standards Board. AASB 15 Revenue from Contracts with Customers, compilation 8, 31 December 2022, 2022c. Retained official text; selected recognition, measurement and contract-balance provisions.

Australian Accounting Standards Board. AASB 17 Insurance Contracts, compiled December 2022, 2022d. URL

https://www.aasb.gov.au/admin/file/content105/c9/AASB17_07-17_COMPdec22_01-23.pdf. Official adopted IFRS 17 text; paragraphs 14–24, 32–38, 53–70A and B126–B127; selected operations only.

Australian Accounting Standards Board. AASB 9 Financial Instruments, compilation 5, 31 December 2022, 2022e. Official compilation, registration F2023C00399. Selected paragraphs on classification, initial measurement, liabilities, effective interest and impairment; dated applicability remains explicit.

Australian Accounting Standards Board. AASB 107 Statement of Cash Flows, compilation 5, 31 December 2023, 2023a. URL https://www.aasb.gov.au/admin/file/content105/c9/AASB107_08-15_COMPjun23_01-24.pdf. Registration F2024C00050; selected interest, tax, noncash and financing reconciliation provisions.

Australian Accounting Standards Board. AASB 112 Income Taxes, compilation 7, 29 September 2023, 2023b. URL https://www.aasb.gov.au/admin/file/content105/c9/AASB112_08-15_COMPsep23_01-23.pdf. Registration F2023C00930; selected recognition, temporary-difference, rate and presentation provisions.

Australian Accounting Standards Board. AASB 16 Leases, compilation 8, 31 December 2023, 2023c. URL https://www.aasb.gov.au/admin/file/content105/c9/AASB16_02-16_COMPnov22_01-24.pdf. Registration F2024C00049. Cover date controls despite the filename; selected lessee provisions.

Egor Babcinețchi, Daniel Pogorevici, Iulia Țăruș, and Rafaela Cerlat. Domain Specific Language for Accounting. Technical University of Moldova student research conference, pp. 342–345, 2022.

Markus Behn, Claudio Daminato, and Carmelo Salleo. A dynamic model of bank behaviour under multiple regulatory constraints. European Central Bank Working Paper 2233, 2019. URL https://doi.org/10.2139/ssrn.3327174. ECB Working Paper 2233; local DOI 10.2866/943267. OpenAlex SSRN record used only for citation metadata.

Eli Bingham, Jonathan P. Chen, Martin Jankowiak, et al. Pyro: Deep Universal Probabilistic Programming. Journal of Machine Learning Research, 20(28):1–6, 2019. URL https://jmlr.org/papers/volume20/18-403/18-403.pdf.

Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, et al. Stan : A Probabilistic Programming Language. Journal of Statistical Software, 76(1), 2017. URL https://www.jstatsoft.org/index.php/jss/article/view/v076i01/v76i01.pdf.

Caterpillar Inc. 2024 Annual Report, 2024. URL https://www.annualreports.com/HostedData/AnnualReportArchive/c/NYSE_CAT_2024.pdf. Company-authored report; Note 1.C inventories and Note 8. Evidence of disclosed company policy, not a substitute for full ASC 330 authority.

Zhiyu Chen, Wenhu Chen, Charese Smiley, et al. FinQA: A Dataset of Numerical Reasoning over Financial Data. Proceedings of EMNLP, 2021. URL https://aclanthology.org/2021.emnlp-main.300.pdf.

Cristian Cosentino, Simone Squillace, and Fabrizio Marozzo. An Auditable LLM-RAG Architecture for Financial Document Intelligence and Decision Support. Future Internet, 18,

article 284, 2026. URL https://mdpi-res.com/d_attachment/futureinternet/futureinternet-18-00284/article_deploy/futureinternet-18-00284.pdf.

Leonardo de Moura and Nikolaj Bjørner. Z3: An Efficient SMT Solver. Proceedings of TACAS, 2008. URL https://link.springer.com/content/pdf/10.1007%2F978-3-540-78800-3_24.pdf.

Docling contributors. Docling source: conversion, page outcomes and pipeline execution. Official source repository, commit eda5b5fc86d10b2b79a900ff3e69fc159d5f6014, 2026. URL https://github.com/docling-project/docling/tree/eda5b5fc86d10b2b79a900ff3e69fc159d5f6014. Inspected document_converter.py, standard_pdf_pipeline.py and base_pipeline.py; retained source copies. Source inspection and isolated method diagnostics, not an installed adapter run.

Janhavi Doshi. A Domain-Specific Language for Accounting. Master thesis, Rochester Institute of Technology, 2024. Title page 2024; original user filename incorrectly suggests 2026.

European Commission. Commission Regulation (EU) 2023/1803 adopting certain international accounting standards, 2023. URL https://eur-lex.europa.eu/eli/reg/2023/1803/oj. Original incorporating text; selected IFRS 2, IFRS 3, IFRS 10, IFRS 13 and IAS 19, IAS 21, IAS 23, IAS 36, IAS 37 passages. Later amendment and target-entity admission are separate.

Financial Accounting Foundation. SEC Reporting Taxonomy: Notice of Authorized Uses, 2026a. URL https://xbrl.fasb.org/terms/SRT_TermsConditions.html. Retrieved 7–8 October 2026. Unversioned page displays copyright through 2027; applicable historical terms remain separate. Authorized-use and third-party conditions inspected; original response retained.

Financial Accounting Foundation. US GAAP Financial Reporting Taxonomy: Notice of Authorized Uses, 2026b. URL https://xbrl.fasb.org/terms/TaxonomiesTermsConditions.html. Retrieved 7–8 October 2026. Unversioned page displays copyright through 2027; the applicable terms vintage for the 2024 schema is not established by this retrieval. Authorized-use and third-party conditions inspected; original response retained.

Financial Accounting Standards Board. Accounting Standards Update No. 2016-13: Financial Instruments—Credit Losses (Topic 326): Measurement of Credit Losses on Financial Instruments, 2016a. URL https://storage.fasb.org/ASU%202016-13.pdf. Official June 2016 amendment communication. Selected paragraphs and amendment markup inspected. The ASU says the Codification, not an ASU, is authoritative; later changes and current applicability are not verified.

Financial Accounting Standards Board. Accounting Standards Update 2016-02, Leases (Topic 842), Section A, 2016b. Retained official historical amendment. The update is not itself authoritative; current Codification access and full applicability remain unresolved.

Raffaella Giacomini and Toru Kitagawa. Robust Bayesian Inference for Set-Identified Models. Econometrica, 89; archived CWP12/20 working version, 2021. URL https://ifs.org.uk/sites/default/files/output_url_files/CW1220-Robust-Bayesian-inference-for-set-identified-models.pdf. CWP12/20, 8 April 2020 working paper underlying 2021 Econometrica article; technical claims anchored to this local version.

Todd J. Green, Grigoris Karvounarakis, and Val Tannen. Provenance semirings. Proceedings of PODS, 2007. URL https://repository.upenn.edu/bitstreams/b598c0a7-0d24-4162-8279-5f51a17d29c2/download.

Grzegorz Halaj. Dynamic balance sheet model with liquidity risk. European Central Bank Working Paper 1896, 2016. URL https://doi.org/10.1142/s0219024916500527. ECB Working Paper 1896, April 2016. Index journal DOI is not asserted to be the version read.

Jennifer A. Hoeting, David Madigan, Adrian E. Raftery, and Chris T. Volinsky. Bayesian Model Averaging - A Tutorial. Statistical Science, 14(4):382–417, 1999. URL https://www.stat.colostate.edu/~jah/papers/statsci.pdf.

HSBC Holdings plc. Annual report and accounts 2023, 2023. URL https://www.hsbc.com/-/files/hsbc/investors/hsbc-results/2023/annual/pdfs/hsbc-holdings-plc/240226-annual-report-and-accounts-2023.pdf. Annual-report year; retained 450-page PDF, cover (physical page 1). Local extraction, render and painting-order inspection, 12 October 2026.

IFRS Foundation. IAS 32 Financial Instruments: Presentation: issued-standard overview, 2026a. URL https://www.ifrs.org/issued-standards/list-of-standards/ias-32-financial-instruments-presentation/. Official source, accessed 2 October 2026.

IFRS Foundation. IFRS 17 Insurance Contracts: issued-standard overview, 2026b. URL https://www.ifrs.org/issued-standards/list-of-standards/ifrs-17-insurance-contracts/. Official source, accessed 2 October 2026.

IFRS Foundation. IFRS 18 Presentation and Disclosure in Financial Statements: issued-standard overview, 2026c. URL https://www.ifrs.org/issued-standards/list-of-standards/ifrs-18-presentation-and-disclosure-in-financial-statements/. Official source, accessed 2 October 2026.

IFRS Foundation. IFRS 9 Financial Instruments: issued-standard overview, 2026d. URL https://www.ifrs.org/issued-standards/list-of-standards/ifrs-9-financial-instruments/. Official source, accessed 2 October 2026.

Jaided AI. EasyOCR 1.7.2: requirements and Reader implementation, 2024. URL https://github.com/JaidedAI/EasyOCR/tree/v1.7.2. Exact tagged source retained and inspected: requirements.txt and easyocr/easyocr.py, module import and Arabic result-ordering branch. Source inspection only; no installed OCR run.

JPMorgan Chase & Co. 2022 Annual Report on Form 10-K, 2023. URL https://www.jpmorganchase.com/content/dam/jpmc/jpmorgan-chase-and-co/investor-relations/documents/quarterly-earnings/2022/4th-quarter/corp-10k-2022.pdf. Issuer-hosted PDF retrieved 10 October 2026. Item 1/1A, parent-funding discussion and Exhibits 21/22.2 inspected; fiscal year ended 31 December 2022; printed pages 1–42 and 304 are identified as omitted from the selected shareholder annual report. Source-layer evidence, not regulator-byte equivalence.

JPMorgan Chase & Co. 2023 Annual Report on Form 10-K, 2024. URL https://www.jpmorganchase.com/content/dam/jpmc/jpmorgan-chase-and-co/investor-relations/documents/quarterly-earnings/2023/4th-quarter/corp-10k-2023.pdf. Issuer-hosted PDF retrieved 10 October 2026. Item 1/1A, parent-funding discussion, Item 8/15 navigation and Exhibits 21/22.2 inspected; fiscal year ended 31 December 2023; printed pages 1–44 and 322 are identified as omitted from the selected shareholder annual report. Item 15's page-156 reference is retained as a checked source navigation defect; source-layer evidence, not regulator-byte equivalence.

JPMorgan Chase & Co. 2024 Annual Report on Form 10-K, 2025. URL https://www.jpmorganchase.com/content/dam/jpmc/jpmorgan-chase-and-co/

investor-relations/documents/quarterly-earnings/2024/4th-quarter/corp-10k-2024.pdf. Issuer-hosted PDF retrieved 10 October 2026. Item 1/1A, parent-funding discussion and Exhibits 21/22.2 inspected; fiscal year ended 31 December 2024; printed pages 1–48 and 334 are identified as omitted from the selected shareholder annual report. Source-layer evidence, not regulator-byte equivalence.

JPMorgan Chase & Co. 2025 Annual Report on Form 10-K, 2026. URL https://www.jpmorganchase.com/content/dam/jpmc/jpmorgan-chase-and-co/investor-relations/documents/quarterly-earnings/2025/4th-quarter/corp-10k-2025.pdf. Issuer-hosted PDF retrieved 10 October 2026. Item 1/1A, parent-funding discussion and Exhibits 21/22.2 inspected; fiscal year ended 31 December 2025; printed pages 1–42 and 328 are identified as omitted from the selected shareholder annual report. Source-layer evidence, not regulator-byte equivalence.

Meir Kriheli and contributors. python-bidi 0.6.11 source distribution, 2026. URL https://pypi.org/project/python-bidi/0.6.11/. Exact source archive verified against the PyPI SHA256. Inspected pyproject.toml, bidi module and wrapper, README, COPYING, COPYING.LESSER, Cargo.toml, Cargo.lock and third-party notice. Candidate release only; no resolved install or distribution clearance.

Minghao Li, Lei Cui, Shaohan Huang, Furu Wei, Ming Zhou, and Zhoujun Li. TableBank - A Benchmark Dataset for Table Detection and Recognition. Proceedings of LREC; arXiv:1903.01949, 2020. Metadata search returned original 2019 preprint; local version title is TableBank: A Benchmark Dataset for Table Detection and Recognition.

Leonid Libkin. Incomplete Data - What Went Wrong, and How to Fix It. Proceedings of PODS, 2014. URL https://homepages.inf.ed.ac.uk/libkin/papers/pods14.pdf. Author PDF footer prints 10.1145/2594538.2594551, which indexes another paper. Title/author and Crossref bibliography resolve this paper to 10.1145/2594538.2594561. The citation uses the corrected metadata and author-hosted text; discrepancy retained.

Thomas Mack. Distribution-free Calculation of the Standard Error of Chain Ladder Reserve Estimates. ASTIN Bulletin, 23(2):213–225, 1993. URL https://www.casact.org/sites/default/files/database/astin_vol23no2_213.pdf. Published 1993 article, pp. 213–225, CAS-hosted copy; equations (1)–(3), Theorems 1–3 and aggregate-error corollary inspected. No full predictive distribution follows from the moment assumptions.

Denis Merigoux, Nicolas Chataing, and Jonathan Protzenko. Catala: a programming language for the law. Proceedings of the ACM on Programming Languages, 5(ICFP), 2021. URL https://arxiv.org/pdf/2103.03198.

Marjan Mernik, Jan Heering, and Anthony M. Sloane. When and how to develop domain-specific languages. ACM Computing Surveys; archived CWI SEN-E0309 (2003), 2005. URL https://ir.cwi.nl/pub/4109/04109D.pdf. Local full text is CWI report SEN-E0309, December 2003; publication 2005. Claims anchored to the archived working paper.

Ministry of Finance of the People's Republic of China. Accounting Standard for Business Enterprises No. 1: Inventories. Chinese text reproduced by Wikisource, 2006a. Articles 1–22 inspected, particularly 14–19. Unofficial reproduction; official origin and applicable-version verification pending; access 2 October 2026.

Ministry of Finance of the People's Republic of China. Accounting Standard for Business Enterprises No. 18: Income Taxes, 2006b. URL

https://kjs.mof.gov.cn/zt/kjzzss/kuaijizhunzeshishi/200806/t20080618_46230.htm. Official page published 9 March 2006; original articles 1–25 inspected. Read with subsequent interpretations; not a consolidated current amendment text. Original Chinese retained; English title translated.

Ministry of Finance of the People's Republic of China. Accounting Standard for Business Enterprises No. 22: Recognition and Measurement of Financial Instruments, 2017. URL https://kjs.mof.gov.cn/zt/kjzzss/kuaijizhunzeshishi/201709/t20170908_2694655.htm. Cai Kuai [2017] No. 7; official page dated 8 September 2017, retrieved 8 October 2026. Selected articles 17, 38–43, 46–63, 72–73, 83–84 inspected. Original Chinese preserved; English title and interpretations translated. The issuance notice's entity-group adoption schedule and a complete subsequent amendment chain remain unverified.

Ministry of Finance of the People's Republic of China. Notice on Issuing the Revised Accounting Standard for Business Enterprises No. 21: Leases, 2018. URL https://kjs.mof.gov.cn/zhengcefabu/201812/t20181213_3092629.htm. Cai Kuai [2018] No. 35, signed 7 December 2018; webpage published 14 December 2018. Notice paragraphs I–III and attached standard articles 1–8, 14–18, 20–23, 25–33, 60–68 inspected. Original Chinese notice and attachment retained; English title translated.

Ministry of Finance of the People's Republic of China. Accounting Standard for Business Enterprises No. 21: Leases, 2019. URL https://kjs.mof.gov.cn/zt/kjzzss/kuaijizhunzeshishi/201910/t20191028_3411190.htm. Official page dated 28 October 2019; selected articles inspected. Read with the 2018 issuance notice. Later amendments and actual entity adoption remain to be established.

Ministry of Finance of the People's Republic of China. Accounting Standard for Business Enterprises No. 25: Insurance Contracts, 2020. URL http://kjs.mof.gov.cn/zt/kjzzss/kuaijizhunzeshishi/202202/t20220224_3790240.htm. Cai Kuai [2020] No. 20; official page published 24 February 2022, retrieved 8 October 2026. Selected recognition, measurement, premium-allocation and disclosure articles inspected. Original Chinese retained; English title and interpretations translated. Article 120 alone does not establish the complete entity adoption schedule, later implementation conditions or amendment chain.

Ministry of Finance of the People's Republic of China. Interpretation No. 16 of the Accounting Standards for Business Enterprises, 2022. URL https://kjs.mof.gov.cn/gongzuotongzhi/202212/P020221213437987613379.pdf. Operative five-page attachment to Cai Kuai [2022] No. 31. Issuance notice signed 30 November 2022 and published on its webpage 13 December 2022. Sections I(a)–(b) and IV govern the single-transaction tax treatment and its application; all five pages inspected. Original Chinese retained; English title translated.

Ministry of Finance of the People's Republic of China, Accounting Department. Questions and Answers on the Notice on Further Implementing the New Insurance Contracts Accounting Standard, 2025. URL http://kjs.mof.gov.cn/zhengcejiedu/202506/t20250612_3965540.htm. Official Chinese explanation dated 12 June 2025 of Cai Kuai [2025] No. 12. All seven answers inspected; the operative notice remains separately unverified. Retrieved 8 October 2026.

Robert A. Nehmer and Derek Robinson. An algebraic model for the representation of accounting systems. Annals of Operations Research, 71:179–198, 1997. URL https://doi.org/10.1023/a:1018915430594.

Judea Pearl. Causal inference in statistics: An overview. Statistics Surveys, 3:96–146, 2009. URL https://ftp.cs.ucla.edu/pub/stat_ser/r350.pdf.

Simon Peyton Jones, Jean-Marc Eber, and Julian Seward. Composing contracts: an adventure in financial engineering. Proceedings of ICFP; author manuscript dated 23 August 2000, 2000. URL https://www.microsoft.com/en-us/research/wp-content/uploads/2016/07/contracts-icfp.pdf.

Kawika Pierson. Operationalizing Accounting Reporting in System Dynamics Models. Systems, 8(1), article 9, 2020. URL https://doi.org/10.3390/systems8010009.

Ping An Insurance (Group) Company of China, Ltd. 2022 Annual Report: Chinese ASBE Edition, 2023. URL https://www.pingan.com/app_upload/images/info/upload/2091cada-04c0-443f-be50-0b1396c10575.pdf. Consolidated balance sheet, printed pp. 146–148; preparation basis, p. 159; Note XIV.2, pp. 315–316. Bibliographic year refers to the subsequent reporting cycle, not a verified publication date. Retained corpus copy revalidated 2 October 2026.

Ping An Insurance (Group) Company of China, Ltd. 2023 Annual Report: Chinese ASBE Edition, 2024. URL https://www.pingan.com/app_upload/images/info/upload/55e017c8-da54-42f5-81cc-b4ff9586b461.pdf. Consolidated balance sheet, printed pp. 141–143; Notes I–III, p. 154; Note IV.34, p. 192. Bibliographic year refers to the subsequent reporting cycle, not a verified publication date. Retained corpus copy revalidated 2 October 2026.

PyO3, Servo, Rust and dependency contributors. Exact Rust sources named by the python-bidi 0.6.11 lockfile. Official crates.io source archives, inspected 11 October 2026, 2026. Fifteen external archive checksums matched to the candidate lock. Includes PyO3 0.29.0, unicode-bidi 0.3.18, target-lexicon 0.13.3 and unicode-ident 1.0.12. Package manifests, licence bodies, LLVM exception, Unicode terms and source-provenance references retained. Exact-version ledger: bidi-direct-rust-terms-20261011 and bidi-remaining-rust-terms-20261011.

Varshini Reddy, Rik Koncel-Kedziorski, Viet Dac Lai, Michael Krumdick, Charles Lovering, and Chris Tanner. DocFinQA: A Long-Context Financial Reasoning Dataset. Proceedings of ACL, short papers, 2024. URL https://aclanthology.org/2024.acl-short.42.pdf.

Hovick Shahnazarian. A Dynamic Microeconometric Simulation Model for Incorporated Businesses. Sveriges Riksbank Occasional Paper 11, 2004.

Brandon Smock, Rohith Pesala, and Robin Abraham. PubTables-1M: Towards comprehensive table extraction from unstructured documents. Proceedings of CVPR; author manuscript, 2022. URL https://arxiv.org/pdf/2110.00061.

Richard Snodgrass and Ilsoo Ahn. Temporal Databases. Computer, 19(9):35–42, 1986. URL https://www2.cs.arizona.edu/~rts/pubs/Computer.pdf. Corrected after title-page inspection: author link Computer.pdf is the 1986 paper, not Jensen and Snodgrass 1999.

Andrew M. Stuart. Inverse problems: A Bayesian perspective. Acta Numerica, 19:451–559, 2010. URL https://authors.library.caltech.edu/records/fk9j9-5tq15/files/stuart15c.pdf.

Jan-Willem van de Meent, Brooks Paige, Hongseok Yang, and Frank Wood. An Introduction to Probabilistic Programming. arXiv:1809.10756; version 2 (2021), 2018. URL https://arxiv.org/pdf/1809.10756. arXiv v2 dated 19 October 2021; first submission 2018; OpenAlex merged DOI belongs to another venue and is not used.

Arie van Deursen, Paul Klint, and Joost Visser. Domain-Specific Languages. ACM SIGPLAN Notices, 35(6):26–36, 2000. URL https://homepages.cwi.nl/~paulk/publications/Sigplan00.pdf.

XBRL International. Open Information Model 1.0: Recommendation, 13 October 2021, 2021. URL https://www.xbrl.org/Specification/oim/REC-2021-10-13/oim-REC-2021-10-13.html. Official source, accessed 2 October 2026.

XBRL International. Copyright Information, 2026. URL https://www.xbrl.org/the-consortium/about/legal/copyright-information/. Retrieved 8 October 2026. Public-document permission and express distinction from the Software Notice inspected; this page does not itself settle all schema-specific terms.

Qianqian Xie, Weiguang Han, Zhengyu Chen, et al. FinBen: A Holistic Financial Benchmark for Large Language Models. NeurIPS Datasets and Benchmarks; arXiv:2402.12659, 2024. URL https://arxiv.org/pdf/2402.12659.

Xinyi Zheng, Douglas Burdick, Lucian Popa, Xu Zhong, and Nancy Xin Ru Wang. Global Table Extractor (GTE): A Framework for Joint Table Identification and Cell Structure Recognition Using Visual Context. Proceedings of WACV, 2021. URL https://openaccess.thecvf.com/content/WACV2021/papers/Zheng_Global_Table_Extractor_GTE_A_Framework_for_Joint_Table_Identification_WACV_2021_paper.pdf.

Fengbin Zhu, Wenqiang Lei, Youcheng Huang, Chao Wang, Shuo Zhang, Jiancheng Lv, Fuli Feng, and Tat-Seng Chua. TAT-QA: A Question Answering Benchmark on a Hybrid of Tabular and Textual Content in Finance. Proceedings of ACL-IJCNLP, 2021. URL https://aclanthology.org/2021.acl-long.254.pdf.